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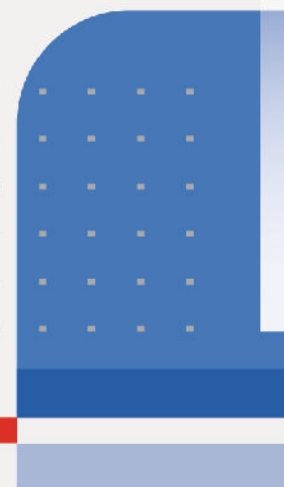
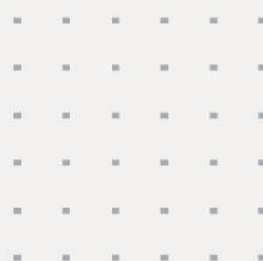
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Network  
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# Enterprise Firewall Administrator Lab Guide

FortiOS 7.6

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5/12/2025

## TABLE OF CONTENTS

|                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| <b>Firmware Version</b>                                                        | <b>7</b>  |
| <b>Network Topology</b>                                                        | <b>8</b>  |
| <b>Lab Device Credentials</b>                                                  | <b>9</b>  |
| <b>Lab 1: Network Security Architecture</b>                                    | <b>10</b> |
| <b>Lab 2: Central Management</b>                                               | <b>11</b> |
| <b>Exercise 1: Running Remote, Device, and Policy Scripts</b>                  | <b>12</b> |
| Network Topology                                                               | 12        |
| Run a Script Using the Remote Method                                           | 12        |
| Run a Script Using the Device Database Method                                  | 16        |
| Run a Script Using the Policy Package or ADOM Database Method                  | 19        |
| <b>Exercise 2: Configuring LTP</b>                                             | <b>22</b> |
| Network Topology                                                               | 22        |
| Configure the Metadata Variables                                               | 22        |
| Add Metadata Variables to the Pre-CLI Template                                 | 24        |
| Add BR2-FGT-1 to FortiManager                                                  | 26        |
| Register BR2-FGT-1 on FortiManager                                             | 32        |
| Add BR3-FGT-1 to FortiManager                                                  | 34        |
| Register BR3-FGT-1 on FortiManager                                             | 38        |
| <b>Lab 3: VLANs and VDOMs</b>                                                  | <b>41</b> |
| <b>Exercise 1: Configuring VDOMs and VLANs on ISFW</b>                         | <b>42</b> |
| Network Topology                                                               | 42        |
| Enable VDOMs                                                                   | 42        |
| Create the Zone1 and Zone2 VDOMs                                               | 43        |
| Create VLAN101 and VLAN102                                                     | 46        |
| Create Inter-VDOM Links                                                        | 49        |
| Create Static Routes in VDOMs                                                  | 53        |
| Install Firewall Policies in VDOMs                                             | 55        |
| Validate the Communication Between HQ-PC-2 and HQ-PC-3                         | 58        |
| Validate the Communication From HQ-PC-2 to HQ-Web-1, the Internet, and 4.2.2.2 | 60        |
| <b>Lab 4: High Availability</b>                                                | <b>62</b> |
| <b>Exercise 1: Configuring VDOM Partitioning</b>                               | <b>63</b> |
| Network Topology                                                               | 63        |
| Verify the HA Status                                                           | 63        |

|                                                                          |            |
|--------------------------------------------------------------------------|------------|
| Configure VDOM Partitioning.....                                         | 64         |
| Analyze Traffic Distribution.....                                        | 67         |
| Perform a Failover and Analyze the Traffic.....                          | 68         |
| <b>Exercise 2: Configuring FGSP.....</b>                                 | <b>71</b>  |
| Network Topology.....                                                    | 71         |
| Configure FGSP.....                                                      | 71         |
| Test ICMP Session Synchronization Between BR1-FGT-1 and BR1-FGT-2.....   | 72         |
| Analyze the Asymmetric Traffic.....                                      | 74         |
| <b>Exercise 3: Encrypting the Session Synchronization.....</b>           | <b>79</b>  |
| Encrypt the Session Synchronization.....                                 | 79         |
| <b>Lab 5: Dynamic Routing Protocols.....</b>                             | <b>82</b>  |
| <b>Exercise 1: Configuring OSPF With ECMP.....</b>                       | <b>83</b>  |
| Configure OSPF on HQ-DCFW.....                                           | 83         |
| Configure OSPF on HQ-ISFW.....                                           | 86         |
| Configure OSPF on HQ-NGFW.....                                           | 89         |
| Check the OSPF Status on the HQ-ISFW Root VDOM.....                      | 92         |
| Enable OSPF ECMP on the HQ-ISFW Root VDOM.....                           | 95         |
| Enable OSPF ECMP on HQ-NGFW.....                                         | 96         |
| Verify OSPF ECMP.....                                                    | 97         |
| Check the OSPF Status on HQ-DCFW and HQ-NGFW.....                        | 98         |
| Check Connectivity.....                                                  | 99         |
| <b>Exercise 2: Configuring BGP.....</b>                                  | <b>100</b> |
| Configure BGP on FortiManager.....                                       | 100        |
| <b>Exercise 3: Configuring a Loopback Interface as a BGP Source.....</b> | <b>108</b> |
| Configure a Loopback Interface as a BGP Source on BR1-FGT-2.....         | 108        |
| Establish the BGP connection.....                                        | 110        |
| <b>Lab 6: Security Profiles.....</b>                                     | <b>112</b> |
| <b>Exercise 1: Solving an IPS False Positive.....</b>                    | <b>113</b> |
| Apply the IPS_Block Security Profile.....                                | 113        |
| Install the Policy.....                                                  | 114        |
| Test Using hping to HQ-Web-1.....                                        | 115        |
| Verify That HQ-DCFW Detects the Jumbo Packets.....                       | 116        |
| Solve the IPS False Positive.....                                        | 118        |
| Install the Policy.....                                                  | 119        |
| Test Using hping to HQ-Web-1.....                                        | 119        |
| Verify That HQ-DCFW Detects the Jumbo Packets.....                       | 120        |
| <b>Exercise 2: Protecting Against Unencrypted Attacks.....</b>           | <b>122</b> |
| Apply the IPS_Monitor Security Profile.....                              | 122        |
| Install the Policy.....                                                  | 123        |
| Access the Website.....                                                  | 124        |
| Simulate the Attack.....                                                 | 124        |



|                                                                       |            |
|-----------------------------------------------------------------------|------------|
| Verify That HQ-DCFW Detects the Attack .....                          | 125        |
| Block an Attack on Unencrypted Traffic .....                          | 126        |
| Install the Policy .....                                              | 128        |
| Simulate the Attack .....                                             | 128        |
| Verify That HQ-DCFW Blocks the Attack .....                           | 128        |
| <b>Exercise 3: Protecting Against Encrypted Attacks .....</b>         | <b>131</b> |
| Create a Dynamic Local Certificate and an SSL/SSH Profile .....       | 131        |
| Block an Attack on Encrypted Traffic .....                            | 133        |
| Verify That HQ-DCFW Drops the Attack .....                            | 134        |
| <b>Lab 7: IPsec VPN (IKEv2) .....</b>                                 | <b>136</b> |
| <b>Exercise 1: Configuring IPsec Templates .....</b>                  | <b>137</b> |
| Configure IPsec Templates .....                                       | 137        |
| Create a Normalized Interface for the Hub .....                       | 148        |
| Configure the Firewall Policies .....                                 | 149        |
| Check the Status of the VPN Tunnels .....                             | 150        |
| Delete the IPsec Tunnels .....                                        | 151        |
| <b>Lab 8: Auto-Discovery VPN .....</b>                                | <b>161</b> |
| <b>Exercise 1: Configuring ADVPN and IBGP .....</b>                   | <b>164</b> |
| Configure IPsec Templates .....                                       | 164        |
| Configure the BGP Templates .....                                     | 171        |
| Install the Policies .....                                            | 175        |
| Bring Up the On-Demand Tunnel .....                                   | 177        |
| <b>Exercise 2: Configuring ADVPN IBGP and EBGp .....</b>              | <b>180</b> |
| Prerequisites .....                                                   | 180        |
| Configure ADVPN IBGP and EBGp .....                                   | 181        |
| Bring Up the On-Demand Tunnel .....                                   | 184        |
| <b>Lab 9: Security Fabric .....</b>                                   | <b>188</b> |
| <b>Exercise 1: Configuring the Security Fabric and SAML SSO .....</b> | <b>189</b> |
| Configure the Security Fabric on HQ-NGFW-1 .....                      | 189        |
| Configure the Security Fabric on HQ-DCFw .....                        | 191        |
| Configure the Security Fabric on HQ-ISFW .....                        | 193        |
| Access Security Fabric Devices With SAML SSO .....                    | 194        |
| <b>Exercise 2: Configuring Automatic Configuration Backup .....</b>   | <b>199</b> |
| Manage Security Fabric Devices on FortiManager .....                  | 199        |
| Configure an Automation on FortiManager .....                         | 200        |
| Verify the Stitch on the Security Fabric Devices .....                | 205        |
| <b>Exercise 3: Configuring the Automation With a Script .....</b>     | <b>208</b> |
| Configure the Automation .....                                        | 208        |
| Test the Automation Stitch and View the Email Alert .....             | 211        |
| <b>Lab 10: Use Cases .....</b>                                        | <b>215</b> |
| <b>Exercise 1: Configuring the HR Network .....</b>                   | <b>219</b> |

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|                                                       |            |
|-------------------------------------------------------|------------|
| Network Topology.....                                 | 219        |
| Requirements.....                                     | 220        |
| Test the Configuration.....                           | 220        |
| <b>Exercise 2: Configuring ADVPN.....</b>             | <b>222</b> |
| Network Topology.....                                 | 222        |
| Requirements.....                                     | 222        |
| Test the Configuration.....                           | 223        |
| <b>Exercise 3: Configuring Automatic Backups.....</b> | <b>226</b> |
| Network Topology.....                                 | 226        |
| Requirements.....                                     | 226        |
| Test the Configuration.....                           | 227        |

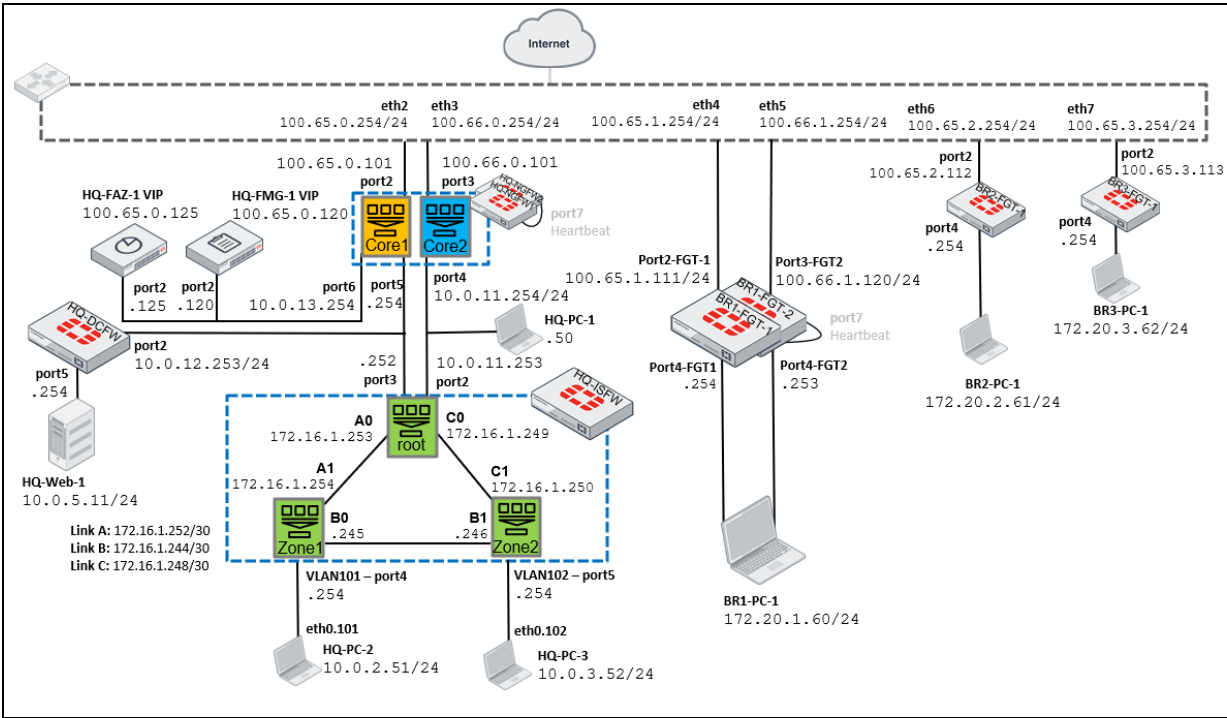
# Firmware Version

The Enterprise Firewall course content is based on the following products and firmware versions:

| Product       | Firmware Version |
|---------------|------------------|
| FortiGate     | 7.6.2            |
| FortiManager  | 7.6.2            |
| FortiAnalyzer | 7.6.2            |

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# Network Topology



## Lab Device Credentials

Below are the credentials for all the devices used in the lab guide.

### VM Usernames and Passwords

| VM        | Username      | Password   |
|-----------|---------------|------------|
| HQ-PC-1   | Administrator | Fortinet1! |
| HQ-PC-2   | Administrator | Fortinet1! |
| HQ-PC-3   | Administrator | Fortinet1! |
| BR1-PC-1  | Administrator | Fortinet1! |
| BR2-PC-1  | Administrator | Fortinet1! |
| BR3-PC-1  | Administrator | Fortinet1! |
| HQ-Web-1  | fortinet      | Fortinet1! |
| HQ-ISFW   | admin         | Fortinet1! |
| HQ-DCFW   | admin         | Fortinet1! |
| HQ-NGFW-1 | admin         | Fortinet1! |
| HQ-NGFW-2 | admin         | Fortinet1! |
| BR1-FGT-1 | admin         | Fortinet1! |
| BR1-FGT-2 | admin         | Fortinet1! |
| BR2-FGT-1 | admin         | Fortinet1! |
| BR3-FGT-1 | admin         | Fortinet1! |
| HQ-FMG-1  | admin         | Fortinet1! |
| HQ-FAZ-1  | admin         | Fortinet1! |

## Lab 1: Network Security Architecture

There is no lab associated with this lesson.

## Lab 2: Central Management

FortiManager is one of the key pieces of an enterprise firewall solution. Without it, managing multiple FortiGate devices would be cumbersome. Using FortiManager, you can centralize the management of all FortiGate devices and create common security policies that multiple devices can easily share. In enterprise networks, FortiManager ADOMs are used to organize your FortiGate devices into groups whose members all share similar security roles and policies.

### Objectives

- Run remote, device, and policy scripts
- Add BR2-FGT-1 and BR3-FGT-1 using the model device and pre-shared key method to deploy them as if you were using the low-touch provisioning (LTP) model

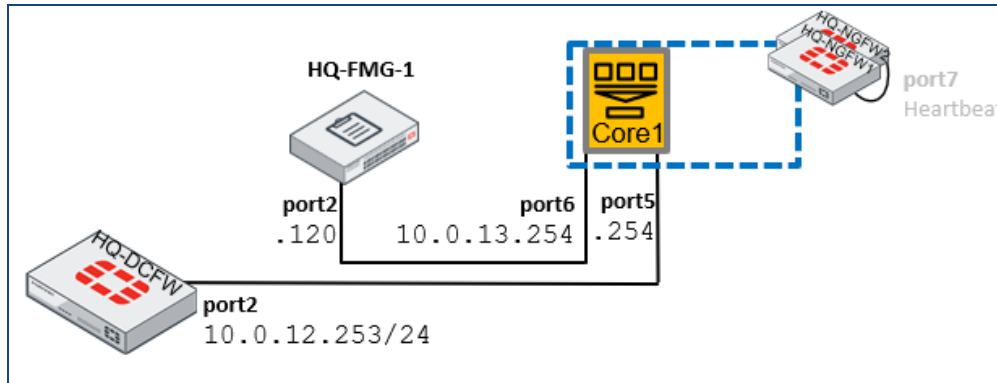
### Time to Complete

Estimated: 55 minutes

## Exercise 1: Running Remote, Device, and Policy Scripts

In this exercise, you will find the differences between running remote, device, and policy scripts.

### Network Topology



### Run a Script Using the Remote Method

You will create a local certificate using the remote FortiGate direct method (using the CLI).

#### To run a script using the remote method to create a local certificate

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW** ADOM.
3. Click **Device Manager > Scripts**.

You will see the scripts that are shown in the following image:



| <input type="checkbox"/> | Name ↕              | Type ↕ | Target ↕                            |
|--------------------------|---------------------|--------|-------------------------------------|
| <b>Script (15)</b>       |                     |        |                                     |
| <input type="checkbox"/> | ACME Certificate    | CLI    | Remote FortiGate Directly (via CLI) |
| <input type="checkbox"/> | ADVPN_Branches      | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | ADVPN_Core1_ex1     | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | ADVPN_Core1_ex2     | CLI    | Remote FortiGate Directly (via CLI) |
| <input type="checkbox"/> | ADVPN_Core2_ex2     | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | BR2-FGT-1_ADVPN_ex2 | CLI    | Remote FortiGate Directly (via CLI) |
| <input type="checkbox"/> | BR3-FGT-1_ADVPN_ex2 | CLI    | Remote FortiGate Directly (via CLI) |
| <input type="checkbox"/> | Firewall rule       | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | HQ_NGFW_ADVPN_ex2   | CLI    | Remote FortiGate Directly (via CLI) |
| <input type="checkbox"/> | Static Route        | CLI    | Device Database                     |
| <input type="checkbox"/> | VPN_Core1           | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | VPN_Spokes          | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | Zone1               | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | Zone2               | CLI    | Policy Package or ADOM Database     |
| <input type="checkbox"/> | root_ISFW           | CLI    | Policy Package or ADOM Database     |

4. Select the **ACME Certificate** checkbox, and then click **Run Script**.

| <a href="#">+ Create New</a>        | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Run Script</a>          | <a href="#">Import CLI</a> |
|-------------------------------------|----------------------|------------------------|-------------------------------------|----------------------------|
| <input type="checkbox"/>            | Name ↕               | Type ↕                 | Target ↕                            |                            |
| <b>Script (3)</b>                   |                      |                        |                                     |                            |
| <input checked="" type="checkbox"/> | ACME Certificate     | CLI                    | Remote FortiGate Directly (via CLI) |                            |

5. In the **Available Entries** field, select the **HQ-DCFW** checkbox, move it to the **Selected Entries** field, and then click **Run Now**.

Run Script - ACME Certificate

Use Shift key for multiple selections and double click for moving one item.

Available Entries (2)

☐ HQ-ISFW (IP: 10.0.12.252, Platform: FortiGate-VM64)
 ☐ HQ-NGFW (IP: 10.0.13.254, Platform: FortiGate-VM64)

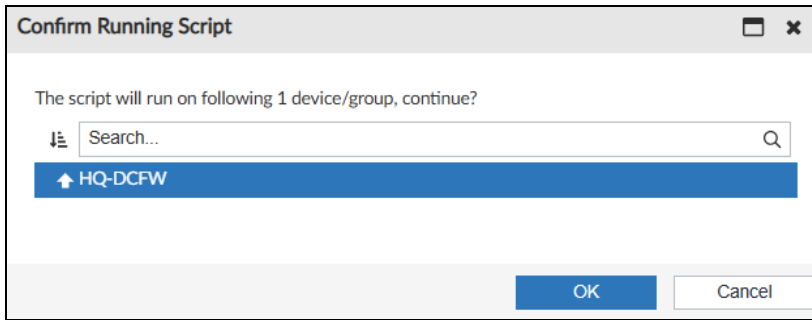
Selected Entries (1)

☒ HQ-DCFW (IP: 10.0.12.253, Platform: FortiGate-VM64)

Run Now

Cancel

6. In the **Confirm Running Script** window, click **OK** to confirm that you want to run the script.



7. Click **Close** to close the **Run Script** window.

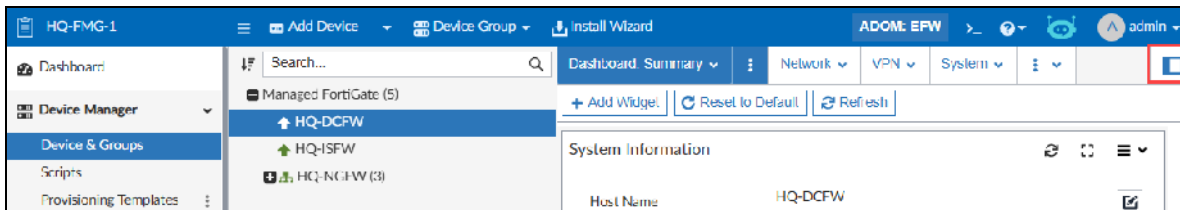


After the configuration is applied directly to your FortiGate, for the purposes of this learning experience, it is recommended that you verify the following configurations:

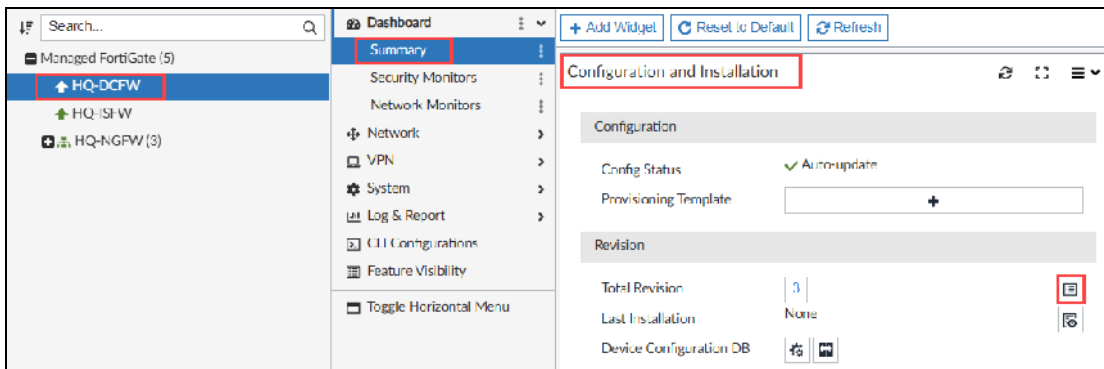
- FortiManager has triggered an automatic retrieval process to receive the configuration from FortiGate.
- The FortiGate device database at the device layer has been updated with the configuration.

You will have a better view of the HQ-DCFW objects that are available in the device layer if you enable the vertical menu.

8. Click **Device Manager > Device & Groups > Managed FortiGate > HQ-DCFW**, and then click the **Toggle Vertical Menu** icon.



9. Click **Device Manager > Device & Groups > Managed FortiGate > HQ-DCFW > Dashboard > Summary**.
10. In the **Configuration and Installation** widget, in the **Revision** section, in the **Total Revision** field, click the **Revision History** icon.



You can see that the **script\_manager** created the last event, which triggered a **Retrieved** operation.

| Configuration Revision History                                                      |                     |      |                |              |          |
|-------------------------------------------------------------------------------------|---------------------|------|----------------|--------------|----------|
| <div> View Config View Install Log Revision Diff Retrieve Config More Search </div> |                     |      |                |              |          |
| ID                                                                                  | Date & Time         | Name | Created by     | Installation | Comments |
| 3                                                                                   | 2025-02-18 04:47:51 |      | script_manager | Retrieved    |          |

Next, you will enable the visibility of HQ-DCFw certificates, so you can check if the configuration has been applied to the device database.

- Click **HQ-DCFw > Feature Visibility**, and then select the **Certificates** checkbox.

Search...

Dashboard
Network
VPN
System
Log & Report
CLI Configurations
Feature Visibility
Toggle Horizontal Menu

Feature Visibility

Global Feature Visibility

Dashboard

Network

Security Profiles

VPN

System

Dashboard

Interfaces
DNS
DNS Service on Interface
Local Out Routing
SD-WAN
Policy Routes
OSPF
Routing Objects

DHCP Servers
DNS Databases
IPAM
Explicit Proxy
Static Routes
RIP
BGP
Multicast

Web Filter Overrides

IPsec Phase 1

IPsec Aggregate

Concentrator

IPsec Phase 2

Manual Key

Virtual Domain

Administrators

Settings

Replacement Messages

FortiGuard

Modem

Alert Email

Global Resources

Admin Profiles

SNMP

Replacement Message Groups

Certificates

Local Profiles

Check All
Reset to Default
OK

- Click **OK**.
- Click **HQ-DCFw > System > Certificates**, and then ensure that the HQ-DCFw `acmetest` certificate is in the certificates section.

| Certificates                                                               |    |                         |                                                                                                                          |
|----------------------------------------------------------------------------|----|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <div> Filter Full in CLI Delete Import View Details Download Search </div> |    |                         |                                                                                                                          |
| Local CA Certificates (0)                                                  |    |                         |                                                                                                                          |
|                                                                            | 1  | Fortinet_CA_SSL         | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = Certificate Authority, CN = FGVN02TM24013503, emailAd...      |
|                                                                            | 2  | Fortinet_CA_Untrusted   | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = Certificate Authority, CN = Fortinet Untrusted CA, emailAd... |
|                                                                            | 3  | acmetest                | C = CA, ST = BC, O = Internet Widgits Pty Ltd, CN = www.acmetest.com                                                     |
| Local Certificates (16)                                                    |    |                         |                                                                                                                          |
|                                                                            | 4  | Fortinet SSL            | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 5  | Fortinet_GUI_Server     | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FortiGate, emailAddress = su...               |
|                                                                            | 6  | Fortinet_SSL_RSA1024    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 7  | Fortinet_SSL_RSA2048    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 8  | Fortinet_SSL_RSA4096    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 9  | Fortinet_SSL_USA1024    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 10 | Fortinet_SSL_USA2048    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 11 | Fortinet_SSL_FCD5A256   | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 12 | Fortinet_SSL_FCD5A384   | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 13 | Fortinet_SSL_FCD5A512   | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 14 | Fortinet_SSL_ED25519    | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 15 | Fortinet_SSL_ED448      | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 16 | Fortinet_Factory        | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 17 | Fortinet_Factory_Backup | C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = FortiGate, CN = FGVN02TM24013503, emailAddress = su...        |
|                                                                            | 18 | Fortinet_Wifi           | C = US, ST = California, L = Sunnyvale, O = Fortinet, Inc., CN = authorizedforintroduction                               |
|                                                                            | 19 | acmetest                | C = CA, ST = BC, O = Internet Widgits Pty Ltd, CN = www.acmetest.com                                                     |

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15

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**Stop and think!**

At this point, the **ACME** certificate has been pushed to HQ-DCFw and its device database has been updated without you having to manually install the database. FortiManager automatically triggers a retrieve after it runs a remote script to update the database. This process is registered in the HQ-DCFw revision history without affecting the status of the device database and policy layer (which were synchronized before you ran the script).

|                          |               |                 |                         |
|--------------------------|---------------|-----------------|-------------------------|
| <input type="checkbox"/> | Device Name ▾ | Config Status ▾ | Policy Package Status ▾ |
| <input type="checkbox"/> | ↑ HQ-DCFw     | ✓ Synchronized  | ✓ DCFw                  |

You should avoid using the remote method to modify, delete, or create objects that are used in firewall policies. Otherwise, you will trigger a conflict status in the policy layer. If you modify firewall objects using a remote script, you should also make the same changes in the policy package that your firewall is using.



You should also log in to the HQ-DCFw GUI or CLI with read-only permissions and validate that the certificate has been installed, so that you can compare this with the scripts that you will be running next.

## Run a Script Using the Device Database Method

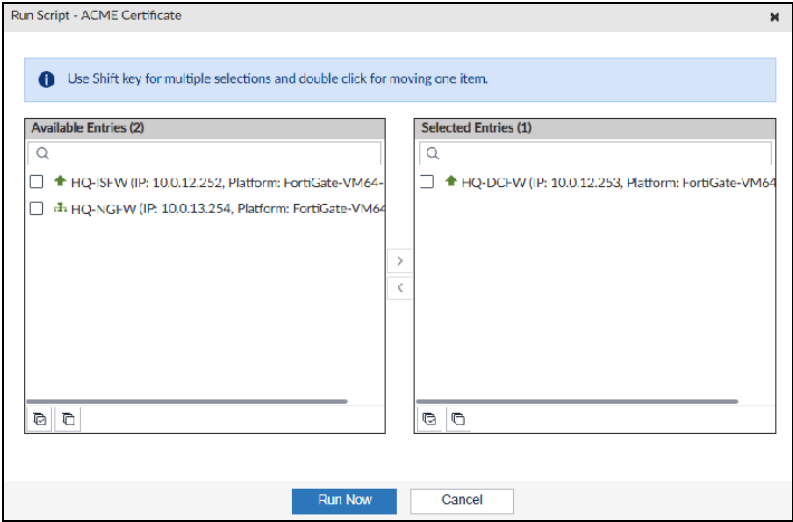
You will create a static route using the device database method.

### To run a script using the device database method

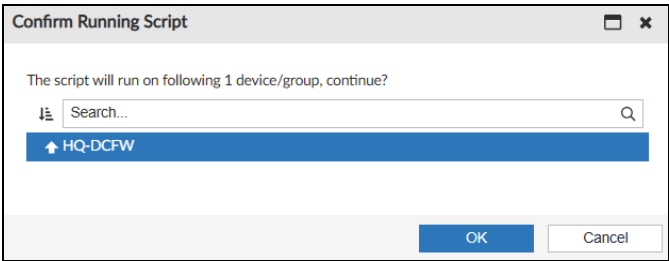
1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW** ADOM.
3. Click **Device Manager > Scripts**.
4. Select the **Static Route** checkbox, and then click **Run Script**.

|                                               |                                     |                                       |                                           |                                                  |
|-----------------------------------------------|-------------------------------------|---------------------------------------|-------------------------------------------|--------------------------------------------------|
| <input type="button" value="+ Create New ▾"/> | <input type="button" value="Edit"/> | <input type="button" value="Delete"/> | <input type="button" value="Run Script"/> | <input type="button" value="Import CLI Script"/> |
| <input type="checkbox"/>                      | Name ▾                              | Type ▾                                | Target ▾                                  | Comments                                         |
| Script (3)                                    |                                     |                                       |                                           |                                                  |
| <input type="checkbox"/>                      | ACME Certificate                    | CLI                                   | Remote FortiGate Directly (via CLI)       |                                                  |
| <input type="checkbox"/>                      | Firewall rule                       | CLI                                   | Policy Package or ADOM Database           |                                                  |
| <input checked="" type="checkbox"/>           | Static Route                        | CLI                                   | Device Database                           |                                                  |

5. In the **Available Entries** field, select the **HQ-DCFw** checkbox, move it to the **Selected Entries** field, and then click **Run Now**.



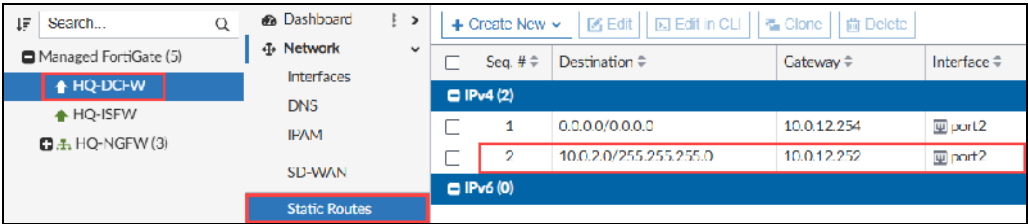
6. In the **Confirm Running Script** window, click **OK** to confirm that you want to run the script.



7. Click **Close** to close the **Run Script** window.
8. Click **Device Manager > Device & Groups > Managed FortiGate**.  
You can see the HQ-DCF-W device database (**Config Status** column) with an orange warning triangle, which indicates that it has been modified.

| <input type="checkbox"/> | Device Name | Config Status |
|--------------------------|-------------|---------------|
| <input type="checkbox"/> | ↑ HQ-DCF-W  | ⚠ Modified    |

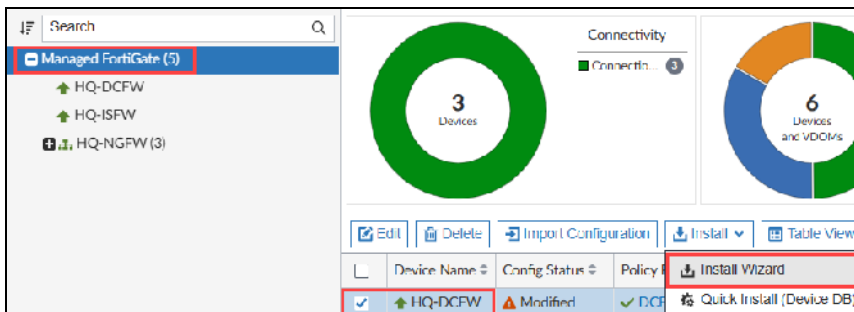
9. Click **Managed FortiGate > HQ-DCF-W > Network > Static Routes**, and then verify that the static route is in the HQ-DCF-W device database.



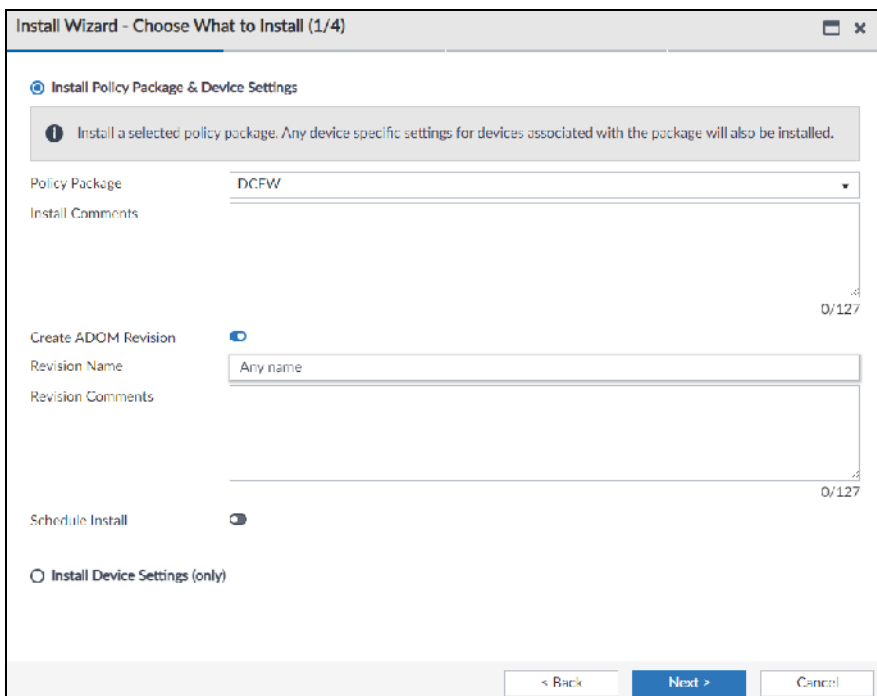
### Stop and think!

At this time, you have not pushed any static routes to FortiGate. Using scripts on the device database is helpful when you want to try new configurations, but you are not ready to push configuration changes to FortiGate. Device layer scripts are used to modify only device-level settings such as interfaces, DNS, static routes, dynamic routes, system settings, log settings, or any other part of the FortiGate configuration that does not affect firewall policy rules.

- Click **Managed FortiGate**, select the **HQ-DCFW** checkbox, click **Install**, and then select **Install Wizard** to install the static route on HQ-DCFW.



- Select **Install Policy Package & Device Settings**, and then in the **Policy Package** field, select **DCFW**.



You can leave the **Create ADOM Revision** setting enabled or disable it. You will not be using this feature in any of the labs, so this setting will not affect the configuration. Also, you can use any name in the **Revision Name** field.

12. Click **Next**.
13. Click **Next**.
14. Click **Install preview** to see what FortiManager will push to HQ-DCFW.
15. Click **Close**.
16. Click **Install**.

#### Stop and think!

Even if you made changes only to the device layer, it is recommended that you install the policy package and device settings at the same time to avoid an unknown policy package status. An unknown policy package status is generated when you run a script in the device layer.

| <input type="checkbox"/> | Device Name ⇅ | Config Status ⇅ | Policy Package Status ⇅ |
|--------------------------|---------------|-----------------|-------------------------|
| <input type="checkbox"/> | ↑ HQ-DCFW     | ✓ Synchronized  | ? DCFW                  |
| <input type="checkbox"/> | ↑ HQ-ISFW     | ✓ Synchronized  | ✓ ISFW                  |

In real-world environments, you must use the install preview option to accept the changes that FortiManager will push.

If there are no changes, FortiManager shows a **No commands to be installed** or **No preview** status.

17. After the installation is done, click **Finish**.  
You should see HQ-DCFW synchronized in both layers.

| <input type="checkbox"/> | Device Name ⇅ | Config Status ⇅ | Policy Package Status ⇅ |
|--------------------------|---------------|-----------------|-------------------------|
| <input type="checkbox"/> | ↑ HQ-DCFW     | ✓ Synchronized  | ✓ DCFW                  |

## Run a Script Using the Policy Package or ADOM Database Method

You will create a firewall policy using the policy package or ADOM database method.

### To run a script using the policy method to create a firewall policy

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW** ADOM.



Before you run the script to create the policy, it is recommended that you view what is in the DCFW policy package, in order to see how the policy script is working and what the firewall policy is creating.

- Click **Policy & Objects > Policy Packages > DCFW > Firewall Policy**.
- Verify that there is one firewall policy in the DCFW policy package.

| # | Name     | From  | To    | Source | Destination |
|---|----------|-------|-------|--------|-------------|
| 1 | Internet | port5 | port2 | 10.0.5 | all         |

Implicit (2/2 Total:1)

- Click **Device Manager > Scripts** to run the policy script to add one firewall policy to the DCFW policy package.
- Select the **Firewall rule** checkbox, and then click **Run Script**.

| Name             | Type | Target                              |
|------------------|------|-------------------------------------|
| ACME Certificate | CLI  | Remote FortiGate Directly (via CLI) |
| Static Route     | CLI  | Device Database                     |
| Firewall rule    | CLI  | Policy Package or ADOM Database     |

- In the **Run script on policy package** field, select **DCFW**, and then click **Run Now**.

Run Script - Firewall rule

Run script on policy package: DCFW

Skip Resolving Variables: ☐

Run Now Cancel

- Click **Close** to close the **Run Script** window.
- Click **Policy & Objects > Policy Packages > DCFW > Firewall Policy**.
- Verify that there are two firewall policies in the DCFW policy package.

| # | Name        | From  | To    | Source | Destination |
|---|-------------|-------|-------|--------|-------------|
| 1 | Internet    | port5 | port2 | 10.0.5 | all         |
| 2 | To HQ-Web-1 | port2 | port5 | all    | 10.0.5      |

Implicit (3/3 Total:1)



Stop and think!

You selected the DCFW policy package previously because there was a firewall policy to create. However, if you need to create only firewall addresses, services, virtual IP addresses, IP pools, security profiles, user, user groups, or anything related to the policy layer, you can select any policy package—for example, the default policy package, which is usually not being used. You should only select a specific policy package if you create a firewall policy.

Running policy scripts is useful when you need to import hundreds of objects into the policy layer database, which saves the effort a policy package import requires when you already have FortiGate devices added to your FortiManager. You can try to generate an object, but make sure that it is not pushed to a FortiGate, and that you do not reference the object in any policy rules.

11. In the drop-down list beside **Install Wizard**, select **Re-install Policy** to reinstall the policy package.

|                      |                      |                   |                        |        |
|----------------------|----------------------|-------------------|------------------------|--------|
| HQ-FMG-1             | Policy Package       | Install Wizard    | ADOM Revisions         | Tools  |
| Dashboard            | Search...            | Re-install Policy | Edit                   | Delete |
| Device Manager       | Core1                |                   |                        |        |
| Policy & Objects     | Core2                |                   |                        |        |
| Policy Packages      | DCFW                 |                   |                        |        |
| Normalized Interface | Firewall Policy      |                   |                        |        |
| Firewall Objects     | Installation Targets |                   |                        |        |
| Security Profiles    | ISFW                 |                   |                        |        |
|                      | default              |                   |                        |        |
|                      |                      | #                 | Name                   | From   |
|                      |                      | 1                 | Internet               | port5  |
|                      |                      | 2                 | To Linux Servers       | port2  |
|                      |                      |                   |                        | port5  |
|                      |                      |                   | Implicit (3/3 Total:1) |        |
|                      |                      | 3                 | Implicit Deny          | any    |

12. Click **OK**.



You can reinstall a policy package after you install the policy package the first time. This applies whether you assign a policy package to FortiGate or you import it, and then install it for the first time.

Also, reinstalling a policy package shows you an installation preview, which you must view if you do not know the changes that FortiManager will push.

13. Click **Install preview** to see what FortiManager will push to HQ-DCFW, and then click **Close**.

You can see that FortiManager will push one firewall policy.

14. Click **Next**.

15. After the installation is done, click **Finish**.

You should see HQ-DCFW synchronized in both layers.

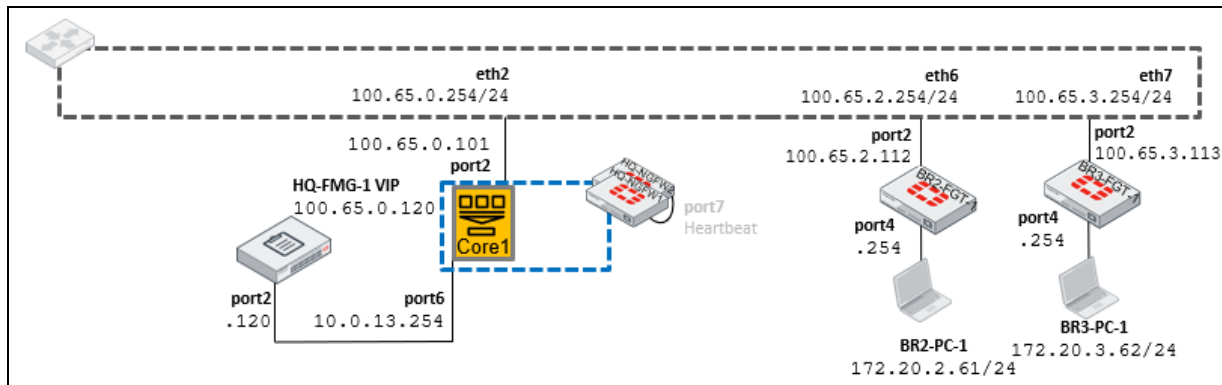
|  |             |                |                       |
|--|-------------|----------------|-----------------------|
|  | Device Name | Config Status  | Policy Package Status |
|  | ↑ HQ-DCFW   | ✓ Synchronized | ✓ DCFW                |

## Exercise 2: Configuring LTP

You will add two FortiGate devices using the model device and link them to FortiManager with the pre-shared key method. The general process is the following:

1. Configure the metadata variables.
2. Configure the pre-CLI template.
3. Add the BR2-FGT-1 model to FortiManager.
4. Add the pre-CLI template and install the policy package.
5. Register BR2-FGT-1 on FortiManager.
6. Add the BR3-FGT-1 model to FortiManager.
7. Add the pre-CLI template and install the policy package.
8. Register BR3-FGT-1 on FortiManager.

### Network Topology



### Configure the Metadata Variables

You will configure the metadata variables.

#### To configure the metadata variables

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW ADOM**.
3. Click **Policy & Objects > Advanced > Metadata Variables > Create New**.
4. In the **Name** field, type `IP_port2`.

Create New Metadata Variables

Name **IP\_port2**

Description

Default Value

Per-Device Mapping >

Revision

Change Note

Revision History

[Revert](#) [View Diff](#) Search...

| <input type="checkbox"/> | Revision # | Changed by | Date/Time | Entry Key | Entry name | Action | Change Note |
|--------------------------|------------|------------|-----------|-----------|------------|--------|-------------|
| No record found.         |            |            |           |           |            |        |             |

0

OK Cancel

5. Click **OK**.

You can see the metadata variables shown in the following image:

| Metadata Variables                                                                            |                     | CLI Configurations |
|-----------------------------------------------------------------------------------------------|---------------------|--------------------|
| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">More</a> |                     |                    |
| <input type="checkbox"/>                                                                      | Name                | Default Value      |
| <input type="checkbox"/>                                                                      | GW                  |                    |
| <input type="checkbox"/>                                                                      | Hostname            |                    |
| <input type="checkbox"/>                                                                      | IP_port1            |                    |
| <input type="checkbox"/>                                                                      | IP_port2            |                    |
| <input type="checkbox"/>                                                                      | IP_port4            |                    |
| <input type="checkbox"/>                                                                      | LAN_BR              |                    |
| <input type="checkbox"/>                                                                      | vm_interface_number | 1                  |



The **GW**, **Hostname**, **IP\_port1**, **IP\_port4**, and **LAN\_BR** variables have been preconfigured for you to save time. At this stage, creating the metadata variables consists of creating an object only. After the FortiGate is added as a model device, you will assign a value to the object.

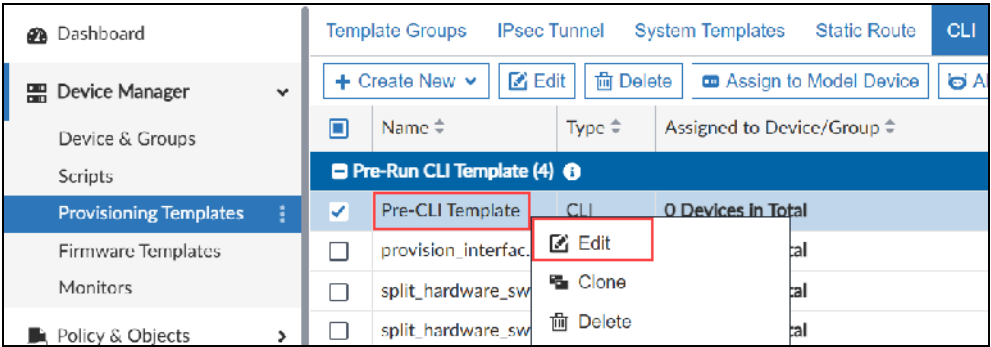
FortiManager creates the **vm\_interface\_number** variable by default, which is useful for creating a number of interfaces on FortiGate VM models.

## Add Metadata Variables to the Pre-CLI Template

You will add metadata variables to the pre-CLI template.

### To add metadata variables to the pre-CLI template

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW ADOM**.
3. Click **Device Manager > Provisioning Templates > CLI**.
4. Expand **Pre-Run CLI Template**, right-click **Pre-CLI Template**, and then select **Edit**.



5. In the **Script Details** section, in the port2 part of the script, at the end of the `set ip` line, type `$`, and then in the list that appears, select `(IP_port2)`.

Edit Pre-Run CLI Template - Pre-CLI Template

Script Details

Validate on change

Validation device platform

FortiGate-VM64

Validate

Format

Revert

Shortcuts

Search...

1 config system interface

2 edit "port2"

3 set mode static

4 set ip \$

5 set allo

6 next

7 edit "po

8 set mode

9 set ip

10 set allo

11 next

12 edit "po

13 set mode

14 set in

(IP\_port2)

Default value:

Description:

(IP\_port4)

Default value:

Description:



The **IP\_port4**, **IP\_port1**, **GW**, and **Hostname** metadata variables have been preconfigured for you in **Pre-CLI Template** to save time.

Your script should look like the following image:

```
1 config system interface
2 edit "port2"
3 set mode static
4 set ip $(IP_port2)
5 set allowaccess ping https ssh fgfm
6 next
7 edit "port4"
8 set mode static
9 set ip $(IP_port4)
10 set allowaccess ping https ssh
11 next
12 edit "port1"
13 set mode static
14 set ip $(IP_port1)
15 set allowaccess ping https ssh http
16 next
17 end
18
19 config router static
20 edit 1
21 set gateway $(GW)
22 set device "port2"
23 next
24 end
25
26 config system global
27 set admin-server-cert "self-sign"
28 set admintimeout 480
29 set hostname $(Hostname)
30 end
31
32 config system automation-stitch
33 edit "Firmware upgrade notification"
34 set status disable
35 next
36 end
37
38 config system central-management
39 set allow-push-firmware disable
40 set allow-remote-firmware-upgrade disable
41 config server-list
42 edit 1
43 set server-type update
44 set server-address 192.168.1.131
45 next
46 edit 2
47 set server-type rating
48 set server-address 192.168.1.132
49 next
50 end
51 set include-default-servers disable
52 end
```



The dollar sign (\$) must precede any metadata variable (enclosed in parentheses). Otherwise, you will receive an error when you try to install them.

6. Click **OK**.

#### Stop and think!

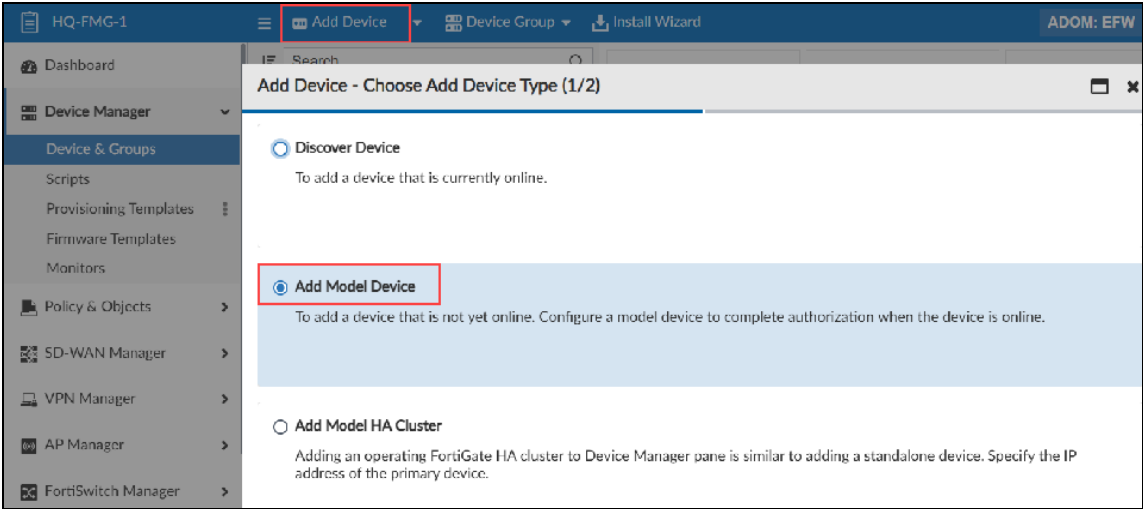
You have created the metadata variables that you will assign when you add new FortiGate devices using the LTP model. Once you add a FortiGate, you must define each value according to each site.

## Add BR2-FGT-1 to FortiManager

You will add BR2-FGT-1 to FortiManager using the model device method.

### To add BR2-FGT-1 to FortiManager as a model device

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW ADOM**.
3. Click **Device Manager > Device & Groups > Add Device**, and then select **Add Model Device**.



4. Configure the following settings:

| Field                 | Value                                                                    |
|-----------------------|--------------------------------------------------------------------------|
| Name                  | BR2-FGT-1                                                                |
| Link Device By        | Pre-shared Key                                                           |
| Pre-shared Key        | 123456789                                                                |
| Device Model          | FortiGate-VM64-KVM                                                       |
| Port Provisioning     | 4                                                                        |
| Pre-Run CLI Template  | Pre-CLI Template                                                         |
| Assign Policy Package | BR                                                                       |
| Metadata Variables    | Click <b>Edit Variable Mapping</b> , and then continue to the next step. |

Add Device - Provide Model Device Info (1/2)

Name

BR2-FGT-1

Link Device By

Serial NumberPre-shared Key

Pre-shared Key

123456789

Use Device Blueprint

Device Model

FortiGate-VM64-KVM

Port Provisioning

4

Has Log Disk

Automatically Link to Real Device

Enforce Firmware Version

7.6 (By Default)

Enforce Device Configuration

Managed by SD-WAN Manager

Add to Device Group

Add to Folder

Fabric Authorization Template

Pre-Run CLI Template

Pre-CLI Template

Assign Policy Package

BR

Provisioning Templates

Metadata Variables

Edit Variable Mapping

Copy Device Dashboard

Click to select

< Back

Next >

Cancel

5. In the **Mapping Value** field, configure the following settings:

When you add a metadata variable, ensure that you click the check mark so that the metadata variable is saved.



| Edit Metadata Variable Mapping |               |               |
|--------------------------------|---------------|---------------|
| #                              | Variable Name | Mapping Value |
| 1                              | \$(GW)        | 100.65.2.254  |
| 2                              | \$(Hostname)  |               |
| 3                              | \$(IP_port1)  |               |
| 4                              | \$(IP_port2)  |               |

| Variable Name | Mapping Value |
|---------------|---------------|
| \$(GW)        | 100.65.2.254  |



| Variable Name | Mapping Value    |
|---------------|------------------|
| \$(Hostname)  | BR2-FGT-1        |
| \$(IP_port1)  | 192.168.1.112/16 |
| \$(IP_port2)  | 100.65.2.112/24  |
| \$(IP_port4)  | 172.20.2.254/24  |
| \$(LAN_BR)    | 172.20.2.0/24    |

Edit Metadata Variable Mapping

Search...

| # | Variable Name           | Mapping Value    | Default Value |  |
|---|-------------------------|------------------|---------------|--|
| 1 | \$(G.W.)                | 100.65.2.254     |               |  |
| 2 | \$(Hostname)            | BR2-FGT-1        |               |  |
| 3 | \$(IP_port1)            | 192.168.1.112/16 |               |  |
| 4 | \$(IP_port2)            | 100.65.2.112/24  |               |  |
| 5 | \$(IP_port4)            | 172.20.2.254/24  |               |  |
| 6 | \$(LAN_BR)              | 172.20.2.0/24    |               |  |
| 7 | \$(vm_interface_number) | 4                | 1             |  |

1 Selected
7

OK Cancel



The **vm\_interface\_number** metadata variable is automatically assigned a mapping value of 4, if you selected **Port Provisioning** in step 4. This way, you do not need to configure it in this section.

- Click **OK**.
- Click **Next**.
- Click **Finish**.

You can see that BR2-FGT-1 is available in the device database.



If your view is different from the following image, you can rearrange the table columns to display the **Policy Package Status** column, using the gear icon in the upper-right corner.

| <input type="checkbox"/> | Device Name ⇅ | Config Status ⇅ | Policy Package Status ⇅ |
|--------------------------|---------------|-----------------|-------------------------|
| <input type="checkbox"/> | BR2-FGT-1     | Unknown         | BR                      |
| <input type="checkbox"/> | HQ-DCFw       | Synchronized    | DCFw                    |
| <input type="checkbox"/> | HQ-ISFW       | Synchronized    | ISFW                    |
| <input type="checkbox"/> | HQ-NGFW       | Synchronized    |                         |

### Stop and think!

BR2-FGT-1 has been added to the FortiManager device database and it has already been assigned a name, pre-shared key, device model, port provisioning, pre-run CLI template, policy package, and metadata variables.

Notice that, at this step, the IP interfaces are configured.

| # | Name  | Type               | Normalized Interface | Addressing Mode | IP/Netmask                 | Access                       |
|---|-------|--------------------|----------------------|-----------------|----------------------------|------------------------------|
| 1 | port1 | Physical Interface | port1                | static          | 192.168.1.112/255.255.0.0  | HTTPS, Ping, SSH, HTTP       |
| 2 | port2 | Physical Interface | port2                | static          | 100.65.2.112/255.255.255.0 | HTTPS, Ping, SSH, FMG-Access |
| 3 | port3 | Physical Interface | port3                | static          | 0.0.0.0/0.0.0.0            |                              |
| 4 | port4 | Physical Interface | port4                | static          | 172.20.2.254/255.255.255.0 | HTTPS, Ping, SSH             |

The static route is configured.

| Seq. # | Destination     | Gateway      | Interface |
|--------|-----------------|--------------|-----------|
| 1      | 0.0.0.0/0.0.0.0 | 100.65.2.254 | port2     |

But there are no firewall policies.

| policy            | firewall policy |
|-------------------|-----------------|
| firewall          |                 |
| DoS-policy        |                 |
| DoS-policy6       |                 |
| interface-policy  |                 |
| interface-policy6 |                 |
| local-in-policy   |                 |
| local-in-policy6  |                 |
| multicast-policy  |                 |
| multicast-policy6 |                 |
| policy            |                 |

### Stop and think!

Pre-run CLI templates are designed for LTP model devices. After you install one for the first time, you will notice that the template is automatically unassigned from the firewall.

| Template Groups              | IPsec Tunnel         | System Templates       | Static Route                           | CLI                                                | Feature Visibility   |
|------------------------------|----------------------|------------------------|----------------------------------------|----------------------------------------------------|----------------------|
| <a href="#">+ Create New</a> | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Assign to Device/Group</a> | <a href="#">AI Assistant</a>                       | <a href="#">More</a> |
| <input type="checkbox"/>     | Name                 | Type                   | Assigned to Device/Group               | Variables                                          |                      |
| Pre-Run CLI Template (4)     |                      |                        |                                        |                                                    |                      |
| <input type="checkbox"/>     | Pre-CLI Template     | CLI                    | 0 Devices in Total                     | GW<br>Hostname<br>IP_port1<br>IP_port2<br>IP_port4 |                      |

CLI templates are assigned and stick until you remove them, but pre-run CLI templates are removed automatically, as you can see in this example.

- Click **Device Manager > Device & Groups > Managed FortiGate**.
- Select **BR2-FGT-1**, click **Install**, and then select **Install Wizard** to install the policy package in the device layer.
- Click **Install Policy Package & Device Settings**.
- In the **Policy Package** field, select **BR**.

Install Wizard - Choose What to Install (1/4)

☒ Install Policy Package & Device Settings

Install a selected policy package. Any device specific settings for devices associated with the package will also be installed.

Policy Package: BR

Install Comments:

0/127

- Click **Next**.
- Click **Next** one more time.
- Click **Install**.
- Click **Finish**.

Now, the BR2-FGT-1 device database also shows one firewall policy besides the host name, IP addresses in port1, port2, and port4, and static route.

Once the FortiGate is connected to FortiManager, the firewall is ready to receive all configurations.

The BR2-FGT-1 configuration should look like the following image:

| <input type="checkbox"/>            | Device Name ▾ | Config Status ▾ | Policy Package Status ▾ |
|-------------------------------------|---------------|-----------------|-------------------------|
| <input checked="" type="checkbox"/> | BR2-FGT-1     | Unknown         | ✓ BR                    |

## Register BR2-FGT-1 on FortiManager

You will register BR2-FGT-1 on FortiManager.

### To register BR2-FGT-1 on FortiManager

1. Log in to the BR2-FGT-1 serial console with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following commands:

```
config system interface
  edit "port2"
    set ip 100.65.2.112 255.255.255.0
    set allowaccess ping https ssh fgfm
  end
config router static
  edit 1
    set device port2
    set gateway 100.65.2.254
  end
```



Notice that after you create the default static route, the following message appears:

The destination is set to 0.0.0.0/0 which means all IP addresses.

```
config system central-management
  set type fortimanager
  set fmg 100.65.0.120
end
```



Notice that after you configure the central-management settings, the following message appears:

The Serial Number for FortiManager is not entered.

If serial number is not set, connection will be set as unverified.

FortiGate can establish a connection to obtain the serial number. Do you want to try to connect now? (y/n)

3. Type y.



After typing `y`, you must not press `Enter`. The following message appears:

Obtained serial number from FortiManager 100.65.0.120 is: FMG-VMTM24012945

Do you confirm that this is the correct serial number? (y/n)

4. Type `y`.



The following message appears:

Successfully registered to FortiManager. This device may need to be authorized on FortiManager.

**Stop and think!**

Notice that this firewall does not have any other configuration except port2 and the FortiManager acting as a local FortiGuard Distribution Server (FDS). In your topology environments, as long as your firewall has internet access through a valid internet connection, so it can validate its FortiGuard license, this is enough since FortiGate must have a valid license to work correctly.

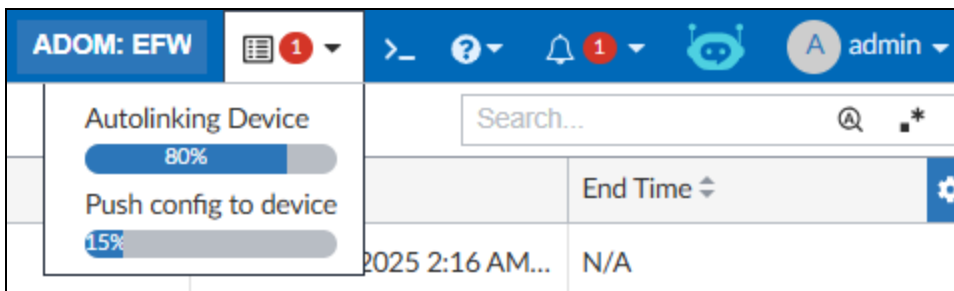
5. Return to the BR2-FGT-1 serial console, and then enter the following command to register FortiGate on FortiManager:

```
execute central-mgmt register-device FMG-VMTM24012945 123456789
```



Ensure that you type the command correctly—if it is not correct, the FortiGate will not connect to FortiManager.

6. Return to the FortiManager GUI, and then in **ADOM EFW**, click **System Settings > Task Monitor**. In the upper-right corner, you can see that the autolink device process started automatically.



You can also see the autolink and installation configuration process running in **ADOM EFW > System Settings > Task Monitor**.

| <a href="#">+ Group Error Devices</a> <a href="#">Delete</a> <a href="#">View Details</a> <a href="#">Show Status</a> |    |                       |                       |           |        |           |      |
|-----------------------------------------------------------------------------------------------------------------------|----|-----------------------|-----------------------|-----------|--------|-----------|------|
| <input type="checkbox"/>                                                                                              | ID | Source                | Description           | User      | Status | Time Used | ADCM |
| <input type="checkbox"/>                                                                                              | 60 | Install Configuration | Push config to device | admin     | 15%    | N/A       | FFW  |
| <input type="checkbox"/>                                                                                              | 59 | Device Manager        | Autolinking Device    | Auto link | 80%    | N/A       | EPW  |

## 7. Return to the BR2-FGT-1 serial console.

You will notice that the host name has changed. You can also verify other configurations like firewall policies.

```
FortiGate-VM64-KVM (central-management) # and
The Serial Number for FortiManager is not entered.
In order to verify identity of FortiManager serial number is needed.
If serial number is not set, connection will be set as unverified.
FortiGate can establish a connection to obtain the serial number now. Do you want to try to connect now? (y/n)y

Obtained serial number from FortiManager 100.65.0.120 is: FMG VMTM24012945
Do you confirm that this is the correct serial number? (y/n)y

Successfully registered to FortiManager. This device may need to be authorized on FortiManager.

FortiGate-VM64-KVM # execute central-mgmt register-device FMG-VMTM24012945 123456789
Start Registering ...
Registering request sent
FortiGate-VM64-KVM #
BR2-FGT-1 #
BR2-FGT-1 #
BR2-FGT-1 #
```



After you register BR2-FGT-1 on FortiManager, it should not take more than 2 minutes to finish the installation process. If you registered the device and you do not see any logs in the task monitor, it means that you should check the spelling of the previous commands, such as the FortiManager serial number or the correct IP address when you typed it directly on the BR2-FGT-1 console.

## Add BR3-FGT-1 to FortiManager

You will register BR3-FGT-1 on FortiManager.

### Take the Expert Challenge!

Add and register BR3-FGT-1 on FortiManager using the following settings:

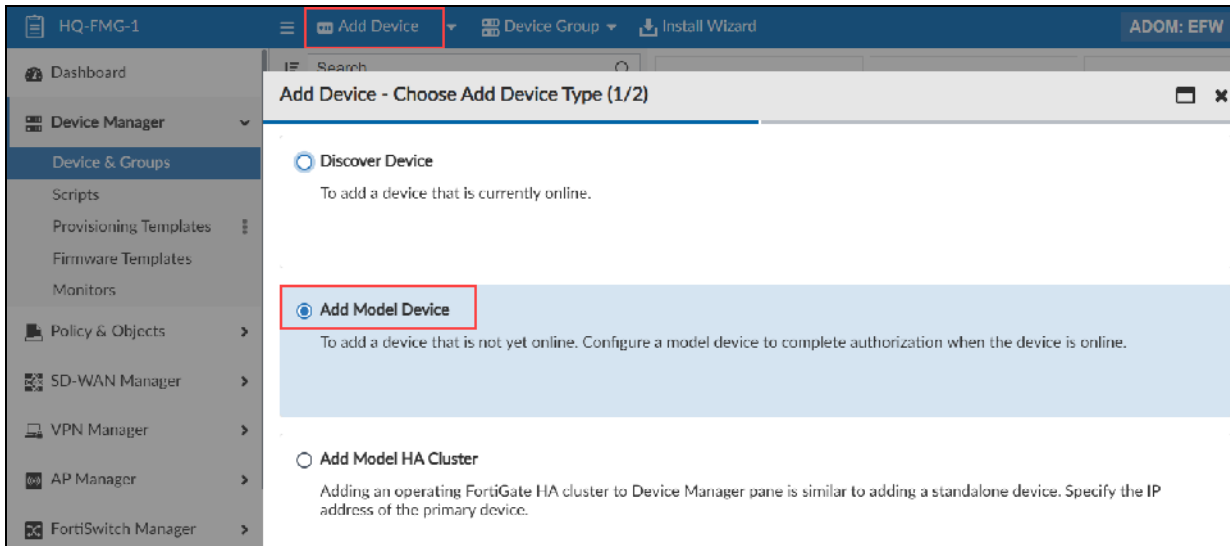
```
$(GW): 100.65.3.254
$(Hostname): BR3-FGT-1
$(IP_port1): 192.168.1.113/16
$(IP_port2): 100.65.3.113/24
$(IP_port4): 172.20.3.254/24
$(LAN_BR): 172.20.3.0/24
Policy Package: BR
```

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see "Configuring LTP" on page 38.

### To register BR3-FGT-1 on FortiManager

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW ADOM**.
3. Click **Device Manager > Device & Groups > Add Device**, and then select **Add Model Device**.



You do not need to create any templates since you have already done this. Now, you will add new branches.

4. Configure the following settings:

| Field                 | Value                                                                    |
|-----------------------|--------------------------------------------------------------------------|
| Name                  | BR3-FGT-1                                                                |
| Link Device By        | Pre-shared Key                                                           |
| Pre-shared Key        | 987654321                                                                |
| Device Model          | FortiGate-VM64-KVM                                                       |
| Port Provisioning     | 4                                                                        |
| Pre-Run CLI Template  | Pre-CLI Template                                                         |
| Assign Policy Package | BR                                                                       |
| Metadata Variables    | Click <b>Edit Variable Mapping</b> , and then continue to the next step. |



Notice that you are using a different pre-shared key. Since you can add multiple firewalls, it is recommended that you use a different password based on the firewall you will be adding.

Add Device - Provide Model Device Info (1/2)

Name

BR3 FGT 1

Link Device By

Serial NumberPre-shared Key

Pre-shared Key

987654321

Use Device Blueprint

Device Model

FortiGate-VM64-KVM

Port Provisioning

4

Has Log Disk

Automatically Link to Real Device

Enforce Firmware Version

7.6 (By Default)

Enforce Device Configuration

Managed by SD-WAN Manager

Add to Device Group

Add to Folder

Fabric Authorization Template

Pre-Run CLI Template

Pre-CLI Template

Assign Policy Package

BR

Provisioning Templates

Metadata Variables

Edit Variable Mapping

Copy Device Dashboard

Click to select

< Back

Next >

Cancel

5. In the **Metadata Variables** field, configure the following settings:

| Variable Name | Mapping Value    |
|---------------|------------------|
| \$(GW)        | 100.65.3.254     |
| \$(Hostname)  | BR3-FGT-1        |
| \$(IP_port1)  | 192.168.1.113/16 |
| \$(IP_port2)  | 100.65.3.113/24  |
| \$(IP_port4)  | 172.20.3.254/24  |
| \$(LAN_BR)    | 172.20.3.0/24    |



### Stop and think!

Think about all the possibilities that you can configure using metadata variables, based on the necessities of your firewalls or sites. At this stage, you have configured three simple parameters: GW, host name, and network segment. However, you can go beyond the dynamic objects you need to use at the beginning of the setup of your LTP implementation and your pre-run CLI template, which are only assigned at the beginning of adding your device to FortiManager.

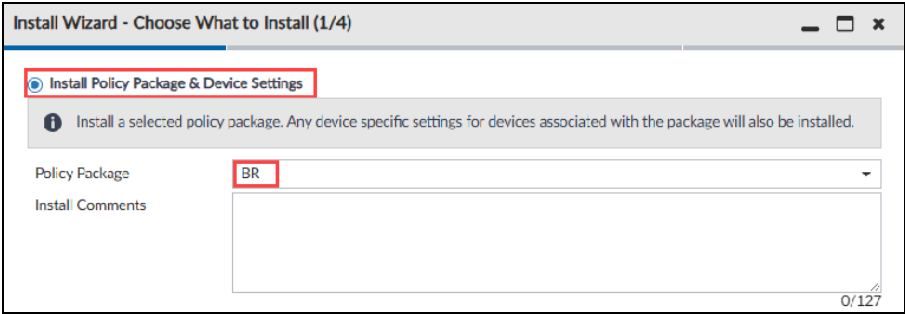
| # | Variable Name           | Mapping Value    | Default Value |
|---|-------------------------|------------------|---------------|
| 1 | \$(GW)                  | 100.65.3.254     |               |
| 2 | \$(Hostname)            | BR3-FGT-1        |               |
| 3 | \$(IP_port1)            | 192.168.1.113/16 |               |
| 4 | \$(IP_port2)            | 100.65.3.113/24  |               |
| 5 | \$(IP_port4)            | 172.20.3.254/24  |               |
| 6 | \$(LAN_BR)              | 172.20.3.0/24    |               |
| 7 | \$(vm_interface_number) | 4                | 1             |

- Click **OK**.
- Click **Next**.
- Click **Finish**.

You can see that BR3-FGT-1 is available in the device database.

|                          | Device Name | Config Status  | Policy Package Status |
|--------------------------|-------------|----------------|-----------------------|
| <input type="checkbox"/> | BR2-FGT-1   | ✓ Auto-update  | ✓ BR                  |
| <input type="checkbox"/> | BR3-FGT-1   | ⚠ Unknown      | ⚠ BR                  |
| <input type="checkbox"/> | HQ-DCFV     | ✓ Synchronized | ✓ DCFV                |
| <input type="checkbox"/> | HQ-ISFW     | ✓ Synchronized | ✓ ISFW                |
| <input type="checkbox"/> | HQ-NGFW     | ✓ Synchronized |                       |

- Click **Device Manager > Device & Groups > Managed FortiGate**.
- Select **BR3-FGT-1**, click **Install**, and then select **Install Wizard** to install the policy package in the device layer.
- Click **Install Policy Package & Device Settings**.
- In the **Policy Package** field, select **BR**.



13. Click **Next**.

If you leave the **BR2-FGT-1** checkbox selected, you will see **No commands to be installed** at the next step.



| Install Preview                     |             |                              |
|-------------------------------------|-------------|------------------------------|
| Policy Package Diff                 |             |                              |
| <input checked="" type="checkbox"/> | Device Name | Status                       |
| <input type="checkbox"/>            | BR2-FGT-1   | No commands to be installed. |
| <input checked="" type="checkbox"/> | BR3-FGT-1   | Copy Only                    |

14. Click **Next**.

15. Click **Install**.

16. Click **Finish**.

Now, the BR3-FGT-1 device database shows the host name, IP addresses in port1, port2, and port4, static route, and one firewall policy.

Once FortiGate is connected to FortiManager, the firewall is ready to receive all configurations.

The BR3-FGT-1 configuration should look like the following image:

| <input type="checkbox"/>            | Device Name | Config Status | Policy Package Status |
|-------------------------------------|-------------|---------------|-----------------------|
| <input type="checkbox"/>            | BR2-FGT-1   | Synchronized  | BR                    |
| <input checked="" type="checkbox"/> | BR3-FGT-1   | Unknown       | BR                    |

## Register BR3-FGT-1 on FortiManager

You will register BR3-FGT-1 on FortiManager.

### To register BR3-FGT-1 on FortiManager

1. Log in to the BR3-FGT-1 serial console with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following commands:

```
config system interface
```

```
edit "port2"
set ip 100.65.3.113 255.255.255.0
set allowaccess ping https ssh fgfm
end
config router static
edit 1
set device port2
set gateway 100.65.3.254
end
config system central-management
set type fortimanager
set fmg 100.65.0.120
end
```

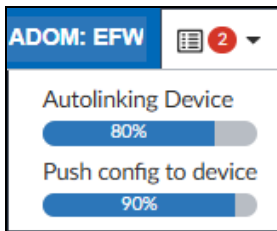
3. Type `y` to obtain the FortiManager serial number.
4. Type `y` to confirm the serial number.
5. Enter the following command to register FortiGate on FortiManager:

```
execute central-mgmt register-device FMG-VMTM24012945 987654321
```



Ensure that you type the command correctly—if it is not correct, FortiGate will not connect to FortiManager.

6. Return to the FortiManager GUI, and then in **ADOM EFW**, click **System Settings > Task Monitor**. In the upper-right corner, you can see that the autolink device process started automatically.



You can also see the autolink and installation configuration process running in **ADOM EFW > System Settings > Task Monitor**.

| <a href="#">+ Group Error Devices</a> <a href="#">Delete</a> <a href="#">View Details</a> <a href="#">Show Status</a> |    |                |                    |           |        |
|-----------------------------------------------------------------------------------------------------------------------|----|----------------|--------------------|-----------|--------|
| <input type="checkbox"/>                                                                                              | ID | Source         | Description        | User      | Status |
| <input type="checkbox"/>                                                                                              | 46 | Device Manager | Autolinking Device | Auto link | 80%    |

7. Return to the BR3-FGT-1 serial console. You will notice that the host name has changed.

```
FortiGate-VM64-KVM # execute central-mgmt register-device FMG-VMTM24012945 987654321
Start Registering ...
Registering request sent
FortiGate-VM64-KVM #
BR3-FGT-1 #
BR3-FGT-1 #
BR3-FGT-1 #
```



After you register BR3-FGT-1 on FortiManager, it should not take more than 2 minutes to finish the installation process. If you registered the device, and you do not see any logs in the task monitor, it means that you should check the spelling of the previous commands, such as the FortiManager serial number or the correct IP address when you typed it directly on the BR3-FGT-1 console.

---

## Lab 3: VLANs and VDOMs

In this lab, you will configure VLANs and enable VDOMs. Then, you will allow inter-VDOM traffic and configure internet access using inter-VDOM routing.

### Objectives

- Configure VLANs on ISFW
- Enable VDOMs on ISFW
- Allow inter-VDOM traffic between the Zone1 VDOM and Zone2 VDOM
- Configure internet access using inter-VDOM routing from the Zone1 VDOM to the root VDOM

### Time to Complete

Estimated: 50 minutes

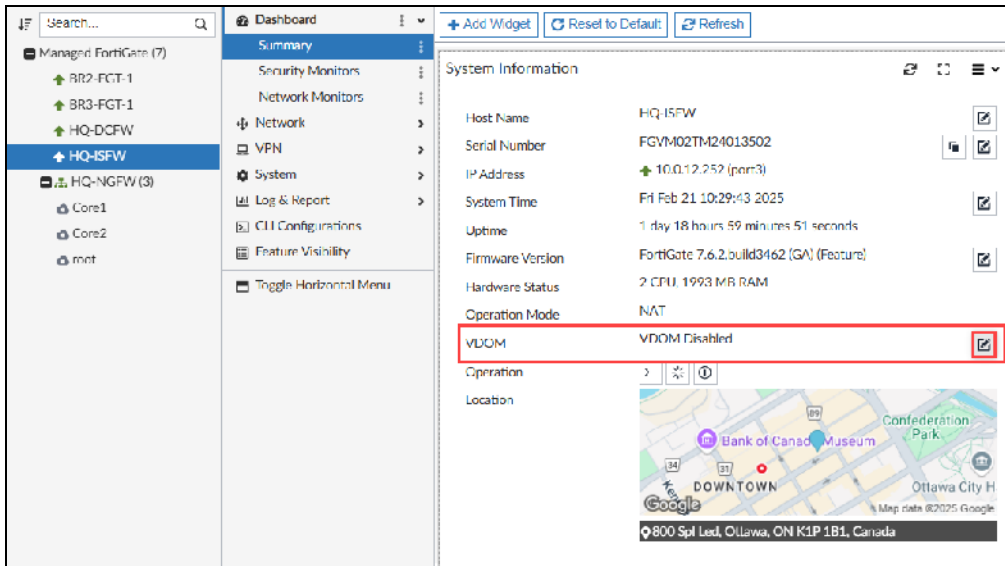
### Prerequisites

Before you begin this lab, you must complete the previous lab. If you haven't done so, tell your instructor.

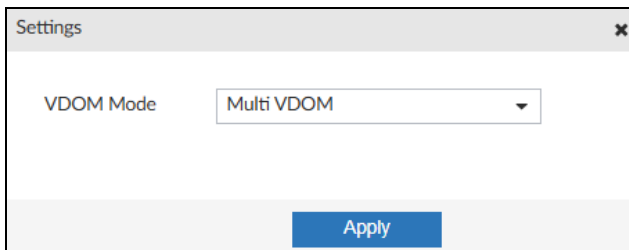


### To enable VDOMs

1. Connect to the FortiManager GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. In the **Select an ADOM** window, select the **EFW ADOM**.
3. Click **Device Manager > Device & Groups > Managed FortiGate > HQ-ISFW > Dashboard > Summary**.
4. In the **System Information** widget, in the **VDOM** field, click the **Edit VDOM** icon.



5. In the **VDOM Mode** field, select **Multi VDOM**, and then click **Apply**.

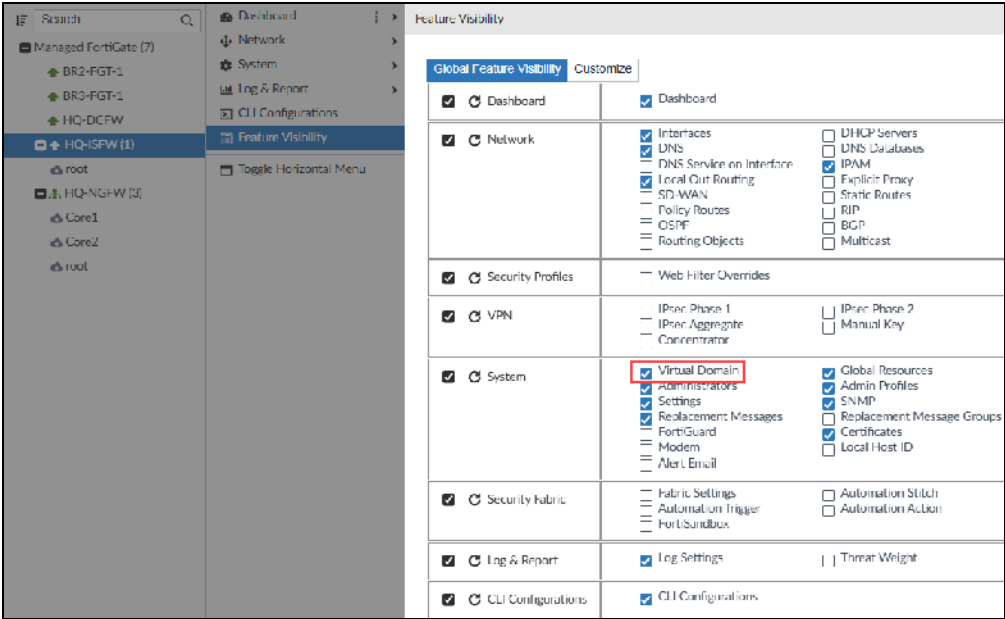


### Create the Zone1 and Zone2 VDOMs

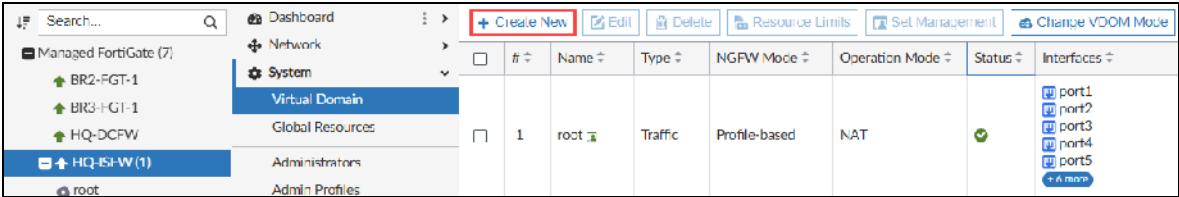
You will create the Zone1 and Zone2 VDOMs on HQ-ISFW.

#### To create the Zone1 VDOM

1. Click **Managed FortiGate > HQ-ISFW > Feature Visibility**.
2. In the **System** section, select the **Virtual Domain** checkbox.



- 3. Click **OK**.
- 4. Click **Managed FortiGate > HQ-ISFW > System > Virtual Domain > Create New**.



- 5. Configure the following settings:

| Field             | Value |
|-------------------|-------|
| VDOM Name         | Zone1 |
| Interface Members | port4 |



Create New Virtual Domain on HQ-ISFW

VDOM Name

Zone1

Description

0/255

Enable

☒

Central SNAT

☐

Operation Mode

NAT

VDOM Type

TrafficAdmin

NGFW Mode

Profile-basedPolicy-based

Interface Members

port4

Click to select

1 entry selected

OK

Cancel

6. Click **OK**.

To create the Zone2 VDOM

- 1. Click **Managed FortiGate > HQ-ISFW > System > Virtual Domain > Create New**.
- 2. Configure the following settings:

| Field             | Value |
|-------------------|-------|
| VDOM Name         | Zone2 |
| Interface Members | port5 |

Create New Virtual Domain on HQ-ISFW

VDOM Name

Zone2

Description

0/255

Enable

☒

Central SNAT

☐

Operation Mode

NAT

VDOM Type

TrafficAdmin

NGFW Mode

Profile-basedPolicy-based

Interface Members

port5

Click to select

1 entry selected

OK

Cancel

3. Click **OK**.

You should now see that the Zone1 and Zone2 VDOMs were created, as shown in the following image:

| <input type="checkbox"/> | # | Name  | Type    | NGFW Mode     | Operation Mode | Status | Interfaces                                                                                                                                                               |
|--------------------------|---|-------|---------|---------------|----------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | 1 | Zone1 | Traffic | Profile-based | NAT            | ✓      | <a href="#">port4</a><br><a href="#">ssl.Zone1 (SSL VPN interface)</a><br><a href="#">nat.Zone1 (SSL VPN interface)</a><br><a href="#">l2t.Zone1 (SSL VPN interface)</a> |
| <input type="checkbox"/> | 2 | Zone2 | Traffic | Profile-based | NAT            | ✓      | <a href="#">port5</a><br><a href="#">ssl.Zone2 (SSL VPN interface)</a><br><a href="#">nat.Zone2 (SSL VPN interface)</a><br><a href="#">l2t.Zone2 (SSL VPN interface)</a> |
| <input type="checkbox"/> | 3 | root  | Traffic | Profile-based | NAT            | ✓      | <a href="#">port1</a><br><a href="#">port2</a><br><a href="#">port3</a><br><a href="#">nat.root</a><br><a href="#">l2t.root</a><br><a href="#">+ 2 more</a>              |

## Create VLAN101 and VLAN102

You will create VLAN101 and VLAN102 on HQ-ISFW.

### To create VLAN101 and VLAN102

1. Click **Managed FortiGate > HQ-ISFW > Network > Interfaces**.
2. Click **Create New**, and then select **Interface**.

Search...

Managed FortiGate (9)

BR2-FGT-1

BR3-FGT-1

HQ-DCFW

HQ-ISFW (3)

root

Zone1

Zone2

Dashboard

Network

Interfaces

DNS

IPAM

Local Out Routing

System

Log & Report

CLI Configurations

Create New

Edit

Edit in CLI

Delete

Where Used

More

Full Screen

Interface

VDOM Link

Virtual Wire Pair

| Type               | Normalized Interface | IP/Netmask                | Virtual Domain |
|--------------------|----------------------|---------------------------|----------------|
| Physical Interface | port1                | 192.168.1.101/255.255.0.0 | root           |
| Physical Interface | port2                | 10.0.11.253/255.255.255.0 | root           |
| Physical Interface | port3                | 10.0.12.252/255.255.255.0 | root           |
| Physical Interface | port4                | 0.0.0.0/0.0.0.0           | Zone1          |
| Physical Interface | port5                | 0.0.0.0/0.0.0.0           | Zone2          |

3. Configure the following settings:



Ensure that you configure the VLAN interface settings correctly because you will use them as normalized interfaces associated between device models and the KVM. If the settings are not spelled or configured correctly, you will have to delete them and create the configuration again.

| Field           | Value   |
|-----------------|---------|
| Interface Name  | VLAN101 |
| Interface       | port4   |
| VLAN ID         | 101     |
| Virtual Domain  | Zone1   |
| Addressing Mode | Manual  |

| Field                        | Value                    |
|------------------------------|--------------------------|
| IP/Netmask                   | 10.0.2.254/255.255.255.0 |
| Administrative Access (IPv4) | PING                     |

Create New Interface

Interface Name

VLAN101

Alias Name

Type

VLAN

VLAN Protocol

802.1Q802.1AD

Interface

port4

VLAN ID

101

VRF ID

0

Virtual Domain

Zone1

Role

LAN

Address

Addressing Mode

Manual

IP/Netmask

10.0.2.254/255.255.255.0

IPv6 Addressing Mode

ManualDHCPDelegated

IPv6 Address/Prefix

:::0

Auto Configure IPv6 Address

DHCPv6 Prefix Delegation

Create Address Object Matching Subnet

Secondary IP Address

Administrative Access

IPv4

☐ HTTPS

☒ PING

☐ SSH

☐ SNMP

☐ HTTP

☐ TELNET

☐ FMG-Access

☐ RADIUS Accounting

☐ Probe Response

☐ FTM

☐ Security Fabric Connection

☐ Speed Test

☐ SCIM

4. Click **OK**.
5. Click **Managed FortiGate > HQ-ISFW > Network > Interfaces**.
6. Click **Create New**, and then select **Interface**.
7. Configure the following settings:

| Field                        | Value                    |
|------------------------------|--------------------------|
| Interface Name               | VLAN102                  |
| Interface                    | port5                    |
| VLAN ID                      | 102                      |
| Virtual Domain               | Zone2                    |
| Addressing Mode              | Manual                   |
| IP/Netmask                   | 10.0.3.254/255.255.255.0 |
| Administrative Access (IPv4) | PING                     |



**Stop and think!**

If your VLAN interface configuration is correct, the **Normalized Interface** page shows that VLAN101 and VLAN102 were automatically assigned as normalized interfaces. This is because VLAN101 and VLAN102 have already been configured as normalized interfaces for you using the per-platform method.

|                      |                              |                      |                                   |                              |                      |                             |
|----------------------|------------------------------|----------------------|-----------------------------------|------------------------------|----------------------|-----------------------------|
| Dashboard            | <a href="#">+ Create New</a> | <a href="#">Edit</a> | <a href="#">Delete</a>            | <a href="#">Collapse All</a> | <a href="#">More</a> | <a href="#">Full Screen</a> |
| Device Manager       | <input type="checkbox"/>     | Name                 | Mapping Rule                      | Mapped Interface/Zone        |                      |                             |
| Policy & Objects     | <input type="checkbox"/>     | VLAN101              |                                   |                              |                      |                             |
| Policy Packages      | <input type="checkbox"/>     |                      | Per-platform (FortiGate-VM64-KVM) | VLAN101                      |                      |                             |
| Normalized Interface | <input type="checkbox"/>     |                      | Default                           | VLAN101                      |                      |                             |
| Firewall Objects     | <input type="checkbox"/>     | VLAN102              |                                   |                              |                      |                             |
| Security Profiles    | <input type="checkbox"/>     |                      | Per-platform (FortiGate-VM64-KVM) | VLAN102                      |                      |                             |
|                      | <input type="checkbox"/>     |                      | Default                           | VLAN102                      |                      |                             |

Create Inter-VDOM Links

You will create inter-VDOM links.

To create inter-VDOM links

- 1. Click **Managed FortiGate > HQ-ISFW > Network > Interfaces**.
- 2. Click **Create New**, and then select **VDOM Link**.

Search...

Managed FortiGate (9)

BR2-FGT-1

BR3-FGT-1

HQ-DCFW

HQ-ISFW (3)

root

Zone1

Zone2

HQ-NGFW (3)

Core1

Dashboard

Network

Interfaces

DNS

IPAM

Local Out Routing

System

Log & Report

CLI Configurations

Feature Visibility

Toggle Horizontal Menu

Create New

Edit

Edit in CLI

Delete

Where Used

Interface

VDOM Link

Virtual Wire Pair

| Type               | IP/Netmask                |
|--------------------|---------------------------|
| Physical Interface | 192.168.1.101/255.255.0.0 |
| Physical Interface | 10.0.11.253/255.255.255.0 |
| Physical Interface | 10.0.12.252/255.255.255.0 |
| Physical Interface | 0.0.0.0/0.0.0.0           |
| VLAN               | 10.0.2.254/255.255.255.0  |
| Physical Interface | 0.0.0.0/0.0.0.0           |
| VLAN               | 10.0.3.254/255.255.255.0  |

- 3. Configure the following settings:

| Field          | Value                        |
|----------------|------------------------------|
| Name           | A                            |
| Virtual Domain | root                         |
| IP/Netmask     | 172.16.1.253/255.255.255.252 |

| Field                 | Value                        |
|-----------------------|------------------------------|
| Administrative Access | PING                         |
| IPv6 Address          | ::0                          |
| Virtual Domain        | Zone1                        |
| IP/Netmask            | 172.16.1.254/255.255.255.252 |
| Administrative Access | PING                         |
| IPv6 Address          | ::0                          |

Create New VDOM Link

Name: A

Interface 0 (A)

Virtual Domain: root

IP/Netmask: 172.16.1.253/255.255.255.252

Administrative Access:

- ☐ HTTPS
- ☐ SNMP
- ☐ FMG-Access
- ☐ FTM
- ☐ SCIM
- ☒ PING
- ☐ HTTP
- ☐ RADIUS Accounting
- ☐ Security Fabric Connection
- ☐ SSH
- ☐ TELNET
- ☐ Probe Response
- ☐ Speed Test

IPv6 Address: ::0

IPv6 Administrative Access:

- ☐ HTTPS
- ☐ SNMP
- ☐ FMG-Access
- ☐ PING
- ☐ HTTP
- ☐ Security Fabric Connection
- ☐ SSH
- ☐ TELNET

Comments: Write a comment

Administrative Status:

---

Interface 1 (A)

Virtual Domain: Zone1

IP/Netmask: 172.16.1.254/255.255.255.252

Administrative Access:

- ☐ HTTPS
- ☐ SNMP
- ☐ FMG-Access
- ☐ FTM
- ☐ SCIM
- ☒ PING
- ☐ HTTP
- ☐ RADIUS Accounting
- ☐ Security Fabric Connection
- ☐ SSH
- ☐ TELNET
- ☐ Probe Response
- ☐ Speed Test

IPv6 Address: ::0

IPv6 Administrative Access:

- ☐ HTTPS
- ☐ SNMP
- ☐ FMG-Access
- ☐ PING
- ☐ HTTP
- ☐ Security Fabric Connection
- ☐ SSH
- ☐ TELNET

OK Cancel

- Click **OK**.
- Click **Managed FortiGate > HQ-ISFW > Network > Interfaces**.
- Click **Create New**, and then select **VDOM Link**.
- Configure the following settings:

| Field | Value |
|-------|-------|
| Name  | B     |

| Field                 | Value                        |
|-----------------------|------------------------------|
| Virtual Domain        | Zone1                        |
| IP/Netmask            | 172.16.1.245/255.255.255.252 |
| Administrative Access | PING                         |
| IPv6 Address          | ::/0                         |
| Virtual Domain        | Zone2                        |
| IP/Netmask            | 172.16.1.246/255.255.255.252 |
| Administrative Access | PING                         |
| IPv6 Address          | ::/0                         |

Create New VDOM Link

Name: B

Interface 0 (B)

Virtual Domain: Zone1

IP/Netmask: 172.16.1.245/255.255.255.252

Administrative Access: ☐ HTTPS ☒ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ RADIUS Accounting ☐ Probe Response ☐ FTM ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

IPv6 Address: ::/0

IPv6 Administrative Access: ☐ HTTPS ☐ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ Security Fabric Connection ☐ TFTP

Comments: Write a comment

Administrative Status: ☒

Interface 1 (B)

Virtual Domain: Zone2

IP/Netmask: 172.16.1.246/255.255.255.252

Administrative Access: ☐ HTTPS ☒ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ RADIUS Accounting ☐ Probe Response ☐ FTM ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

IPv6 Address: ::/0

IPv6 Administrative Access: ☐ HTTPS ☐ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ Security Fabric Connection ☐

OK Cancel

- Click **OK**.
- Click **Managed FortiGate > ISFW > Network > Interfaces**.
- Click **Create New**, and then select **VDOM Link**.
- Configure the following settings:

| Field                 | Value                        |
|-----------------------|------------------------------|
| Name                  | C                            |
| Virtual Domain        | root                         |
| IP/Netmask            | 172.16.1.249/255.255.255.252 |
| Administrative Access | PING                         |
| IPv6 Address          | ::/0                         |
| Virtual Domain        | Zone2                        |
| IP/Netmask            | 172.16.1.250/255.255.255.252 |
| Administrative Access | PING                         |
| IPv6 Address          | ::/0                         |

Create New VDOM Link

Name: C

Interface 0 (C)

Virtual Domain: root

IP/Netmask: 172.16.1.249/255.255.255.252

Administrative Access: ☐ HTTPS ☒ PING ☐ SSH ☐ SNMP ☐ TELNET ☐ FMG-Access ☐ RADIUS Accounting ☐ Probe Response ☐ FIM ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

IPv6 Address: ::/0

IPv6 Administrative Access: ☐ HTTPS ☐ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

Comments: Write a comment

Administrative Status: ☒

Interface 1 (C)

Virtual Domain: Zone2

IP/Netmask: 172.16.1.250/255.255.255.252

Administrative Access: ☐ HTTPS ☒ PING ☐ SSH ☐ SNMP ☐ TELNET ☐ FMG-Access ☐ RADIUS Accounting ☐ Probe Response ☐ FIM ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

IPv6 Address: ::/0

IPv6 Administrative Access: ☐ HTTPS ☐ PING ☐ SSH ☐ SNMP ☐ HTTP ☐ TELNET ☐ FMG-Access ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

OK Cancel

12. Click **OK**.

13. Click **Network > Interfaces**.

Your interface configuration should look like the following image:



Dashboard
Network
Interfaces
DNS
IPAM
Local Out Routing
System
Log & Report
CLI Configurations
Feature Visibility
Toggle Horizontal Menu

Create New
Edit
Edit in CLI
Delete
Where Used
More
Full Screen

|               | #  | Name | Type      | IP/Netmask                   | Virtual Domain | Normalized Interface |
|---------------|----|------|-----------|------------------------------|----------------|----------------------|
| Physical (5)  |    |      |           |                              |                |                      |
| Aggregate (1) |    |      |           |                              |                |                      |
| Tunnel (9)    |    |      |           |                              |                |                      |
| VDOM Link (6) |    |      |           |                              |                |                      |
|               | 16 | A0   | VDOM Link | 172.16.1.253/255.255.255.252 | root           | A0                   |
|               | 17 | A1   | VDOM Link | 172.16.1.254/255.255.255.252 | Zone1          | A1                   |
|               | 18 | B0   | VDOM Link | 172.16.1.245/255.255.255.252 | Zone1          | B0                   |
|               | 19 | B1   | VDOM Link | 172.16.1.246/255.255.255.252 | Zone2          | B1                   |
|               | 20 | C0   | VDOM Link | 172.16.1.249/255.255.255.252 | root           | C0                   |
|               | 21 | C1   | VDOM Link | 172.16.1.250/255.255.255.252 | Zone2          | C1                   |

## Stop and think!

You have created inter-VDOM links, which act like physical interfaces between your virtual firewalls.

Also, the A0, A1, B0, B1, C0, and C1 normalized interfaces have been created for you. Therefore, FortiManager automatically associates the normalized interfaces with the interfaces you have created.

## Create Static Routes in VDOMs

You will create static routes in the Zone1, Zone2, and root VDOMs.

## To create static routes in the Zone1 VDOM

1. Click **Managed FortiGate > HQ-ISFW > Zone1 > Network > Static Routes**.
2. Click **Create New**, and then select **IPv4 Static Route**.
3. Configure the following settings:

| Field           | Value           |
|-----------------|-----------------|
| Destination     | 0.0.0.0/0.0.0.0 |
| Gateway Address | 172.16.1.253    |
| Interface       | A1              |

4. Click **OK**.

You will see the following message:

```
Using default destination route 0.0.0.0 0.0.0.0. Are you sure you want to
inject default routes?
```

5. Click **OK**.
6. Click **Managed FortiGate > HQ-ISFW > Zone1 > Network > Static Routes**.
7. Click **Create New**, and then select **IPv4 Static Route**.
8. Configure the following settings:

| Field           | Value                  |
|-----------------|------------------------|
| Destination     | 10.0.3.0/255.255.255.0 |
| Gateway Address | 172.16.1.246           |
| Interface       | B0                     |

9. Click **OK**.

You should see the static routes shown in the following image in the Zone1 VDOM:

| Seq. # | Destination            | Gateway      | Interface | Distance | Priority |
|--------|------------------------|--------------|-----------|----------|----------|
| 1      | 0.0.0.0/0.0.0.0        | 172.16.1.253 | A1        | 10       | 1        |
| 2      | 10.0.3.0/255.255.255.0 | 172.16.1.246 | B0        | 10       | 1        |

**Stop and think!**

You will push these changes once you assign the policy package to all VDOMs. This way, you will save time by installing these changes once instead of twice. If you do it the other way, you would install the device settings, and then install the policy packages.

**To create a static route in the Zone2 VDOM**

1. Click **Managed FortiGate > HQ-ISFW > Zone2 > Network > Static Routes**.
2. Click **Create New**, and then select **IPv4 Static Route**.
3. Configure the following settings:

| Field           | Value                  |
|-----------------|------------------------|
| Destination     | 10.0.2.0/255.255.255.0 |
| Gateway Address | 172.16.1.245           |
| Interface       | B1                     |

4. Click **OK**.

You should see the static route shown in the following image in the Zone2 VDOM:

| Seq. # | Destination            | Gateway      | Interface | Distance | Priority |
|--------|------------------------|--------------|-----------|----------|----------|
| 1      | 10.0.2.0/255.255.255.0 | 172.16.1.245 | B1        | 10       | 1        |

### To create a static route in the root VDOM

- Click **Managed FortiGate > HQ-ISFW > root > Network > Static Routes**.
- Click **Create New**, and then select **IPv4 Static Route**.
- Configure the following settings:

| Field           | Value                  |
|-----------------|------------------------|
| Destination     | 10.0.2.0/255.255.255.0 |
| Gateway Address | 172.16.1.254           |
| Interface       | A0                     |

- Click **OK**.

You should see the static routes shown in the following image in the root VDOM:

|                       |  |                   |  |              |  |                         |  |              |  |           |  |          |  |
|-----------------------|--|-------------------|--|--------------|--|-------------------------|--|--------------|--|-----------|--|----------|--|
| Search...             |  | Dashboard         |  | + Create New |  | Edit                    |  | Edit in CLI  |  | Clone     |  | Delete   |  |
| Managed FortiGate (9) |  | Network           |  | Seq. #       |  | Destination             |  | Gateway      |  | Interface |  | Distance |  |
| BR2-FGT-1             |  | Interfaces        |  | 1            |  | 0.0.0.0/0.0.0.0         |  | 10.0.12.254  |  | port3     |  | 10       |  |
| BR3-FGT-1             |  | Local Out Routing |  | 2            |  | 10.0.5.0/255.255.255.0  |  | 10.0.12.253  |  | port3     |  | 10       |  |
| HQ-DCFW               |  | Static Routes     |  | 3            |  | 4.2.2.2/255.255.255.255 |  | 10.0.11.254  |  | port2     |  | 10       |  |
| HQ-ISFW (3)           |  | System            |  | 4            |  | 10.0.2.0/255.255.255.0  |  | 172.16.1.254 |  | A0        |  | 10       |  |
| mnt                   |  | Log & Report      |  |              |  |                         |  |              |  |           |  |          |  |



You could install device settings only to all VDOMs. But, since you still need to push policy changes, in this example, you will add the policies, and then install the policy package and device settings.

## Install Firewall Policies in VDOMs

You will run a script on the **Policy Package** or **ADOM Database** target for the **root\_ISFW** policy package. Then, you will install the associated policy packages in the Zone1, Zone2, and root VDOMs.

### To run a script on the Policy Package or ADOM Database target for the root\_ISFW policy package

- Click **Device Manager > Scripts**.
- Select the **root\_ISFW** checkbox, and then click **Run Script**.

|                        |  |                  |  |      |  |                                     |  |            |  |                   |  |              |  |      |  |
|------------------------|--|------------------|--|------|--|-------------------------------------|--|------------|--|-------------------|--|--------------|--|------|--|
| Dashboard              |  | + Create New     |  | Edit |  | Delete                              |  | Run Script |  | Import CLI Script |  | AI Assistant |  | More |  |
| Device Manager         |  | Name             |  | Type |  | Target                              |  |            |  | Comments          |  |              |  |      |  |
| Device & Groups        |  | Script (4)       |  |      |  |                                     |  |            |  |                   |  |              |  |      |  |
| Scripts                |  | ACME Certificate |  | CLI  |  | Remote FortiGate Directly (via CLI) |  |            |  |                   |  |              |  |      |  |
| Provisioning Templates |  | Firewall rule    |  | CLI  |  | Policy Package or ADOM Database     |  |            |  |                   |  |              |  |      |  |
| Firmware Templates     |  | Static Route     |  | CLI  |  | Device Database                     |  |            |  |                   |  |              |  |      |  |
| Monitors               |  | root_ISFW        |  | CLI  |  | Policy Package or ADOM Database     |  |            |  |                   |  |              |  |      |  |

3. In the **Run script on policy package** field, select **root\_ISFW**, and then click **Run Now**.



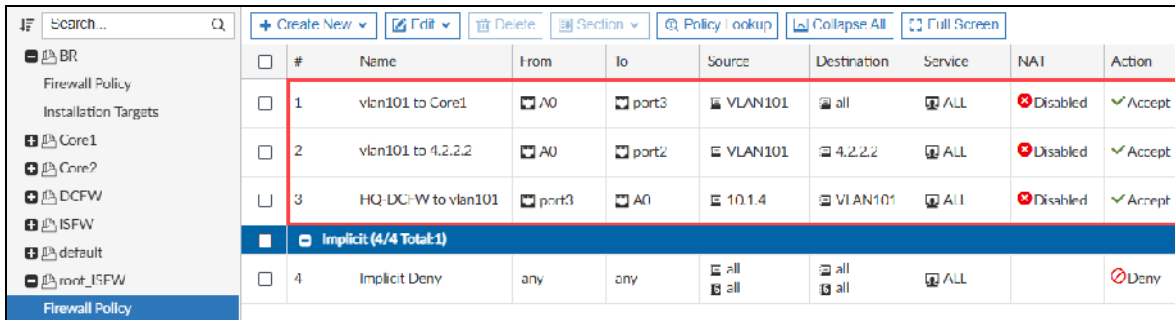
Run Script - root\_ISFW

Run script on policy package: **root\_ISFW**

Skip Resolving Variables: ☐

**Run Now** **Cancel**

4. Click **Close** to close the **Run Script** window.
5. Click **Policy & Objects > Policy Packages > root\_ISFW > Firewall Policy** to validate the firewall policies created in the **root\_ISFW** policy package.



| #                             | Name               | From  | To    | Source  | Destination | Service | NAT      | Action |
|-------------------------------|--------------------|-------|-------|---------|-------------|---------|----------|--------|
| 1                             | vlan101 to Core1   | A0    | port3 | VLAN101 | all         | ALL     | Disabled | Accept |
| 2                             | vlan101 to 4.2.2.2 | A0    | port2 | VLAN101 | 4.2.2.2     | ALL     | Disabled | Accept |
| 3                             | HQ-DC:W to vlan101 | port3 | A0    | 10.1.4  | VLAN101     | ALL     | Disabled | Accept |
| <b>Implicit (4/4 Total:1)</b> |                    |       |       |         |             |         |          |        |
| 4                             | Implicit Deny      | any   | any   | all     | all         | ALL     |          | Deny   |

6. Click **Install Wizard** to install policies in the root VDOM.
7. In the **Choose What to Install (1/4)** window, select the **root\_ISFW** policy package, and then click **Next**.

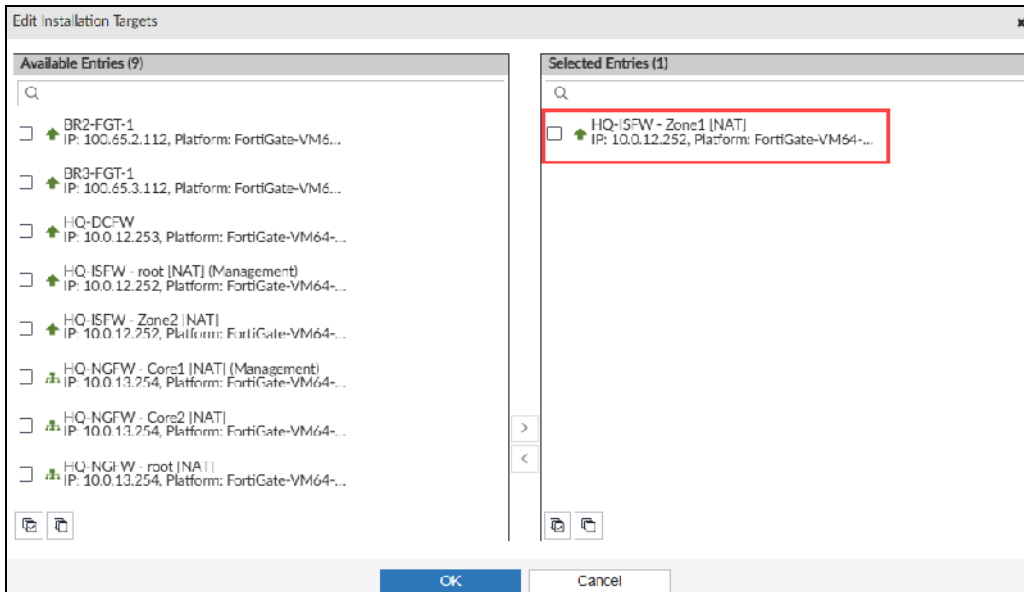


Because this is the first time installation is being done from FortiManager to these VDOMs, you will see more than the policies being pushed to the VDOMs. After you install this the first time, consecutive installations should show only the changes you applied to the policy package or device database installations.

8. In the **Select Devices to Install (root\_ISFW) (2/4)** window, ensure that the **root [NAT] (Management)** VDOM is selected, and then click **Next**.
9. In the **Validate Devices (root\_ISFW) (3/4)** window, click **Install**.
10. In the **Installation Progress (root\_ISFW) (4/4)** window, click **Finish**.

### To configure firewall policies in the Zone1 VDOM

1. Click **Policy & Objects > Policy Packages > Zone1 > Installation Targets > Edit**.
2. In the **Available Entries** field, select the **HQ-ISFW - Zone1** checkbox, move it to the **Selected Entries** field, and then click **OK**.



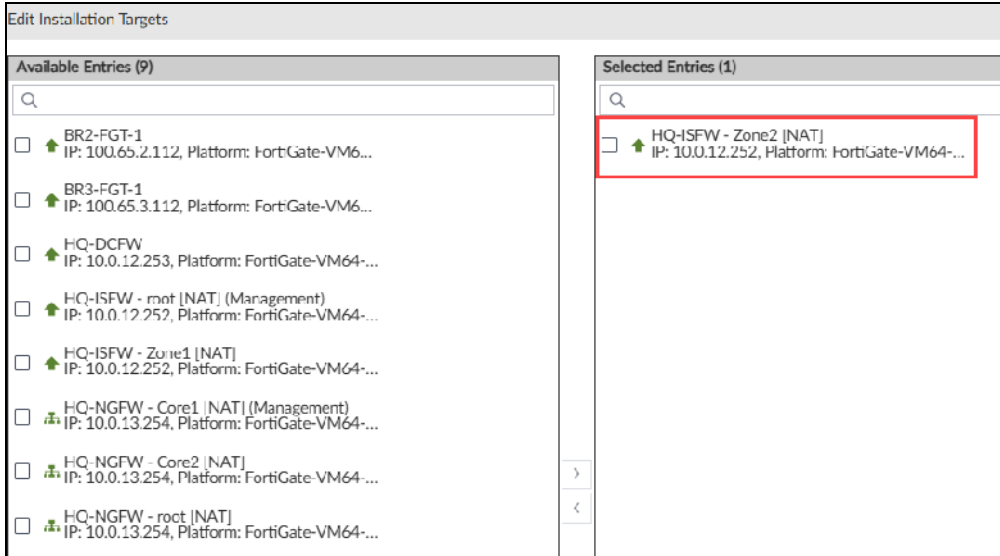
3. Click **Zone1 > Firewall Policy**, and then verify the firewall policies that will be installed.

| <div> <div>IF Search...</div> <div> <div>+ Create New</div> <div>Edit</div> <div>Delete</div> <div>Section</div> <div>Policy Lookup</div> <div>Collapse All</div> <div>Full Screen</div> </div> </div> |                        |                    |         |         |         |             |         |          |        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|---------|---------|---------|-------------|---------|----------|--------|--|
|                                                                                                                                                                                                        | #                      | Name               | From    | To      | Source  | Destination | Service | NAT      | Action |  |
|                                                                                                                                                                                                        | 1                      | HQ-DCFV to vlan101 | A1      | VLAN101 | 10.0.5  | VLAN101     | ALL     | Disabled | Accept |  |
|                                                                                                                                                                                                        | 2                      | vlan101 to vln102  | VLAN101 | B0      | VLAN101 | VLAN102     | ALL     | Disabled | Accept |  |
|                                                                                                                                                                                                        | 3                      | vln102 to vln101   | B0      | VLAN101 | VLAN102 | VLAN101     | ALL     | Disabled | Accept |  |
|                                                                                                                                                                                                        | 4                      | vln101 to root     | VLAN101 | A1      | VLAN101 | all         | ALL     | Disabled | Accept |  |
|                                                                                                                                                                                                        | Implicit (5/5 Total:1) |                    |         |         |         |             |         |          |        |  |
|                                                                                                                                                                                                        | 5                      | Implicit: Deny     | any     | any     | all     | all         | ALL     |          | Deny   |  |

4. Click **Install Wizard** to install the policies in the Zone1 VDOM.
5. In the **Choose What to Install (1/4)** window, select the **Zone1** policy package, and then click **Next**.
6. In the **Select Devices to Install (Zone1) (2/4)** window, ensure that the **Zone1 [NAT]** VDOM is selected, and then click **Next**.
7. In the **Validate Devices (Zone1) (3/4)** window, click **Install**.
8. In the **Installation Progress (Zone1) (4/4)** window, click **Finish**.

## To configure firewall policies in the Zone2 VDOM

1. Click **Policy & Objects > Policy Packages > Zone2 > Installation Targets > Edit**.
2. In the **Available Entries** field, select the **HQ-ISFW - Zone2** checkbox, move it to the **Selected Entries** field, and then click **OK**.



- Click **Zone2 > Firewall Policy**, and then verify the firewall policies that will be installed.

Search

Create New

Edit

Delete

Section

Policy Lookup

Collapse All

Full Screen

BR

Core1

Core2

DCFV

ISFW

Zone1

Zone2

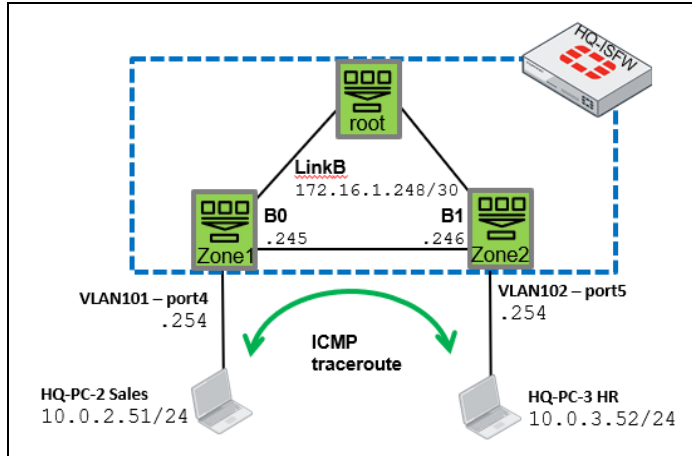
Firewall Policy

|                          | # | Name               | From    | To      | Source  | Destination | Service | NAT      | Action |
|--------------------------|---|--------------------|---------|---------|---------|-------------|---------|----------|--------|
| <input type="checkbox"/> | 1 | vlan101 to vlan102 | B1      | VLAN102 | VLAN101 | VLAN102     | ALL     | Disabled | Accept |
| <input type="checkbox"/> | 2 | vlan102 to vlan101 | VLAN102 | B1      | VLAN102 | all         | ALL     | Disabled | Accept |
| Implicit (3/3 Total)     |   |                    |         |         |         |             |         |          |        |
| <input type="checkbox"/> | 3 | Implicit Deny      | any     | any     | all     | all         | All     |          | Deny   |

- Click **Install Wizard** to install the policies in the Zone2 VDOM.
- In the **Choose What to Install (1/4)** window, select the **Zone2** policy package, and then click **Next**.
- In the **Select Devices to Install (Zone2) (2/4)** window, ensure that the **Zone2 [NAT]** VDOM is selected, and then click **Next**.
- In the **Validate Devices (Zone2) (3/4)** window, click **Install**.
- In the **Installation Progress (Zone2) (4/4)** window, click **Finish**.

## Validate the Communication Between HQ-PC-2 and HQ-PC-3

You will validate the communication between HQ-PC-2 and HQ-PC-3.



### To validate the communication from HQ-PC-3 to HQ-PC-2

1. Connect to HQ-PC-3 with the following credentials:
  - Username: Administrator
  - Password: Fortinet1!
2. Open a terminal window.
3. Enter the following command to ping HQ-PC-2:  
`ping 10.0.2.51`
4. Enter the following command to validate the traceroute to HQ-PC-2:  
`traceroute 10.0.2.51`

You should see the following jumps:

```
1 10.0.3.254
2 172.16.1.245
3 10.0.2.51
```

### To validate the communication from HQ-PC-2 to HQ-PC-3

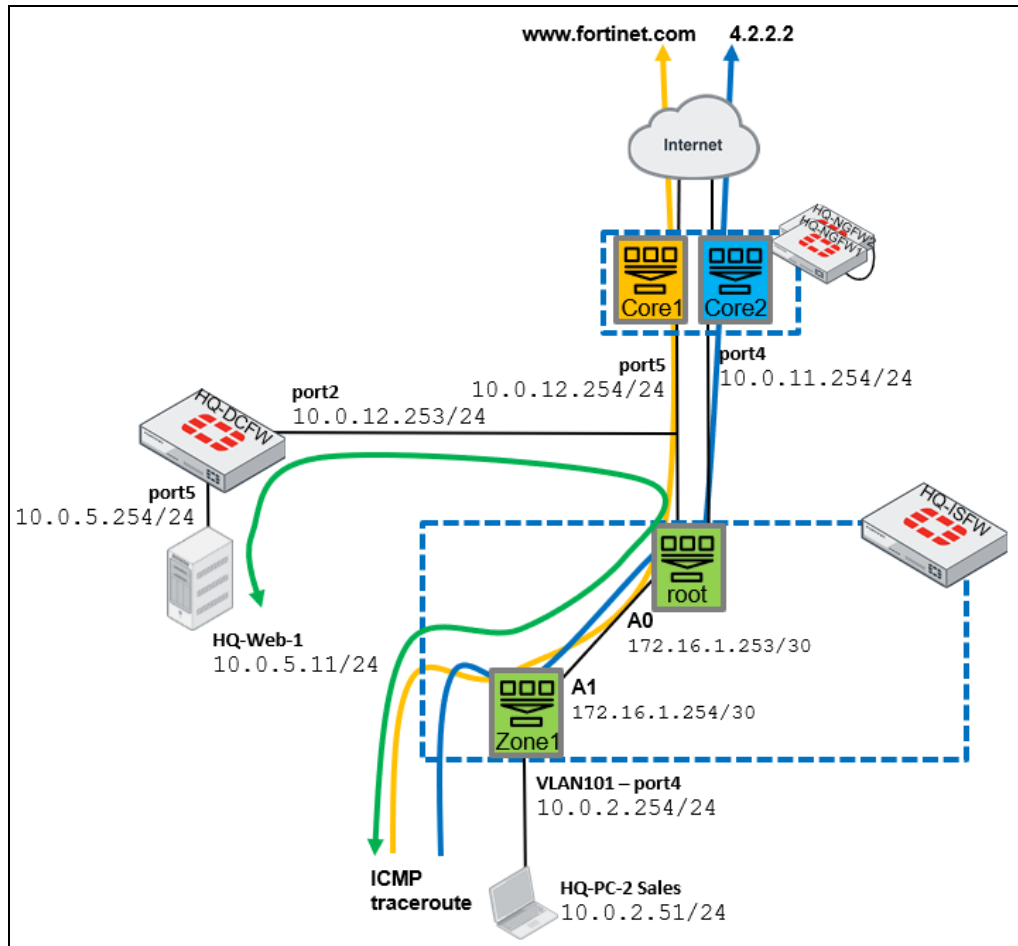
1. Connect to HQ-PC-2 with the following credentials:
  - Username: Administrator
  - Password: Fortinet1!
2. Open a terminal window.
3. Enter the following command to ping HQ-PC-3:  
`ping 10.0.3.52`
4. Enter the following command to validate the traceroute to HQ-PC-3:  
`traceroute 10.0.3.52`

You should see the following jumps:

```
1 10.0.2.254
2 172.16.1.246
3 10.0.3.52
```

## Validate the Communication From HQ-PC-2 to HQ-Web-1, the Internet, and 4.2.2.2

You will validate that HQ-PC-2 can communicate with HQ-Web-1 and the internet. You will notice in the traceroute the IP address of the virtual interface—172.16.1.253—that was created using the inter-VDOM link.



### To validate the communication from HQ-PC-2 to HQ-Web-1, the internet, and 4.2.2.2

1. Continuing on HQ-PC-2, enter the following commands to ping HQ-Web-1, the internet (www.fortinet.com), and 4.2.2.2:  

```
ping 10.0.5.11
```

```
ping www.fortinet.com
```

```
ping 4.2.2.2
```
2. Enter the following commands to validate the traceroute to HQ-Web-1, the internet (www.fortinet.com), and 4.2.2.2:  

```
traceroute 10.0.5.11
```

```
traceroute www.fortinet.com
```



```
traceroute 4.2.2.2
```

You should see the following jumps for 10.0.5.11:

```
1 10.0.2.254
2 172.16.1.253
3 10.0.12.253
4 10.0.5.11
```

You should see the following jumps for `www.fortinet.com`:

```
1 10.0.2.254
2 172.16.1.253
3 10.0.12.254
4 ...
```

You should see the following jumps for 4.2.2.2:

```
1 10.0.2.254
2 172.16.1.253
3 10.0.11.254
4 ...
```



Notice that the root VDOM will send the traffic to either HQ-DCFW (10.0.12.253), Core1 (10.0.12.254) or Core2 (10.0.11.254), depending on the destination IP address.

## Lab 4: High Availability

In this lab, you will configure VDOM partitioning. Then, you will configure a FortiGate Session Life Support Protocol (FGSP) cluster inspecting asymmetric traffic with a layer 3 connection, and encrypt the session synchronization.

### Objectives

- Configure virtual clustering on HQ-NGFW-1 and HQ-NGFW-2
- Analyze session handling in VDOM partitioning
- Configure an FGSP cluster with BR1-FGT-1 and BR1-FGT-2 over a layer 3 connection
- Analyze asymmetric traffic across the FGSP cluster
- Configure encryption for the session synchronization

### Time to Complete

Estimated: 55 minutes

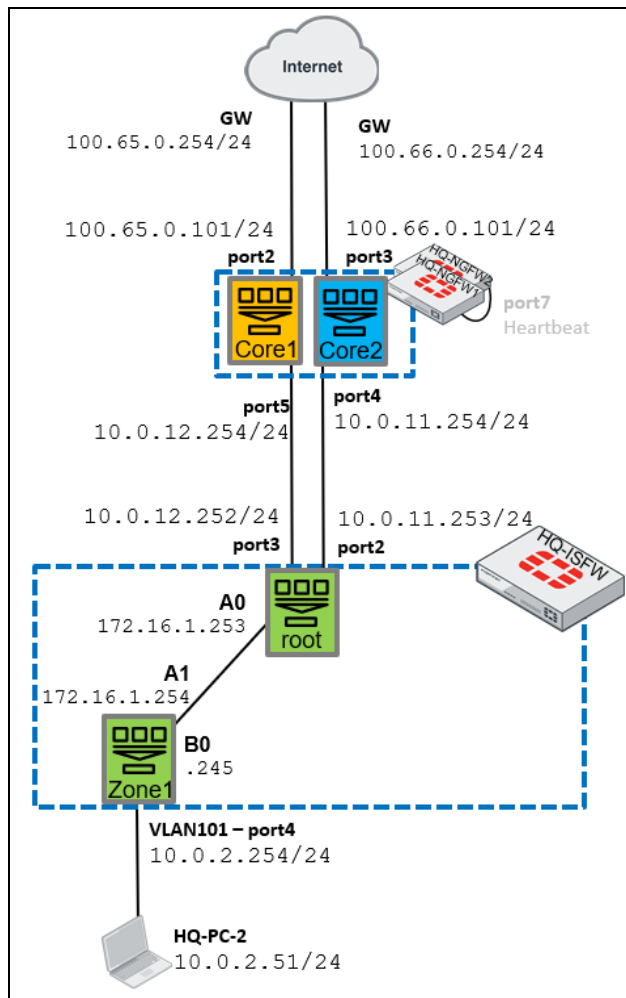
### Prerequisites

Before you begin this lab, you must complete the previous lab. If you haven't done so, tell your instructor.

## Exercise 1: Configuring VDOM Partitioning

In this exercise, you will configure a virtual cluster between HQ-NGFW-1 and HQ-NGFW-2. HA is configured in active-passive mode between HQ-NGFW-1 and HQ-NGFW-2. Using VDOM partitioning, you will configure virtual clustering between the Core1 and Core2 VDOMs in a way that traffic for Core1 will be processed by HQ-NGFW-1, and traffic for Core2 will be processed by HQ-NGFW-2.

### Network Topology

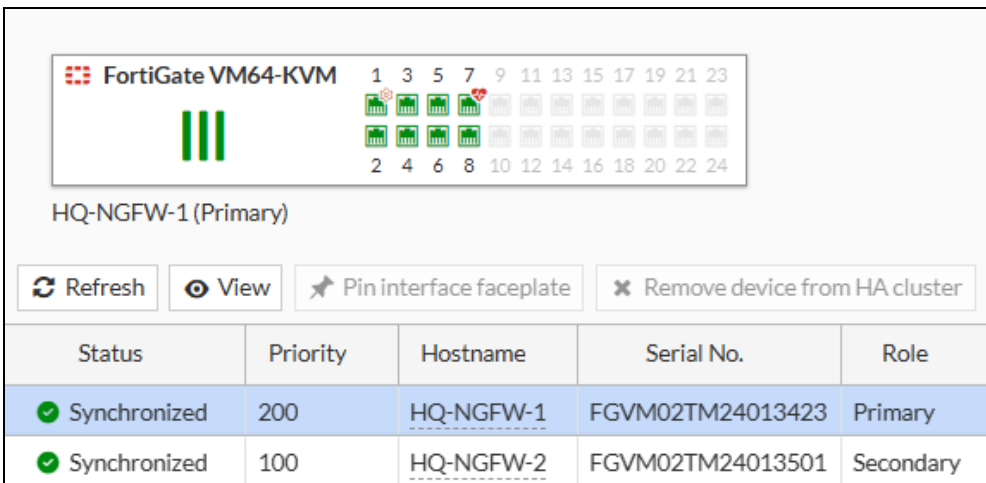


### Verify the HA Status

Before you configure VDOM partitioning, you will check the HA synchronization status between HQ-NGFW-1 and HQ-NGFW-2.

## To check the HA status

1. Connect to the HQ-NGFW-1 GUI.
2. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
3. Click **Login Read-Only**.
4. Click **System > HA**, and then analyze the information displayed.



Two green check marks indicate both devices are synchronized with each other. HQ-NGFW-1 is acting as the primary device and HQ-NGFW-2 is acting as a secondary device.



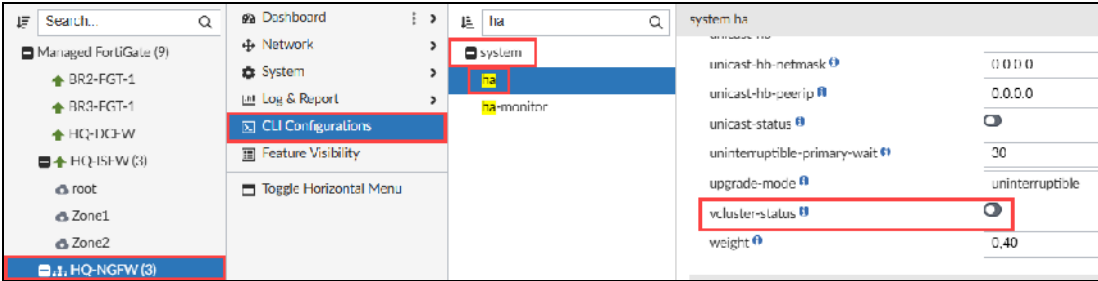
Ensure that both devices are in sync before moving to the next task.

## Configure VDOM Partitioning

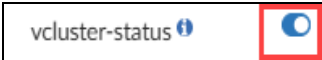
You will configure Core1 for virtual cluster 1 and Core2 for virtual cluster 2 using FortiManager.

## To configure VDOM partitioning

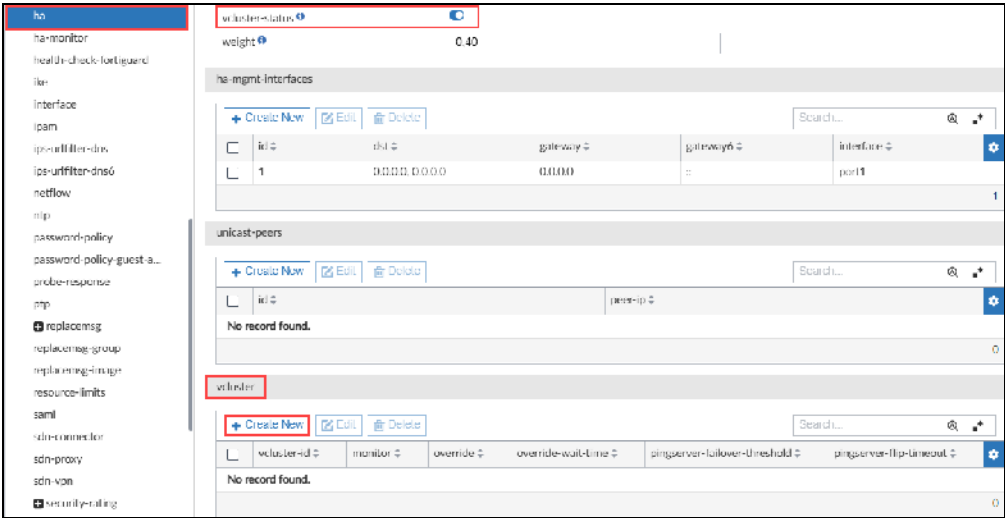
1. Connect to the FortiManager GUI.
2. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
3. Click **EFW**.
4. Click **Device Manager > Device & Groups**.
5. Click **Managed FortiGate > HQ-NGFW > CLI Configurations**.
6. Expand **system**.



- 7. Click **ha**.
- 8. Enable **vcluster-status**.



- 9. In the **vcluster** section, click **Create New**.



- 10. Configure the first virtual cluster with the following settings:

| Field       | Value       |
|-------------|-------------|
| vcluster-id | 1           |
| priority    | 200         |
| vdom        | root, Core1 |

- 11. Click **OK** to save the VDOMs in virtual cluster 1.
- 12. Click **OK** to save the virtual cluster settings.
- 13. In the **vcluster** section, click **Create New**.
- 14. Configure the second virtual cluster with the following settings:

| Field       | Value |
|-------------|-------|
| vcluster-id | 2     |
| priority    | 50    |
| vdom        | Core2 |

15. Click **OK** to save the VDOM in virtual cluster 2.

16. Click **OK** to save the virtual cluster settings.

The configuration should be similar to the following example:

| vcluster                                                                                                                                                                                                                                                            |             |         |                                                                                           |          |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------------------------------------------------------------------------------------------|----------|-------------|
| <div> <div>+ Create New</div> <div>Edit</div> <div>Delete</div> <div>Search...  </div> </div> |             |         |                                                                                           |          |             |
| <input type="checkbox"/>                                                                                                                                                                                                                                            | vcluster-id | monitor | override                                                                                  | priority | vdom        |
| <input type="checkbox"/>                                                                                                                                                                                                                                            | 1           |         |  disable | 200      | Core1, root |
| <input type="checkbox"/>                                                                                                                                                                                                                                            | 2           |         |  disable | 50       | Core2       |



You might need to click the gear icon in the upper-right corner to enable the **priority** and **vdom** columns. Once they appear in the table, move them to the left.

17. Click **Apply**.

18. In the menu at the top, click **Install Wizard**.

19. Select **Install Device Settings (only)**.

20. Click **Next**.

21. Confirm that **HQ-NGFW** is selected, and then click **Next**.

22. Click **Install**.

23. Wait until the installation finishes, and then click **Finish**.

### To verify the virtual clustering status

1. Return to the HQ-NGFW-1 GUI, and then click **System > HA**.

Your HA configuration should look like the following example:

| Status              | Priority | Hostname  | Serial No.       | Virtual Domains | Role      |
|---------------------|----------|-----------|------------------|-----------------|-----------|
| Virtual cluster 1 2 |          |           |                  |                 |           |
| Synchronized        | 200      | HQ-NGFW-1 | FGVM02TM24013423 | Core1<br>root   | Primary   |
| Synchronized        | 100      | HQ-NGFW-2 | FGVM02TM24013501 | Core1<br>root   | Secondary |
| Virtual cluster 2 2 |          |           |                  |                 |           |
| Synchronized        | 50       | HQ-NGFW-1 | FGVM02TM24013423 | Core2           | Secondary |
| Synchronized        | 128      | HQ-NGFW-2 | FGVM02TM24013501 | Core2           | Primary   |

You should see HQ-NGFW-1 as the primary for virtual cluster 1 and HQ-NGFW-2 as the primary for virtual cluster 2.

### Stop and think!

How did HQ-NGFW-2 become the primary for virtual cluster 2?

The default priority for virtual clusters is 128. On HQ-NGFW-1, you configured this priority as 50, which is lower than the default value. Because the priority is the deciding factor in selecting a primary device in this setup, the device with the highest priority is elected as the primary for the cluster.

## Analyze Traffic Distribution

You will generate traffic by running a continuous ping from HQ-PC-2, and then use the sniffer to analyze traffic distribution.

### To generate traffic

1. On HQ-PC-2, open a terminal session, and then enter the following command to start a continuous ping to 8.8.8.8:  
`ping 8.8.8.8`
2. Open a new terminal session, and then enter the following command to start a continuous ping to 4.2.2.2:  
`ping 4.2.2.2`
3. Keep both terminal windows open and leave the pings running.

### To analyze traffic for virtual cluster 1

1. Connect over SSH to HQ-NGFW-1.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Enter the following commands to view the active ICMP sessions on Core1:

```
config vdom
edit Core1
get system session list | grep icmp
```

You should see the session information for the continuous ping going to 8.8.8.8.

```
HQ-NGFW-1 (Core1) # get system session list | grep icmp
icmp      59      10.0.2.51:1      100.65.0.101:5118 8.8.8.8:8
```

### To analyze traffic for virtual cluster 2

1. Connect over SSH to HQ-NGFW-1.
2. Log in with the following credentials:
  - Username: admin
  - Password: password
3. Enter the following commands to view the active ICMP sessions on Core2:

```
config vdom
edit Core2
get system session list | grep icmp
```

#### Stop and think!

Why do you not see traffic destined for 4.2.2.2?

You are sending 4.2.2.2 traffic through Core2, which belongs to virtual cluster 2. HQ-NGFW-2 is the primary device for virtual cluster 2. You must connect to HQ-NGFW-2 to see traffic destined for 4.2.2.2.

4. Continuing in the same SSH session, enter the following commands to connect to HQ-NGFW-2:

```
end
config global
execute ha manage 0 admin
```

5. At the prompt, type the password `Fortinet1!`, and then press `Enter`.
6. Enter the following commands to view the active ICMP sessions on Core2:

```
config vdom
edit Core2
get system session list | grep icmp
```

You should see session information for the continuous ping going to 4.2.2.2.

```
HQ-NGFW-2 (Core2) # get system session list | grep icmp
icmp      59      10.0.2.51:2      100.66.0.101:5119 4.2.2.2:8
```

7. Close the SSH window.

## Perform a Failover and Analyze the Traffic

You will perform a manual failover for virtual cluster 2 by increasing the priority for HQ-NGFW-1. Then, you will analyze the traffic.

### To perform a manual failover

1. Continuing on the FortiManager GUI, click **Device Manager > Device & Groups**.
2. Click **Managed FortiGate > HQ-NGFW > CLI Configurations**.
3. Expand **system**.
4. Click **ha**.



- In the **vcluster** section, select the checkbox for the row with a value of **2** in the **vcluster-id** column, and then click **Edit**.

| vcluster                                                               |             |         |          |          |             |
|------------------------------------------------------------------------|-------------|---------|----------|----------|-------------|
| <div> <div>+ Create New</div> <div>Edit</div> <div>Delete</div> </div> |             |         |          |          |             |
| <input type="checkbox"/>                                               | vcluster-id | monitor | override | priority | vdom        |
| <input type="checkbox"/>                                               | 1           |         | disable  | 200      | Core1, root |
| <input checked="" type="checkbox"/>                                    | 2           |         | disable  | 50       | Core2       |

- In the **priority** field, type 150.
- Click **OK**.
- Click **Apply**.
- In the menu at the top, click **Install Wizard**.
- Select **Install Device Settings (only)**.
- Click **Next**.
- Confirm that **HQ-NGFW** is selected, and then click **Next**.
- Click **Install**.
- Wait until the installation finishes, and then click **Finish**.

### To verify the failover

- Continuing on the HQ-NGFW-1 GUI, click **System > HA**.

After you change the priority, you should see HQ-NGFW-1 as the primary for both virtual clusters.

| Virtual cluster 1 2 |     |           |                  |               |           |
|---------------------|-----|-----------|------------------|---------------|-----------|
| Synchronized        | 200 | HQ-NGFW-1 | FGVM02TM24013423 | Core1<br>root | Primary   |
| Synchronized        | 100 | HQ-NGFW-2 | FGVM02TM24013501 | Core1<br>root | Secondary |
| Virtual cluster 2 2 |     |           |                  |               |           |
| Synchronized        | 150 | HQ-NGFW-1 | FGVM02TM24013423 | Core2         | Primary   |
| Synchronized        | 128 | HQ-NGFW-2 | FGVM02TM24013501 | Core2         | Secondary |

### To analyze traffic after a failover

- Connect over SSH to HQ-NGFW-1.
- Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
- Enter the following commands to view the active ICMP sessions on Core2:

```
config vdom
edit Core2
get system session list | grep icmp
```

You should see that the session for 4.2.2.2 is now active on NGFW-1.

```
HQ-NGFW-1 (Core2) # get system session list | grep icmp  
icmp    59      10.0.2.51:2      100.66.0.101:5119 4.2.2.2:8
```



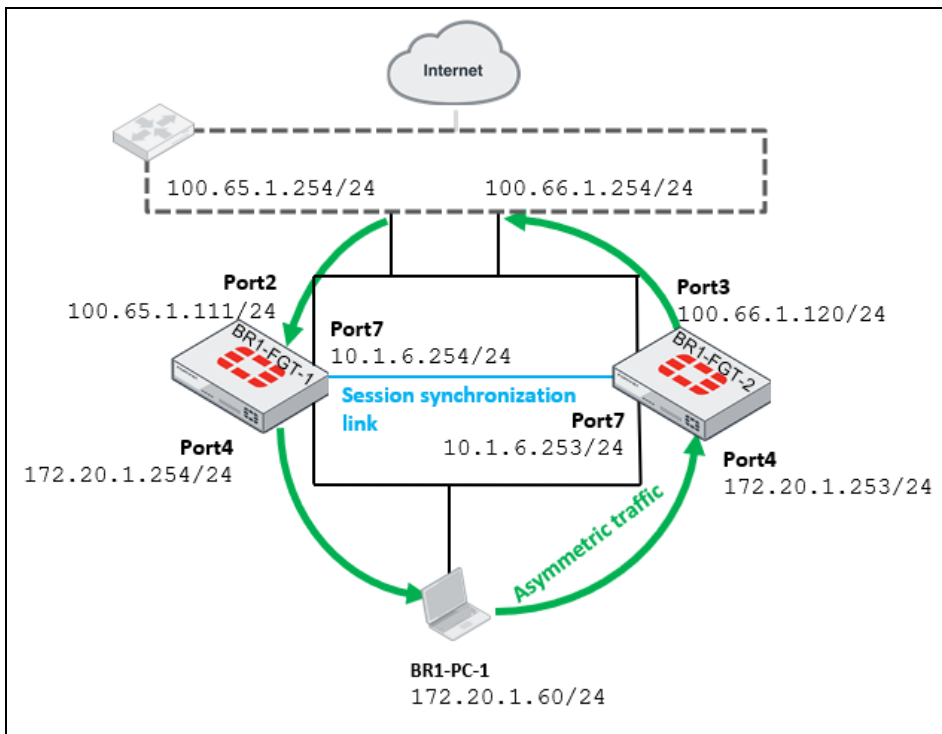
After the failover, HQ-NGFW-1 handles all the traffic for the Core1 and Core2 VDOMs.

4. Close the SSH window.
5. On HQ-PC-2, close the terminal windows.

## Exercise 2: Configuring FGSP

In this exercise, you will configure FGSP on BR1-FGT-1 and BR1-FGT-2, which handle asymmetric traffic. Port7 on BR1-FGT-1 and port7 on BR1-FGT-2 are both in the same LAN subnet. You will apply the session synchronization using a layer 3 connection and analyze the asymmetric traffic.

### Network Topology



### Configure FGSP

Before you test the session synchronization, you will configure FGSP between BR1-FGT-1 and BR1-FGT-2.

#### To configure FGSP on BR1-FGT-1

1. Connect over SSH to BR1-FGT-1.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Enter the following CLI commands to enable FGSP on BR1-FGT-1:

```
config system standalone-cluster
set standalone-group-id 5
```

```
set group-member-id 1
config cluster-peer
edit 1
set peerip 10.1.6.253
next
end
end
```

4. Type `y`.
5. Leave the SSH session open.

### To configure FGSP on BR1-FGT-2

1. Connect over SSH to BR1-FGT-2.
2. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
3. Enable FGSP on BR1-FGT-2 following the procedure you used for BR1-FGT-1 and using the following information:

| Command                          | Value      |
|----------------------------------|------------|
| <code>standalone-group-id</code> | 5          |
| <code>group-member-id</code>     | 2          |
| <code>peerip</code>              | 10.1.6.254 |



The `standalone-group-id` must be the same value for all members. The ID can be a value from 1–255. The `group-member-id` can be a value from 1–15 and must be different for each member in the same group.

## Test ICMP Session Synchronization Between BR1-FGT-1 and BR1-FGT-2

You will test ICMP session synchronization.

### To enable ICMP session synchronization

1. Connect over SSH to BR1-FGT-1.
2. Enter the following CLI commands to enable ICMP session synchronization on BR1-FGT-1:

```
config system ha
set session-pickup enable
set session-pickup-connectionless enable
end
```
3. Leave the SSH session open.

### To test ICMP session synchronization

1. On BR1-PC-1, open a terminal window, type `ping 35.183.176.123`, and then press `Enter`.
2. Do not close the terminal window.

3. Connect over SSH to BR1-FGT-2.
4. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
5. Enter the following command to view the session:  
`get system session list | grep icmp`  
The result should be similar to the following output:

```
BR1-FGT-2 # get sys session list | grep icmp
icmp      59      172.20.1.60:6      -      35.183.176.123:8 -
```

6. Connect over SSH to BR1-FGT-1.
7. Enter the following command to view the session:  
`get system session list | grep icmp`  
The result should be similar to the following output:

```
BR1-FGT-1 # get system session list | grep icmp

BR1-FGT-1 #
```

#### Stop and think!

Why is the ICMP session not synchronized on BR1-FGT-1?

You must configure ICMP synchronization on both members.

FGSP does not synchronize the configuration by default. Here, the traffic is asymmetric and therefore the configuration, like the firewall policy, is asymmetric also. Therefore, you cannot use standalone configuration synchronization.

### To enable ICMP session synchronization on BR1-FGT-2

1. Connect over SSH to BR1-FGT-2.
2. Enter the following CLI commands to enable ICMP session synchronization on Remote-2:  

```
config system ha
    set session-pickup enable
    set session-pickup-connectionless enable
end
```
3. Leave the SSH session open.

### To confirm ICMP session synchronization

1. On BR1-PC-1, open a terminal window, type `ping www.fortinet.com`, and then press Enter.  
The result should be similar to the following output:

```
$ ping www.fortinet.com
PING ipv6.lb-2.ca-central-1.aws.waas-online.net (35.182.106.253) 56(84) bytes of
data.
64 bytes from ec2-35-182-106-253.ca-central-1.compute.amazonaws.com (35.182.106.253):
icmp_seq=1 ttl=50 time=15.2 ms
```

```
64 bytes from ec2-35-182-106-253.ca-central-1.compute.amazonaws.com (35.182.106.253):
icmp_seq=2 ttl=50 time=15.6 ms
64 bytes from ec2-35-182-106-253.ca-central-1.compute.amazonaws.com (35.182.106.253):
icmp_seq=3 ttl=50 time=15.9 ms
```

2. Connect over SSH to BR1-FGT-1.
3. Enter the following command to view the session:  
# get system session list | grep icmp  
The result should be similar to the following output:

```
BR1-FGT-1 # get system session list | grep icmp
icmp      59      172.20.1.60:8      -      35.182.106.253:8 -
```

4. On BR1-PC-1, close the terminal sessions.

## Analyze the Asymmetric Traffic

You will confirm the asymmetric path that FGSP allowed.

### To confirm the asymmetric path through the FGSP cluster

1. Connect over SSH to BR1-FGT-1.
2. Enter the following CLI command to sniff the HTTPS traffic:  
diagnose sniffer packet port2 'port 443' 4 6
3. Connect over SSH to BR1-FGT-2.
4. Enter the following CLI command to sniff the HTTPS traffic:  
diagnose sniffer packet port3 'port 443' 4 6
5. On BR1-PC-1, open a browser, and then enter the following URL:  
www.fortinet.com

6. Return to the BR1-FGT-1 SSH session.  
The result should be similar to the following output:

```
BR1-FGT-1 # diag sniff packet port2 'port 443' 4 6
Using Original Sniffing Mode
interfaces=[port2]
filters=[port 443]
3.084637 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: syn 4250996217 ack 501154715
3.084637 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: syn 4250996217 ack 501154715
3.094267 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: ack 501155393
3.094267 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: ack 501155393
3.095256 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996218 ack 501155393
3.095256 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996218 ack 501155393
3.103939 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996478 ack 501155473
3.103939 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996478 ack 501155473
3.104037 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: ack 501158369
3.104037 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: ack 501158369
3.120030 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996765 ack 501159082
3.120030 port2 -- 35.183.176.123.443 -> 172.20.1.60.60942: psh 4250996765 ack 501159082
```

7. Return to the BR1-FGT-2 SSH session.  
The result should be similar to the following output:

```
BR1-FGT-2 # diag sniff packet port3 'port 443' 4 6
Using Original Sniffing Mode
interfaces=[port3]
filters=[port 443]
9.136477 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: syn 501154714
9.136477 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: syn 501154714
9.143396 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: ack 4250996218
9.143396 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: ack 4250996218
9.146520 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: psh 501154715 ack 4250996218
9.146520 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: psh 501154715 ack 4250996218
9.153661 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: ack 4250996478
9.153661 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: ack 4250996478
9.155772 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: psh 501155393 ack 4250996478
9.155772 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: psh 501155393 ack 4250996478
9.156184 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: 501155473 ack 4250996478
9.156184 port3 -- 172.20.1.60:60942 -> 35.183.176.123:443: 501155473 ack 4250996478
```



The sniffer trace confirms that the traffic is exiting from port3 of BR1-FGT-2 and the return traffic is entering from port2 of BR1-FGT-1.

### Stop and think!

The firewall policy configured on BR1-FGT-1 allows only sessions from port4 to port2.

| Policy          | Source       | Destination | NAT      | Security Profiles | Log |
|-----------------|--------------|-------------|----------|-------------------|-----|
| port4 → port2 1 | Internet (1) | Internal    | Disabled | ssl no-inspection | UTM |
| Implicit 1      |              |             |          |                   |     |

How does BR1-FGT-1 allow the return traffic from port2?

FGSP synchronizes the session over the layer 3 connection, which means port7 on BR1-FGT-1 and BR1-FGT-2.

### To confirm the TCP session with asymmetric traffic that FGSP allowed

1. Connect to BR1-PC-1.
2. Open a browser, and then enter the following URL:  
www.fortinet.com
3. Return to the BR1-FGT-1 SSH session, and then enter the following command to view the session information:  
get system session list | grep 443  
The result should be similar to the following output:

```
BR1-FGT-1 # get system session list | grep 443
tcp      3593    172.20.1.60:57598 -          35.182.106.253:443 -
tcp      3593    172.20.1.60:57600 -          35.182.106.253:443 -
```

4. In the BR1-FGT-2 SSH session, enter the same command to view the session information:

```
get system session list | grep 443
```

The result should be similar to the following output:

```
BR1-FGT-2 # get system session list | grep 443
tcp      3596    172.20.1.60:57598 -          35.182.106.253:443 -
tcp      3596    172.20.1.60:57600 -          35.182.106.253:443 -
```

Both FortiGate devices show the same TCP session, allowing the asymmetric traffic with FGSP.



The above session lists are extracts because the lists have many entries.

You can use `http://portquiz.net:8080` instead of `www.fortinet.com`. You can use the `get system session list | grep 8080` command on the BR1-FGT-1 and BR1-FGT-2 SSH sessions to list the corresponding TCP sessions. Alternatively, you may use the HQ-FMG-1 VIP address (100.65.0.120) and the `get system session list | grep 100.65.0.120` command on the BR1-FGT-1 and BR1-FGT-2 SSH sessions.

If the session does not appear in the list, refresh the browser page.

### To confirm the TCP session FGSP synchronized

1. Connect to BR1-PC-1.
2. Open a browser, and then enter the following URL:  
`http://portquiz.net:8080`
3. Return to the BR1-FGT-1 SSH session, and then enter the following command to view the session information:

```
diagnose sys session filter dport 8080
diagnose sys session list
```

The result should be similar to the following output:



```
BR1-FGT-1 # diagnose sys session list

session info: proto=6 proto_state=01 duration=3 expire=3596 timeout=3600 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=2:0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=dirty may_dirty synses
statistic(bytes/packets/allow_err): org=0/0/0 reply=50/1/1 tuples=2
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 15/0
origin->sink: org pre->post, reply pre->post dev=6->5/4->6 gw=0.0.0.0/0.0.0.0
hook-pre dir=org act=noop 172.20.1.60:47702->35.180.139.74:8080(0.0.0.0:0)
hook-post dir=reply act=noop 35.180.139.74:8080->172.20.1.60:47702(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uid_idx=0 auth_info=0 chk_client_info=0 vd=0
serial=0000ec96 tos=ff/ff app_list=0 app=0 url_cat=0
rpidb_link_id=00000000 ngfwid=n/a
npu_state=0x2000100
no_ofld_reason: npu-flag-off

session info: proto=6 proto_state=01 duration=3 expire=3596 timeout=3600 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=2:0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=dirty may_dirty synses
statistic(bytes/packets/allow_err): org=0/0/0 reply=112/2/1 tuples=2
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 20/0
origin->sink: org pre->post, reply pre->post dev=6->5/4->6 gw=0.0.0.0/0.0.0.0
hook-pre dir=org act=noop 172.20.1.60:47702->35.180.139.74:8080(0.0.0.0:0)
hook-post dir=reply act=noop 35.180.139.74:8080->172.20.1.60:47702(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uid_idx=0 auth_info=0 chk_client_info=0 vd=0
serial=0000ec95 tos=ff/ff app_list=0 app=0 url_cat=0
rpidb_link_id=00000000 ngfwid=n/a
npu_state=0x2000100
no_ofld_reason: npu-flag-off
total session: 2
```

4. Return to the BR1-FGT-2 SSH session, and then enter the following command to view the session information:

```
diagnose sys session filter dport 8080
diagnose sys session list
```

The result should be similar to the following output:

```
BR1-FGT-2 # diagnose sys session list

session info: proto=6 proto_state=04 duration=7 expire=117 timeout=3600 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=2:0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=may_dirty synsed
statistic(bytes/packets/allow_err): org=164/3/1 reply=0/0/0 tuples=2
tx speed(Bps/kbps): 22/0 rx speed(Bps/kbps): 0/0
origin->sink: org pre->post, reply pre->post dev=6->5/5->6 gw=100.66.1.254/0.0.0.0
hook-pre dir=org act=noop 172.20.1.60:47702->35.180.139.74:8080(0.0.0.0:0)
hook-post dir=reply act=noop 35.180.139.74:8080->172.20.1.60:47702(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uid_idx=16318 auth_info=0 chk_client_info=0 vd=0
serial=0000ec96 tos=ff/ff app_list=0 app=0 url_cat=0
rpidb_link_id=00000000 ngfwid=n/a
npu_state=0x2000100
no_ofld_reason: npu-flag-off

session info: proto=6 proto_state=01 duration=7 expire=3597 timeout=3600 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=2:0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=may_dirty synsed
statistic(bytes/packets/allow_err): org=513/3/1 reply=0/0/0 tuples=2
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 0/0
origin->sink: org pre->post, reply pre->post dev=6->5/5->6 gw=100.66.1.254/0.0.0.0
hook-pre dir=org act=noop 172.20.1.60:47702->35.180.139.74:8080(0.0.0.0:0)
hook-post dir=reply act=noop 35.180.139.74:8080->172.20.1.60:47702(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uid_idx=16318 auth_info=0 chk_client_info=0 vd=0
serial=0000ec95 tos=ff/ff app_list=0 app=0 url_cat=0
rpidb_link_id=00000000 ngfwid=n/a
npu_state=0x2000100
no_ofld_reason: npu-flag-off
total session: 2
```



The BR1-FGT-2 session list shows a `synced` flag, which means that BR1-FGT-2 has synchronized these sessions with its peer (BR1-FGT-1).

It is confirmed in the BR1-FGT-1 session list, which shows a `syn_ses` flag.

## Exercise 3: Encrypting the Session Synchronization

In this exercise, you will examine the protocol that handles the layer 3 session synchronization. Then, you will configure the encryption of the session synchronization.

### Encrypt the Session Synchronization

You will verify the protocol that handles the session synchronization, and then configure the encryption.

#### To verify the protocol handling the session synchronization

1. Connect over SSH to BR1-FGT-1.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Enter the following CLI command:  
`diagnose sniffer packet port7 '' 1 6`  
The result should be similar to the following output:

```
BR1-FGT-1 # diagnose sniffer packet port7 '' 1 6
Using Original Sniffing Mode
interfaces=[port7]
filters=[]
0.191925 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.191925 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.202884 10.1.6.253.708 -> 10.1.6.254.708: udp 76
0.202884 10.1.6.253.708 -> 10.1.6.254.708: udp 76
0.393658 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.393658 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.404756 10.1.6.253.708 -> 10.1.6.254.708: udp 76
0.404756 10.1.6.253.708 -> 10.1.6.254.708: udp 76
0.595400 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.595400 10.1.6.254.708 -> 10.1.6.253.708: udp 76
0.596203 10.1.6.253.708 -> 10.1.6.254.708: udp 76
0.596203 10.1.6.253.708 -> 10.1.6.254.708: udp 76
```



By default, the session synchronization over a layer 3 connection is encapsulated in UDP packets.

#### To configure session synchronization encryption

1. Continuing in the BR1-FGT-1 SSH session, enter the following commands to enable encryption:

```
config system standalone-cluster
  set encryption enable
  set psksecret fortinet
end
```

2. Connect over SSH to BR1-FGT-2.

3. Log in with the following credentials:

- Username: admin
- Password: Fortinet1!

4. Enter the following commands to enable encryption:

```
config system standalone-cluster
  set encryption enable
  set psksecret fortinet
end
```

### To verify session synchronization encryption

1. Continuing in the BR1-FGT-2 SSH session, enter the following command:

```
diagnose sniffer packet port7 '' 1 6
```

The result should be similar to the following output:

```
BR1-FGT-2 # diagnose sniffer packet port7 '' 1 6
Using Original Sniffing Mode
interfaces=[port7]
filters=[]
0.160407 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x62)
0.160407 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x62)
0.191142 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x62)
0.191142 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x62)
0.362605 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x63)
0.362605 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x63)
0.383270 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x63)
0.383270 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x63)
0.564333 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x64)
0.564333 10.1.6.254 -> 10.1.6.253: ESP spi=0xb3555032, seq=0x64)
0.585133 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x64)
0.585133 10.1.6.253 -> 10.1.6.254: ESP spi=0x5fa4c1d1, seq=0x64)
```

2. Enter the following command:

```
diagnose vpn tunnel list
```

The result should be similar to the following output:

```
BRI-FGT-2 # diag vpn tunnel list
list all ipsec tunnel in vd 0

name= SESSYMC_1 ver=1 serial=1 10.1.6.25d:e>10.1.6.25d:e nexthop=0.0.0.0 tun id=10.1.6.25d tun idb=:10.1.6.25d status=up dst mtu=1500 weight=1
bound_if=0 real_if=0 lgyu-static/1 tun-1ncf mode=auto/1 encap=none/552 options[0228]-npu frag rfc run_state=0 role=primary accept_traffic=1 overlay=ic-0
proxyid_num=1 child_num=0 refcnt=4 ilast=0 elast=0 ad=0
stat: rxp=802 txp=804 rxb=80388 txb=153528
dpd: mode=on demand on=1 status=ok idlc=2000000 retry=3 count=0 seqno=8
nat: mode=none draft=0 interval=0 remote_port=0
fec: express=0 ingress=0
proxyid=SESSYMC_1 proto=0 op=1 ref=5 serial=1 auto negotiate
enc: 0:10.1.6.253-10.1.6.253:0
dst: 0:10.1.6.254 10.1.6.254:0
SA: rel=3 options=38203 type=00 solt=0 mtu=1438 expire=42770/00 replaywin=2048
seqno=325 on=0 replaywin_lactseq=00000000 qat=0 pkty=0 hash_onch=on=1
life: type=01 bytes=0/0 timeout=42930/43200
der: spi=h755012 esp=aes key=16 bc4aa71a9c95121b5a6ba85422d27a7
ah-shal key=20 tc7r4u4u5u0ba73b10du1708e2002669521f979
enc: spi=yfa4a1d1 esp=aes key=16 9c7bdc0c0a1145a8c8784c10448c7471
ah-shal key=20 0e5c360c440061de116d04b92f0d5b37c7f6792
dec:pkts/bytes=802/80388, enc:pkts/bytes=804/142272
npu_flag=00 npu_lgyu=10.1.6.254 npu_id=10.1.6.253 npu_solid=0 der_npuid=0 enc_npuid=0
```



The session synchronization over layer 3 is now encapsulated in ESP packets and therefore encrypted in the `SESSYNC_1` IPsec tunnel.

- 3.** Close the SSH windows.

## Lab 5: Dynamic Routing Protocols

In this lab, on FortiManager, you will configure the FortiGate devices to use OSPF as the dynamic routing protocol for the enterprise network. You will also configure BGP on NGFW and use a loopback interface as a BGP source on BR1-FGT-2.

### Objectives

- Use OSPF to dynamically distribute the routes in an enterprise network
- Configure OSPF equal-cost multi-path (ECMP)
- Configure BGP using a FortiManager BGP template
- Implement a loopback interface as a BGP source

### Time to Complete

Estimated: 60 minutes

### Prerequisites

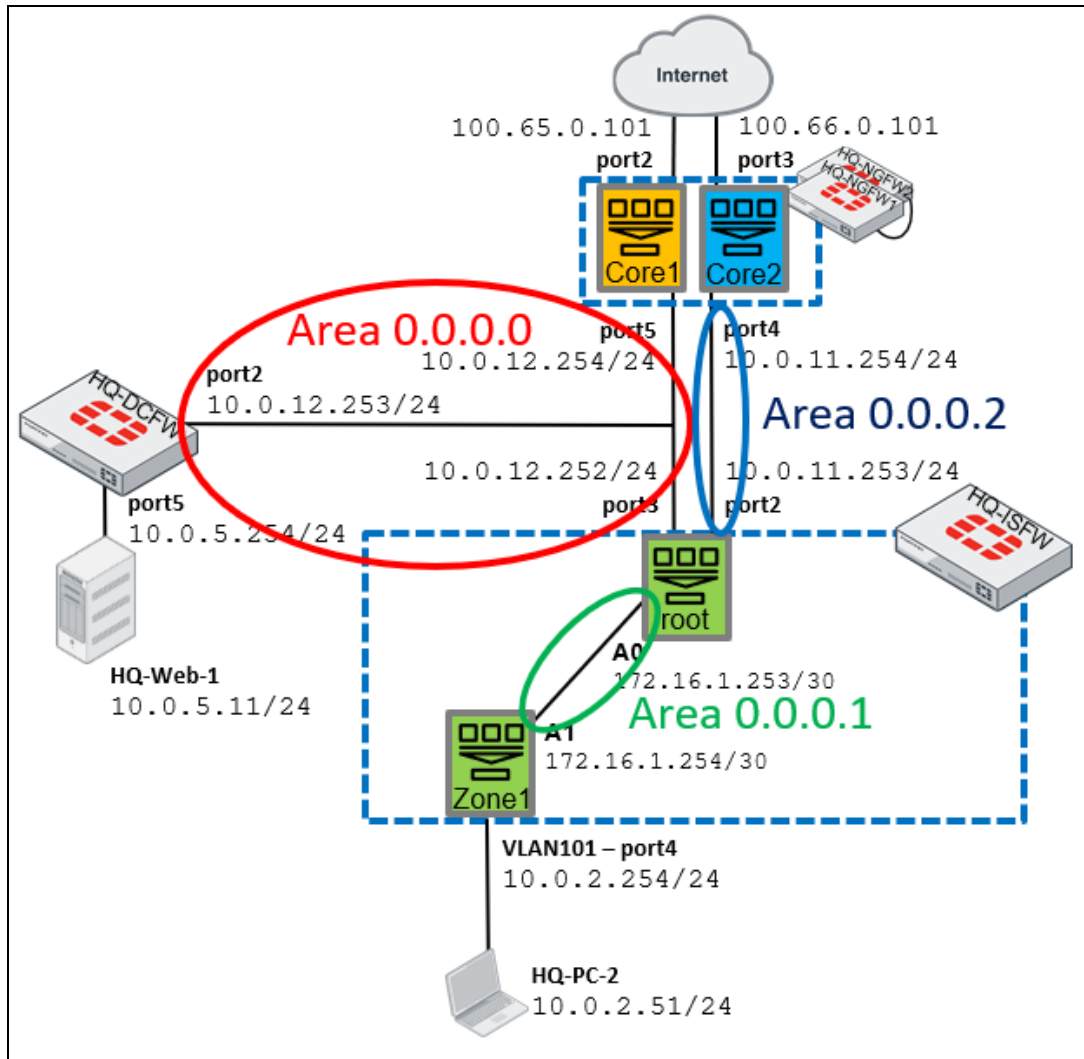
Before you begin this lab, you must complete the previous lab. If you haven't done so, tell your instructor.

### Which Network Segment Will You Work On?

On FortiManager, you will configure OSPF between the HQ-ISFW, HQ-DCFW, and HQ-NGFW FortiGate devices. Then, you will enable BGP between HQ-NGFW Core1 and Core2 using FortiManager, and configure BGP on BR1-FGT-2 using a loopback interface.

## Exercise 1: Configuring OSPF With ECMP

In this exercise, you will configure OSPF on the three FortiGate devices that are part of the hub network: HQ-ISFW, HQ-DCFW, and HQ-NGFW. The objective is to remove all static routes from the three firewalls and use only OSPF to route traffic internally. You will use three OSPF areas and enable the **rfc1583-compatible** setting to provide OSPF ECMP for external routes.

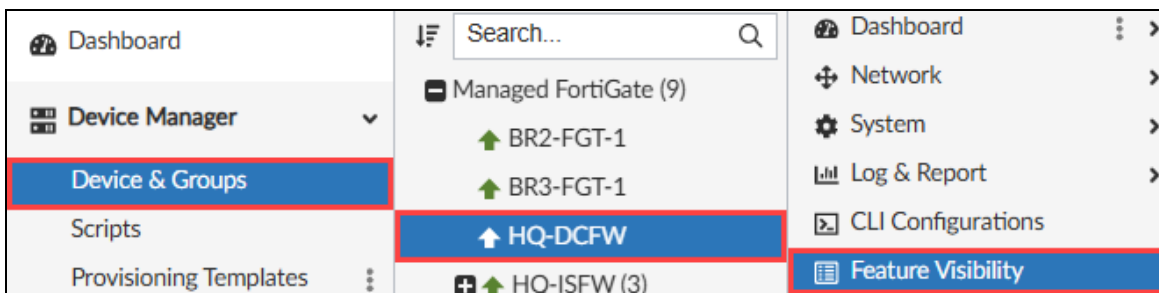


### Configure OSPF on HQ-DCFW

First, you will configure OSPF on HQ-DCFW. Next, you will remove the static routes. Finally, you will install the changes.

### To configure OSPF on HQ-DCFW

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Device Manager**.
4. Click **Device & Groups > Managed FortiGate > HQ-DCFW > Feature Visibility**.



5. On the **Global Feature Visibility** tab, in the **Network** section, click **OSPF**.



By default, the OSPF settings are hidden on the FortiManager GUI.

6. Click **OK**.
7. Click **Device & Groups > Managed FortiGate > HQ-DCFW > Network > OSPF**.
8. Configure the following settings:

| Field     | Value                        |
|-----------|------------------------------|
| Router ID | 0.0.0.2                      |
| Areas     | 0.0.0.0                      |
| Networks  | 10.0.5.0/24 and 10.0.12.0/24 |

The configuration should look like the following image:



OSPF

Router ID 0.0.0.2

Areas

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Area    | Type    | Authentication |
|--------------------------|---|---------|---------|----------------|
| <input type="checkbox"/> | 1 | 0.0.0.0 | Regular | None           |

Networks

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Network                 | Area    |
|--------------------------|---|-------------------------|---------|
| <input type="checkbox"/> | 1 | 10.0.5.0/255.255.255.0  | 0.0.0.0 |
| <input type="checkbox"/> | 2 | 10.0.12.0/255.255.255.0 | 0.0.0.0 |

9. Click **Apply**.

### To remove the static routes

- Continuing on the FortiManager GUI, click **Network > Static Routes**.
- Select the checkbox for the static route used to route internal traffic (don't select the default routes), and then click **Delete**.

| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Edit in CLI</a> <a href="#">Clone</a> <span style="border: 1px solid red; padding: 2px;"><a href="#">Delete</a></span> |        |                        |             |           |          |          |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------|-------------|-----------|----------|----------|--------|
| <input type="checkbox"/>                                                                                                                                                             | Seq. # | Destination            | Gateway     | Interface | Distance | Priority | Status |
| IPv4 (2)                                                                                                                                                                             |        |                        |             |           |          |          |        |
| <input type="checkbox"/>                                                                                                                                                             | 1      | 0.0.0.0/0.0.0.0        | 10.0.12.254 | port2     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                                                                                                  | 2      | 10.0.2.0/255.255.255.0 | 10.0.12.252 | port2     | 10       | 1        | Enable |

3. Click **OK** to confirm.

### To install the configuration changes

- Continuing on the FortiManager GUI, click **Install Wizard**.
- Verify that **Install Device Settings (only)** is selected, and then click **Next**.
- Verify that only **HQ-DCFW** is selected, and then click **Next**.
- Click **Install**.

5. Wait until the installation finishes.
6. Click **Finish**.

## Configure OSPF on HQ-ISFW

You will configure OSPF on the HQ-ISFW Zone1 VDOM and root VDOM through FortiManager. Try to do this yourself using the procedure you followed to configure OSPF on HQ-DCFW.

### To configure OSPF on the HQ-ISFW Zone1 VDOM

1. Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-ISFW > Zone1**.
2. Click **Network > OSPF**.
3. Configure the following settings:

| Field     | Value                                                                |
|-----------|----------------------------------------------------------------------|
| Router ID | 0.0.0.4                                                              |
| Areas     | 0.0.0.1 (with the type <b>STUB</b> )                                 |
| Networks  | 172.16.1.252/30 (for area 0.0.0.1)<br>10.0.2.0/24 (for area 0.0.0.1) |

The configuration should look like the following image:

OSPF

Router ID 0.0.0.4

Areas

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Area    | Type | Authentication |
|--------------------------|---|---------|------|----------------|
| <input type="checkbox"/> | 1 | 0.0.0.1 | STUB | None           |

Networks

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Network                      | Area    |
|--------------------------|---|------------------------------|---------|
| <input type="checkbox"/> | 1 | 172.16.1.252/255.255.255.252 | 0.0.0.1 |
| <input type="checkbox"/> | 2 | 10.0.2.0/255.255.255.0       | 0.0.0.1 |



The type of area is **STUB** because there is no further routing expected with the Zone2 VDOM.

- Click **Apply**.
- Click **Network > Static Routes**.
- Delete the static route used to route internal traffic (don't delete the default route).

| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Edit in CLI</a> <a href="#">Clone</a> <a href="#" style="border: 1px solid red;">Delete</a> |        |                        |              |           |          |          |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------|--------------|-----------|----------|----------|--------|
| <input type="checkbox"/>                                                                                                                                  | Seq. # | Destination            | Gateway      | Interface | Distance | Priority | Status |
| IPv4 (2)                                                                                                                                                  |        |                        |              |           |          |          |        |
| <input type="checkbox"/>                                                                                                                                  | 1      | 0.0.0.0/0.0.0.0        | 172.16.1.253 | A1        | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                                                                       | 2      | 10.0.3.0/255.255.255.0 | 172.16.1.246 | B0        | 10       | 1        | Enable |

- Install the configuration changes for the HQ-ISFW Zone1 VDOM.

### To configure OSPF on the HQ-ISFW root VDOM

- Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-ISFW > root > Network > OSPF**.
- Configure the following settings:

| Field     | Value                                   |
|-----------|-----------------------------------------|
| Router ID | 0.0.0.3                                 |
| Areas     | 0.0.0.0 (with the type <b>Regular</b> ) |
|           | 0.0.0.1 (with the type <b>STUB</b> )    |
|           | 0.0.0.2 (with the type <b>Regular</b> ) |
| Networks  | 10.0.12.0/24 (for area 0.0.0.0)         |
|           | 172.16.1.252/30 (for area 0.0.0.1)      |
|           | 10.0.11.0/24 (for area 0.0.0.2)         |

The configuration should look like the following image:

OSPF

Router ID

0.0.0.3

Areas

+ Create New

Edit

Delete

| <input type="checkbox"/> | # | Area    | Type    | Authentication |
|--------------------------|---|---------|---------|----------------|
| <input type="checkbox"/> | 1 | 0.0.0.0 | Regular | None           |
| <input type="checkbox"/> | 2 | 0.0.0.1 | STUB    | None           |
| <input type="checkbox"/> | 3 | 0.0.0.2 | Regular | None           |

Networks

+ Create New

Edit

Delete

| <input type="checkbox"/> | # | Network                      | Area    |
|--------------------------|---|------------------------------|---------|
| <input type="checkbox"/> | 1 | 10.0.12.0/255.255.255.0      | 0.0.0.0 |
| <input type="checkbox"/> | 2 | 172.16.1.252/255.255.255.252 | 0.0.0.1 |
| <input type="checkbox"/> | 3 | 10.0.11.0/255.255.255.0      | 0.0.0.2 |



The HQ-ISFW root VDOM router is connected to multiple areas and is then an area border router (ABR).

3. Click **Apply**.
4. Delete the two static routes used to route internal traffic (don't delete the default route or the route to 4.2.2.2/255.255.255.255).

| <div><div>+ Create New</div><div>Edit</div><div>Edit in CLI</div><div>Clone</div><div>Delete</div></div> |        |                         |              |           |          |          |        |
|----------------------------------------------------------------------------------------------------------|--------|-------------------------|--------------|-----------|----------|----------|--------|
| <input type="checkbox"/>                                                                                 | Seq. # | Destination             | Gateway      | Interface | Distance | Priority | Status |
| IPv4 (4)                                                                                                 |        |                         |              |           |          |          |        |
| <input type="checkbox"/>                                                                                 | 1      | 0.0.0.0/0.0.0.0         | 10.0.12.254  | port3     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                      | 2      | 10.0.5.0/255.255.255.0  | 10.0.12.253  | port3     | 10       | 1        | Enable |
| <input type="checkbox"/>                                                                                 | 3      | 4.2.2.2/255.255.255.255 | 10.0.11.254  | port2     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                      | 4      | 10.0.2.0/255.255.255.0  | 172.16.1.254 | A0        | 10       | 1        | Enable |

5. Install the configuration changes for the HQ-ISFW root VDOM.

## Configure OSPF on HQ-NGFW

You will configure OSPF on the HQ-NGFW Core1 VDOM and Core2 VDOM through FortiManager. Try to do this yourself using the procedure you followed to configure OSPF on HQ-DCFw and HQ-ISFW.

### To configure OSPF on the Core1 VDOM

1. Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-NGFW > Core1 > Network > OSPF**.
2. Configure the following settings:

| Field     | Value        |
|-----------|--------------|
| Router ID | 0.0.0.5      |
| Areas     | 0.0.0.0      |
| Networks  | 10.0.12.0/24 |

The configuration should look like the following image:

OSPF

Router ID 0.0.0.5

Areas

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Area    | Type    | Authentication |
|--------------------------|---|---------|---------|----------------|
| <input type="checkbox"/> | 1 | 0.0.0.0 | Regular | None           |

Networks

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | # | Network                 | Area    |
|--------------------------|---|-------------------------|---------|
| <input type="checkbox"/> | 1 | 10.0.12.0/255.255.255.0 | 0.0.0.0 |

- In the **Redistribute** section, select the **static** checkbox, and then click **Edit**.



The Core1 VDOM has a static route preconfigured to reach the IP address 100.75.5.1. You will enable the static route redistribution to import it into OSPF.

- Enable **Status**, and then click **OK**.

The configuration should look like the following image:

Redistribute

☒ Edit

| <input type="checkbox"/>            | # | Name      | Status                                     | Metric | Route Map | Metric-Type | Tag |
|-------------------------------------|---|-----------|--------------------------------------------|--------|-----------|-------------|-----|
| <input type="checkbox"/>            | 1 | hgp       | <input type="checkbox"/> Disable           | 0      |           | 2           | 0   |
| <input type="checkbox"/>            | 2 | connected | <input type="checkbox"/> Disable           | 0      |           | 2           | 0   |
| <input type="checkbox"/>            | 3 | isis      | <input type="checkbox"/> Disable           | 0      |           | 2           | 0   |
| <input type="checkbox"/>            | 4 | rip       | <input type="checkbox"/> Disable           | 0      |           | 2           | 0   |
| <input checked="" type="checkbox"/> | 5 | static    | <input checked="" type="checkbox"/> Enable | 0      |           | 2           | 0   |

1 Selected

- Click **Apply**.
- Delete the two static routes used to route internal traffic (don't delete the default route or the route to 100.75.5.1/255.255.255.255).

| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Edit in CLI</a> <a href="#">Clone</a> <a href="#">Delete</a> |        |                           |              |           |          |          |        |
|----------------------------------------------------------------------------------------------------------------------------|--------|---------------------------|--------------|-----------|----------|----------|--------|
| <input type="checkbox"/>                                                                                                   | Seq. # | Destination               | Gateway      | Interface | Distance | Priority | Status |
| IPv4 (4)                                                                                                                   |        |                           |              |           |          |          |        |
| <input type="checkbox"/>                                                                                                   | 1      | 0.0.0.0/0.0.0.0           | 100.65.0.254 | port2     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                                        | 2      | 10.0.2.0/255.255.255.0    | 10.0.12.252  | port5     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                                        | 3      | 10.0.5.0/255.255.255.0    | 10.0.12.253  | port5     | 10       | 1        | Enable |
| <input type="checkbox"/>                                                                                                   | 4      | 100.75.5.1/255.255.255... | 100.65.0.254 | port2     | 10       | 1        | Enable |

7. Install the configuration changes for the HQ-NGFW Core1 VDOM.

### To configure OSPF on the Core2 VDOM

1. Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-NGFW > Core2 > Network > OSPF**.
2. Configure the following settings:

| Field     | Value                           |
|-----------|---------------------------------|
| Router ID | 0.0.0.6                         |
| Areas     | 0.0.0.2                         |
| Networks  | 10.0.11.0/24 (for area 0.0.0.2) |

The configuration should look like the following image:

OSPF

Router ID

0.0.0.6

Areas

[+ Create New](#)
[Edit](#)
[Delete](#)

| <input type="checkbox"/> | # | Area    | Type    | Authentication |
|--------------------------|---|---------|---------|----------------|
| <input type="checkbox"/> | 1 | 0.0.0.2 | Regular | None           |

Networks

[+ Create New](#)
[Edit](#)
[Delete](#)

| <input type="checkbox"/> | # | Network                 | Area    |
|--------------------------|---|-------------------------|---------|
| <input type="checkbox"/> | 1 | 10.0.11.0/255.255.255.0 | 0.0.0.2 |

3. In the **Redistribute** section, select the **static** checkbox, and then click **Edit**.



The same static route to reach the IP address 100.75.5.1 is preconfigured in the Core2 VDOM. You will enable the static route redistribution to import it into OSPF.

4. Enable **Status**, and then click **OK**.

The configuration should look like the following image:

| Redistribute                             |           |         |        |           |             |     |  |
|------------------------------------------|-----------|---------|--------|-----------|-------------|-----|--|
| <input checked="" type="checkbox"/> F-OF |           |         |        |           |             |     |  |
| #                                        | Name      | Status  | Metric | Route Map | Metric-Type | Tag |  |
| 1                                        | hqn       | Disable | 0      |           | 2           | 0   |  |
| 2                                        | connected | Disable | 0      |           | 2           | 0   |  |
| 3                                        | isis      | Disable | 0      |           | 2           | 0   |  |
| 4                                        | rip       | Disable | 0      |           | 2           | 0   |  |
| 5                                        | static    | Enable  | 0      |           | 2           | 0   |  |

5. Click **Apply**.
6. Delete the static route used to route internal traffic (don't delete the default route or the route to 100.75.5.1/255.255.255.255).

| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Edit in CLI</a> <a href="#">Clone</a> <a href="#">Delete</a> |        |                            |              |           |          |          |        |
|----------------------------------------------------------------------------------------------------------------------------|--------|----------------------------|--------------|-----------|----------|----------|--------|
|                                                                                                                            | Seq. # | Destination                | Gateway      | Interface | Distance | Priority | Status |
| IPv4 (3)                                                                                                                   |        |                            |              |           |          |          |        |
| <input type="checkbox"/>                                                                                                   | 1      | 0.0.0.0/0.0.0.0            | 100.66.0.254 | port3     | 10       | 1        | Enable |
| <input checked="" type="checkbox"/>                                                                                        | 2      | 10.0.2.0/255.255.255.0     | 10.0.11.253  | port4     | 10       | 1        | Enable |
| <input type="checkbox"/>                                                                                                   | 3      | 100.75.5.1/255.255.255.255 | 100.66.0.254 | port3     | 10       | 1        | Enable |

7. Install the configuration changes for the HQ-NGFW Core2 VDOM.

## Check the OSPF Status on the HQ-ISFW Root VDOM

You will run OSPF diagnostic commands on the HQ-ISFW root VDOM to verify OSPF operation.

### To check the OSPF status on the HQ-ISFW root VDOM

1. Connect over SSH to HQ-ISFW.
2. Log in with the following credentials:

- Username: admin
- Password: Fortinet1!

3. Enter the following commands:

```
config vdom
edit root
get router info ospf neighbor
```

You should see four neighbors: HQ-DCFV, the HQ-NGFW Core1 and Core2 VDOMs, and the HQ-ISFW Zone1 VDOM. The State column should display Full.



```
HQ-ISFW (root) # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
0.0.0.6        1   Full/Backup     00:00:36   10.0.11.254  port2
0.0.0.2        1   Full/DR         00:00:39   10.0.12.253  port3
0.0.0.5        1   Full/DROther    00:00:38   10.0.12.254  port3
0.0.0.4        1   Full/ -         00:00:39   172.16.1.254 A0
```

**Stop and think!**

Can you identify from this output what the designated router (DR) is?

The `State` of the DR is displayed as `Full/DR`. If neither of the routers in the backbone area display this state, it means that the DR is the local FortiGate. In this case, the DR is HQ-DCFW.

4. Enter the following command:

```
get router info ospf status
```

The output should be similar to the following output:

```
HQ-ISFW (root) # get router info ospf status
Routing Process "ospf 0" with ID 0.0.0.3
Process uptime is 35 minutes
Process bound to VRF default
Conforms to RFC2328, and RFC1583Compatibility flag is disabled
Supports only single TOS(TOS0) routes
Supports opaque LSA
This router is an ABR ABR Type is Standard (RFC2328)
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Refresh timer 10 secs
Number of incoming current DD exchange neighbors 0/5
Number of outgoing current DD exchange neighbors 0/5
Number of external LSA 2. Checksum 0x008991
Number of opaque AS LSA 0. Checksum 0x000000
Number of non-default external LSA 2
External LSA database is unlimited.
Number of LSA originated 20
Number of LSA received 22
Number of areas attached to this router: 3
Area 0.0.0.0 (BACKBONE)
  Number of interfaces in this area is 1(1)
  Number of fully adjacent neighbors in this area is 2
  Area has no authentication
  SPF algorithm last executed 00:01:35.360 ago
  SPF algorithm executed 7 times
  Number of LSA 8. Checksum 0x0246e2
Area 0.0.0.1 (Stub)
  Number of interfaces in this area is 1(1)
  Number of fully adjacent neighbors in this area is 1
  Number of fully adjacent virtual neighbors through this area is 0
  Area has no authentication
  SPF algorithm last executed 00:34:46.350 ago
  SPF algorithm executed 3 times
  Number of LSA 6. Checksum 0x03ea13
  Shortcutting mode: Disabled, S-bit consensus: ok
Area 0.0.0.2
  Number of interfaces in this area is 1(1)
  Number of fully adjacent neighbors in this area is 1
  Number of fully adjacent virtual neighbors through this area is 0
  Area has no authentication
  SPF algorithm last executed 00:01:06.360 ago
  SPF algorithm executed 6 times
  Number of LSA 8. Checksum 0x049dd9
```



This confirms that the HQ-ISFW root VDOM is connected to three OSPF areas and is therefore an ABR. You can also see that OSPF ECMP (defined with the RFC1583Compatibility flag) is disabled by default.

5. Enter the following command:

```
get router info routing-table all
```

You should see that the ISFW root VDOM has learned the routes to the 10.0.5.0/24 and 10.0.2.0/24 subnets through OSPF.

```
HQ-ISFW (root) # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*    0.0.0.0/0 [10/0] via 10.0.12.254, port3, [1/0]
S     4.2.2.2/32 [10/0] via 10.0.11.254, port2, [1/0]
O     10.0.2.0/24 [110/101] via 172.16.1.254, A0, 02:58:25, [1/0]
O     10.0.5.0/24 [110/2] via 10.0.12.253, port3, 02:58:25, [1/0]
C     10.0.11.0/24 is directly connected, port2
C     10.0.12.0/24 is directly connected, port3
O E2  100.75.5.1/32 [110/10] via 10.0.11.254, port2, 02:24:43, [1/0]
C     172.16.1.248/30 is directly connected, C0
C     172.16.1.252/30 is directly connected, A0
C     192.168.0.0/16 is directly connected, port1
```

#### Stop and think!

Why is the external route to 100.75.5.1/32 available only through the NGFW Core2 VDOM, meaning 10.0.11.254?

This is because the OSPF ECMP **rfc1583-compatible** setting is disabled by default.

## Enable OSPF ECMP on the HQ-ISFW Root VDOM

You will enable and install OSPF ECMP on the HQ-ISFW root VDOM.

### To enable OSPF ECMP on the HQ-ISFW root VDOM

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Device Manager**.
4. Click **Device & Groups**.
5. In the **Managed FortiGate** list, click **HQ-ISFW > root > Network > OSPF**.
6. Click **Advanced Options**.
7. Enable **rfc1583-compatible**.

The configuration should look like the following image:

Advanced Options ▾

|                              |                                     |     |
|------------------------------|-------------------------------------|-----|
| abr-type ⓘ                   | standard                            | ✕ ▾ |
| auto-cost-ref-bandwidth ⓘ    | 1000                                |     |
| bfd ⓘ                        | <input type="checkbox"/>            |     |
| default-metric ⓘ             | 10                                  |     |
| distribute-list-in ⓘ         | Click to select                     | ▾   |
| distribute-route-map-in ⓘ    | Click to select                     | ▾   |
| log-neighbour-changes ⓘ      | <input checked="" type="radio"/>    |     |
| lsa-refresh-interval ⓘ       | 5                                   |     |
| restart-mode ⓘ               | none                                | ✕ ▾ |
| restart-on-topology-change ⓘ | <input type="checkbox"/>            |     |
| restart-period ⓘ             | 120                                 |     |
| rfc1583-compatible ⓘ         | <input checked="" type="checkbox"/> |     |
| spf-timers ⓘ                 | 5                                   | ✕   |
|                              | 10                                  | ✕   |

8. Click **Apply**.

### To install the configuration changes

1. Continuing on the FortiManager GUI, in the menu at the top, click **Install Wizard**.
2. Verify that **Install Device Settings (only)** is selected, and then click **Next**.
3. Verify that only **HQ-ISFW** and **root [NAT] (Management)** are selected, and then click **Next**.
4. Click **Install**.
5. Wait until the installation finishes.
6. Click **Finish**.

## Enable OSPF ECMP on HQ-NGFW

You will enable and install OSPF ECMP on the HQ-NGFW Core1 and Core2 VDOMs.

### Take the Expert Challenge!

On FortiManager, enable and install OSPF ECMP on the HQ-NGFW Core1 and Core2 VDOMs using the procedure you followed to enable the **rfc1583-compatible** setting on the HQ-ISFW root VDOM.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Check Connectivity on page 99](#).

### To enable OSPF ECMP on the HQ-NGFW Core1 VDOM

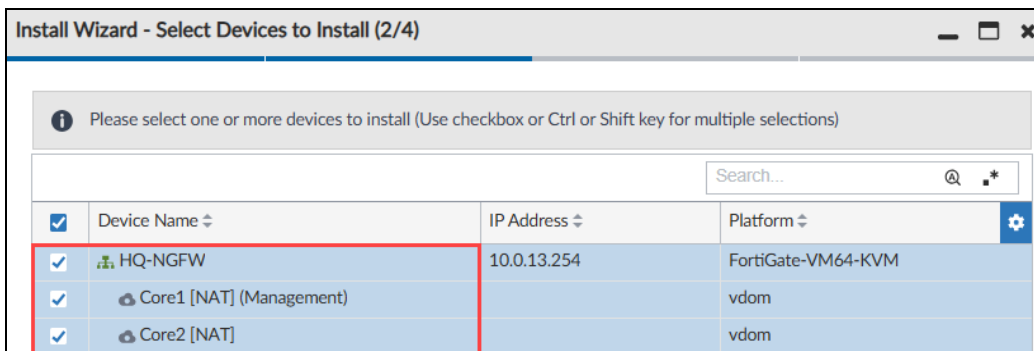
1. Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-NGFW > Core1 > Network > OSPF**.
2. Click **Advanced Options**.
3. Enable **rfc1583-compatible**.
4. Click **Apply**.

### To enable OSPF ECMP on the HQ-NGFW Core2 VDOM

1. Continuing on the FortiManager GUI, in the **Managed FortiGate** list, click **HQ-NGFW > Core2 > Network > OSPF**.
2. Click **Advanced Options**.
3. Enable **rfc1583-compatible**.
4. Click **Apply**.

### To install the configuration changes

1. Continuing on the FortiManager GUI, in the menu at the top, click **Install Wizard**.
2. Verify that **Install Device Settings (only)** is selected, and then click **Next**.
3. Verify that only the **HQ-NGFW, Core1 [NAT]**, and **Core2 [NAT]** checkboxes are selected, and then click **Next**.



4. Click **Install**.
5. Wait until the installation finishes.
6. Click **Finish**.

## Verify OSPF ECMP

You will verify that the HQ-ISFW root VDOM has multiple routes to the same destination with different next hops.

### To verify OSPF ECMP on the HQ-ISFW root VDOM

1. Connect over SSH to HQ-ISFW.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Enter the following commands:

```
config vdom
edit root
```

```
get router info ospf status
```

The output should be similar to the following output:

```
HQ-ISFW (root) # get router info ospf status
Routing Process "ospf 0" with ID 0.0.0.3
Process uptime is 5 minutes
Process bound to VRF default
Conforms to RFC2328, and RFC1583Compatibility flag is enabled
```

4. Enter the following command:

```
get router info routing-table all
```

The output should be similar to the following output:

```
HQ-ISFW (root) # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*    0.0.0.0/0 [10/0] via 10.0.12.254, port3, [1/0]
S     4.2.2.2/32 [10/0] via 10.0.11.254, port2, [1/0]
O     10.0.2.0/24 [110/101] via 172.16.1.254, A0, 00:07:29, [1/0]
O     10.0.5.0/24 [110/2] via 10.0.12.253, port3, 00:07:29, [1/0]
C     10.0.11.0/24 is directly connected, port2
C     10.0.12.0/24 is directly connected, port3
O E2  100.75.5.1/32 [110/10] via 10.0.12.254, port3, 00:03:07, [1/0]
      [110/10] via 10.0.11.254, port2, 00:03:07, [1/0]
C     172.16.1.248/30 is directly connected, C0
C     172.16.1.252/30 is directly connected, A0
C     192.168.0.0/16 is directly connected, port1
```



With OSPF ECMP enabled, the HQ-ISFW root VDOM can now load balance the traffic to 100.75.5.1/32 with the two routes of the same cost installed with the OSPF protocol.

## Check the OSPF Status on HQ-DCFW and HQ-NGFW

You will run OSPF diagnostic commands on HQ-DCFW and HQ-NGFW to verify OSPF operation.

### Take the Expert Challenge!

1. Connect over SSH to HQ-DCFW.
2. Enter the following commands to verify OSPF operation on HQ-DCFW:
  - `get router info ospf neighbor`
  - `get router info routing-table all`
3. Connect over SSH to HQ-NGFW.
4. Enter the following commands to verify OSPF operation on the HQ-NGFW Core1 and Core2 VDOMs:
  - `get router info ospf neighbor`
  - `get router info routing-table all`

After you complete the challenge, see [Check Connectivity](#) on page 99.

## Check Connectivity

You will confirm that the FortiGate devices are routing traffic correctly by running a ping from HQ-PC-2 to HQ-Web-1.

### To check connectivity

1. On HQ-PC-2, open a terminal window.
2. Run a ping to HQ-Web-1 (10.0.5.11).

The ping should succeed, which confirms that the FortiGate devices are correctly routing the traffic between the 10.0.2.0/24 and 10.0.5.0/24 subnets.
3. Run a traceroute to `www.fortinet.com` and `4.2.2.2`:

```
traceroute www.fortinet.com
```

```
traceroute 4.2.2.2
```

The traceroute should succeed, which confirms that the OSPF protocol is correctly handling your internal enterprise network routing.
4. Close the terminal windows.

## Exercise 2: Configuring BGP

HQ-NGFW has two connections to the internet—one using port2 in the Core1 VDOM and the other using port3 in the Core2 VDOM. You will configure BGP on HQ-NGFW so Core1 and Core2 become BGP peers.

### Configure BGP on FortiManager

You will create metadata variables and use them in a BGP template. Then, you will install the template in the two VDOMs, Core1 and Core2.

#### To create metadata variables

1. Log in to the FortiManager GUI with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
2. Click **EFW**.
3. Click **Policy & Objects > Advanced**.
4. Click **Metadata Variables**, and then click **Create New**.
5. In the **Name** field, type `Neighbor_IP`.
6. Click **Per-Device Mapping**, and then click **Create New**.
7. In the **Mapped Device** field, select **HQ-NGFW [Core1]**.
8. In the **Value** field, type `100.66.0.101`.
9. Click **OK**.
10. Create a second **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core2] |
| Value         | 100.65.0.101    |

11. Click **OK**.  
The configuration should look like the following image:



Name

Neighbor\_IP

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

| <input type="checkbox"/> | Mapped Device   | Value        |
|--------------------------|-----------------|--------------|
| <input type="checkbox"/> | HQ-NGFW [Core1] | 100.66.0.101 |
| <input type="checkbox"/> | HQ-NGFW [Core2] | 100.65.0.101 |

- 12. Click **OK**.
- 13. Click **Create New** to create a second metadata variable.
- 14. In the **Name** field, type `Router_ID`.
- 15. Click **Per-Device Mapping**, and then click **Create New**.
- 16. Create a **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core1] |
| Value         | 172.16.1.1      |

- 17. Create a second **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core2] |
| Value         | 172.16.2.1      |

The configuration should look like the following image:

Create New Metadata Variables

Name

Router\_ID

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

| <input type="checkbox"/> | Mapped Device   | Value      |
|--------------------------|-----------------|------------|
| <input type="checkbox"/> | HQ-NGFW [Core1] | 172.16.1.1 |
| <input type="checkbox"/> | HQ-NGFW [Core2] | 172.16.2.1 |

18. Click **OK**.
19. Click **Create New** to create a third metadata variable.
20. In the **Name** field, type **AS**.
21. Click **Per-Device Mapping**, and then click **Create New**.
22. Create a **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core1] |
| Value         | 65100           |

23. Create a second **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core2] |
| Value         | 65200           |

The configuration should look like the following image:

Name

AS

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

|                          |                 |       |
|--------------------------|-----------------|-------|
| <input type="checkbox"/> | Mapped Device   | Value |
| <input type="checkbox"/> | HQ-NGFW [Core1] | 65100 |
| <input type="checkbox"/> | HQ-NGFW [Core2] | 65200 |

24. Click **OK**.
25. Click **Create New** to create a fourth metadata variable.
26. In the **Name** field, type `Remote_AS`.
27. Click **Per-Device Mapping**, and then click **Create New**.
28. Create a **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core1] |
| Value         | 65200           |

29. Create a second **Per-Device Mapping** with the following settings:

| Field         | Value           |
|---------------|-----------------|
| Mapped Device | HQ-NGFW [Core2] |
| Value         | 65100           |

The configuration should look like the following image:

Create New Metadata Variables

Name **Remote\_AS**

Description

Default Value

Per-Device Mapping ▾

+ Create New Edit Delete

| <input type="checkbox"/> | Mapped Device ▾ | Value ▾ |
|--------------------------|-----------------|---------|
| <input type="checkbox"/> | HQ-NGFW [Core1] | 65200   |
| <input type="checkbox"/> | HQ-NGFW [Core2] | 65100   |

30. Click **OK**.

### To display the BGP settings on the FortiManager GUI

1. Continuing on the FortiManager GUI, click **Device Manager > Provisioning Templates**.



By default, the BGP settings are hidden on the FortiManager GUI. You will enable them in the **Feature Visibility** section.

2. Click **Feature Visibility**.
3. Select the **BGP** checkbox.

Feature Visibility

Assistant More ▾

variables ▾

Feature Visibility

☒ Template Groups

☒ IPsec Tunnel ☒ BGP

4. Click **OK**.

### To configure BGP

1. Continuing on the FortiManager GUI, in the menu at the top, click **BGP**.
2. Click **Create New**.

3. Configure the following settings:

| Field     | Value         |
|-----------|---------------|
| Name      | BGP_NGFW      |
| Local As  | \$(AS)        |
| Router ID | \$(Router_ID) |

4. Create a neighbor with the following settings:

| Field         | Value           |
|---------------|-----------------|
| IP            | \$(Neighbor_IP) |
| Remote AS     | \$(Remote_AS)   |
| Connect Timer | 120             |



When you type \$ in the **Local AS**, **Router ID**, **IP**, **Remote AS** field, you can then select **(AS)**, **(Router\_ID)**, **(Neighbor\_IP)**, and **(Remote\_AS)** in the metadata variables list that appears.

5. Expand **Advanced Options**, and then enable **ebgp-enforce-multihop**.

Edit Neighbor

Advanced Options

activate-evpn
☐

activate-vpnv4
☐

activate-vpnv6
☐

additional-path-vpnv4

disable

additional-path-vpnv5

disable

additional-path6

disable

adv-additional-path

2

adv-additional-path-vpnv4

2

adv-additional-path-vpnv6

2

adv-additional-path6

2

allow-as-in-enable-evpn
☐

allow-as-in-enable-vpnv4
☐

description

distribute-list-in-vpnv4

Click to select

distribute-list-in-vpnv6

Click to select

distribute-list-out-vpnv4

Click to select

distribute-list-out-vpnv6

Click to select

dont-capability-negotiate
☐

ebgp-enforce-multihop
☒



Because Core1 and Core2 are not directly connected, you must enable multihop in the BGP configuration.

6. Click **OK**.  
The configuration should look like the following image:

|                                                                                   |                 |               |
|-----------------------------------------------------------------------------------|-----------------|---------------|
| Name                                                                              | BGP_NGFW        |               |
| Description                                                                       |                 |               |
| Local As                                                                          | \$(AS)          |               |
| Router ID                                                                         | \$(Router_ID)   |               |
| Neighbors                                                                         |                 |               |
| <div><div>+ Create New</div><div>Edit</div><div>Delete</div><div>More</div></div> |                 |               |
| <input type="checkbox"/>                                                          | IP              | Remote AS     |
| <input type="checkbox"/>                                                          | \$(Neighbor_IP) | \$(Remote_AS) |

7. Click **OK**.  
8. Select the **BGP\_NGFW** checkbox, and then click **Assign to Device/Group**.

|                                     |                        |                          |                        |              |      |
|-------------------------------------|------------------------|--------------------------|------------------------|--------------|------|
| + Create New                        | Edit                   | Delete                   | Assign to Device/Group | AI Assistant | More |
| <input checked="" type="checkbox"/> | Name                   | Assigned to Device/Group |                        |              |      |
| <input type="checkbox"/>            | HUB_BGP_Recommended    | 0 Devices in Total       |                        |              |      |
| <input type="checkbox"/>            | BRANCH_BGP_Recommended | 0 Devices in Total       |                        |              |      |
| <input checked="" type="checkbox"/> | BGP_NGFW               | 0 Devices in Total       |                        |              |      |

9. Move **HQ-NGFW [Core1]** and **HQ-NGFW [Core2]** from the **Available Entries** field to the **Selected Entries** field, and then click **OK**.

### To install the BGP configuration

1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. Verify that **Install Device Settings (only)** is selected, and then click **Next**.
3. Verify that the **HQ-NGFW**, **Core1 [NAT]**, and **Core2 [NAT]** checkboxes are selected, and then click **Next**.
4. Click **Install**.
5. Wait until the installation finishes.
6. Click **Finish**.



You can check the installation and BGP status in the two VDOMs—Core1 and Core 2—using the following commands:

```
get router info bgp summary  
get router info bgp neighbors
```

## Exercise 3: Configuring a Loopback Interface as a BGP Source

In this exercise, you will create a loopback interface and configure BGP on BR1-FGT-2, and then establish a BGP connection with the ISP router.

### Configure a Loopback Interface as a BGP Source on BR1-FGT-2

Since FortiManager does not manage BR1-FGT-2, you must perform the loopback and BGP configurations on BR1-FGT-2.

#### To configure the loopback interface as the BGP source

1. Connect over SSH to BR1-FGT-2.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Enter the following commands to create the loopback interface:

```
config system interface
  edit Loopback_BR
  set vdom root
  set ip 100.75.5.1 255.255.255.0
  set allowaccess ping
  set type loopback
  next
end
```

4. Continuing in the same SSH session, enter the following commands to configure BGP:

```
config router bgp
  set as 65300
  set router-id 172.75.5.1
  config neighbor
    edit 100.66.0.101
      set remote-as 65200
      set update-source Loopback_BR
    next
  end
end
```

#### To update the BGP configuration on HQ-NGFW Core2

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Device Manager > Provisioning Templates > BGP**.



- 4. Select the **BGP\_NGFW** checkbox, and then click **Edit**.
- 5. Create a new neighbor with the following settings:

| Field         | Value      |
|---------------|------------|
| IP            | 100.75.5.1 |
| Remote AS     | 65300      |
| Connect Timer | 120        |

- 6. Expand **Advanced Options**, and then enable **ebgp-enforce-multihop**.
- 7. Click **OK**.

The configuration should look like the following image:

Name

BGP\_NGFW

Description

Local As

\$(AS)

Router ID

\$(Router\_ID)

Neighbors

+ Create New

Edit

Delete

More

|                          |                 |               |
|--------------------------|-----------------|---------------|
| <input type="checkbox"/> | IP              | Remote AS     |
| <input type="checkbox"/> | \$(Neighbor_IP) | \$(Remote_AS) |
| <input type="checkbox"/> | 100.75.5.1      | 65300         |

- 8. Click **OK**.

To install the BGP configuration on Core2

- 1. Select the **BGP\_NGFW** checkbox, and then click **Assign to Device/Group**.
- 2. Move **HQ-NGFW-1[Core1]** from the **Selected Entries** field to the **Available Entries** field, and then click **OK**.
- 3. Click **Install Wizard**.
- 4. Verify that **Install Device Settings (only)** is selected, and then click **Next**.
- 5. Verify that only the **HQ-NGFW** and **Core2 [NAT]** checkboxes are selected, and then click **Next**.
- 6. Click **Install**.
- 7. Wait until the installation finishes.
- 8. Click **Finish**.

## Establish the BGP connection

You will verify the BGP status, finalize the BGP configuration, and establish the BGP connection.

### To verify the BGP status

1. Connect over SSH to BR1-FGT-2.
2. Log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!

3. Enter the following command:

```
get router info bgp neighbors
```

You should see that the BGP neighbor is not directly connected.

```
BR1-FGT-2 # et router info bgp neighbors
VRF 0 neighbor table:
BGP neighbor is 100.66.0.101, remote AS 65200, local AS 65300, external link
BGP version 4, remote router ID 0.0.0.0
BGP state = Idle
Not directly connected EBGP
Last read          , hold time is 180, keepalive interval is 60 seconds
Configured hold time is 180, keepalive interval is 60 seconds
Received 0 messages, 0 notifications, 0 in queue
Sent 0 messages, 0 notifications, 0 in queue
Route refresh request: received 0, sent 0
NLRI treated as withdraw: 0
Minimum time between advertisement runs is 30 seconds
Update source is Loopback_BR
```

### Stop and think!

Why is the peer not directly connected?

Because a loopback interface adds one hop, you must always enable multihop in the BGP configuration that includes a loopback interface.

### To finalize the BGP configuration on BR-1-FGT-2

1. Continuing in the same SSH session, enter the following commands to finalize the BGP configuration:

```
config router bgp
config neighbor
edit 100.66.0.101
set ebgp-enforce-multihop enable
next
end
end
```

2. Enter the following command:

```
get router info bgp summary
```

You should see that the BGP connection is established.

```
BR1-FGT-2 # get router info bgp summary

VRF 0 BGP router identifier 172.75.5.1, local AS number 65300
BGP table version is 1
0 BGP AS-PATH entries
0 BGP community entries

Neighbor      V      AS MsgRcvd MsgSent   TblVer   InQ OutQ Up/Down State/PfxRcd
100.66.0.101  4      65200      3       3         0    0    0 00:01:21         0

Total number of neighbors 1
```



You may see the BGP state showing `Connect`. You can then enter the `get router info bgp summary` command again to confirm that the BGP connection is established.

## Lab 6: Security Profiles

In this lab, you will learn how to solve false positive scenarios in your network. Then, you will learn how to protect against PHP code injection attacks over encrypted and unencrypted traffic.

### Objectives

- Use IPS profiles with clients and servers as targets to protect your network resources
- Use IPS profiles to protect against PHP code injection attacks over unencrypted protocols
- Configure a certificate signed by an internal CA to block code injection occurrences over an encrypted protocol

### Time to Complete

Estimated: 55 minutes

### Which Network Segment Will You Work On?

You will modify the existing IPS profiles from FortiManager and push the changes to HQ-DCFV.

### Prerequisites

Before you begin this lab, you must complete the previous lab. If you haven't done so, tell your instructor.

## Exercise 1: Solving an IPS False Positive

In this exercise, you will learn how to solve false positive scenarios in your network. You will configure IPS profiles with clients and servers as targets to protect your network resources.

### Apply the IPS\_Block Security Profile

You will modify the existing DCFW policy in the FortiManager policy package to apply the **IPS\_Block** profile.

#### To apply the security profile

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Policy & Objects > Policy Packages**.
4. Click **DCFW > Firewall Policy**.
5. Select the checkbox for the **To HQ-Web-1** policy, and then click **Edit > Edit**.

|                                     |            |             |       |       |        |  |         |  |        |
|-------------------------------------|------------|-------------|-------|-------|--------|--|---------|--|--------|
|                                     | Create New |             | Edit  |       | Delete |  | Section |  | Policy |
| <input type="checkbox"/>            | #          | Name        | From  | To    |        |  |         |  |        |
| <input type="checkbox"/>            | 1          | Internet    | port5 | port2 |        |  |         |  |        |
| <input checked="" type="checkbox"/> | 2          | To HQ-Web-1 | port2 | port5 |        |  |         |  |        |

6. Configure the following settings:

| Field                   | Value                          |
|-------------------------|--------------------------------|
| Inspection Mode         | Flow-based                     |
| Security Profile Status | Enable                         |
| Profile Type            | Use Standard Security Profiles |
| IPS                     | IPS_Block                      |
| SSL/SSH Inspection      | certificate-inspection         |

The configuration should look like the following example:

Firewall/Network Options

Inspection Mode **Flow-based** Proxy-based

NAT ☐

Protocol Options **PROT** default +

Disclaimer Options

Display Disclaimer ☐

Security Profiles

**Security Profile Status** ☒

Profile Type **Use Standard Security Profiles** Use Security Profile Group

|                     |                                   |   |
|---------------------|-----------------------------------|---|
| AntiVirus Profile   |                                   | + |
| Web Filter Profile  |                                   | + |
| DNS Filter          |                                   | + |
| Application Control |                                   | + |
| IPS                 | <b>IPS</b> IPS_Block              | x |
|                     |                                   | + |
| File Filter         |                                   | + |
| Email Filter        |                                   | + |
| DLP Profile         |                                   | + |
| Virtual Patching    |                                   | + |
| VOIP                |                                   | + |
| ICAP                |                                   | + |
| SSL/SSH Inspection  | <b>SSL</b> certificate-inspection |   |
|                     |                                   | + |

7. Click **OK**.

## Install the Policy

You will install the policy and object changes on HQ-DCFw.

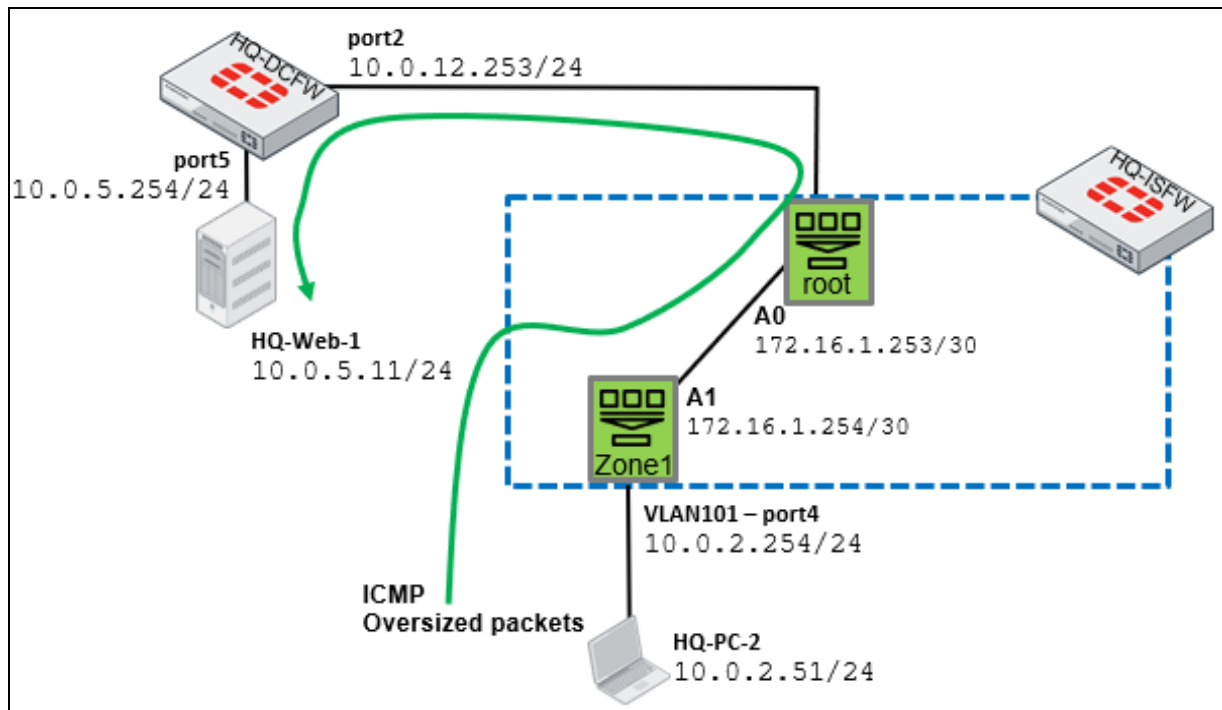
### To install the policy

1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. In the **Install Policy Package & Device Settings** window, confirm that the **DCFw** policy package is selected.
3. Click **Next**.
4. Confirm that **HQ-DCFw** is selected, and then click **Next**.
5. Click **Install Preview** to see the changes that will be applied to the FortiGate.

6. On the **Install Preview** page, click **Close**.
7. Click **Install**.
8. Wait until the installation finishes.
9. Click **Finish**.

## Test Using hping to HQ-Web-1

You will confirm that HQ-DCFW is not allowing ICMP oversized packets.



### To test using hping to HQ-Web-1

1. Connect to HQ-PC-2.
2. On HQ-PC-2, open a terminal window, and then enter the following command to send a jumbo packet to the Linux server:  

```
sudo hping3 -1 -d 654001 -c 4 10.0.5.11
```
3. When you are prompted for the student account, enter the password `Fortinet1!`.



`hping3` is the program that is used to generate and send the ICMP packets. `-1` instructs `hping3` to use ICMP for sending packets. `-d 654001` sets the ICMP packet size to 654001 bytes, a size that exceeds the maximum transmission unit (MTU). `-c 4` instructs `hping3` to send 4 ICMP packets. `10.0.5.11` is the destination IP address for the ICMP packets.

HQ-Web-1 fails to receive packets from HQ-PC-2.

```
Administrator@ubuntu-2204-desktop:~$ sudo hping3 -1 -d 654001 -c 4 10.0.5.11
[sudo] password for Administrator:
HPING 10.0.5.11 (eth0.101 10.0.5.11): icmp mode set, 28 headers + 64177 data bytes

--- 10.0.5.11 hping statistic ---
4 packets transmitted, 0 packets received 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
Administrator@ubuntu-2204-desktop:~$
```

You can see that HQ-DCFW is blocking the jumbo packets, which means that suspected risky traffic is being blocked according to the assigned IPS profile.

## Verify That HQ-DCFW Detects the Jumbo Packets

You will verify that HQ-DCFW detects jumbo packets as a possible attack.

### To verify the HQ-DCFW logs

1. Log in to the FortiAnalyzer GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Log View > Logs > Fortinet Logs**.
4. Select **Security > Intrusion Prevention**.

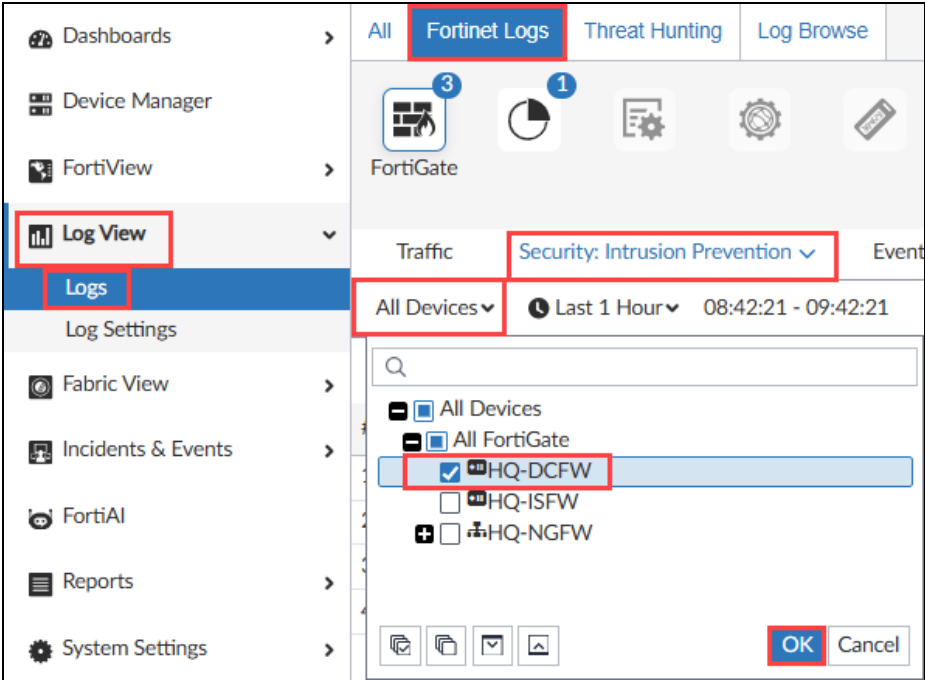


If the **Security** section does not appear, you can click **Log View > Logs** again to refresh the page.

The log may then take a few minutes to appear.

5. In the **All Devices** drop-down list, select the **HQ-DCFW** checkbox, and then click **OK**.





6. Double-click one of the dropped **PING** logs.
- You will notice that the traffic has been blocked by the HQ-DCFW firewall policy with the policy ID 2 and attack name **ICMP.Oversized.Packet**. Optionally, you can click the attack name to see more details.

| # | Date/Time    | Device ID | Severity | Destination IP | Action  | Service |  |
|---|--------------|-----------|----------|----------------|---------|---------|--|
| 1 | 2025-02-28 0 | FGVM02TM  | low      | 10.0.5.11      | dropped | PING    |  |
| 2 | 2025-02-28 0 | FGVM02TM  | low      | 10.0.5.11      | dropped | PING    |  |
| 3 | 2025-02-28 0 | FGVM02TM  | low      | 10.0.5.11      | dropped | PING    |  |
| 4 | 2025-02-28 0 | FGVM02TM  | low      | 10.0.5.11      | dropped | PING    |  |

Action

Action

dropped

Policy ID

2

Policy UUID

c78fe35c-f4eb-51ef-0c56-d1aab3ac408e

Threat

32768

Application

Profile

IPS\_Block

Protocol

1

Service

PING

Data

Archive

42991624:0 (63 KB)

Threat

Attack ID

12880

Attack Name

ICMP.Oversized.Packet

Direction

outgoing

ICMP Code

0x00

ICMP ID

0x3c85

ICMP Type

0x08

Incident Serial No.

42991624

Reference

<https://fortiguard.fortinet.com/encyclopedia/ips/12880>



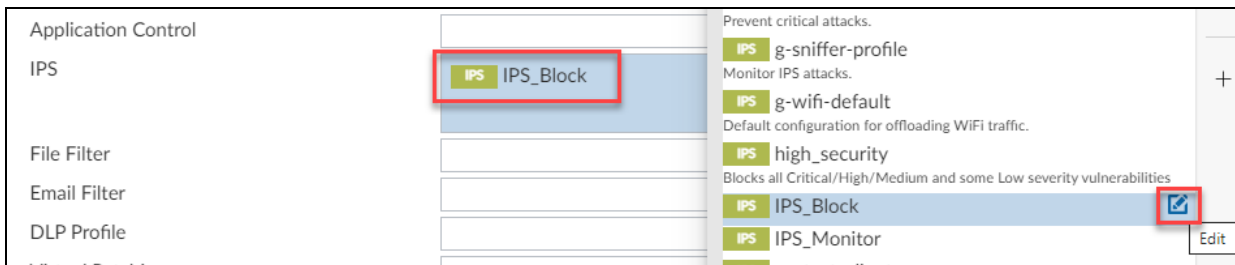
This example of jumbo ICMP packets illustrates a hypothetical scenario where HQ-PC-2 is trying to transfer, connect, or sync an application to HQ-Web-1, and FortiGate mistakenly flags it as intrusive. Such occurrences are common when you use IPS profiles that are aimed at blocking. However, not all flagged traffic is genuinely malicious. It is important to consider the possibility of false positives when you block IPS traffic.

## Solve the IPS False Positive

You will modify the **IPS\_Block** profile to allow false positive jumbo packets.

### To modify and install the IPS\_Block profile

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Policy & Objects > Policy Packages**.
4. Click **DCFW > Firewall Policy**.
5. Select the checkbox for the **To HQ-Web-1** policy, and then click **Edit > Edit**.
6. In the **IPS** field, click **IPS\_Block**, and then click the **Edit** icon.



7. In the **IPS Signatures and Filters** section, click **Create New**.
8. In the **Type** field, select **Signature**.
9. Configure the following settings:

| Field          | Value                 |
|----------------|-----------------------|
| Action         | Monitor               |
| Packet logging | Enable                |
| Status         | Enable                |
| Signatures     | ICMP.Oversized.Packet |

10. In the **Signatures** field, click **Add Signature**.
11. In the search field, search for **ICMP.Oversized.Packet**.
12. Select the **ICMP.Oversized.Packet** signature, and then click **Use Selected Signatures**.

13. Click **OK**.
14. Move the **ICMP.Oversized.Packet** signature to the top of the table.

| IPS Signatures and Filters                                                                                                                      |                          |               |            |         |                |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|------------|---------|----------------|---------|
| <div> <span>+ Create New</span> <span>Edit</span> <span>Delete</span> <span>Move Up</span> <span>Move Down</span> <span>Search...</span> </div> |                          |               |            |         |                |         |
| <input type="checkbox"/>                                                                                                                        | Details                  | CVE-ID        | Exempt IPs | Action  | Packet Logging | Status  |
| <input checked="" type="checkbox"/>                                                                                                             | ICMP.Oversized.Packet    | CVE-1999-0128 | 0          | Monitor | Enabled        | Enabled |
| <input type="checkbox"/>                                                                                                                        | TGT server<br>TGT client |               | 0          | Block   | Enabled        | Enabled |
| 1 Selected                                                                                                                                      |                          |               |            |         |                | 2       |

15. Click **OK** to save the changes.
16. Click **Close** to close the **Select Entries** window.
17. Click **OK** to save the changes to the policy.

## Install the Policy

You will install the policy and object changes on HQ-DCFW.

### To install the policy

1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. In the **Install Policy Package & Device Settings** window, confirm that the **DCFW** policy package is selected.
3. Click **Next**.
4. Confirm that **HQ-DCFW** is selected, and then click **Next**.
5. Click **Install Preview** to see the changes that will be applied to the FortiGate.
6. On the **Install Preview** page, click **Close**.
7. Click **Install**.
8. Click **Finish**.

## Test Using hping to HQ-Web-1

You will confirm that HQ-DCFW is allowing ICMP oversized packets.

### To test using hping to HQ-Web-1

1. Connect to HQ-PC-2.
2. On HQ-PC-2, open a terminal window, and then enter the following command to send a jumbo packet to HQ-Web-1:  

```
sudo hping3 -1 -d 654001 -c 4 10.0.5.11
```
3. When you are prompted for the student account, enter the password `Fortinet1!`.  
HQ-Web-1 is receiving packets from HQ-PC-2.

```
Administrator@ubuntu-2204-desktop:~$ sudo hping3 -1 -d 654001 -c 4 10.0.5.11
[sudo] password for Administrator:
HPING 10.0.5.11 (eth0.101 10.0.5.11): icmp mode set, 28 headers + 64177 data bytes
len=1500 ip=10.0.5.11 ttl=61 DF id=43100 icmp_seq=0 rtt=38.4 ms
len=1500 ip=10.0.5.11 ttl=61 DF id=43110 icmp_seq=1 rtt=5.3 ms
len=1500 ip=10.0.5.11 ttl=61 DF id=43213 icmp_seq=2 rtt=12.3 ms
len=1500 ip=10.0.5.11 ttl=61 DF id=43409 icmp_seq=3 rtt=10.1 ms

--- 10.0.5.11 hping statistic ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 5.3/16.5/38.4 ms
Administrator@ubuntu-2204-desktop:~$
```

The HQ-DCFW firewall now detects the new signature oversized packet as the action monitor, so it allows the traffic and also generates a log.

## Verify That HQ-DCFW Detects the Jumbo Packets

You will verify that HQ-DCFW detects the jumbo packets and allows the traffic.

### To verify the HQ-DCFW logs

1. Log in to the FortiAnalyzer GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Log View > Logs > Fortinet Logs**.
4. Select **Security > Intrusion Prevention**.
5. In the **All Devices** drop-down list, select the **HQ-DCFW** checkbox, and then click **OK**.
6. Double-click one of the detected **PING** logs.

You will notice that the HQ-DCFW firewall policy with the policy ID **2** and attack name **ICMP.Oversized.Packet** allowed the traffic, and the action is **detected**.



If the log with the **detected** action does not appear, you can click **Log View > Logs** again to refresh the page.

Traffic

Security: Intrusion Prevention

Event

HQ-DCFW

Last 1 Hour

06:31:23 - 07:31:23

Full Screen

Create Custom View

Refresh

More

Filter Mode

Add Filter

| # | Date/Time    | Device ID | Severity | Destination IP | Action   | Service |
|---|--------------|-----------|----------|----------------|----------|---------|
| 1 | 2025-03-01 0 | FGVM02TM  | low      | 10.0.5.11      | detected | PING    |
| 2 | 2025-03-01 0 | FGVM02TM  | low      | 10.0.5.11      | detected | PING    |
| 3 | 2025-03-01 0 | FGVM02TM  | low      | 10.0.5.11      | detected | PING    |
| 4 | 2025-03-01 0 | FGVM02TM  | low      | 10.0.5.11      | detected | PING    |

Action

Action

detected

Policy ID

2

Policy UUID

c78fe35c-f4eb-51ef-0c56-d1aab3ac

Threat

408e32768

Application

Profile

IPS\_Block

Protocol

1

Service

PING

Data

Archive

42991632:0 (63 KB)

Threat

Attack ID

12880

Attack Name

ICMP.Oversized.Packet

Direction

incoming

ICMP Code

0x00

ICMP ID

0xc19f

ICMP Type

0x00

Incident Serial No.

42991632

Reference

https://fortiguard.fortinet.com/encyclopedia/ips/12880



If you encounter scenarios where an IPS signature blocks your native traffic, and an application that you are using is considered hostile, you should set the application signature to **monitor**, at the top of your rules.

## Exercise 2: Protecting Against Unencrypted Attacks

In this exercise, you will learn how to protect against a PHP code injection attack that is acting as an unencrypted protocol. You will also learn to identify the issue, and then rectify the issue using an IPS profile.

### Apply the IPS\_Monitor Security Profile

You will modify the existing DCFW policy in the FortiManager policy package to apply the **IPS\_Monitor** profile.

#### To apply the security profile

1. Log in to the FortiManager GUI with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
2. Click **EFW**.
3. Click **Policy & Objects > Policy Packages**.
4. Click **DCFW > Firewall Policy**.
5. Select the checkbox for the **To HQ-Web-1** policy, and then click **Edit > Edit**.
6. Configure the following settings:

| Field              | Value                  |
|--------------------|------------------------|
| IPS                | IPS_Monitor            |
| SSL/SSH Inspection | certificate-inspection |

The configuration should look like the following example:

Security Profiles

Security Profile Status

Profile Type

Use Standard Security Profiles

Use Security Profile Group

AntiVirus Profile

+

Web Filter Profile

+

DNS Filter

+

Application Control

+

IPS

IPS

IPS\_Monitor

+

×

File Filter

+

Email Filter

+

DLP Profile

+

Virtual Patching

+

VOIP

+

ICAP

+

SSL/SSH Inspection

SSL

certificate-inspection

+

7. Click **OK**.

Install the Policy

You will install the policy and object changes on HQ-DCFW.

To install the policy

1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. In the **Install Policy Package & Device Settings** window, confirm that the **DCFW** policy package is selected.
3. Click **Next**.
4. Confirm that **HQ-DCFW** is selected, and then click **Next**.
5. Click **Install Preview** to see the changes that will be applied to the FortiGate.
6. On the **Install Preview** page, click **Close**.
7. Click **Install**.
8. Wait until the installation finishes.
9. Click **Finish**.

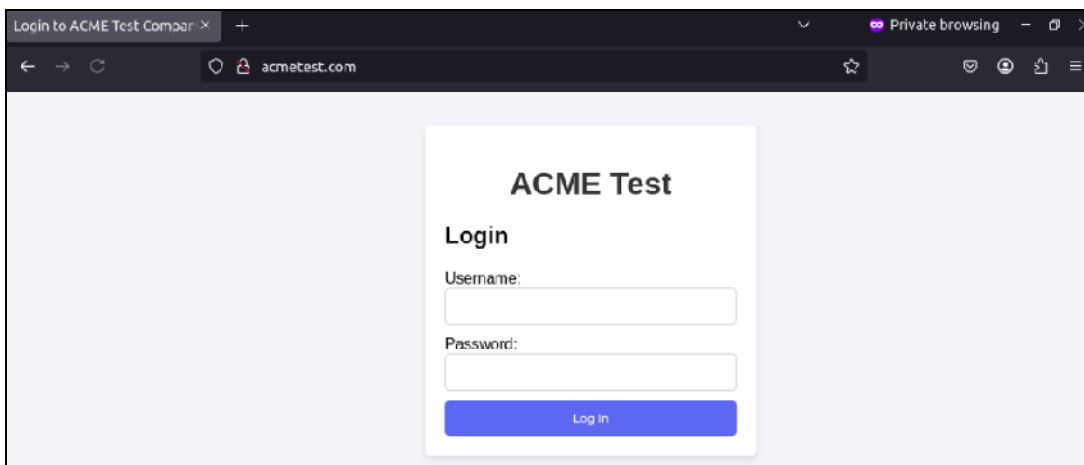
## Access the Website

You will test to see if the `http://acmetest.com` website is accessible, which is hosted on HQ-Web-1.

### To access the website

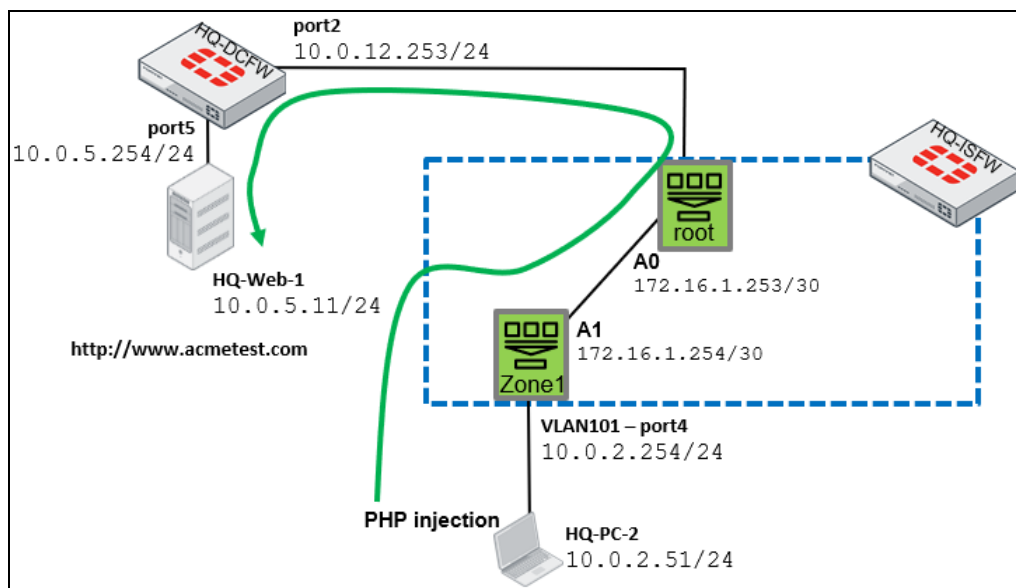
1. Connect to HQ-PC-2.
2. Open a browser, and then enter the `http://acmetest.com` URL.

You should be able to access the website, which is hosted on HQ-Web-1.



## Simulate the Attack

You will simulate an attack to access the files on HQ-Web-1 using a URL with a PHP code injection.



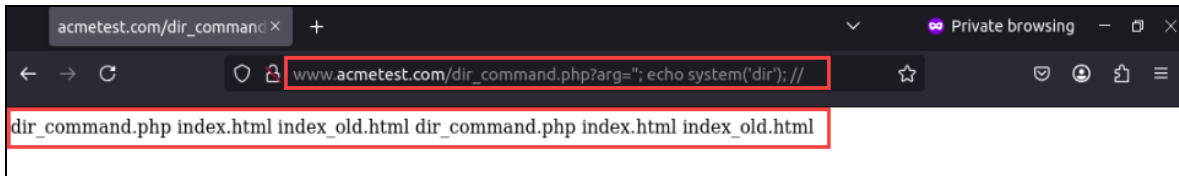


### To simulate the attack

1. Connect to HQ-PC-2.
2. Navigate to **Desktop > Resources > Enterprise-FW**.
3. Open the `PHP injection.txt` file.
4. In the **Exercise 2** section, copy the URL.

The URL is `http://www.acmetest.com/dir_command.php?arg="; echo system('dir'); //`.

5. Open a browser, open a new private window, and then paste the URL.

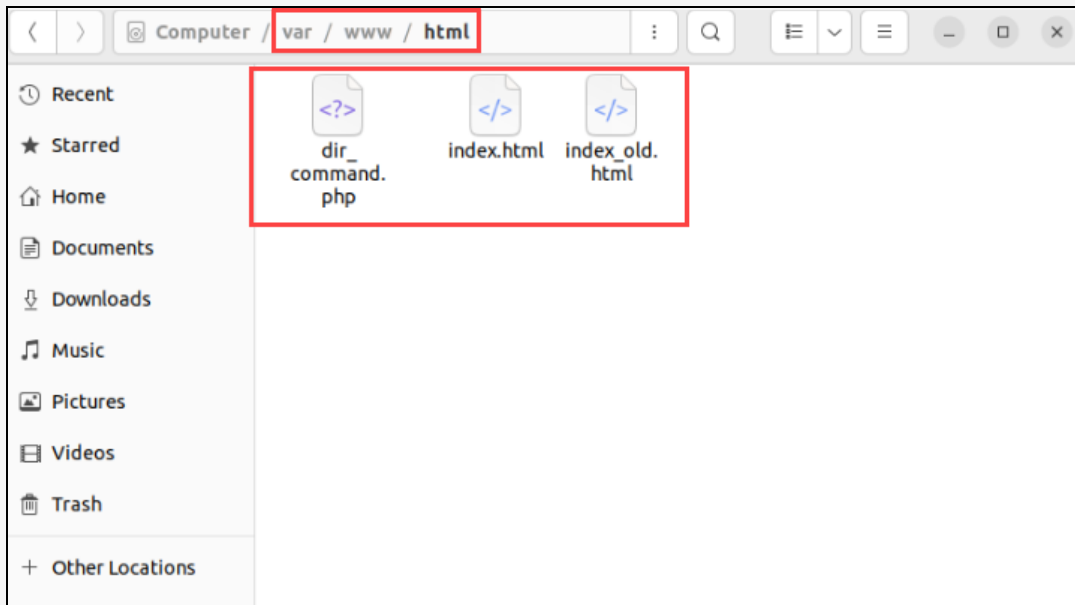


### Stop and think!

This HTTP URL triggers the `dir` command on HQ-Web-1. For example, if you are an administrator user, accessing the URL in an external browser displays the following files:

```
dir_command.php
index.html
index_old.html
```

These files are located in the following directory on HQ-Web-1: `Other Locations/Computer/var/www/html`.



### Verify That HQ-DCFW Detects the Attack

You will verify that HQ-DCFW detects the attack.

### To verify the HQ-DCFW logs

1. Log in to the FortiAnalyzer GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Log View > Logs > Fortinet Logs**.
4. Select **Security > Intrusion Prevention**.
5. In the **All Devices** drop-down list, select the **HQ-DCFW** checkbox, and then click **OK**.
6. Double-click one of the detected **HTTP** logs.

You will notice that the HQ-DCFW firewall policy with the policy ID **2** and attack name **PHP.URI.Code.Injection** identified the traffic. Optionally, you can click the attack name to see more details.

| # | Date/Time    | Device ID | Severity | Destination IP | Action   | Service | User | country | Count |  |
|---|--------------|-----------|----------|----------------|----------|---------|------|---------|-------|--|
| 1 | 2025-03-01 0 | FGVM02TME | high     | 10.0.5.11      | detected | HTTP    |      |         |       |  |
| 2 | 2025-03-01 0 | FGVM02TME | high     | 10.0.5.11      | detected | HTTP    |      |         |       |  |
| 3 | 2025-03-01 0 | FGVM02TME | high     | 10.0.5.11      | dropped  | HTTP    |      |         |       |  |

|                    |                                               |
|--------------------|-----------------------------------------------|
| <b>Action</b>      | detected                                      |
| Policy ID          | 2                                             |
| Policy UUID        | 278625c74ee51e40c56d1aab3ac                   |
| Threat             | 400e R192                                     |
| <b>Application</b> |                                               |
| Profile            | IPS_Monitor                                   |
| Protocol           | 6                                             |
| Service            | HTTP                                          |
| URI                | /cli_command.php?arg=echo system%0A           |
| <b>Data</b>        |                                               |
| Archive            | 42991639.0 (510 B)                            |
| <b>Threat</b>      |                                               |
| Attack ID          | 15403                                         |
| Attack Name        | PHP.URI.Code.Injection                        |
| Direction          | outgoing                                      |
| Incident Serial No | 42991639                                      |
| Reference          | https://brave.dumps.com/en/fortinet/ips/15403 |
| Severity           | high                                          |

The attacker was able to successfully run the CLI command on HQ-Web-1 (HTTP) because the IPS profile is set to monitor only.



This PHP command, or instruction, clearly demonstrates how, without having access to the remote server, there could potentially be an attempt to exploit a security vulnerability known as remote code execution (RCE). Specifically, the URL includes a query parameter that injects PHP code. This type of code injection is malicious because attackers can use any type of CLI command without having administrative credentials for HQ-Web-1.

## Block an Attack on Unencrypted Traffic

You will learn how to modify the **IPS\_Monitor** profile to block PHP code injection attacks.

### To modify and install the IPS\_Block profile

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!

- Click **EFW**.
- Click **Policy & Objects > Policy Packages**.
- Click **DCFW > Firewall Policy**.
- Select the checkbox for the **To HQ-Web-1** policy, and then click **Edit > Edit**.
- In the **IPS** field, click **IPS\_Monitor**, and then click the **Edit** icon.



- In the **IPS Signatures and Filters** section, click **Create New**.
- Configure the following settings:

| Field          | Value                  |
|----------------|------------------------|
| Type           | Signature              |
| Action         | Block                  |
| Packet logging | Enable                 |
| Status         | Enable                 |
| Signatures     | PHP.URI.Code.Injection |

- Click **OK**.
- Move the **PHP.URI.Code.Injection** signature to the top of the table.

| IPS Signatures and Filters                                                                                                          |                          |        |            |         |                |         |   |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------|------------|---------|----------------|---------|---|
| <div> <div>+ Create New</div> <div>Edit</div> <div>Delete</div> <div>Move Up</div> <div>Move Down</div> <div>Search...</div> </div> |                          |        |            |         |                |         |   |
| <input type="checkbox"/>                                                                                                            | Details                  | CVE-ID | Exempt IPs | Action  | Packet Logging | Status  |   |
| <input type="checkbox"/>                                                                                                            | PHP.URI.Code.Injection   | N/A    | 0          | Block   | Enabled        | Enabled |   |
| <input type="checkbox"/>                                                                                                            | TGT server<br>TGT client |        | 0          | Monitor | Enabled        | Enabled |   |
|                                                                                                                                     |                          |        |            |         |                |         | 2 |

- Click **OK**.
- Click **Close** to close the **Select Entries** window.
- Click **OK** to save the changes to the policy.

## Install the Policy

You will install the policy and object changes on HQ-DCFW.

### To install the policy

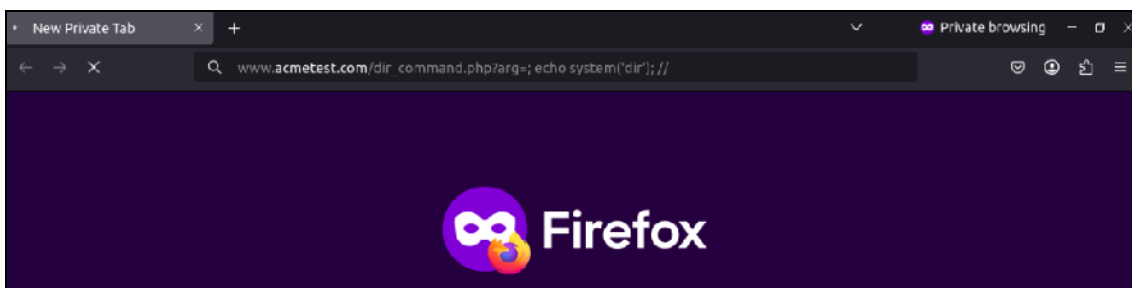
1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. In the **Install Policy Package & Device Settings** window, confirm that the **DCFW** policy package is selected.
3. Click **Next**.
4. Confirm that **HQ-DCFW** is selected, and then click **Next**.
5. Click **Install Preview** to see the changes that will be applied to the FortiGate.
6. On the **Install Preview** page, click **Close**.
7. Click **Install**.
8. Click **Finish**.

## Simulate the Attack

You will simulate an attack to access the files on HQ-Web-1 using a URL with a PHP code injection.

### To simulate the attack

1. Connect to HQ-PC-2.
2. Navigate to **Desktop > Resources > Enterprise-FW**.
3. Open the `PHP injection.txt` file.
4. In the **Exercise 2** section, copy the URL.  
The URL is `http://www.acmetest.com/dir_command.php?arg="; echo system('dir'); //`.
5. Open a browser, open a new private window, and then paste the URL.  
This time, you can see that the attacker cannot access the HQ-Web-1 file.



## Verify That HQ-DCFW Blocks the Attack

You will verify that HQ-DCFW blocks the attack.

### To verify the HQ-DCFW logs

1. Log in to the FortiAnalyzer GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Log View > Logs > Fortinet Logs**.
4. Select **Security > Intrusion Prevention**.
5. In the **All Devices** drop-down list, select the **HQ-DCFW** checkbox, and then click **OK**.
6. Double-click one of the **HTTP** dropped logs.

You will notice that the HQ-DCFW firewall policy with the policy ID **2** and attack name **PHP.URI.Code.Injection** blocked the traffic. Optionally, you can click the attack name to see more details.

| # | Date/Time           | Device ID | Severity | Destination IP | Action   | Service |
|---|---------------------|-----------|----------|----------------|----------|---------|
| 1 | 2025-03-01 01:00:00 | FGVM02TM  | high     | 10.0.5.11      | dropped  | HTTP    |
| 2 | 2025-03-01 01:00:00 | FGVM02TM  | high     | 10.0.5.11      | detected | HTTP    |

**Action**  
Action: dropped  
Policy ID: 2  
Policy UUID: c78fc35c-f4eb-51ef-0c56-d1aab3ac408e  
Threat: 8192  
**Application**  
Profile: IPS\_Monitor  
Protocol: 6  
Service: HTTP  
URI: /dir\_command.php?arg=; echo system(d  
**Data**  
Archive: 42991642.0 (510 B)  
**Threat**  
Attack ID: 15463  
Attack Name: PHP.URI.Code.Injection  
Direction: outgoing  
Incident Serial No.: 42991642  
Reference: <https://fortiguard.fortinet.com/encyclopedia/ips/15463>  
Severity: high

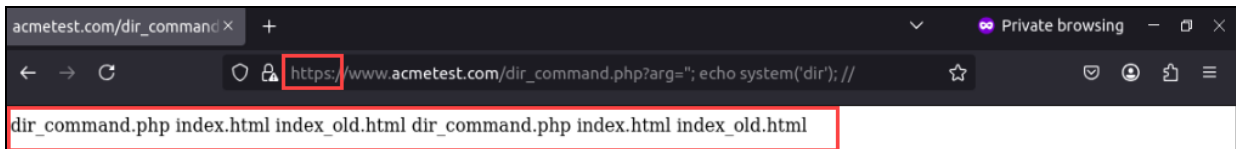
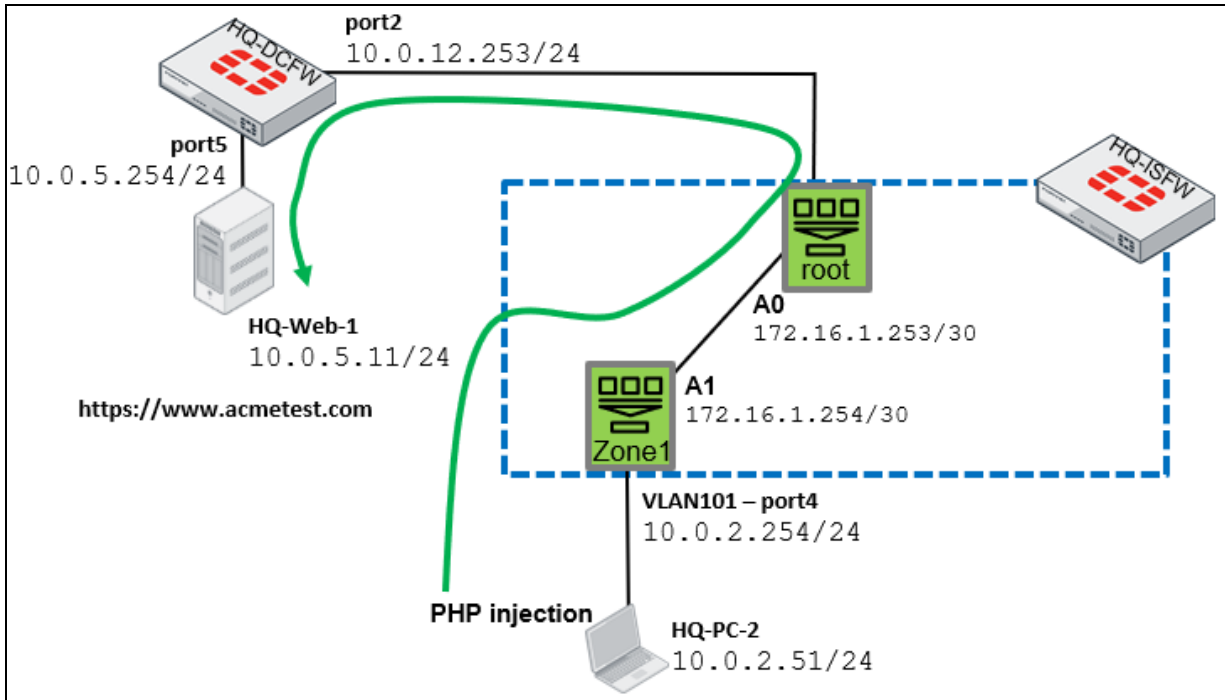


When an attack like this occurs, one recommendation from the FortiGuard Labs Threat Encyclopedia is to block it. Ensure that you add a signature at the top of the signature list to prioritize blocking this attack. When the PHP URI code is considered malicious, the firewall will drop packets, and the browser will show a **connection reset** error because the code can no longer pass through the firewall.

#### Stop and think!

In this exercise, you have used only unencrypted traffic (HTTP) to access HQ-Web-1. What happens if a code injection occurs over an encrypted protocol? Will a policy with an IPS profile and certificate inspection be sufficient to analyze HTTPS traffic? In this scenario, FortiGate will detect HTTPS traffic, but it will not detect an attack. FortiGate will not decrypt packets to analyze if there is a possible attack.

If you use HTTPS (encrypted) traffic, HQ-DCFW will not be able to detect the attack. The attacker will be able to access the files on HQ-Web-1, as shown in the following example:



When you receive the **Potential Security Risk Ahead** warning, click **Advanced > Accept the Risk and Continue** to access [www.acmetest.com](https://www.acmetest.com) (unsafe).

### Stop and think!

This HTTPS URL triggers the `dir` command on HQ-Web-1. Why didn't HQ-DCFW block the HTTPS attack?

The **PHP.URI.Code.Injection** signature is set to **block** in the profile, but HQ-DCFW cannot open the packet because the traffic is encrypted, and certificate inspection on its own does not help HQ-DCFW to see the encrypted traffic.

## Exercise 3: Protecting Against Encrypted Attacks

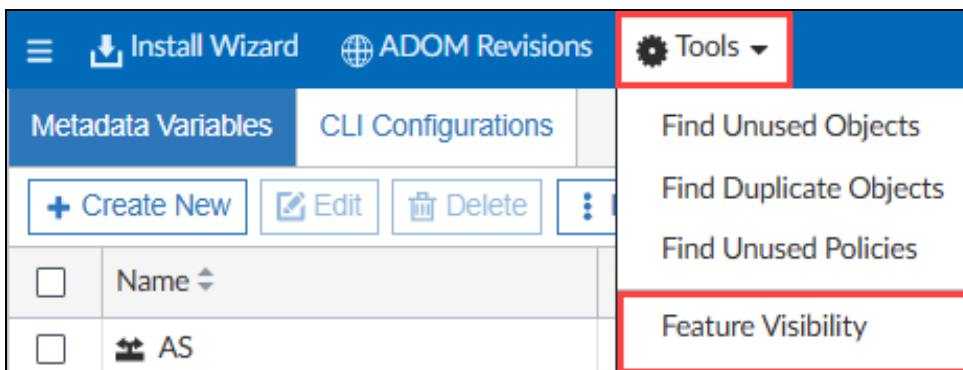
In this exercise, you will learn how to simulate a PHP code injection attack over an encrypted protocol. You will use a certificate signed by an internal CA to block code injection occurrences over an encrypted protocol.

### Create a Dynamic Local Certificate and an SSL/SSH Profile

You will learn how to block encrypted packets on FortiGate. You must use an SSL protection server on your SSL/SSH profile.

#### To create a dynamic local certificate

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Policy & Objects > Advanced**.
4. Click **Tools > Feature Visibility**.



5. In the **Advanced** section, enable **Dynamic Local Certificate**.
6. Click **OK**.
7. Click **Dynamic Local Certificate > Create New**.  
Configure the following settings:

| Field              | Value        |
|--------------------|--------------|
| Name               | Enable_HTTPS |
| Per-Device Mapping | HQ-DCFw      |
| Local Certificate  | acmetest     |



The acmetest certificate that you are using in this exercise was created in Lab 2, Exercise 1.

8. Click **OK**.
  9. Click **OK** again to save the changes.
- The configuration should look like the following example:

| Metadata Variables       |                 | CLI Configurations |  | Dynamic Local Certificate |  |
|--------------------------|-----------------|--------------------|--|---------------------------|--|
| + Create New             |                 | Edit               |  | Delete                    |  |
|                          |                 |                    |  | More ▾                    |  |
|                          |                 |                    |  | Full Screen               |  |
| <input type="checkbox"/> | Name ▾          |                    |  | Description ▾             |  |
| <input type="checkbox"/> | Enable_HTTPS    |                    |  |                           |  |
| <input type="checkbox"/> | Fortinet_CA_SSL |                    |  |                           |  |

### To create an SSL/SSH inspection profile

1. Click **Policy & Objects > Security Profiles > SSL/SSH Inspection**.

| Protocol Options | ⋮ ▾                         |
|------------------|-----------------------------|
|                  | SSL/SSH Inspection          |
|                  | Application Signatures      |
|                  | IPS Signatures              |
|                  | Virtual Patching Signatures |

2. Click **Create New**.
- Configure the following settings:

| Field                    | Value                 |
|--------------------------|-----------------------|
| Name                     | HTTPS_Profile         |
| Enable SSL Inspection of | Protecting SSL Server |
| Server Certificate       | Enable_HTTPS          |

The configuration should look like the following example:



Create New SSL/SSH Inspection Profile

Name: **HTTPS\_Profile**

Comments:

SSL Inspection Options

Enable SSL Inspection of Server Certificate: **Protecting SSL Server**

Search: **Enable\_HTTPS Local Certificate**

3. Click **OK** to save the profile.

### Stop and think!

Why did you have to dynamically map the local certificate? When you have a local certificate, you cannot use it directly in the policy layer. The local certificate exists only in the device layer. To use the device layer local certificate in the policy layer, you must use dynamic mapping.

## Block an Attack on Encrypted Traffic

You will learn how to modify the policy to use the certificate that you created in the previous task to block a PHP code injection attack on encrypted traffic.

### To modify and install the policy

1. Continuing on the FortiManager GUI, click **Policy & Objects > Policy Packages**.
2. Click **DCFw > Firewall Policy**.
3. Select the **To HQ-Web-1** policy, and then click **Edit > Edit**.
4. In the **Inspection Mode** field, select **Proxy-based**.
5. Click **SSL/SSH Inspection > certificate-inspection**.
6. Select **HTTPS\_Profile > Close**.
7. Click **OK** to save the changes.

The policy configuration should look like the following example:

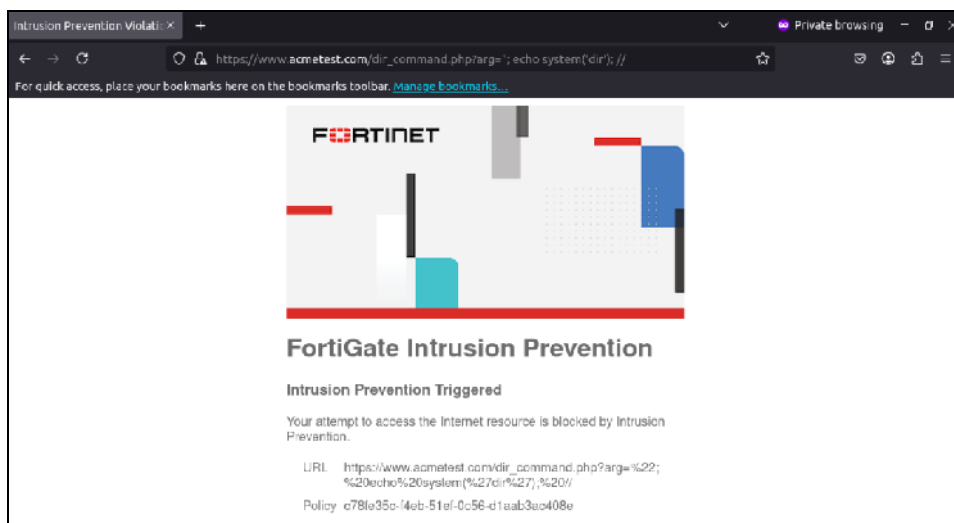
| # | Name        | From  | To    | Source | Destination | Service | NAT      | Action | Schedule | Security Profiles                    |
|---|-------------|-------|-------|--------|-------------|---------|----------|--------|----------|--------------------------------------|
| 1 | Internet    | port5 | port2 | 10.0.5 | all         | ALL     | Disabled | Accept | always   | SSL no-inspection default            |
| 2 | To HQ-Web-1 | port2 | port5 | all    | 10.0.5      | ALL     | Disabled | Accept | always   | IPS IPS_Monitor<br>SSL HTTPS_Profile |

8. Continuing on the FortiManager GUI, click **Install Wizard**.
9. In the **Install Policy Package & Device Settings** window, confirm that the **DCFw** policy package is selected.
10. Click **Next**.

11. Confirm that **HQ-DCFW** is selected, and then click **Next**.
12. Click **Install Preview** to see the changes that will be applied to the FortiGate.
13. On the **Install Preview** page, click **Close**.
14. Click **Install**.
15. Click **Finish**.

### To simulate the attack again with the server certificate

1. Connect to HQ-PC-2.
2. Navigate to **Desktop > Resources > Enterprise-FW**.
3. Open the `PHP injection.txt` file.
4. In the **Exercise 3** section, copy the URL.  
The URL is `https://www.acmetest.com/dir_command.php?arg="; echo system('dir'); //`.
5. Open a browser, open a new incognito window, and then paste the URL.  
You can see that HQ-DCFW is blocking the attack.



## Verify That HQ-DCFW Drops the Attack

You will verify that HQ-DCFW drops the attack.

### To verify the HQ-DCFW logs

1. Log in to the FortiAnalyzer GUI with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
2. Click **EFW**.
3. Click **Log View > Logs**.
4. Select **Security > Intrusion Prevention**.
5. In the **All Devices** drop-down list, select the **HQ-DCFW** checkbox, and then click **OK**.
6. Double-click one of the **HTTPS** dropped logs.

You will notice that the HQ-DCFW firewall policy with the policy ID **2** and attack name **PHP.URI.Code.Injection** blocked the traffic. Optionally, you can click the attack name to see more details.

| Device ID | Severity | Destination IP | Action   | Service | User | country | Count |  |
|-----------|----------|----------------|----------|---------|------|---------|-------|--|
| FGVM02TM  | High     | 10.0.5.11      | dropped  | HTTPS   |      |         |       |  |
| FGVM02TM  | High     | 10.0.5.11      | dropped  | HTTP    |      |         |       |  |
| FGVM02TM  | High     | 10.0.5.11      | detected | HTTP    |      |         |       |  |
| FGVM02TM  | High     | 10.0.5.11      | detected | HTTP    |      |         |       |  |
| FGVM02TM  | High     | 10.0.5.11      | detected | HTTP    |      |         |       |  |
| FGVM02TM  | High     | 10.0.5.11      | dropped  | HTTP    |      |         |       |  |

Action

Action

dropped

Policy ID

2

Policy UUID

c78fe95c-f4e0-51ef-0c56-d1a03ac408e

Threat

81192

Application

Profile

IPS\_Monitor

Protocol

6

Service

HTTPS

URL

/dir\_command.php?arg="; echo system'd

Data

Archive

42991653:0 (605 B)

Threat

Attack ID

15463

Attack Name

PHP.URI.Code.Injection

Direction

outgoing

Incident Serial No.

42991653

Reference

https://fortiguard.fortinet.com/encyclopedia/ips/15463

Severity

high

Stop and think!

Why does HQ-DCFW block the attack this time?

With the internally signed CA certificate, HQ-DCFW can inspect the HTTPS traffic and identify the attack.

## Lab 7: IPsec VPN (IKEv2)

In this lab, you will configure a hub-and-spoke VPN network using the FortiManager IPsec templates.

### Objectives

- Configure a hub-and-spoke topology using IPsec templates on FortiManager
- Use automatic routing to interconnect remote network segments
- Use a zone interface to group spoke interfaces in a hub
- Use a custom configuration of the VPN IPsec templates
- Identify and apply the process to delete IPsec templates

### Time to Complete

Estimated: 45 minutes

### Prerequisites

Before you begin this lab, you must complete the previous lab. If you haven't done so, tell your instructor.

## Exercise 1: Configuring IPsec Templates

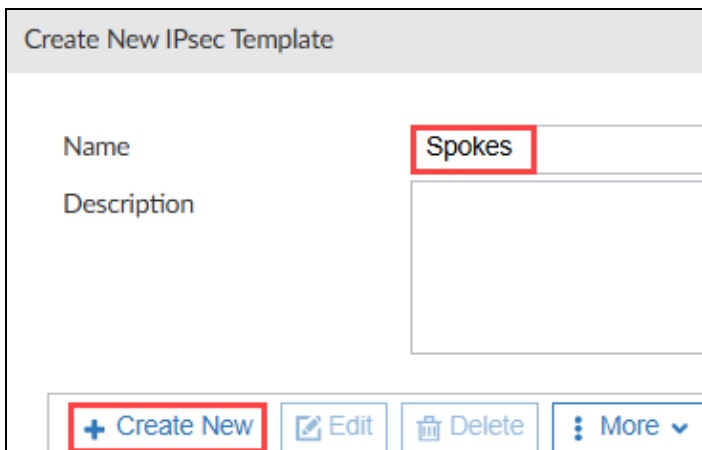
In this exercise, you will configure a hub-and-spoke topology using IPsec templates.

### Configure IPsec Templates

You will configure IPsec templates for a hub-and-spoke topology.

#### To configure the spokes IPsec template

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Device Manager > Provisioning Templates > IPsec Tunnel > Create New**.
4. In the **Name** field, type *Spokes*.



5. Click **Create New**, and then configure the following settings:

| Field            | Value                    |
|------------------|--------------------------|
| Tunnel Name      | To_hub                   |
| IP Address       | 100.65.0.101             |
| Interface        | port2                    |
| Remote Subnet    | 10.0.12.0/255.255.255.0  |
| Pre-shared Key   | 123456789                |
| Phase 1 Proposal | <b>AES256 and SHA256</b> |

| Field                 | Value                                                         |
|-----------------------|---------------------------------------------------------------|
| Diffie-Hellman Groups | 20                                                            |
| Phase 2 selectors     | Click <b>Create New</b> , and then continue to the next step. |

Create New IPsec Tunnel

Tunnel Name

🔍 To\_hub

Network ▾

Routing

Manual Automatic

IP Version

IPv4 IPv6

Remote Gateway

Static IP Address Dialup User Dynamic DNS

IP Address

🔍 100.65.0.101

Interface

🔍 port2

Local Gateway

Main Interface IP Specify

Mode Config

☐

Transport ⓘ

UDP Auto TCP Encapsulation

Use Fortinet Encapsulation ⓘ

☐

NAT Traversal

Enable Disable

Keepalive Frequency

10

Dead Peer Detection

Disable On Idle On Demand

DPD Retry Count

3

DPD Retry Interval

20 Seconds ✕ +

Forward Error Correction

☐ Egress ☐ Ingress

Remote Subnet

🔍 10.0.12.0/255.255.255.0 ✕ +

FEC Health Check

🔍

Advanced Network Settings >

Authentication ▾

Authentication Method

Pre-shared Key Signature

Pre-shared Key

123456789 ✕ 🔒

IKE

Version 1 Version 2

XAUTH

Disable Client Server

Phase 1 Proposal ▾

Encryption - Authentication

Encryption Authentication Action

AES256 SHA256 ✕ +

Diffie-Hellman Groups

☐ 1 ☐ 2 ☐ 5

☐ 14 ☐ 15 ☐ 16

☐ 17 ☐ 18 ☐ 19

☒ 20 ☐ 21 ☐ 27

☐ 28 ☐ 29 ☐ 30

☐ 31 ☐ 32



Make sure that the **FEC Health Check** field remains empty.

Stop and think!

Make sure that you distinguish names. You have created the spokes template with the name **Spokes**, but the interfaces are named **To\_hub**. This will help you recognize interfaces that are connected to other peers. When you go to BR2-FGT-1, you will find the hub interface, which will refer to the IPsec VPN connected to the hub.

6. Configure the following settings:

| Field            | Value         |
|------------------|---------------|
| Name             | PH2_to_hub    |
| Proposal         | aes256-sha256 |
| Advanced Options | dhgrp 20      |

Create New Phase 2 Interface

Name

PH2\_to\_hub

Keep Alive

☐

Key Life

43200

Proposal

aes256-sha256

1 entry selected

Advanced Options

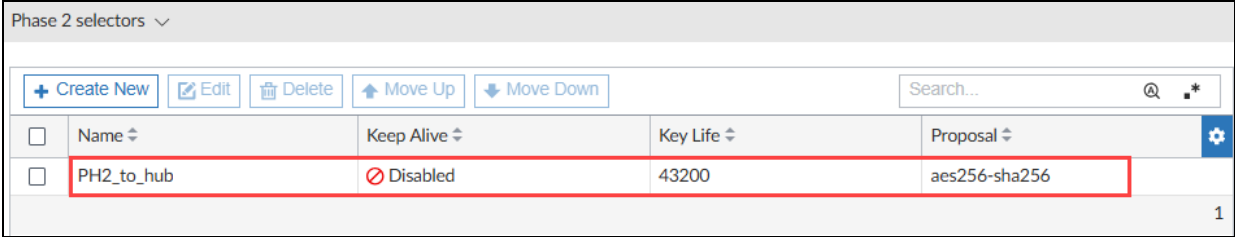
dhgrp

☐ 1 ☐ 2 ☐ 5 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☒ 20 ☐ 21 ☐ 27 ☐ 28 ☐ 29 ☐ 30 ☐ 31 ☐ 32



You will find multiple advanced options for IPsec phase2. You can press **Ctrl+F** to find dhgrp.

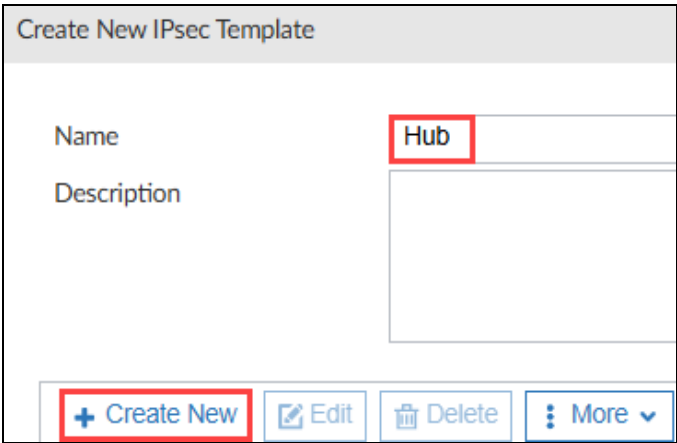
7. Click **OK**.



- 8. Click **OK**.
- 9. Click **OK**.

To configure the hub IPsec template

- 1. Continuing on the FortiManager GUI, click **Create New** to create a second IPsec template.
- 2. In the **Name** field, type `Hub`.



- 3. Click **Create New**, and then configure the following settings:

| Field                 | Value                                                         |
|-----------------------|---------------------------------------------------------------|
| Tunnel Name           | To_BR2-FGT-1                                                  |
| IP Address            | 100.65.2.112                                                  |
| Interface             | port2                                                         |
| Remote Subnet         | 172.20.2.0/255.255.255.0                                      |
| Pre-shared Key        | 123456789                                                     |
| Proposal              | <b>AES256 and SHA256</b>                                      |
| Diffie-Hellman Groups | 20                                                            |
| Phase 2 selectors     | Click <b>Create New</b> , and then continue to the next step. |



Tunnel Name

Network ▾

Routing

IP Version

Remote Gateway

IP Address

Interface

Local Gateway

Mode Config ☐

Transport

Use Fortinet Encapsulation ☐

NAT Traversal

Keepalive Frequency

Dead Peer Detection

DPD Retry Count

DPD Retry Interval  Seconds

Forward Error Correction ☐ Egress ☐ Ingress

Remote Subnet

FEC Health Check

Advanced Network Settings >

Authentication ▾

Authentication Method

Pre-shared Key

IKE

XAUTH

Phase 1 Proposal ▾

| Encryption - Authentication | Encryption                                                                                                                                                                                       | Authentication                                                                                                                                                                        | Action                                                                                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <input type="button" value="AES256"/>                                                                                                                                                            | <input type="button" value="SHA256"/>                                                                                                                                                 | <input type="button" value="x"/> <input type="button" value="+"/>                                                                                      |
| Diffie-Hellman Groups       | <input type="checkbox"/> 1<br><input type="checkbox"/> 14<br><input type="checkbox"/> 17<br><input checked="" type="checkbox"/> 20<br><input type="checkbox"/> 28<br><input type="checkbox"/> 31 | <input type="checkbox"/> 2<br><input type="checkbox"/> 15<br><input type="checkbox"/> 18<br><input type="checkbox"/> 21<br><input type="checkbox"/> 29<br><input type="checkbox"/> 32 | <input type="checkbox"/> 5<br><input type="checkbox"/> 16<br><input type="checkbox"/> 19<br><input type="checkbox"/> 27<br><input type="checkbox"/> 30 |

### Stop and think!

Make sure to distinguish names. You have created the hub template with the name **Hub**, but the interface for BR2-FGT-1 is **To\_BR2-FGT-1** and BR3-FGT-1 will be **To\_BR3-FGT-1**. This will help you recognize interfaces that are connected to the spokes.

#### 4. Configure the following settings:

| Field            | Value            |
|------------------|------------------|
| Name             | PH2_to_BR2-FGT-1 |
| Proposal         | aes256-sha256    |
| Advanced Options | dhgrp 20         |

Create New Phase 2 Interface

Name

Keep Alive ☐

Key Life

Proposal

Click to select 1 entry selected

Advanced Options ▾

dhgrp ☐ 1 ☐ 2 ☐ 5 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☒ 20 ☐ 21 ☐ 27 ☐ 28 ☐ 29 ☐ 30 ☐ 31 ☐ 32

5. Click **OK**.

Phase 2 selectors ▾

[+ Create New](#) [Edit](#) [Delete](#) [Move Up](#) [Move Down](#)

| <input type="checkbox"/> | Name             | Keep Alive | Key Life | Proposal      |   |
|--------------------------|------------------|------------|----------|---------------|---|
| <input type="checkbox"/> | PH2_to_BR2-FGT-1 | Disabled   | 43200    | aes256-sha256 | 1 |

6. Click **OK** again.

7. Click **Create New**, and then configure the following settings:

Name

Description

[+ Create New](#) [Edit](#) [Delete](#) [More ▾](#)

| <input type="checkbox"/> | Name         | Type   | Outgoing Interface |
|--------------------------|--------------|--------|--------------------|
| <input type="checkbox"/> | To_BR2-FGT-1 | Static | port2              |

| Field                 | Value                                                         |
|-----------------------|---------------------------------------------------------------|
| Tunnel Name           | To_BR3-FGT-1                                                  |
| IP Address            | 100.65.3.113                                                  |
| Interface             | port2                                                         |
| Remote Subnet         | 172.20.3.0/255.255.255.0                                      |
| Pre-shared Key        | 123456789                                                     |
| Proposal              | <b>AES256</b> and <b>SHA256</b>                               |
| Diffie-Hellman Groups | 20                                                            |
| Phase 2 selectors     | Click <b>Create New</b> , and then continue to the next step. |

Tunnel Name

To\_BR3-FGT-1

Network

Routing

ManualAutomatic

IP Version

IPv4IPv6

Remote Gateway

Static IP AddressDialup UserDynamic DNS

IP Address

100.65.3.113

Interface

port2

Local Gateway

Main Interface IPSpecify

Mode Config

Transport

UDPAutoTCP Encapsulation

Use Fortinet Encapsulation

NAT Traversal

EnableDisable

Keepalive Frequency

10

Dead Peer Detection

DisableOn IdleOn Demand

DPD Retry Count

3

DPD Retry Interval

20Seconds

Forward Error Correction

Egress

Ingress

Remote Subnet

172.20.3.0/255.255.255.0

FEC Health Check

Advanced Network Settings

Authentication

Authentication Method

Pre-shared KeySignature

Pre-shared Key

123456789

IKE

Version 1Version 2

XAUTH

DisableClient Server

Phase 1 Proposal

| Encryption - Authentication | Encryption                                                                                                                                                                                                                                                             | Authentication | Action                            |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------|
|                             | AES256                                                                                                                                                                                                                                                                 | SHA256         | <div><div></div><div></div></div> |
| Diffie-Hellman Groups       | <div><div><div>1</div><div>14</div><div>17</div><div>20</div><div>28</div><div>31</div></div><div><div>2</div><div>15</div><div>18</div><div>21</div><div>29</div><div>32</div></div><div><div>5</div><div>16</div><div>19</div><div>27</div><div>30</div></div></div> |                |                                   |

8. Configure the following settings:

| Field | Value            |
|-------|------------------|
| Name  | PH2_to_BR3-FGT-1 |

| Field            | Value         |
|------------------|---------------|
| Proposal         | aes256-sha256 |
| Advanced Options | dhgrp 20      |

Create New Phase 2 Interface

Name

PH2\_to\_BR3-FGT-1

Keep Alive

☐

Key Life

43200

Proposal

aes256-sha256

1 entry selected

Advanced Options

dhgrp

☐ 1 ☐ 2 ☐ 5 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☒ 19 ☒ 20 ☐ 21 ☐ 27 ☐ 28 ☐ 29 ☐ 30 ☐ 31 ☐ 32

9. Click **OK**.

Phase 2 selectors

+ Create New

Edit

Delete

Move Up

Move Down

Search...

| <input type="checkbox"/> | Name             | Keep Alive | Key Life | Proposal      |
|--------------------------|------------------|------------|----------|---------------|
| <input type="checkbox"/> | PH2_to_BR3-FGT-1 | Disabled   | 43200    | aes256-sha256 |

1

10. Click **OK**.

Name

Hub

Description

+ Create New

Edit

Delete

More

| <input type="checkbox"/> | Name         | Type   | Outgoing Interface |
|--------------------------|--------------|--------|--------------------|
| <input type="checkbox"/> | To_BR2-FGT-1 | Static | port2              |
| <input type="checkbox"/> | To_BR3-FGT-1 | Static | port2              |

11. Click **OK**.

### Stop and think!

Make sure to distinguish names. You have created the hub template with the name **Hub**, but the interface for BR2-FGT-1 is **To\_BR2-FGT-1** and BR3-FGT-1 is **To\_BR3-FGT-1**. This will help you recognize interfaces from Core1 that are connected to each spoke.

12. On the **IPsec Tunnel** tab, select the **Hub** checkbox, and then click **Assign to Device/Group** to assign the **Hub** IPsec template.

| Template Groups                                                                                                                                                   | IPsec Tunnel               | BGP | System Templates | Static Route | CLI | Feature Visibility |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----|------------------|--------------|-----|--------------------|
| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">Assign to Device/Group</a> <a href="#">AI Assistant</a> <a href="#">More</a> |                            |     |                  |              |     |                    |
| <input checked="" type="checkbox"/>                                                                                                                               | Name                       |     |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | HUB_IPsec_Recommended      |     |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | BRANCH_IPsec_Recommended   |     |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | IPsec_Fortinet_Recommended |     |                  |              |     |                    |
| <input checked="" type="checkbox"/>                                                                                                                               | Hub                        |     |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | Spokes                     |     |                  |              |     |                    |

13. In the **Available Entries** field, select the **HQ-NGFW [Core1]** checkbox, and then move it to the **Selected Entries** field.
14. Click **OK**.
15. On the **IPsec Tunnel** tab, select the **Spokes** checkbox, and then click **Assign to Device/Group** to assign the **Spokes** IPsec template.
16. In the **Available Entries** field, select the **BR2-FGT-1** and **BR3-FGT-1** checkboxes, and then move them to the **Selected Entries** field.
17. Click **OK**.

| Template Groups                                                                                                                                                   | IPsec Tunnel               | BGP                                                                                          | System Templates | Static Route | CLI | Feature Visibility |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------|------------------|--------------|-----|--------------------|
| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">Assign to Device/Group</a> <a href="#">AI Assistant</a> <a href="#">More</a> |                            |                                                                                              |                  |              |     |                    |
| <input checked="" type="checkbox"/>                                                                                                                               | Name                       | Assigned to Device/Group                                                                     |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | HUB_IPsec_Recommended      | 0 Devices in Total                                                                           |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | BRANCH_IPsec_Recommended   | 0 Devices in Total                                                                           |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | IPsec_Fortinet_Recommended | 0 Devices in Total                                                                           |                  |              |     |                    |
| <input type="checkbox"/>                                                                                                                                          | Hub                        | 1 Device in Total <a href="#">View Details &gt;</a><br>HQ-NGFW [Core1]                       |                  |              |     |                    |
| <input checked="" type="checkbox"/>                                                                                                                               | Spokes                     | 2 Devices in Total <a href="#">View Details &gt;</a><br>BR2-FGT-1 [root]<br>BR3-FGT-1 [root] |                  |              |     |                    |

You can see what the template will install if you right-click the template, and then select **Preview CLI Configuration**.



| Name                                                | Assigned to Device/Group                                                                     |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> HUB_IPsec_Recommended      | 0 Devices In Total                                                                           |
| <input type="checkbox"/> BRANCH_IPsec_Recommended   | 0 Devices In Total                                                                           |
| <input type="checkbox"/> IPsec_Fortinet_Recommended | 0 Devices In Total                                                                           |
| <input checked="" type="checkbox"/> Hub             | 1 Device In Total <a href="#">View Details &gt;</a><br>HQ-NGFW [Core1]                       |
| <input type="checkbox"/> Spokes                     | 2 Devices In Total <a href="#">View Details &gt;</a><br>BR2-FGT-1 [root]<br>BR3-FGT-1 [root] |

Edit  
Clone  
Delete  
Assign to Device/Group  
**Preview CLI Configuration**  
Activate

### Stop and think!

You are installing the following settings:

```
config firewall address
config vpn ipsec phase1-interface
config system interface
config vpn ipsec phase2-interface
config router static
```

If you need to delete your IPsec template, you must consider the objects that the template has created. FortiManager cannot delete the objects that the template created because there are some dependencies.

- Click **Device Manager > Device & Groups > Managed FortiGate**, and then select the **Core1**, **BR2-FGT-1**, and **BR3-FGT-1** checkboxes.
- Click **Install > Install Wizard**.
- Verify that **Install Device Settings (only)** is selected, and then click **Next**.
- Verify that **BR2-FGT-1**, **BR3-FGT-1**, **HQ-NGFW**, and **Core1** are selected, and then click **Next**.
- Click **Install**.

### Stop and think!

Notice that once you have installed the templates, FortiManager can automatically detect normalized interfaces that IPsec templates create.

| Managed FortiGate (7)  | Network           | Name         | Type   | IP/Network         | Virtual Domain             | Normalized Interface |
|------------------------|-------------------|--------------|--------|--------------------|----------------------------|----------------------|
| <div>BR2-FGT-1</div>   | Interfaces        | Physical (4) |        |                    |                            |                      |
| <div>BR3-FGT-1</div>   | DNS               | 1            | port1  | Physical interface | 192.168.1.112/255.255.0.0  | root port1           |
| <div>HQ-NGFW</div>     | IPAM              | 2            | port2  | Physical interface | 100.65.2.112/255.255.255.0 | root port2           |
| <div>HQ-IPFW (3)</div> | Local Out Routing | To_Hub       | Tunnel | 0.0.0.0/0.0.0.0    | root                       | To_Hub               |

This normalized interface (**To\_hub**) was created for you before you started this lab. But, consider that every time you use the IPsec template, the interfaces should be mapped to normalized interfaces. You can use per device mapping or per platform mapping. If you use per platform mapping, ensure that the spelling of the platform is correct because it is case sensitive. For example, **To\_hub** will not be the same as **To\_Hub**.

Create a Normalized Interface for the Hub

You will create a normalized interface for the hub, so that you can manipulate the same firewall policies for all hubs that you have in your topology.

To create a zone interface for the hub

- 1. Continuing on the FortiManager GUI, click **Policy & Objects > Normalized Interface > Create New**.
- 2. In the **Name** field, type `Zone_spokes`.
- 3. In the **Per-Platform Mapping** section, click **Create New**.
- 4. In the **Match Platform** field, select **FortiGate-VM64-KVM**, and then in the **Mapped Interface Name** field, type `Zone_spokes`.
- 5. Click **OK**.

Name

Zone\_spokes

Description

Color



Change

Wildcard

☐

Per-Platform Mapping

+ Create New

Edit

Delete

| <input type="checkbox"/> | Name               | Device Interface Name |
|--------------------------|--------------------|-----------------------|
| <input type="checkbox"/> | FortiGate-VM64-KVM | Zone_spokes           |

- 6. Click **OK**.
- 7. Click **Device Manager > Device & Groups**.
- 8. Click **Core1 > Network > Interfaces**.
- 9. Click **Create New > Device Zone**.
- 10. Configure the following settings:

| Field            | Value                         |
|------------------|-------------------------------|
| Zone Name        | Zone_spokes                   |
| Interface Member | To_BR2-FGT-1 and To_BR3-FGT-1 |

- 11. Click **OK**.



Search...

Managed FortiGate (9)

BR2-FGT-1

BR3-FGT-1

HQ-DCFV

HQ-ISP-W (3)

root

Zone1

Zone2

HQ-NGFW (3)

Core1

Core2

root

Dashboard

Network

Interfaces

Local Out Routing

Static Routes

OSPF

System

Log & Report

CLI Configurations

Feature Visibility

Toggle Horizontal Menu

Create New

Edit

Edit in CLI

Delete

Where Used

More

Full Screen

|                 | # | Name                          | Type               | Members                      | Normalized Interface |
|-----------------|---|-------------------------------|--------------------|------------------------------|----------------------|
| Physical (3)    |   |                               |                    |                              |                      |
|                 | 1 | port2                         | Physical Interface |                              | port2                |
|                 |   | To_BR2-FGT-1                  | Tunnel             |                              |                      |
|                 |   | To_BR3-FGT-1                  | Tunnel             |                              |                      |
|                 | 2 | port5                         | Physical Interface |                              | port5                |
|                 | 3 | port6                         | Physical Interface |                              | port6                |
| Tunnel (3)      |   |                               |                    |                              |                      |
|                 | 4 | nat.Core1                     | Tunnel             |                              |                      |
|                 | 5 | l2l.Core1                     | Tunnel             |                              |                      |
|                 | 6 | ssl.Core1 (SSL VPN Interface) | Tunnel             |                              |                      |
| Zone (1)        |   |                               |                    |                              |                      |
|                 | / | Zone spokes                   | Zone               | To_BR3-FGT-1<br>To_BR2-FGT-1 | Zone spokes          |
| SD-WAN Zone (1) |   |                               |                    |                              |                      |
|                 | 8 | virtual-wan-link              | SD-WAN Zone        |                              |                      |

### Stop and think!

Using a zone in the hub makes it much faster to configure firewall policies for spokes, but everything depends on your requirements.

## Configure the Firewall Policies

After you install the VPN configuration on all FortiGate devices, the interfaces are available to configure the firewall policies.

### To configure the firewall policies on Core1

1. Continuing on the FortiManager GUI, click **Device Manager > Scripts**.
2. Click the **VPN\_Core1** checkbox, and then click **Run Script**.
3. Select the **Core1** policy package, and then click **Run Now**.



You are installing two firewall policies that use the `Zone_spokes` normalized interface.

4. Click **Close**.
5. Click **Device Manager > Device & Groups > Managed FortiGate**.
6. Select the **Core1** checkbox, and then click **Install Wizard**.
7. Select **Install Policy Package & Device Settings**.
8. In the **Policy Package** field, select **Core1**.
9. Click **Next**.
10. Click **Next**.
11. Click **Install**.
12. Click **Finish**.

### To configure the firewall policies on the spokes

1. Continuing on the FortiManager GUI, click **Device Manager > Scripts**.
2. Click the **VPN\_Spokes** checkbox, and then click **Run Script**.
3. Select the **BR** policy package, and then click **Run Now**.



You are installing two firewall policies that use the **To\_hub** normalized interface.

4. Click **Close**.
5. Click **Device Manager > Device & Groups > Managed FortiGate**.
6. Select the **BR2-FGT-1** and **BR3-FGT-1** checkboxes, and then click **Install Wizard**.
7. Select **Install Policy Package & Device Settings**.
8. In the **Policy Package** field, select **BR**.
9. Click **Next**.
10. Click **Next**.
11. Click **Install**.
12. Click **Finish**.

## Check the Status of the VPN Tunnels

You will check the status of the VPN tunnels.

### To check the status of the VPN tunnels

1. Continuing on the FortiManager GUI, click **Device Manager > Monitors > VPN Monitor**.
2. Enable **Show Table**.

The table should contain four VPN tunnels.

| Device          | Name         | Type      | Remote Gateway | Peer ID      | Incoming Data | Outgoing Data | Phase 1      | Phase 2 Selectors |
|-----------------|--------------|-----------|----------------|--------------|---------------|---------------|--------------|-------------------|
| BR2-FGT-1[root] | To_hub       | automatic | 100.65.0.101   | 100.65.0.101 | 0.00 KB       | 0.00 KB       | To_hub       | PH2_to_hub        |
| BR3-FGT-1[root] | To_hub       | automatic | 100.65.0.101   | 100.65.0.101 | 0.00 KB       | 0.00 KB       | To_hub       | PH2_to_hub        |
| HQ-NGFW[Core1]  | To_BR2-FGT-1 | automatic | 100.65.2.112   | 100.65.2.112 | 0.00 KB       | 0.00 KB       | To_BR2-FGT-1 | PH2_to_BR2-FGT-1  |
| HQ-NGFW[Core1]  | To_BR3-FGT-1 | automatic | 100.65.3.113   | 100.65.3.113 | 0.00 KB       | 0.00 KB       | To_BR3-FGT-1 | PH2_to_BR3-FGT-1  |



In **VPN Monitor**, the message **Internal Server Error** appears instead of the map because HQ-FMG-1 is in a closed environment without internet access.

## Delete the IPsec Tunnels

Unsetting the template does not delete the configuration in the device layer and on the FortiGate. In this case, you must remove the associated firewall policies where you reference IPsec interfaces and objects that were generated in the IPsec templates. The general steps are:

1. Delete the policies in the policy and device layers.
2. Delete the zone interface (only in **Core1**).
3. Delete the static routes in the device layer.
4. Delete the firewall addresses in the device layer.
5. Delete phase 2 in the device layer.
6. Delete phase 1 in the device layer.

### To unassign IPsec templates

1. Continuing on the FortiManager GUI, click **Device Manager > Provisioning Templates > IPsec Tunnel**.
2. Unassign the **Hub** and **Spokes** IPsec templates.

|                                                                                   |                            |                      |                        |                                        |              |                              |                          |                      |
|-----------------------------------------------------------------------------------|----------------------------|----------------------|------------------------|----------------------------------------|--------------|------------------------------|--------------------------|----------------------|
| Template Groups                                                                   |                            | IPsec Tunnel         | BGP                    | System Templates                       | Static Route | CLI                          | Feature Visibility       |                      |
| <a href="#">+ Create New</a>                                                      |                            | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Assign to Device/Group</a> |              | <a href="#">AI Assistant</a> |                          | <a href="#">More</a> |
|  | Name                       |                      |                        |                                        |              |                              | Assigned to Device/Group |                      |
| <input type="checkbox"/>                                                          | HUB_IPsec_Recommended      |                      |                        |                                        |              |                              | 0 Devices in Total       |                      |
| <input type="checkbox"/>                                                          | BRANCH_IPsec_Recommended   |                      |                        |                                        |              |                              | 0 Devices in Total       |                      |
| <input type="checkbox"/>                                                          | IPsec_Fortinet_Recommended |                      |                        |                                        |              |                              | 0 Devices in Total       |                      |
| <input checked="" type="checkbox"/>                                               | Hub                        |                      |                        |                                        |              |                              | 0 Devices in Total       |                      |
| <input checked="" type="checkbox"/>                                               | Spokes                     |                      |                        |                                        |              |                              | 0 Devices in Total       |                      |

3. Delete the **Hub** and **Spokes** IPsec templates.

| Template Groups              |                                                                                                                | IPsec Tunnel               | BGP                      | System Templates                         | Static Route | CLI                            | Feature Visibility       |
|------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------|------------------------------------------|--------------|--------------------------------|--------------------------|
| <a href="#">+ Create New</a> |                                                                                                                | <a href="#">✎ Edit</a>     | <a href="#">🗑 Delete</a> | <a href="#">🗨 Assign to Device/Group</a> |              | <a href="#">🤖 AI Assistant</a> | <a href="#">⋮ More ▾</a> |
| <input type="checkbox"/>     | Name ↕                                                                                                         | Assigned to Device/Group ↕ |                          |                                          |              |                                |                          |
| <input type="checkbox"/>     |  HUB_IPsec_Recommended      | 0 Devices in Total         |                          |                                          |              |                                |                          |
| <input type="checkbox"/>     |  BRANCH_IPsec_Recommended   | 0 Devices in Total         |                          |                                          |              |                                |                          |
| <input type="checkbox"/>     |  IPsec_Fortinet_Recommended | 0 Devices in Total         |                          |                                          |              |                                |                          |

### To delete the IPsec tunnels on Core1

1. Continuing on the FortiManager GUI, click **Policy & Objects > Policy Packages > Core1 > Firewall Policy**.
2. Expand **Core1**, and then click **Firewall Policy**.
3. Delete the policies that are using the **Zone\_spokes** interface in the policy layer.

Delete the IPsec Tunnels

|                      |                                     |      |                |             |               |
|----------------------|-------------------------------------|------|----------------|-------------|---------------|
| Search...            | + Create New                        | Edit | Delete         | Section     | Policy Lookup |
| BR                   | #                                   | Name | From           | To          |               |
| Firewall Policy      | <input type="checkbox"/>            | 1    | Internet       | port5       | port2         |
| Installation Targets | <input type="checkbox"/>            | 2    | FAZ_and_FMG... | port2       | port6         |
| Core1                | <input type="checkbox"/>            | 3    | FAZ_FMG_Acc... | port5       | port6         |
| Firewall Policy      | <input checked="" type="checkbox"/> | 4    | Hub to Spokes  | port2       | Zone_spokes   |
| Installation Targets | <input checked="" type="checkbox"/> | 5    | Spokes to Hub  | Zone_spokes | port2         |
| Core2                | Implicit (6/6 Total:1)              |      |                |             |               |
| DCFV                 | <input type="checkbox"/>            | 6    | Implicit Deny  | any         | any           |
| ISFW                 |                                     |      |                |             |               |
| Zone1                |                                     |      |                |             |               |
| Zone2                |                                     |      |                |             |               |
| default              |                                     |      |                |             |               |
| root_ISFW            |                                     |      |                |             |               |

- Click **OK**.
- Click **Device manager > Device & Groups > Managed FortiGate > HQ-NGFW > Core1 > CLI Configurations > firewall > policy**.
- Click **Core1 > CLI Configurations > firewall > policy**.

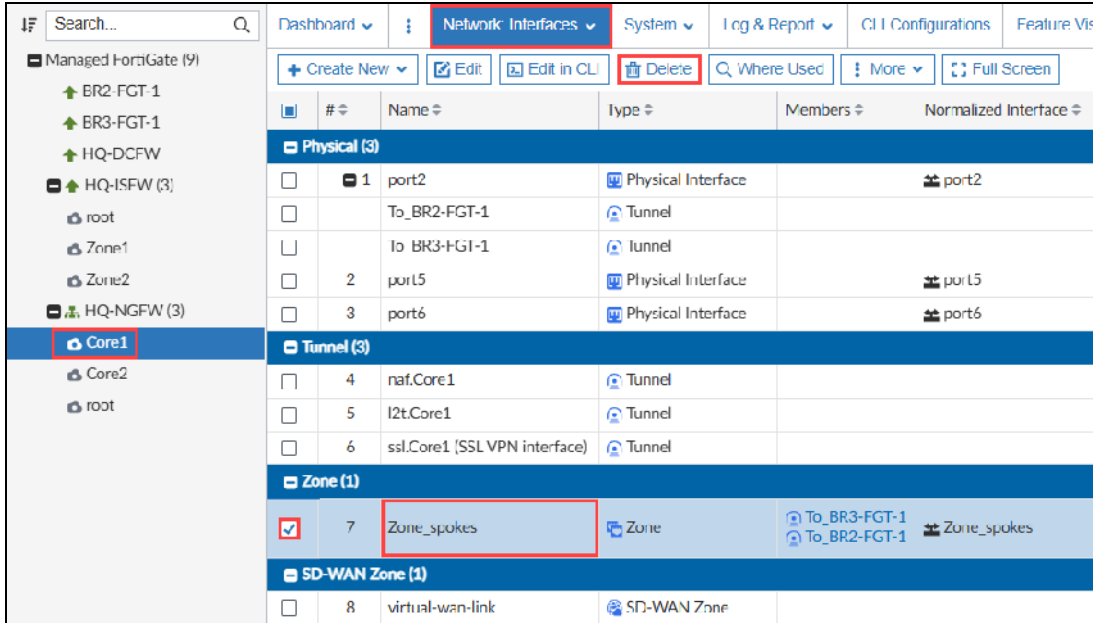


Enable `srcintf` and `dstintf` in the engine icon tool, located in the upper-right corner, so that you can easily identify policies that are using a specific interface. Then, move the `srcintf` and `dstintf` columns to the left.

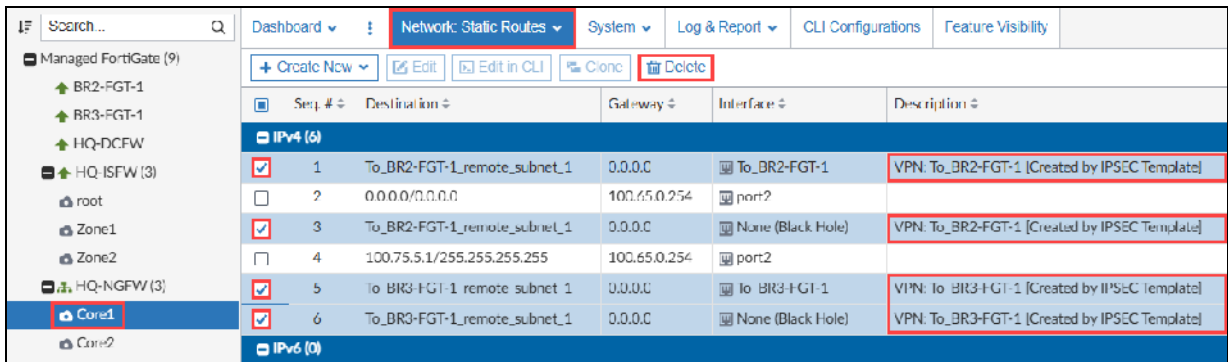
- Delete the same policies you deleted in the policy layer for **Core1**—the ones that use the **Zone\_spokes** interface.

|                       |                   |                                     |              |              |                    |                    |
|-----------------------|-------------------|-------------------------------------|--------------|--------------|--------------------|--------------------|
| Search...             | Dashboard         | Network                             | System       | Log & Report | CLI Configurations | Feature Visibility |
| Managed FortiGate (9) | policy            | firewall policy                     | + Create New | Edit         | Delete             |                    |
| BR2-FGT-1             | firewall          | policyid                            | dstintf      | srcintf      |                    |                    |
| BR3-FGT-1             | DoS-policy        | 1                                   | port2        | port5        |                    |                    |
| HQ-DCFV               | DoS-policy6       | <input type="checkbox"/>            | 3            | port6        | port2              |                    |
| HQ-ISFW (3)           | interface-policy  | <input type="checkbox"/>            | 4            | port6        | port5              |                    |
| root                  | interface-policy6 | <input checked="" type="checkbox"/> | 5            | Zone_spokes  | port2              |                    |
| Zone1                 | local-in-policy   | <input checked="" type="checkbox"/> | 6            | port2        | Zone_spokes        |                    |
| Zone2                 | local-in-policy6  |                                     |              |              |                    |                    |
| HQ-NGFW (3)           | multicast-policy  |                                     |              |              |                    |                    |
| Core1                 | multicast-policy6 |                                     |              |              |                    |                    |
| Core2                 | policy            |                                     |              |              |                    |                    |

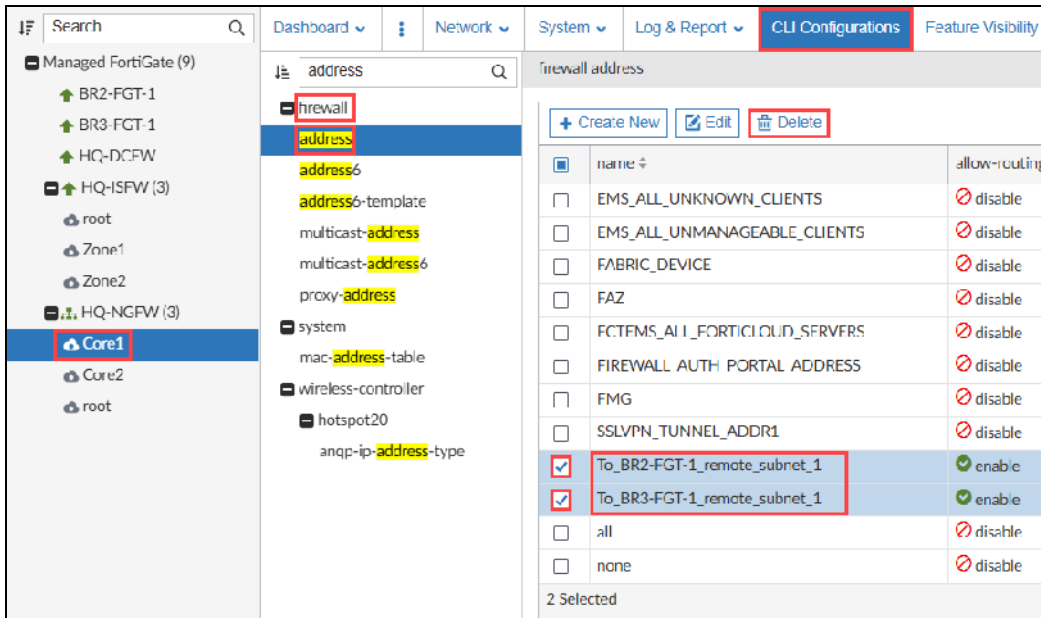
- Click **OK**.
- Click **Core1 > Network > Interfaces**.
- Select the **Zone\_spokes** checkbox, and then click **Delete**.



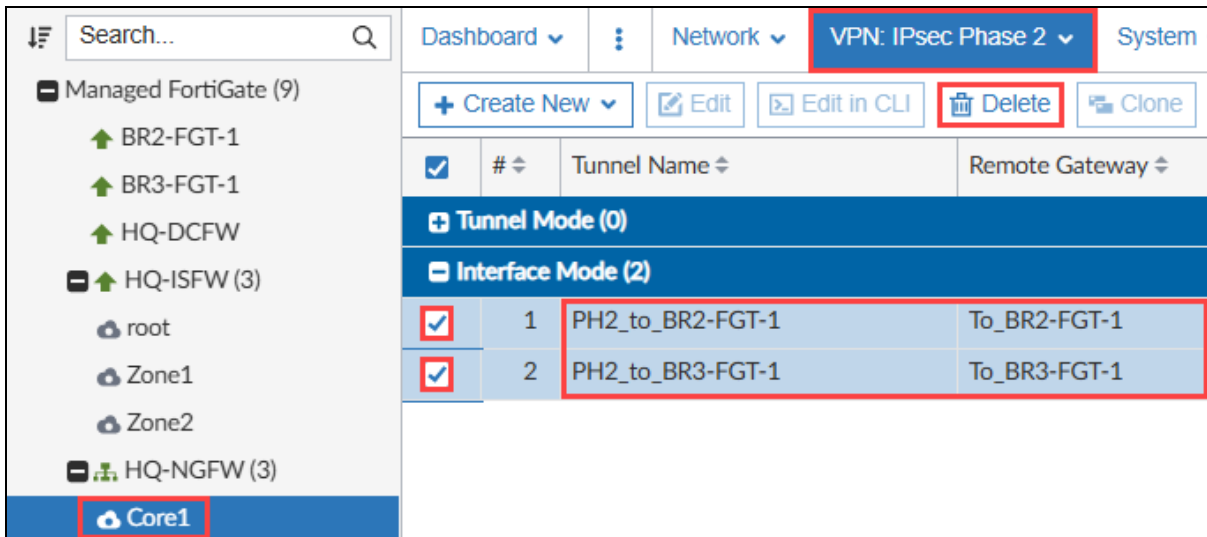
- Click **OK**.
- Click **Core1 > Network > Static Routes**, select all **Static routes** that the IPsec template generated, and then click **Delete**.



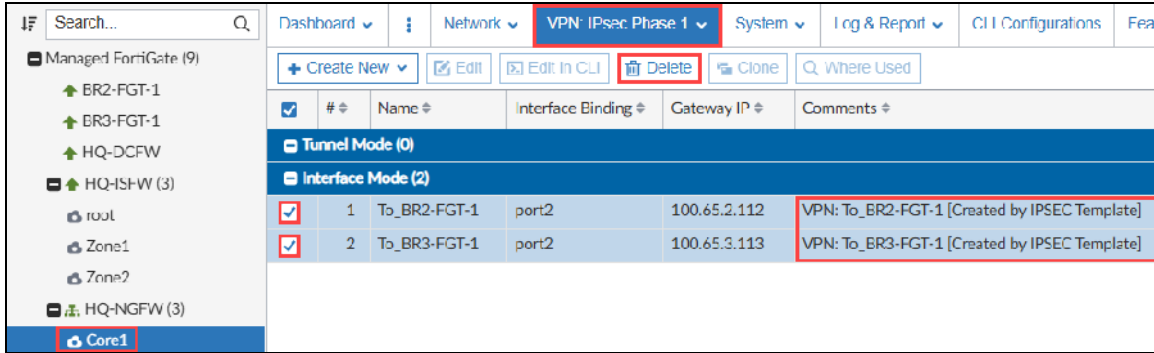
- Click **OK**.
- Click **Core1 > CLI Configurations > firewall > address**.
- Select the **To\_BR2-FGT-1\_remote\_subnet\_1** and **To\_BR3-FGT-1\_remote\_subnet\_1** checkboxes, and then click **Delete**.



16. Click **OK**.
17. Click **Core1 > VPN > IPsec Phase 2**.
18. Select the **IPsec Phase 2** that FortiManager created, and then click **Delete**.



19. Click **OK**.
20. Click **Core1 > VPN > IPsec Phase 1**.
21. Select the **IPsec Phase 1** that the IPsec template created, and then click **Delete**.



22. Click **OK**.
23. Click **Device Manager > Device & Groups > Managed FortiGate**, and then select the **Core1** checkbox.
24. Click **Install > Re-install Policy**.
25. Click **OK**.
26. Click **Next**, and then click **Finish**.



Make sure you have deleted all previous objects. Otherwise, when you reinstall, you will receive an error that the policy package cannot be installed.

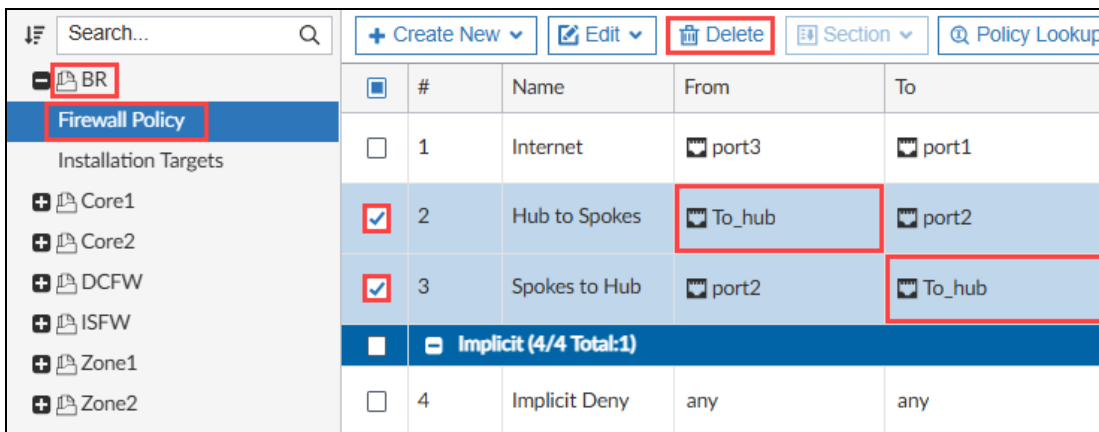
### Take the Expert Challenge!

Delete all objects that IPsec templates installed on BR2-FGT-1 and BR3-FGT-2 and reinstall the **BR** policy package on both firewalls. There should be no errors at the end of your installation.

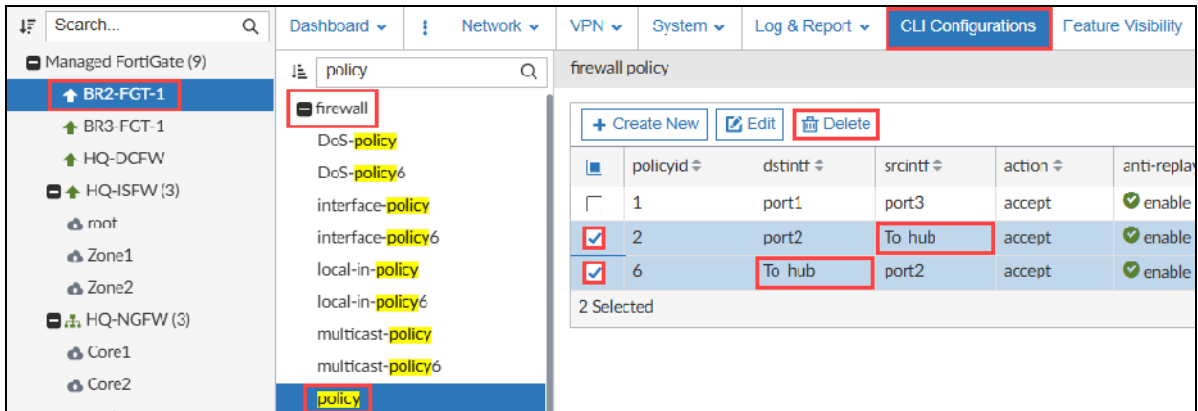
If you require assistance, or to verify your work, use the step-by-step instructions that follow.

### To delete the IPsec template on BR2-FGT-1

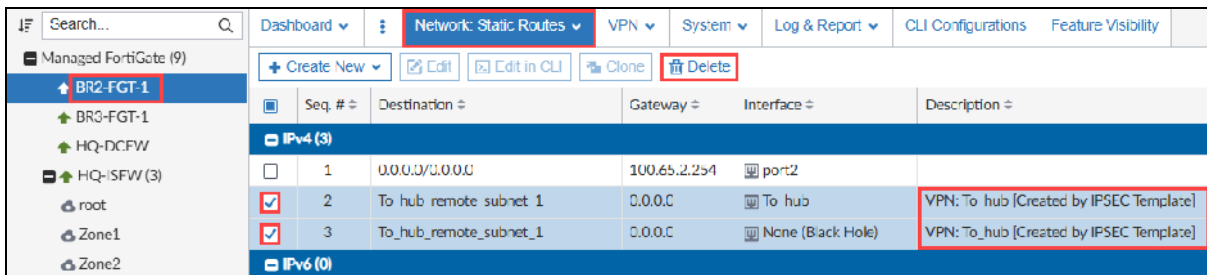
1. Continuing on the FortiManager GUI, click **Policy & Objects > Policy Packages > BR > Firewall Policy**.
2. Delete the policies in the policy layer that are using the **To\_hub** interface.



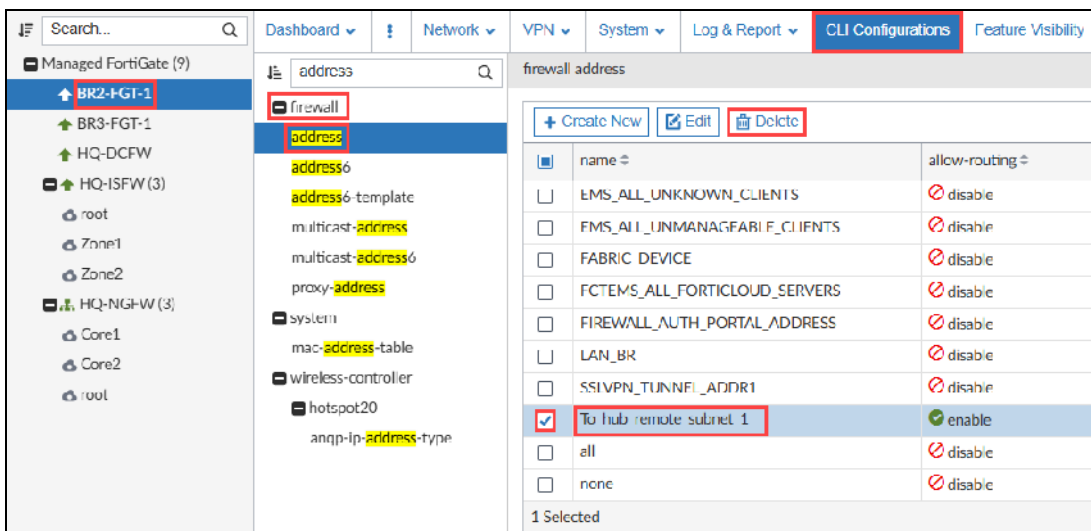
3. Click **OK**.
4. Click **Device Manager > Device & Groups > Managed FortiGate**.
5. Click **BR2-FGT-1 > CLI Configurations > firewall > policy**.
6. Delete the policies in the device layer that are using the **To\_hub** interface.



7. Click **OK**.
8. Click **Network > Static Routes**.
9. Delete the static routes that the IPsec template generated in the device layer.

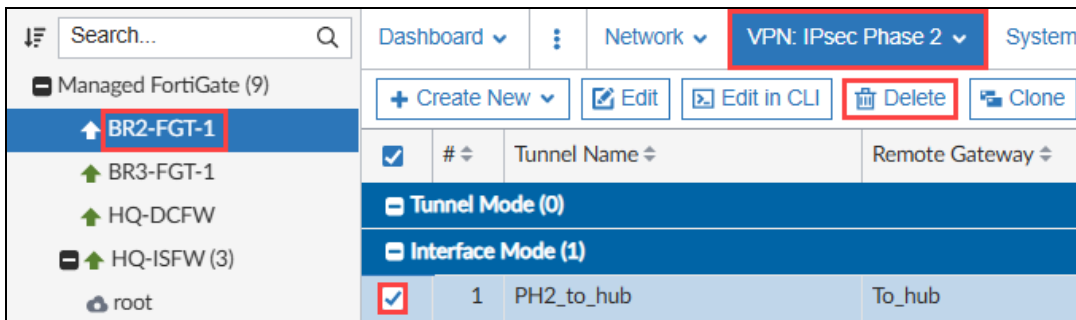


10. Click **OK**.
11. Click **CLI Configurations > firewall > address**.
12. Delete the **To\_hub\_remote\_subnet\_1** address in the device layer.

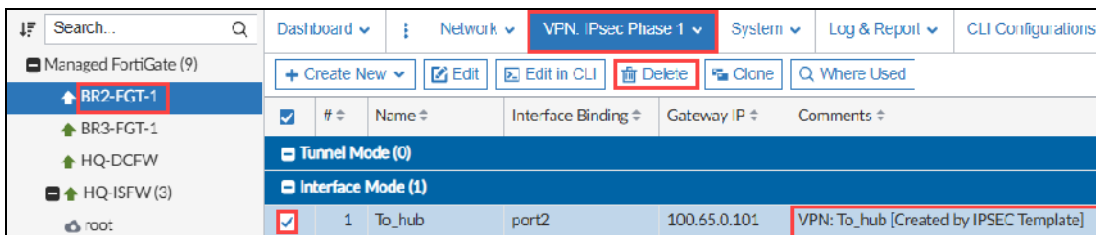




13. Click **OK**.
14. Click **VPN > IPsec Phase 2**.
15. Delete the IPsec Phase 2 that FortiManager created.



16. Click **OK**.
17. Click **VPN > IPsec Phase 1**.
18. Delete the IPsec Phase 1 that the IPsec template created.



19. Click **OK**.
20. Click **Device Manager > Device & Groups > Managed FortiGate**, and then select the **BR2-FGT-1** checkbox.
21. Click **Install > Re-install Policy**.
22. Click **OK**.
23. Click **Next**, and then click **Finish**.

### To delete the IPsec template on BR3-FGT-1

1. Continuing on the FortiManager GUI, click **Device Manager > Device & Groups > Managed FortiGate**.
2. Click **BR3-FGT-1 > CLI Configurations > firewall > policy**.
3. Delete the policies in the device layer that are using the **To\_hub** interface.

Managed FortiGate (9)

- BR2-FGT-1
- BR3-FGT-1**
- HQ-DCFV
- HQ-ISFW (3)
  - root
  - Zone1
  - Zone2
- HQ-NGFW (3)
  - Core1
  - Core2

policy

- firewall**
- DoS-policy
- DoS-policy6
- interface-policy
- interface-policy6
- local-in-policy
- local-in-policy6
- multicast-policy
- multicast-policy6
- policy**

firewall policy

+ Create New Edit Delete

|                                     | policyid | dstintf       | srcintf       | action |
|-------------------------------------|----------|---------------|---------------|--------|
| <input type="checkbox"/>            | 1        | port1         | port3         | accept |
| <input checked="" type="checkbox"/> | 2        | port2         | <b>To_hub</b> | accept |
| <input checked="" type="checkbox"/> | 6        | <b>To_hub</b> | port2         | accept |

2 Selected

- Click **OK**.



You do not need to delete the firewall policies in the policy layer because you have deleted policies in the **BR** policy package in previous steps.

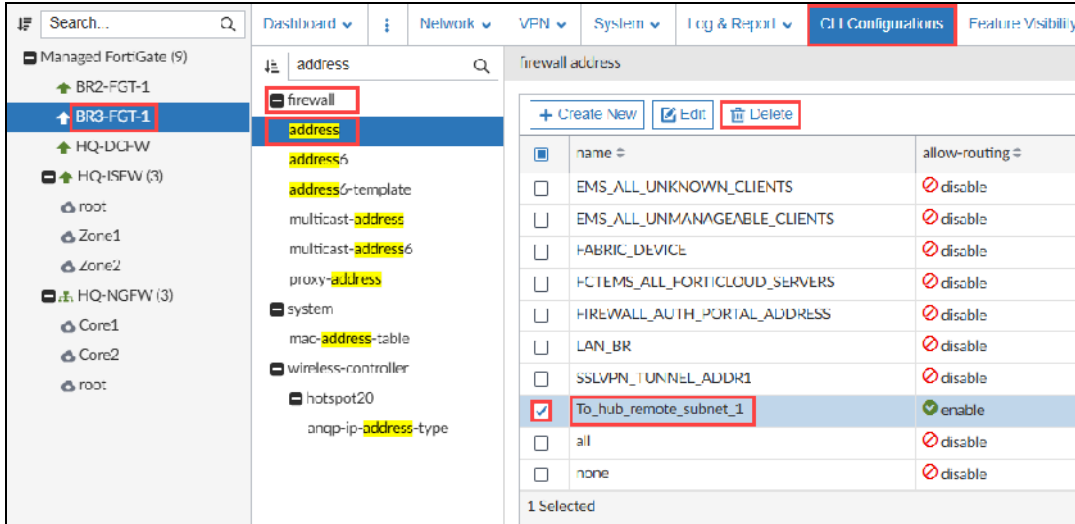
- Click **Network > Static Routes**.
- Delete the static routes that the IPsec template created.

Network Static Routes

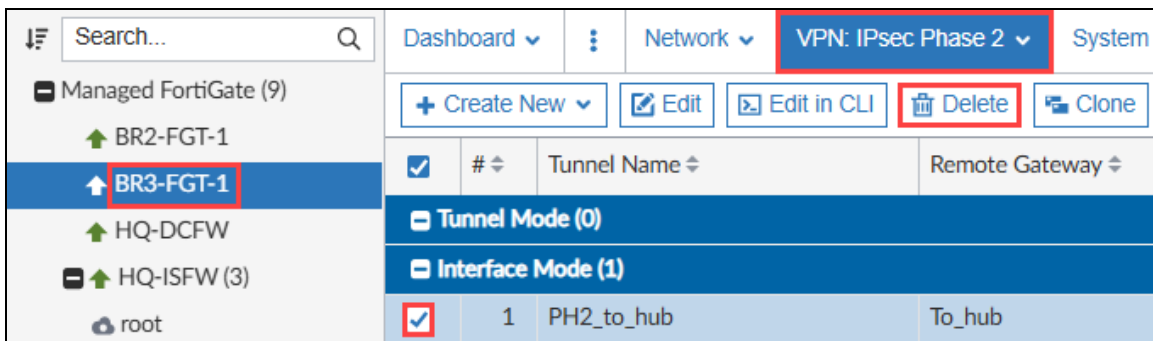
+ Create New Edit Edit in CLI Clone Delete

|                                     | Seq. # | Destination            | Gateway      | Interface         | Description                                    |
|-------------------------------------|--------|------------------------|--------------|-------------------|------------------------------------------------|
| <b>IPv4 (3)</b>                     |        |                        |              |                   |                                                |
| <input type="checkbox"/>            | 1      | 0.0.0.0/0.0.0.0        | 100.65.3.254 | port2             |                                                |
| <input checked="" type="checkbox"/> | 2      | To_hub_remote_subnet_1 | 0.0.0.0      | To_hub            | <b>VPN: To_hub [Created by IPSEC Template]</b> |
| <input checked="" type="checkbox"/> | 3      | To_hub_remote_subnet_1 | 0.0.0.0      | None (Black Hole) | <b>VPN: To_hub [Created by IPSEC Template]</b> |
| <b>IPv6 (0)</b>                     |        |                        |              |                   |                                                |

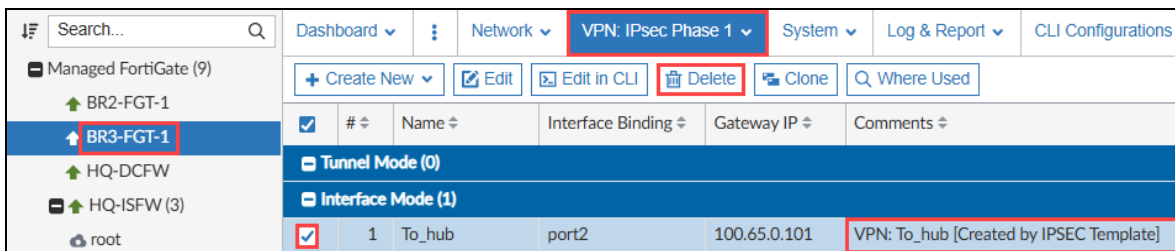
- Click **OK**.
- Click **CLI Configurations > firewall > address**.
- Delete the **To\_hub\_remote\_subnet\_1** address in the device layer.



10. Click **OK**.
11. Click **VPN > IPsec Phase 2**.
12. Delete the IPsec Phase 2 that FortiManager created.



13. Click **OK**.
14. Click **VPN > IPsec Phase 1**.
15. Delete the IPsec Phase 1 that the IPsec template created.



16. Click **OK**.
17. Click **Device Manager > Device & Groups > Managed FortiGate**, and then select the **BR3-FGT-1** checkbox.
18. Click **Install > Re-install Policy**.
19. Click **OK**.
20. Click **Next**, and then click **Finish**.



You can connect to the **Core1**, **BR2-FGT-1**, and **BR3-FGT-1** GUI to confirm that all objects related to the IPsec template have been deleted.

---

## Lab 8: Auto-Discovery VPN

You will modify the IPsec VPN configuration to enable auto-discovery VPN (ADVPN). You will create an on-demand tunnel between the two spokes. You will configure IBGP with route reflector enabled on the hub device to manage the routing.

### Objectives

- Configure ADVPN using IPsec templates in a hub-and-spoke topology using IBGP as dynamic routing
- Configure ADVPN using IPsec templates in a dual hub-and-spoke topology using IBGP and EBGP as dynamic routing

### Time to Complete

Estimated: 45 minutes

### Prerequisites

Before you begin this lab, you must force HQ-NGFW-1 to be the primary FortiGate. Then, you must restore the initial configuration files on FortiManager, BR2-FGT-1, BR3-FGT-1, and HQ-NGFW-1. The configuration files are located on HQ-PC-1.

#### To force HQ-NGFW-1 to be the primary FortiGate

1. Connect over SSH to HQ-NGFW-1, and then log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. On the CLI, enter the following commands:

```
config global
  config system ha
    config vcluster
      edit 1
        set override enable
      next
      edit 2
        set override enable
      next
    end
  end
end
```

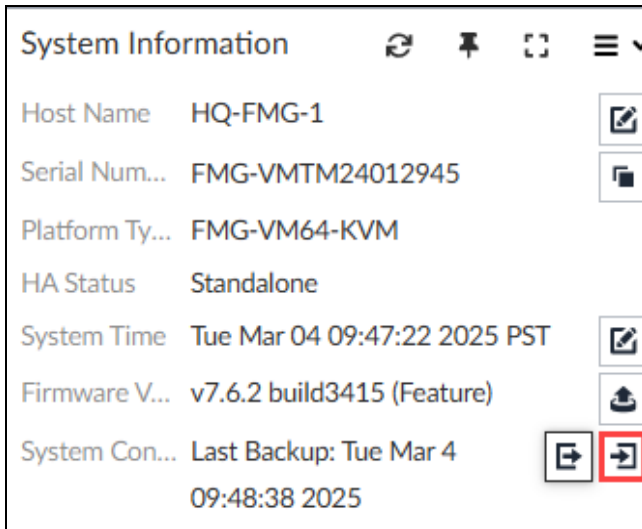


You must restore the configuration files in the recommended order (FortiManager, BR2-FGT-1, BR3-FGT-1, and then HQ-NGFW-1) to keep the synchronization with the FortiManager at the end.

---

### To restore the FortiManager configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the HQ-FMG-1 GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. Click **root**.
4. In the **System Information** widget, in the **System Configuration** field, click the **Restore** icon.



5. In the **Backup File** field, click **Add Files**.
6. Click **Desktop > Resources > Enterprise-FW > ADVPN > Starting\_Point**, select HQ-FMG-1\_ADVPN\_initial.dat, and then click **Select**.
7. In the **Password** field, type Fortinet1!.
8. Click **OK**.

### To restore the BR2-FGT-1 configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the BR2-FGT-1 GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
4. Click **Local PC**, and then click **Upload**.
5. Click **Desktop > Resources > Enterprise-FW > ADVPN > Starting\_Point**, select BR2-FGT-1\_ADVPN\_initial.conf, and then click **Select**.
6. Click **OK**.
7. Click **OK** to reboot.

### To restore the BR3-FGT-1 configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the BR3-FGT-1 GUI, and then log in with the following credentials:

- Username: admin
  - Password: Fortinet1!
3. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
  4. Click **Local PC**, and then click **Upload**.
  5. Click **Desktop > Resources > Enterprise-FW > ADVPN > Starting\_Point**, select BR3-FGT-1\_ADVPN\_initial.conf, and then click **Select**.
  6. Click **OK**.
  7. Click **OK** to reboot.

### To restore the HQ-NGFW-1 configuration file

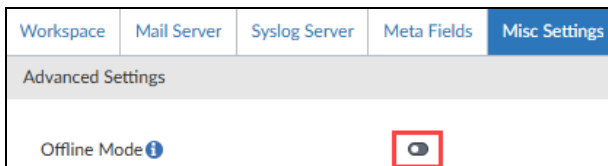
1. On HQ-PC-1, open a browser.
2. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
3. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
4. Click **Local PC**, and then click **Upload**.
5. Click **Desktop > Resources > Enterprise-FW > ADVPN > Starting\_Point**, select HQ-NGFW-1\_ADVPN\_initial.conf, and then click **Select**.
6. Click **OK**.
7. Click **OK** to reboot.



After the reboot, you can verify that HQ-NGFW-1 remains the primary FortiGate.

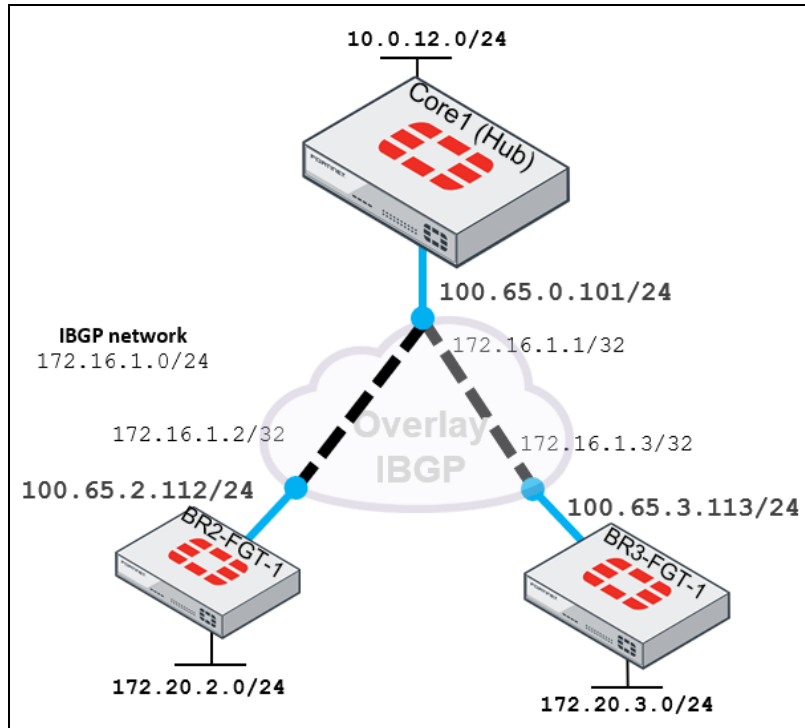
### To disable offline mode on FortiManager

1. Connect to the FortiManager GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **root**.
3. Click **System Settings > Advanced > Misc Settings**.
4. Disable **Offline Mode**.



## Exercise 1: Configuring ADVPN and IBGP

In this exercise, you will configure ADVPN in a hub-and-spoke topology. You will use Core1 as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes.



### Configure IPsec Templates

You will configure IPsec templates for Core1, BR2-FGT-1, and BR3-FGT-1.

#### To configure the metadata variables

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Policy & Objects > Advanced > Metadata Variables**.
4. Click **Create New**, and then in the **Name** field, type `Tunnel_IP`.
5. In the **Per-Device Mapping** section, configure the following settings:



| Field              | Value                      |
|--------------------|----------------------------|
| BR2-FGT-1 [global] | 172.16.1.2/255.255.255.255 |
| BR3-FGT-1 [global] | 172.16.1.3/255.255.255.255 |

Name

Tunnel\_IP

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

|                          |                    |                            |
|--------------------------|--------------------|----------------------------|
| <input type="checkbox"/> | Mapped Device      | Value                      |
| <input type="checkbox"/> | BR2-FGT-1 [global] | 172.16.1.2/255.255.255.255 |
| <input type="checkbox"/> | BR3-FGT-1 [global] | 172.16.1.3/255.255.255.255 |

6. Click **OK**.
7. Click **Create New**, and then in the **Name** field, type `Local_ID`.
8. In the **Per-Device Mapping** section, configure the following settings:

| Field              | Value     |
|--------------------|-----------|
| BR2-FGT-1 [global] | BR2-FGT-1 |
| BR3-FGT-1 [global] | BR3-FGT-1 |

Name

Local\_ID

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

| <input type="checkbox"/> | Mapped Device      | Value     |
|--------------------------|--------------------|-----------|
| <input type="checkbox"/> | BR2-FGT-1 [global] | BR2-FGT-1 |
| <input type="checkbox"/> | BR3-FGT-1 [global] | BR3-FGT-1 |

- 9. Click **OK**.
- 10. Click **Policy & Objects > Advanced > Metadata Variables**.
- 11. Select **Router\_ID**, and then click **Edit**.
- 12. In the **Per-Device Mapping** section, configure the following settings:

| Field              | Value      |
|--------------------|------------|
| BR2-FGT-1 [global] | 172.16.1.2 |
| BR3-FGT-1 [global] | 172.16.1.3 |

Name

Router\_ID

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

| <input type="checkbox"/> | Mapped Device      | Value      |
|--------------------------|--------------------|------------|
| <input type="checkbox"/> | HQ-NGFW [Core1]    | 172.16.1.1 |
| <input type="checkbox"/> | HQ-NGFW [Core2]    | 172.16.2.1 |
| <input type="checkbox"/> | BR2-FGT-1 [global] | 172.16.1.2 |
| <input type="checkbox"/> | BR3-FGT-1 [global] | 172.16.1.3 |

13. Click **OK**.

To configure the IPsec template for Core1

- 1. Continuing on the FortiManager GUI, click **Device Manager**.
- 2. Click **Provisioning Templates**, and then select **IPsec Tunnel**.
- 3. Right-click the **HUB\_IPsec\_Recommended** template, and then select **Activate**.

Template Groups

IPsec Tunnel

BGP

System Templates

Static Route

CLI

Feature Visibility

+ Create New

Edit

Delete

Assign to Device/Group

AI Assistant

More

| <input type="checkbox"/>            | Name                       | Assigned to Device/Group |
|-------------------------------------|----------------------------|--------------------------|
| <input checked="" type="checkbox"/> | HUB_IPsec_Recommended      | 0 Devices in Total       |
| <input type="checkbox"/>            | BRANCH_IPsec_Recommended   | 0 Devices in Total       |
| <input type="checkbox"/>            | IPsec_Fortinet_Recommended | 0 Devices in Total       |

Edit

Clone

Delete

Assign to Device/Group

Preview CLI Configuration

Activate

- 4. Configure the following settings:

| Field              | Value         |
|--------------------|---------------|
| Template Name      | Hub_overlay1  |
| Enable ADVPN       | Enabled       |
| Outgoing Interface | port2         |
| IPv4 Start IP      | 172.16.1.2    |
| IPv4 End IP        | 172.16.1.254  |
| IPv4 Netmask       | 255.255.255.0 |
| Pre-shared Key     | 123456789     |

Activate HUB\_IPsec\_Recommended

Template Name: Hub\_overlay1

Enable ADVPN: ☒

VPN1 ▾

Outgoing Interface: port2

IPv4 Start IP: 172.16.1.2

IPv4 End IP: 172.16.1.254

IPv4 Netmask: 255.255.255.0

Pre-shared Key: 123456789

- Click **OK**.
- Right-click the **Hub\_overlay1** template, and then select **Edit**.
- Right-click the **VPN1** template, and then select **Edit**.
- Disable **Mode Config**.
- In the **Transport** field, select **UDP**.

Mode Config: ☐

Transport: UDP Auto TCP Encapsulation

NAT Traversal: Enable Disable Forced

- In the **Tunnel Interface Setup** section, configure the following settings:

| Field     | Value                      |
|-----------|----------------------------|
| IP        | 172.16.1.1/255.255.255.255 |
| Remote IP | 172.16.1.254/255.255.255.0 |

Tunnel Interface Setup ▾

|           |                                                         |
|-----------|---------------------------------------------------------|
| IP        | <input type="text" value="172.16.1.1/255.255.255.255"/> |
| Remote IP | <input type="text" value="172.16.1.254/255.255.255.0"/> |

- Click **OK**.
- Click **OK**.
- Right-click the **Hub\_overlay1** template, and then select **Assign to Device/Group**.
- In the **Available Entries** field, select the **HQ-NGFW [Core1]** checkbox, and then move it to the **Selected Entries** field.
- Click **OK**.

### To configure the IPsec template for the branches

- Continuing on the FortiManager GUI, right-click the **BRANCH\_IPsec\_Recommended** template, and then select **Activate**.
- Configure the following settings:

| Field              | Value             |
|--------------------|-------------------|
| Template Name      | Branches_overlay1 |
| Enable ADVPN       | Enabled           |
| Outgoing Interface | port2             |
| Local ID           | \$(Local_ID)      |
| Remote Gateway     | 100.65.0.101      |
| Pre-shared Key     | 123456789         |

Activate BRANCH\_IPsec\_Recommended

|               |                                  |
|---------------|----------------------------------|
| Template Name | Branches_overlay1                |
| Enable ADVPN  | <input checked="" type="radio"/> |

HUB1-VPN1 ▾

|                    |              |
|--------------------|--------------|
| Outgoing Interface | port2        |
| Local ID           | \$(Local_ID) |
| Remote Gateway     | 100.65.0.101 |
| Pre-shared Key     | 123456789    |

- Click **OK**.
- Right-click the **Branches\_overlay1** template, and then select **Edit**.
- Right-click **HUB1-VPN1**, and then select **Edit**.
- Disable **Mode Config**.
- In the **Transport** field, select **UDP**.

|               |                            |
|---------------|----------------------------|
| Mode Config   | <input type="radio"/>      |
| Transport ⓘ   | UDP Auto TCP Encapsulation |
| NAT Traversal | Enable Disable Forced      |

- In the **Tunnel Interface Setup** section, configure the following settings:

| Field     | Value                    |
|-----------|--------------------------|
| IP        | \$(Tunnel_IP)            |
| Remote IP | 172.16.1.1/255.255.255.0 |

Tunnel Interface Setup ▾

|           |                          |
|-----------|--------------------------|
| IP        | \$(Tunnel_IP)            |
| Remote IP | 172.16.1.1/255.255.255.0 |

- Click **OK**.
- Click **OK**.

- 11. Right-click the **Branches\_overlay1** template, and then select **Assign to Device/Group**.
- 12. In the **Available Entries** field, select the **BR2-FGT-1** and **BR3-FGT-1** checkboxes, and then move them to the **Selected Entries** field.
- 13. Click **OK**.

| Name ⇅                     | Assigned to Device/Group ⇅                                                                              |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| HUB_IPsec_Recommended      | 0 Devices in Total                                                                                      |
| BRANCH_IPsec_Recommended   | 0 Devices in Total                                                                                      |
| IPsec_Fortinet_Recommended | 0 Devices in Total                                                                                      |
| Branches_overlay1          | 2 Devices in Total <a href="#">View Details &gt;</a> <div> BR2-FGT-1 [root]<br/> BR3-FGT-1 [root]</div> |
| Hub_overlay1               | 1 Device in Total <a href="#">View Details &gt;</a> <div> HQ-NGFW [Core1]</div>                         |

To install the device settings

- 1. Continuing on the FortiManager GUI, click **Install Wizard**.
- 2. In the **Install Wizard** window, select **Install Device Settings (only)**, and then click **Next**.
- 3. Ensure that **BR2-FGT-1**, **BR3-FGT-1**, **HQ-NGFW**, and **Core1** are selected.
- 4. Click **Next**.
- 5. Click **Install**.
- 6. Click **Finish**.

Configure the BGP Templates

You will create BGP templates for the FortiGate devices.

To configure the BGP template for Core1

- 1. Continuing on the FortiManager GUI, click **Device Manager**.
- 2. Click **Provisioning Templates**, and then select **BGP**.



If the **BGP** section is not available, you can enable **BGP** in the **Feature Visibility** section.

- 3. Click **Create New**.
- 4. Configure the following settings:

| Field     | Value        |
|-----------|--------------|
| Name      | Hub_overlay1 |
| Local As  | 65100        |
| Router ID | 172.16.1.1   |

5. Create a **Neighbor Group** with the following settings:

| Field                         | Value    |
|-------------------------------|----------|
| Name                          | Overlay1 |
| Remote AS                     | 65100    |
| Route Reflector Client (IPv4) | Enable   |
| Next Hop Self (IPv4)          | Enable   |

6. Click **OK**.

Name

Hub\_overlay1

Description

Local As

65100

Router ID

172.16.1.1

Neighbors

+ Create New

Edit

Delete

More

☐

IP

Remote AS

No record found.

Neighbor Group

+ Create New

Edit

Delete

More

☐

Name

Remote AS

☐

Overlay1

65100

7. Create a **Neighbor Range** with the following settings:



| Field               | Value                    |
|---------------------|--------------------------|
| Prefix              | 172.16.1.0/255.255.255.0 |
| Neighbor Group      | Overlay1                 |
| Max Neighbor Number | 2                        |

8. Click **OK**.
9. In the **Networks** field, type 10.0.12.0/255.255.255.0.

The screenshot displays the FortiManager GUI for configuring a BGP template. It is divided into three main sections: 'Neighbor Range', 'IPv6 Neighbor Range', and 'Networks'.  
 - The 'Neighbor Range' section contains a table with columns: Prefix, Neighbor Group, and Maximum Neighbor Number. A single entry is listed: Prefix '172.16.1.0/255.255.255.0', Neighbor Group 'Overlay1', and Maximum Neighbor Number '2'.  
 - The 'IPv6 Neighbor Range' section shows a message 'No record found.'  
 - The 'Networks' section has a search bar where '10.0.12.0/255.255.255.0' has been entered.

10. Click **OK**.
11. Right-click the **Hub\_overlay1** template, and then select **Assign to Device/Group**.
12. In the **Available Entries** field, select the **HQ-NGFW [Core1]** checkbox, and then move it to the **Selected Entries** field.
13. Click **OK**.

### To configure the BGP template for the branches

1. Continuing on the FortiManager GUI, click **Create New**.

| Field     | Value             |
|-----------|-------------------|
| Name      | Branches_overlay1 |
| Local As  | 65100             |
| Router ID | \$(Router_ID)     |

2. Create a **Neighbor** with the following settings:

| Field         | Value      |
|---------------|------------|
| IP            | 172.16.1.1 |
| Remote AS     | 65100      |
| Connect Timer | 120        |

3. Click **OK**.

|                                                                                                  |                   |
|--------------------------------------------------------------------------------------------------|-------------------|
| Name                                                                                             | Branches_overlay1 |
| Description                                                                                      |                   |
| Local As                                                                                         | 65100             |
| Router ID                                                                                        | \$(Router_ID)     |
| Neighbors                                                                                        |                   |
| <div> <span>+ Create New</span> <span>Edit</span> <span>Delete</span> <span>More ▾</span> </div> |                   |
| <input type="checkbox"/>                                                                         | IP ▾              |
| <input type="checkbox"/>                                                                         | Remote AS ▾       |
| <input type="checkbox"/>                                                                         | 172.16.1.1        |
| <input type="checkbox"/>                                                                         | 65100             |

4. In the **Networks** field, type \$(LAN\_BR) .

|          |            |
|----------|------------|
| Networks | IP/Netmask |
|          | \$(LAN_BR) |

- Click **OK**.
- Right-click the **Branches\_overlay1** template, and then select **Assign to Device/Group**.
- In the **Available Entries** field, select the **BR2-FGT-1** and **BR3-FGT-1** checkboxes, and then move them to the **Selected Entries** field.
- Click **OK**.

| <input type="checkbox"/>            | Name ⇅                 | Assigned to Device/Group ⇅                                                                   |
|-------------------------------------|------------------------|----------------------------------------------------------------------------------------------|
| <input type="checkbox"/>            | HUB_BGP_Recommended    | 0 Devices in Total                                                                           |
| <input type="checkbox"/>            | BRANCH_BGP_Recommended | 0 Devices in Total                                                                           |
| <input type="checkbox"/>            | BGP_NGFW               | 1 Device in Total <a href="#">View Details &gt;</a><br>HQ-NGFW [Core2]                       |
| <input checked="" type="checkbox"/> | Branches_overlay1      | 2 Devices in Total <a href="#">View Details &gt;</a><br>BR2-FGT-1 [root]<br>BR3-FGT-1 [root] |
| <input checked="" type="checkbox"/> | Hub_overlay1           | 1 Device in Total <a href="#">View Details &gt;</a><br>HQ-NGFW [Core1]                       |

To install the device settings

1. Continuing on the FortiManager GUI, click **Install Wizard**.
2. In the **Install Wizard** window, select **Install Device Settings (only)**, and then click **Next**.
3. Verify that the **BR2-FGT-1**, **BR3-FGT-1**, **HQ-NGFW**, and **Core1** checkboxes are selected.
4. Click **Next**.
5. Click **Install**.
6. Click **Finish**.

Install the Policies

You will run scripts to create policies in the **Core1** and **BR** policy packages. Then, you will install the policies.

To run the scripts

1. Continuing on the FortiManager GUI, click **Device Manager > Scripts**.
2. Select the **ADVPN\_Core1\_ex1** checkbox, and then click **Run Script**.
3. Select the **Core1** policy package, and then click **Run Now**.
4. Click **Close** to close the **Run Script** window.

You have created three policies in the **Core1** policy package.

| <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">Section</a> <a href="#">Policy Lookup</a> |   |                  |       |       |
|--------------------------------------------------------------------------------------------------------------------------------|---|------------------|-------|-------|
| <input type="checkbox"/>                                                                                                       | # | Name             | From  | To    |
| <input type="checkbox"/>                                                                                                       | 1 | Internet         | port5 | port2 |
| <input type="checkbox"/>                                                                                                       | 2 | FAZ_and_FMG_VIPs | port2 | port6 |
| <input type="checkbox"/>                                                                                                       | 3 | FAZ_FMG_Access   | port5 | port6 |
| <input type="checkbox"/>                                                                                                       | 4 | LAN to IPsec     | port5 | VPN1  |
| <input type="checkbox"/>                                                                                                       | 5 | IPsec to LAN     | VPN1  | port5 |
| <input type="checkbox"/>                                                                                                       | 6 | IPsec to IPsec   | VPN1  | VPN1  |

- Click **Device Manager > Scripts**.
- Select the **ADVPN\_Branches** checkbox, and then click **Run Script**.
- Select the **BR** policy package, and then click **Run Now**.
- Click **Close** to close the **Run Script** window.

You have created two policies in the **BR** policy package.

| <a href="#">Search...</a> <a href="#">+ Create New</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">Section</a> <a href="#">Policy Lookup</a> |   |            |           |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------|-----------|-----------|
| <div> <div>BR</div> <div>Firewall Policy</div> <div>Installation Targets</div> <div>Core1</div> <div>Core2</div> <div>DCFw</div> <div>ISFW</div> </div>  |   |            |           |           |
| <input type="checkbox"/>                                                                                                                                 | # | Name       | From      | To        |
| <input type="checkbox"/>                                                                                                                                 | 1 | Internet   | port3     | port1     |
| <input type="checkbox"/>                                                                                                                                 | 2 | LAN to Hub | port4     | HUB1-VPN1 |
| <input type="checkbox"/>                                                                                                                                 | 3 | Hub to LAN | HUB1-VPN1 | port4     |

### To install the policies

- Continuing on the FortiManager GUI, click **Device Manager > Device & Groups > Managed FortiGate**.
- Select the **BR2-FGT-1**, **BR3-FGT-1**, and **Core1** checkboxes.
- Click **Install > Reinstall policy**.
- Click **OK**.
- Verify that **Core1**, **BR2-FGT-1**, and **BR3-FGT-1** are selected.
- Click **Next**.
- Click **Finish**.

## Bring Up the On-Demand Tunnel

Before you generate traffic to trigger the on-demand tunnel, it is a good idea to verify that the IPsec VPN tunnels are up and the BGP route databases are in sync.

### To verify the IPsec tunnel status

1. Continuing on the FortiManager GUI, click **Device Manager**.
2. Click **Monitors**, and then select **VPN Monitor**.
3. Click **Show Table** to enable the table.

| Device          | Name      | Type      | Remote Gateway | Peer ID      | Incoming Data | Outgoing Data | Phase 1   | Phase 2 Selectors |
|-----------------|-----------|-----------|----------------|--------------|---------------|---------------|-----------|-------------------|
| BR2-FGT-1[root] | HUB1-VPN1 | automatic | 100.65.0.101   | 100.65.0.101 | 0.00 KB       | 0.00 KB       | HUB1-VPN1 | HUB1-VPN1         |
| BR3-FGT-1[root] | HUB1-VPN1 | automatic | 100.65.0.101   | 100.65.0.101 | 0.00 KB       | 0.00 KB       | HUB1-VPN1 | HUB1-VPN1         |
| HQ-NGFW[Core1]  | VPN1_0    | dialup    | 100.65.3.113   | BR3-FGT-1    | 0.00 KB       | 0.00 KB       | VPN1_0    | VPN1              |
| HQ-NGFW[Core1]  | VPN1_1    | dialup    | 100.65.2.112   | BR2-FGT-1    | 0.00 KB       | 0.00 KB       | VPN1_1    | VPN1              |

### To check the BGP routes

1. Log in to the HQ-NGFW (Core1) CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following commands:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```



BGP may take about 1 minute to establish and advertise the routes. You can enter the `get router info routing-table bgp` command again to refresh the BGP routing table.

### Stop and think!

Core1 has learned BGP network segments through IBGP (AS 65100). You should see the network segment 172.20.2.0/24 (from BR2-FGT-1) and 172.20.3.0/24 (from BR3-FGT-1).

```
HQ-NGFW-1 (Core1) # get router info routing-table bgp
Routing table for VRF=0
B    172.20.2.0/24 [200/0] via 172.16.1.2 (recursive is directly connected, VPN1), 00:01:54, [1/0]
B    172.20.3.0/24 [200/0] via 172.16.1.3 (recursive is directly connected, VPN1), 00:01:53, [1/0]
```

Notice that the tunnels are connected to IP addresses 100.65.2.112 (BR2-FGT-1) and 100.65.3.113 (BR3-FGT-1).

```
HQ-NGFW-1 (Core1) # get vpn ipsec tunnel summary
'VPN1_0' 100.65.3.113:0 selectors(total,up): 1/1 rx(pkt,err): 15/0 tx(pkt,err): 14/0
'VPN1_1' 100.65.2.112:0 selectors(total,up): 1/1 rx(pkt,err): 15/0 tx(pkt,err): 14/0
```

3. Log in to the BR2-FGT-1 CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

4. Enter the following commands:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
BR2-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B    10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:07:40, [1/0]
B    172.20.3.0/24 [200/0] via 172.16.1.3 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:07:11, [1/0]

BR2-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 26/0 tx(pkt,err): 27/2
```

5. Log in to the BR3-FGT-1 CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

6. Enter the following commands:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
BR3-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B    10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:11:14, [1/0]
B    172.20.2.0/24 [200/0] via 172.16.1.2 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:11:14, [1/0]

BR3-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 33/0 tx(pkt,err): 34/2
```

## To bring up the on-demand tunnel

1. Return to the BR2-FGT-1 CLI, and then enter the following commands:

```
execute ping-options source 172.20.2.254
execute traceroute-options source 172.20.2.254
```

2. Enter the following commands to trigger the on-demand tunnel:

```
execute ping 172.20.3.254
execute traceroute 172.20.3.254
```

```
BR2-FGT-1 # execute ping 172.20.3.254
PING 172.20.3.254 (172.20.3.254): 56 data bytes
64 bytes from 172.20.3.254: icmp_seq=0 ttl=254 time=4.9 ms
64 bytes from 172.20.3.254: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.20.3.254: icmp_seq=2 ttl=255 time=1.2 ms
64 bytes from 172.20.3.254: icmp_seq=3 ttl=255 time=1.3 ms
64 bytes from 172.20.3.254: icmp_seq=4 ttl=255 time=1.3 ms

--- 172.20.3.254 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.2/2.0/4.9 ms

BR2-FGT-1 # execute traceroute 172.20.3.254
traceroute to 172.20.3.254 (172.20.3.254), 32 hops max, 3 probe packets per hop, 84 byte packets
 1 172.20.3.254 1.340 ms 6.837 ms 0.821 ms
```

## To validate the on-demand tunnel

1. Return to the BR2-FGT-1 and BR3-FGT-1 CLIs, and then enter the following command:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

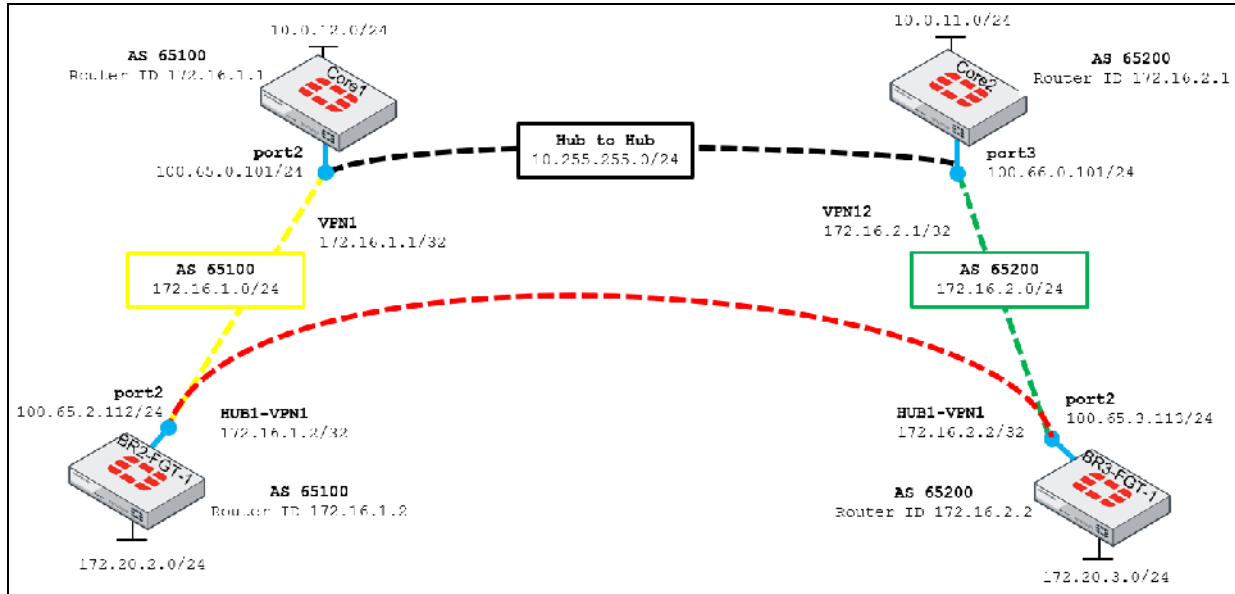
```
BR2-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B      10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:14:14, [1/0]
B      172.20.3.0/24 [200/0] via 172.16.1.3 (recursive is directly connected, HUB1-VPN1_0) 00:00:26, [1/0]

BR2-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1_0' 100.65.3.113:0 selectors(total,up): 2/1 rx(pkt,err): 7/0 tx(pkt,err): 7/2
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 42/0 tx(pkt,err): 43/2
```

The ADVPN tunnel has been created with the IP address 100.65.3.113 and the next hop is 172.16.1.3.

## Exercise 2: Configuring ADVPN IBGP and EBGP

In this exercise, you will configure ADVPN IBGP and EBGP.



### Prerequisites

You must unassign the IPsec templates from the previous exercise.

#### To delete the IPsec templates

1. Log in to the FortiManager GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **EFW**.
3. Click **Device Manager > Provisioning Templates > IPsec Tunnel**.
4. Select, and then right-click the **Hub\_overlay1** and **Branches\_overlay1** templates.
5. Click **Delete**.

#### To delete the BGP templates

1. Continuing on the FortiManager GUI, click **Device Manager > Provisioning Templates > BGP**.
2. Select, and then right-click the **Hub\_overlay1** and **Branches\_overlay1** templates.
3. Click **Delete**.
4. Click **OK**.



## Configure ADVPN IBGP and EBGP

You will run a script to enable ADVPN IBGP and EBGP on Core1, Core2, BR2-FGT-1, and BR3-FGT-1.

### To run the script on HQ-NGFW, BR2-FGT-1, and BR3-FGT-1

1. Continuing on the FortiManager GUI, click **Device Manager > Scripts**.



You may edit the **HQ\_NGFW\_ADVPN\_ex2**, **BR2-FGT-1\_ADVPN\_ex2**, and **BR3-FGT-1\_ADVPN\_ex2** scripts to view the configurations applied.

2. Right-click the **HQ\_NGFW\_ADVPN\_ex2** script, and then click **Run Script**.
3. In the **Available Entries** field, select the **HQ-NGFW** checkbox, and then move it to the **Selected Entries** field.
4. Click **Run Now**.
5. Click **OK**.
6. Click **Close**.
7. Right-click the **BR2-FGT-1\_ADVPN\_ex2** script, and then click **Run Script**.
8. In the **Available Entries** field, select the **BR2-FGT-1** checkbox, and then move it to the **Selected Entries** field.
9. Click **Run Now**.
10. Click **OK**.
11. Click **Close**.
12. Right-click the **BR3-FGT-1\_ADVPN\_ex2** script, and then click **Run Script**.
13. In the **Available Entries** field, select the **BR3-FGT-1** checkbox, and then move it to the **Selected Entries** field.
14. Click **Run Now**.
15. Click **OK**.
16. Click **Close**.

### To assign the per device mapping

1. Continuing on the FortiManager GUI, click **Policy & Objects > Normalized Interface**.
2. In the upper-right corner, in the search field, search for **VPN1**.
3. Right-click the **VPN1** interface, and then click **Edit**.
4. In the **Per-Device Mapping** field, click **Create New**.
5. Configure the following settings:

| Field                 | Value           |
|-----------------------|-----------------|
| Mapped Device         | HQ-NGFW [Core1] |
| Mapped Interface Name | VPN1            |

6. Click **OK**.
7. In the **Per-Device Mapping** field, click **Create New**.
8. Configure the following settings:

| Field                 | Value           |
|-----------------------|-----------------|
| Mapped Device         | HQ-NGFW [Core2] |
| Mapped Interface Name | VPN12           |

9. Click **OK**.

Edit Normalized Interface - VPN1

Name

Description

Color  [Change](#)

Wildcard ☐

Per-Platform Mapping >

Per-Device Mapping v

[+ Create New](#) [Edit](#) [Delete](#)

| <input type="checkbox"/> | Mapped Device   | Details | Type   | Addressing Mode | IP/Netmask                 |
|--------------------------|-----------------|---------|--------|-----------------|----------------------------|
| <input type="checkbox"/> | HQ-NGFW [Core1] | VPN1    | Tunnel | Manual          | 172.16.1.1/255.255.255.255 |
| <input type="checkbox"/> | HQ-NGFW [Core2] | VPN12   | Tunnel | Manual          | 172.16.2.1/255.255.255.255 |

10. Click **OK**.
11. Click **Policy & Objects > Normalized Interface**.
12. Search for **HTH**.
13. Right-click the **HTH** interface, and then click **Edit**.
14. In the **Per-Device Mapping** field, click **Create New**.
15. Configure the following settings:

| Field                 | Value           |
|-----------------------|-----------------|
| Mapped Device         | HQ-NGFW [Core1] |
| Mapped Interface Name | hub_to_hub      |

16. Click **OK**.
17. In the **Per-Device Mapping** field, click **Create New**.
18. Configure the following settings:

| Field                 | Value           |
|-----------------------|-----------------|
| Mapped Device         | HQ-NGFW [Core2] |
| Mapped Interface Name | hub_to_hub2     |

19. Click **OK**.

Edit Normalized Interface - HTH

Name **HTH**

Description

Color  **Change**

Wildcard 

Per-Platform Mapping >

Per-Device Mapping v

**+ Create New**   Search...

| <input type="checkbox"/> | Mapped Device                                                                                       | Details                                                                                         | Type   | Addressing Mode | IP/Netmask                   |
|--------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|-----------------|------------------------------|
| <input type="checkbox"/> |  HQ-NGFW [Core1] |  hub_to_hub  | Tunnel | Manual          | 10.255.255.1/255.255.255.255 |
| <input type="checkbox"/> |  HQ-NGFW [Core2] |  hub_to_hub2 | Tunnel | Manual          | 10.255.255.2/255.255.255.255 |

20. Click **OK**.

### To create policies on Core1 and Core2

1. Continuing on the FortiManager GUI, click **Device Manager > Scripts**.



You may edit the **ADVPN\_Core1\_ex2** and **ADVPN\_Core2\_ex2** scripts to view the configurations applied.

- Right-click the **ADVPN\_Core1\_ex2** script, and then click **Run Script**.
- In the **Run script on policy package** field, select **Core1**.
- Click **Run Now**.
- Click **Close**.
- Right-click the **ADVPN\_Core2\_ex2** script, and then click **Run Script**.
- In the **Run script on policy package** field, select **Core2**.
- Click **Run Now**.
- Click **Close**.

### To install policies on Core1, Core2, BR2-FGT-1, and BR3-FGT12

1. Continuing on the FortiManager GUI, click **Device Manager > Device & Groups > Managed FortiGate**.
2. Select the **Core1** and **Core2** checkboxes.
3. Click **Install**, and then click **Reinstall policy**.
4. Verify that **Core1** and **Core2** are selected, and then click **Next**.
5. Click **Finish**.
6. Select the **BR2-FGT-1** and **BR3-FGT-1** checkboxes, and then click **Install Wizard**.
7. In the **Choose What To Install (1/4)** window, select **Install Policy Package & Device Settings**.
8. In the **Policy Package** field, select **BR**, and then click **Next**.
9. In the **Select Devices to Install (BR) (2/4)** window, click **Next**.
10. In the **Validate Devices (BR) (3/4)** window, make sure that **BR2-FGT-1** and **BR3-FGT-1** are selected, and then click **Install**.
11. In the **Installation progress (4/4)** window, click **Finish**.
12. Click **Device Manager > Scripts**.
13. Right-click the **HQ\_NGFW\_reinitialize** script, and then click **Run Script**.
14. In the **Available Entries** field, select the **HQ-NGFW** checkbox, and then move it to the **Selected Entries** field.
15. Click **Run Now**.
16. Click **OK**.
17. Click **Close**.



After updating the configurations from the previous exercise, you must reboot HQ-NGFW before checking ADVPN.

## Bring Up the On-Demand Tunnel

Before you generate traffic to trigger the on-demand tunnel, it is a good idea to verify that the VPN IPsec tunnels are up and the BGP route databases are in sync.

### To verify the status of the IPsec tunnels check the BGP routes

1. Log in to the HQ-NGFW (Core1) CLI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Enter the following commands:

```
config vdom
edit Core1
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
HQ-NGFW-1 (Core1) # get router info routing-table bgp
Routing table for VRF=0
B      10.0.11.0/24 [20/0] via 10.255.255.2 (recursive via hub_to_hub tunnel 100.66.0.101), 00:02:35, [1/0]
B      172.20.2.0/24 [200/0] via 172.16.1.2 (recursive is directly connected, VPN1), 00:02:14, [1/0]
B      172.20.3.0/24 [20/0] via 172.16.2.2 (recursive via hub_to_hub tunnel 100.66.0.101), 00:02:08, [1/0]

HQ-NGFW-1 (Core1) # get vpn ipsec tunnel summary
'VPN1_0' 100.65.2.112:0 selectors(total,up): 1/1 rx(pkt,err): 16/0 tx(pkt,err): 15/0
'hub_to_hub' 100.66.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 21/0 tx(pkt,err): 22/1
```

Notice that you do not need to change VDOMs to use the `get`, `show`, `diagnose`, or `execute` commands. You can see more details in the following image:



```
HQ-NGFW-1 (Core1) # sudo
<global/vdom-name>  global or virtual domain name
global
Core1
Core2
root

HQ-NGFW-1 (Core1) # sudo Core2
get          Get dynamic and system information.
show         Show configuration.
diagnose     Diagnose facility.
execute      Execute static commands.
```

3. Enter the following commands:

```
sudo Core2 get router info routing-table bgp
sudo Core2 get vpn ipsec tunnel summary
```

```
HQ-NGFW-1 (Core1) # sudo Core2 get router info routing-table bgp
Routing table for VRF=0
B      10.0.12.0/24 [20/0] via 10.255.255.1 (recursive via hub_to_hub2 tunnel 100.65.0.101), 00:00:18, [1/0]
B      172.20.2.0/24 [20/0] via 172.16.1.2 (recursive via hub_to_hub2 tunnel 100.65.0.101), 00:08:28, [1/0]
B      172.20.3.0/24 [200/0] via 172.16.2.2 (recursive is directly connected, VPN12), 00:08:40, [1/0]

HQ-NGFW-1 (Core1) # sudo Core2 get vpn ipsec tunnel summary
'hub_to_hub2' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 38/0 tx(pkt,err): 37/1
'VPN12_0' 100.65.3.113:0 selectors(total,up): 1/1 rx(pkt,err): 31/0 tx(pkt,err): 30/0
```

4. Log in to the BR2-FGT-1 CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

5. Enter the following commands:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
BR2-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B      10.0.11.0/24 [200/0] via 10.255.255.2 (recursive via 100.65.2.254, port2), 00:11:44, [1/0]
B      10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:03:45, [1/0]
B      172.20.3.0/24 [200/0] via 172.16.2.2 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:11:17, [1/0]

BR2-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 38/0 tx(pkt,err): 39/7
```

6. Log in to the BR3-FGT-1 CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

7. Enter the following commands:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
BR3-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B   10.0.11.0/24 [200/0] via 172.16.2.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:13:05, [1/0]
B   10.0.12.0/24 [200/0] via 10.255.255.1 (recursive via 100.65.3.254, port2), 00:13:05, [1/0]
B   172.20.2.0/24 [200/0] via 172.16.1.2 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:10:57, [1/0]

BR3-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 39/0 tx(pkt,err): 40/0
```



BGP may take around one minute to establish and advertise the routes. You can type again `get router info routing-table bgp` to refresh the BGP routing table.

## To bring up the on-demand tunnel

1. Return to the BR2-FGT-1 CLI, and then enter the following commands:

```
execute ping-options source 172.20.2.254
execute traceroute-options source 172.20.2.254
```

2. Enter the following commands to trigger the on-demand tunnel:

```
execute ping 172.20.3.254
execute traceroute 172.20.3.254
```

```
BR2-FGT-1 # execute ping 172.20.3.254
PING 172.20.3.254 (172.20.3.254): 56 data bytes
64 bytes from 172.20.3.254: icmp_seq=0 ttl=253 time=6.9 ms
64 bytes from 172.20.3.254: icmp_seq=1 ttl=255 time=1.6 ms
64 bytes from 172.20.3.254: icmp_seq=2 ttl=255 time=1.3 ms
64 bytes from 172.20.3.254: icmp_seq=3 ttl=255 time=1.4 ms
64 bytes from 172.20.3.254: icmp_seq=4 ttl=255 time=1.6 ms

--- 172.20.3.254 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.3/2.5/6.9 ms

BR2-FGT-1 # execute traceroute 172.20.3.254
traceroute to 172.20.3.254 (172.20.3.254), 32 hops max, 3 probe packets per hop, 84 byte packets
 1 172.20.3.254 1.983 ms 1.157 ms 0.862 ms
```

## To validate the on-demand tunnel

1. Return to the BR2-FGT-1 and BR3-FGT-1 CLIs, and then enter the following command:

```
get router info routing-table bgp
get vpn ipsec tunnel summary
```

```
BR2-FGT-1 # get router info routing-table bgp
Routing table for VRF=0
B   10.0.11.0/24 [200/0] via 10.255.255.2 (recursive via 100.65.2.254, port2), 00:19:19, [1/0]
B   10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:11:50, [1/0]
B   172.20.3.0/24 [200/0] via 172.16.2.2 (recursive via HUB1-VPN1 tunnel 100.65.3.113), 00:01:56, [1/0]

BR2-FGT-1 # get vpn ipsec tunnel summary
'HUB1-VPN1_0' 100.65.3.113:0 selectors(total,up): 2/1 rx(pkt,err): 7/0 tx(pkt,err): 7/0
'HUB1-VPN1' 100.65.0.101:0 selectors(total,up): 1/1 rx(pkt,err): 5/0 tx(pkt,err): 58/0
```

The ADVPN tunnel has been created with the IP address 100.65.3.113 and the next hop is 172.16.2.2, using a different network segment from AS 65100.

## Lab 9: Security Fabric

In this lab, first, you will learn how to configure the Fortinet Security Fabric to enable SAML SSO. Next, you will configure automation for configuration backups. Finally, you will also create a Security Fabric automation to run CLI scripts if a configuration change occurs.

### Objectives

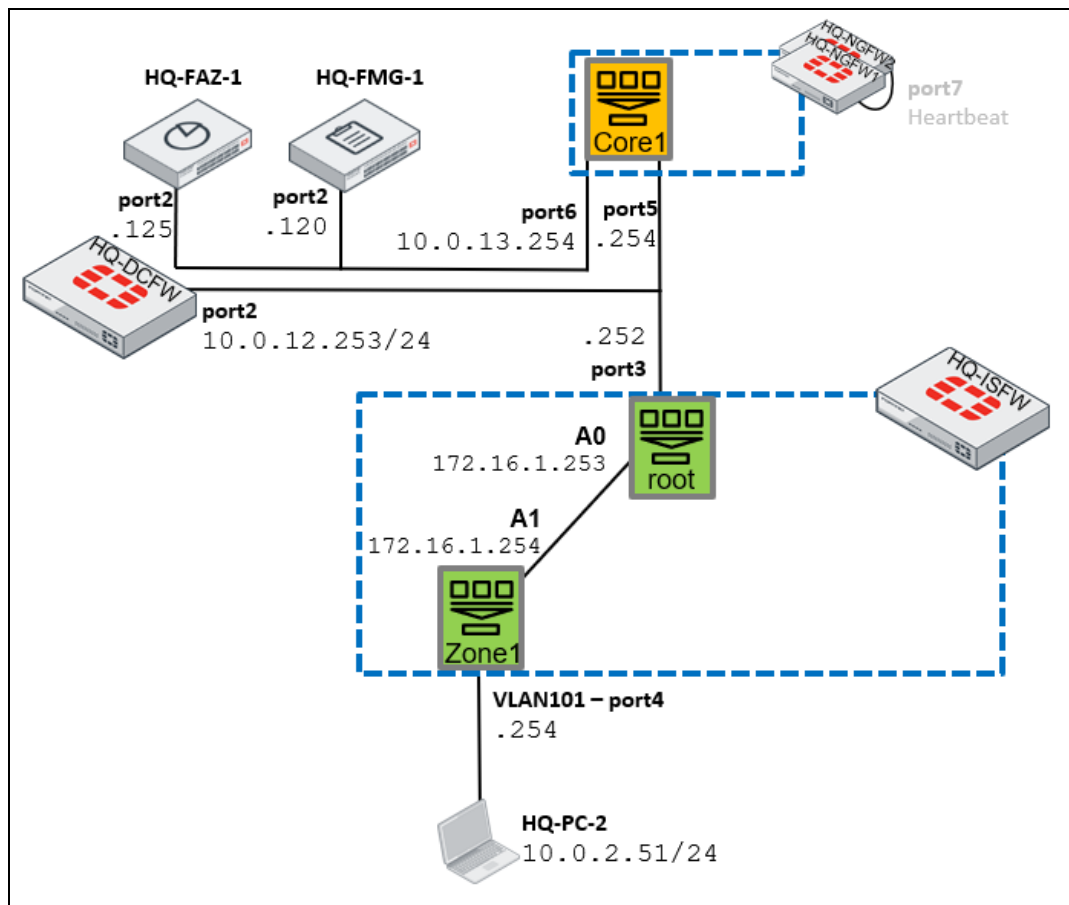
- Use the Security Fabric to enable SAML SSO on all devices
- Use Security Fabric automation for automatic configuration backups
- Use Security Fabric automation to generate CLI scripts

### Time to Complete

Estimated: 55 minutes

### Which Network Segment Will You Work On?

In this lab, you will configure the HQ-ISFW, HQ-DCFw, and HQ-NGFW FortiGate devices.





## Exercise 1: Configuring the Security Fabric and SAML SSO

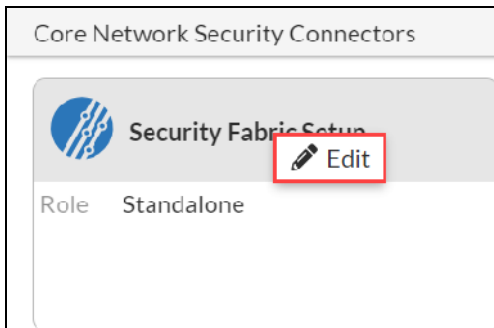
In this exercise, you will configure the Security Fabric with SAML SSO in the lab network.

### Configure the Security Fabric on HQ-NGFW-1

You will configure the root FortiGate in the Security Fabric to be the identity provider (IdP).

#### To configure the Security Fabric on HQ-NGFW-1

1. Log in to the HQ-NGFW-1 GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **Login Read-Write**, and then click **Yes**.
3. Click **Security Fabric > Fabric Connectors**.
4. In the **Core Network Security Connectors** section, click **Security Fabric Setup**, and then click **Edit**.



5. In the **Security Fabric Settings** window, in the **Security Fabric role** field, select **Serve as Fabric Root**.
6. In the **Allow other Security Fabric devices to join** field, select **port5**.
7. Click **Close**.
8. In the **Fabric name** field, type `fortinet`.
9. Enable **SAML Single Sign-On**.
10. In the **IdP certificate** field, select **Fortinet\_CA\_SSL**.
11. In the **IdP address** field, select **Specify**, and then in the field, type `10.0.12.254:443`.

Security Fabric role

Standalone **Serve as Fabric Root** Join Existing Fabric

Allow other Security Fabric devices to join

☒ **port5** X

Fabric name

**fortinet**

FortiCloud account enforcement ⓘ

☒

Allow downstream device REST API access ⓘ

☐

Fabric global object ⚠️

☒

Management IP/FQDN ⓘ

Use WAN IP Specify

Management port

Use Admin Port **Specify**

443

SAML SSO Settings

**SAML Single Sign-On ⓘ** ☒ [Advanced Options](#)

Mode

Identity Provider (IdP)

IdP certificate

**Fortinet\_CA\_SSL** Download

IdP address ⓘ

Use Management IP/FQDN **Specify**

**10.0.12.254:443** Use current browser address

⚠️ IdP address is different from management IP/FQDN.

12. Click **OK**.

To configure an administrator for SAML SSO on HQ-NGFW-1

- 1. Continuing on the HQ-NGFW-1 GUI, click **System > Administrators**.
- 2. Click **Create new**, and then select **Administrator**.
- 3. Configure the following settings:

| Field                 | Value                |
|-----------------------|----------------------|
| Username              | AdminSSO             |
| password              | password             |
| Administrator profile | super_admin_readonly |

- 4. Click **OK**.  
The result should look similar to the following image:

| + Create new Q Search    |          |               |                      |       |                           |                |
|--------------------------|----------|---------------|----------------------|-------|---------------------------|----------------|
| <input type="checkbox"/> | Name     | Trusted Hosts | Profile              | Type  | Two-factor Authentication | Virtual Domain |
| System Administrator 2   |          |               |                      |       |                           |                |
| <input type="checkbox"/> | admin    |               | super_admin          | Local | Disabled                  | Global         |
| <input type="checkbox"/> | AdminSSO |               | super_admin_readonly | Local | Disabled                  | Global         |

## Configure the Security Fabric on HQ-DCFw

You will configure HQ-DCFw as one of the branches of the Security Fabric tree with SAML SSO.

### To enable the Security Fabric on HQ-DCFw

- Log in to the HQ-DCFw GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
- Click **Login Read-Write**, and then click **Yes**.
- Click **Security Fabric > Fabric Connectors**.
- In the **Core Network Security Connectors** section, click **Security Fabric Setup**, and then click **Edit**.
- In the **Security Fabric Settings** section, configure the following settings:

| Field                                       | Value                |
|---------------------------------------------|----------------------|
| Security Fabric role                        | Join Existing Fabric |
| Allow other Security Fabric devices to join | port2                |
| Upstream FortiGate IP/FQDN                  | 10.0.12.254          |
| Management IP/FQDN                          | 10.0.12.253          |
| Management port                             | 443                  |
| Default admin profile                       | super_admin          |

Security Fabric role: Standalone | Serve as Fabric Root | **Join Existing Fabric**

Allow other Security Fabric devices to join: ☒ **port2** X

Upstream FortiGate IP/FQDN: **10.0.12.254**

Allow downstream device REST API access: ☐

Management IP/FQDN: Use WAN IP | **Specify**

**10.0.12.253**

Management port: Use Admin Port | **Specify**

**443**

SAML SSO Settings

SAML Single Sign-On: Auto | Manual

Advanced Options

Mode: Pending

Default login page: Normal | Single Sign-On

Default admin profile: **super\_admin**

6. Click **OK**.
7. Click **OK** to confirm.

### To authorize HQ-DCFW on HQ-NGFW-1

1. Return to the HQ-NGFW-1 GUI, and then click **System > Firmware & Registration**.
2. Click the device with the **Unauthorized** status, and then select **Authorize**.

|                                     | Device                  | Status           | Registration Status |
|-------------------------------------|-------------------------|------------------|---------------------|
| <input type="checkbox"/>            | HQ-NGFW-1               | Online           | Registered          |
| <input checked="" type="checkbox"/> | <b>FGVM02TM24013503</b> | Unauthorized     | Unavailable         |
| Upgrade                             |                         | <b>Authorize</b> |                     |

HQ-NGFW-1 lists the HQ-DCFW as part of the Security Fabric. It may take a few minutes to authorize. You may need to refresh the page.

|                          | Device    | Status | Registration Status |
|--------------------------|-----------|--------|---------------------|
| <input type="checkbox"/> | HQ-NGFW-1 | Online | Registered          |
| <input type="checkbox"/> | HQ-DCFW   | Online | Registered          |

## Configure the Security Fabric on HQ-ISFW

You must configure the Security Fabric on HQ-ISFW with SAML SSO and authorize it on HQ-NGFW-1.

### Take the Expert Challenge!

Configure the Security Fabric on HQ-ISFW using the information in the following table, and then authorize HQ-ISFW on HQ-NGFW-1:

| Field                                       | Value                |
|---------------------------------------------|----------------------|
| Security Fabric role                        | Join Existing Fabric |
| Allow other Security Fabric devices to join | port2                |
| Upstream FortiGate IP/FQDN                  | 10.0.12.254          |
| Management IP/FQDN                          | 10.0.2.254           |
| Management port                             | 443                  |
| Default admin profile                       | super_admin_readonly |

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Access Security Fabric Devices With SAML SSO on page 194](#).

### To enable the Security Fabric on HQ-ISFW

1. Log in to the HQ-ISFW GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **Login Read-Write**, and then click **Yes**.
3. Click **Security Fabric > Fabric Connectors**.
4. In the **Security Fabric Settings** section, configure the following settings:

| Field                                       | Value                |
|---------------------------------------------|----------------------|
| Security Fabric role                        | Join Existing Fabric |
| Allow other Security Fabric devices to join | port2                |
| Upstream FortiGate IP/FQDN                  | 10.0.12.254          |
| Management IP/FQDN                          | 10.0.2.254           |
| Management port                             | 443                  |
| Default admin profile                       | super_admin_readonly |

Security Fabric role

Standalone

Serve as Fabric Root

Join Existing Fabric

Allow other Security Fabric devices to join

port2

+

×

Upstream FortiGate IP/FQDN

10.0.12.254

Allow downstream device REST API access

Management IP/FQDN

Use WAN IP

Specify

10.0.2.254

Management port

Use Admin Port

Specify

443

SAML SSO Settings

SAML Single Sign-On

Auto

Manual

Advanced Options

Mode

Pending

Default login page

Normal

Single Sign-On

Default admin profile

super\_admin\_readonly

5. Click **OK**.
6. Click **OK** to confirm.

To authorize HQ-ISFW on HQ-NGFW-1

1. Return to the HQ-NGFW-1 GUI, and then click **System > Firmware & Registration**.
2. Click the device with the **Unauthorized** status, and then select **Authorize**.

The result should look similar to the following image:

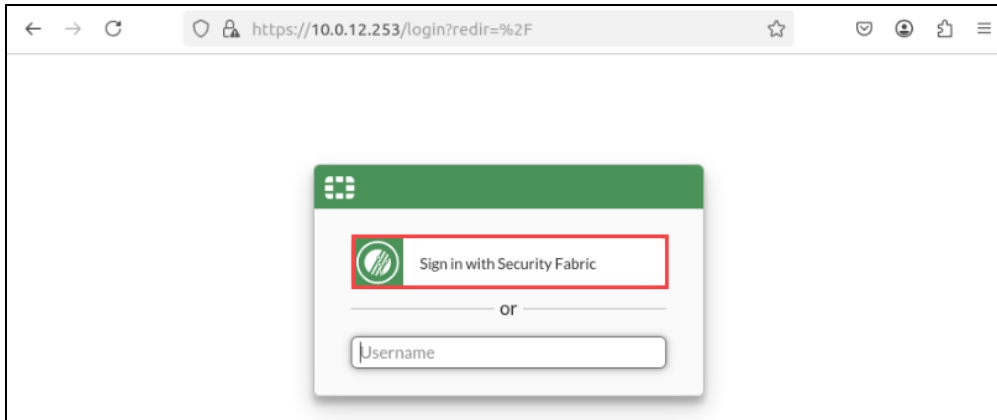
|                          | Device    | Status | Registration Status |
|--------------------------|-----------|--------|---------------------|
| <input type="checkbox"/> | HQ-NGFW-1 | Online | Registered          |
| <input type="checkbox"/> | HQ-ISFW   | Online | Registered          |
| <input type="checkbox"/> | HQ-DCFW   | Online | Registered          |

Access Security Fabric Devices With SAML SSO

You will use SAML SSO to access HQ-DCFW and HQ-ISFW.

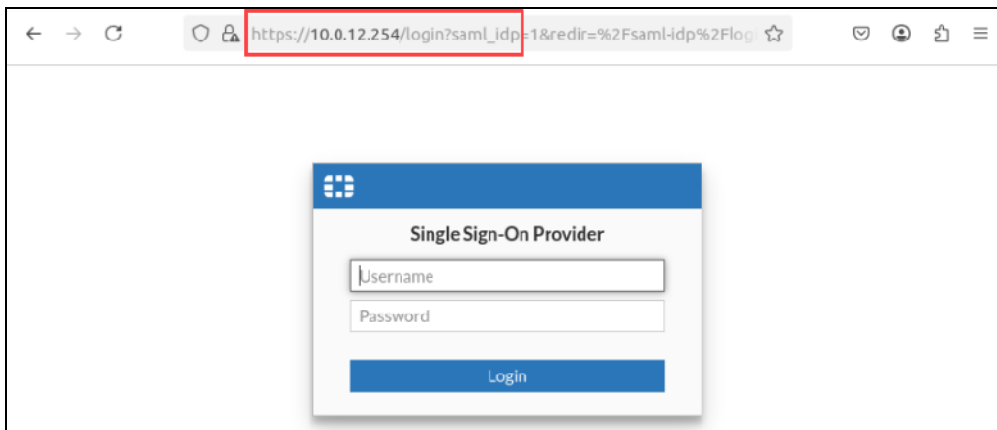
### To access HQ-DCFW using SAML SSO

1. Connect to the HQ-PC-2 VM.
2. Open a browser, and then enter `https://10.0.12.253`.

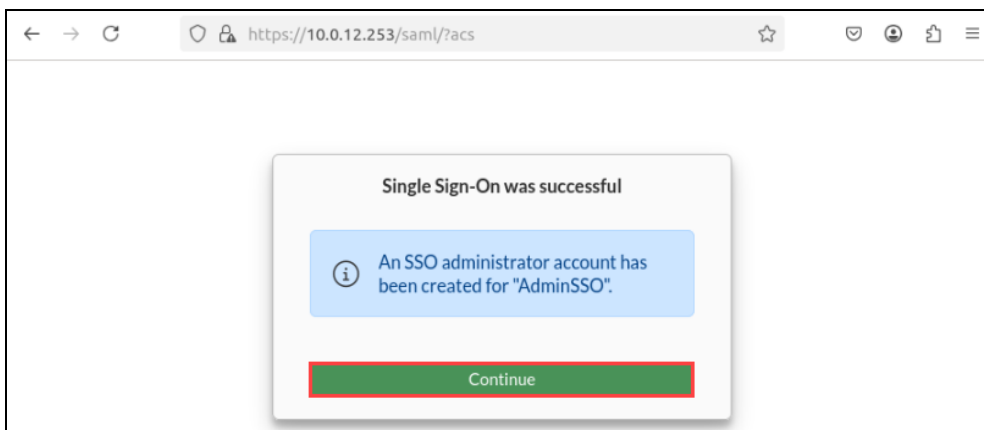


3. Click **Sign in with Security Fabric**.

You are redirected to the IdP, which is HQ-NGFW-1, to log in with SSO provider credentials.

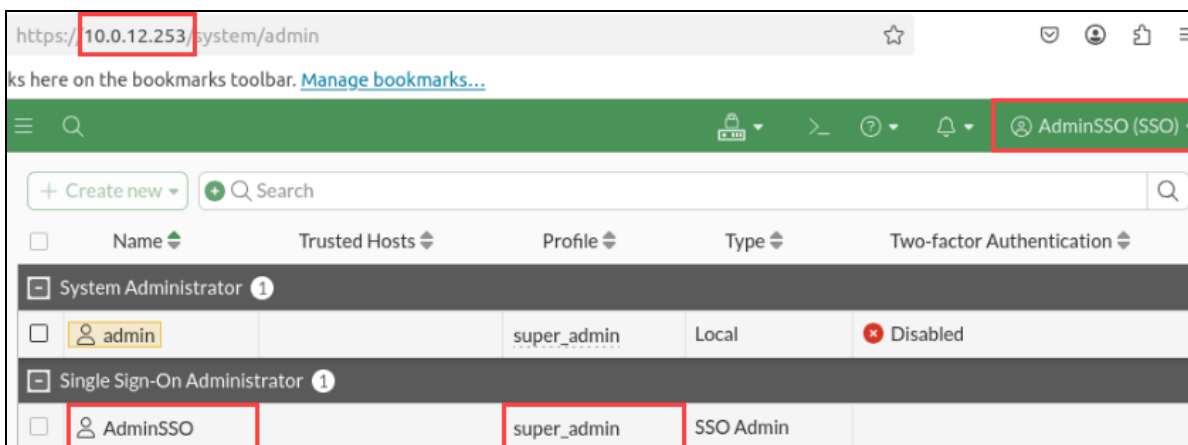


4. Log in with the following credentials:
  - Username: AdminSSO
  - Password: password
5. Click **Login**, and then click **Continue**.



6. Click **Login Read-Only**.
7. Click **Later**, and then click **OK**.
8. Click **System > Administrators**.

Your configuration should be similar to the following image:



You are logged in to HQ-DCFV at the IP address 10.0.12.253 with the newly created SSO administrator, **AdminSSO**.

### Stop and think!

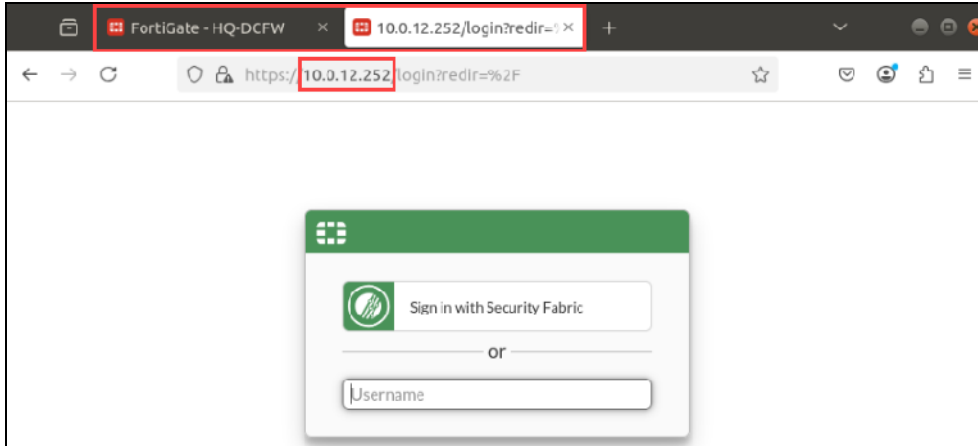
On the root FortiGate, HQ-NGFW-1, you configured the **AdminSSO** administrator with the **super\_admin\_readonly** profile. Why is HQ-DCFV showing **AdminSSO** with the **super\_admin** profile?

When you configured the Security Fabric on HQ-DCFV, you selected **super\_admin** in the **Default admin profile** field. This configuration on the downstream FortiGate takes precedence over the configuration from the root FortiGate.

### To access HQ-ISFW using SAML SSO

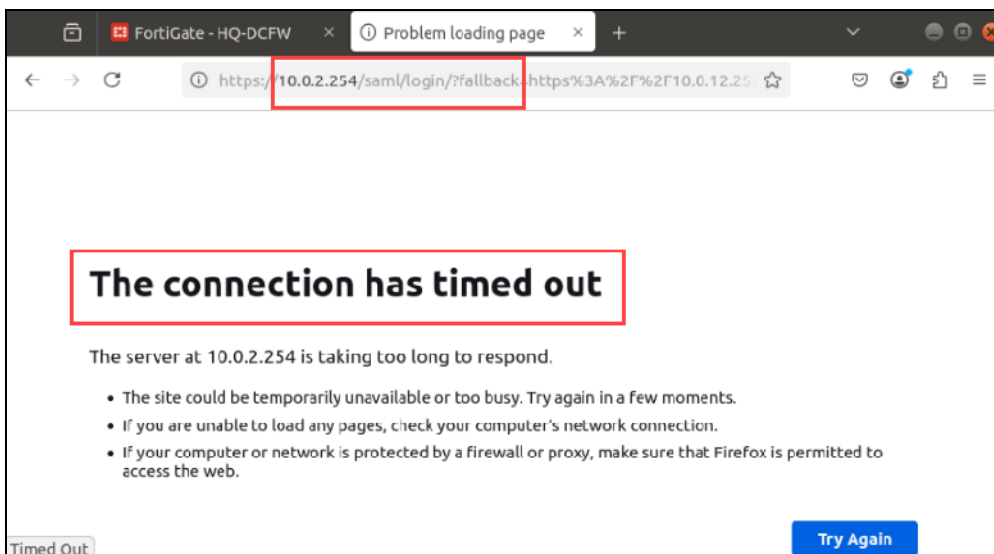
1. On the HQ-PC-2 VM, open a new browser tab, and then enter `https://10.0.12.252`.





2. Click **Sign in with Security Fabric**.

After less than a minute, the result should be similar to the following image:



**Stop and think!**

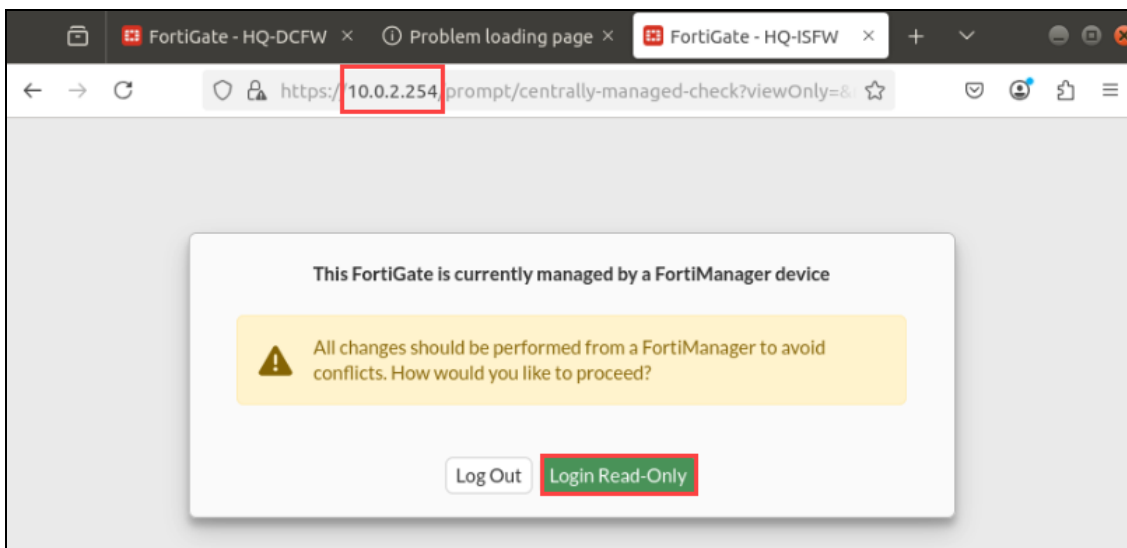
Why is HQ-ISFW not reachable?

In the Security Fabric configuration on HQ-ISFW, you specified the management IP address as 10.0.2.254. After Security Fabric SAML SSO, this IP address must be reachable, along with the corresponding administrative access.

3. Log in to the HQ-ISFW GUI with the following credentials:
  - Username: admin
  - Password: Fortinet1!
4. Click **Login**.
5. Click **Login Read-Write**, and then click **Yes**.
6. Click **Network > Interfaces**, and then click **+** to expand **port4**.

| Name               | Type               | Members | IP/Netmask                | Administrative access                                      |
|--------------------|--------------------|---------|---------------------------|------------------------------------------------------------|
| fortilink          | 802.3ad Aggregate  |         | Dedicated to FortiSwitch  | PING<br>Security Fabric Connection                         |
| Physical Interface |                    |         |                           |                                                            |
| port1              | Physical Interface |         | 192.168.1.101/255.255.0.0 | PING<br>HTTPS<br>SSH<br>HTTP                               |
| port2              | Physical Interface |         | 10.0.11.253/255.255.255.0 | PING<br>HTTPS<br>SSH<br>HTTP<br>Security Fabric Connection |
| port3              | Physical Interface |         | 10.0.12.252/255.255.255.0 |                                                            |
| port4              | Physical Interface |         | 0.0.0.0/0.0.0.0           |                                                            |
| VLAN101            | VLAN               |         | 10.0.2.254/255.255.255.0  | PING                                                       |

7. Select **VLAN101**, and then click **Edit**.
8. In the **Administrative Access** section, enable **HTTPS**.
9. Click **OK**.
10. In the upper-right corner, click **admin**, and then click **Logout**.
11. On the HQ-PC-2 VM, open a new browser tab, and then enter `https://10.0.12.252`.
12. Click **Sign in with Security Fabric**.
13. Click **Continue**.



### Stop and think!

Why is the **Login Read-Write** option not available?

This is because you selected the **super\_admin\_readonly** administration profile in the Security Fabric configuration on HQ-ISFW.

14. On the HQ-PC-2 VM, close the browser tabs.

## Exercise 2: Configuring Automatic Configuration Backup

You will configure a Security Fabric automation on FortiManager to create automatic configuration backups.

### Manage Security Fabric Devices on FortiManager

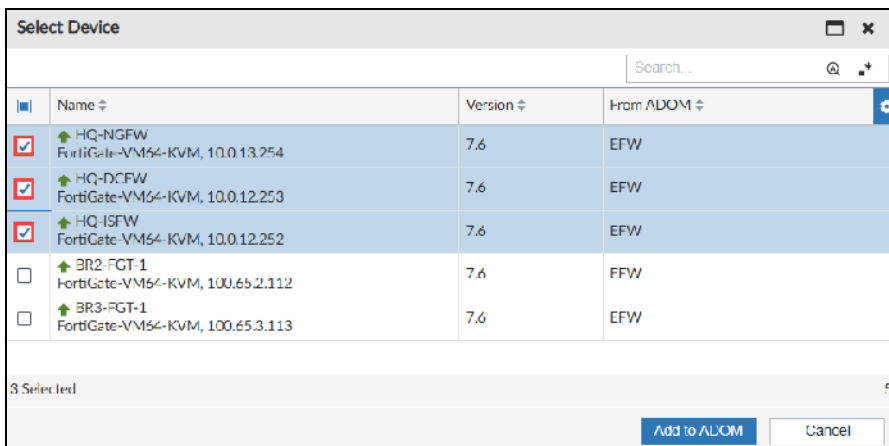
On FortiManager, you will create the Fabric ADOM and move the Security Fabric devices into it.

#### To create the Fabric ADOM

1. Log in to the FortiManager GUI with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Click **Create New ADOM**.
3. In the **Name** field, type `Fabric_ADOM`.
4. In the **Type** field, select **Fabric**.
5. In the **Devices** field, click **Select Device**.



6. Select the checkboxes for **HQ-NGFW**, **HQ-ISFW**, and **HQ-DCFW**, and then click **Add to ADOM**.  
The result should be similar to the following image:

The screenshot shows the 'Create ADOM' window in FortiManager. The 'Name' field contains 'Fabric\_ADOM' and the 'Type' is set to 'Fabric'. The 'Description' field is empty. The 'Devices' section shows a list of three devices: HQ-NCFW, HQ-DCFWD, and HQ-ISFW, all of which are selected. The 'Mode' is set to 'Normal' and the 'Backup' checkbox is checked. The 'OK' button is highlighted in red.

| Name     | IP Address  | Platform           |
|----------|-------------|--------------------|
| HQ-NCFW  | 10.0.13.254 | FortiGate-VM64-KV1 |
| HQ-DCFWD | 10.0.12.253 | FortiGate-VM64-KV1 |
| HQ-ISFW  | 10.0.12.252 | FortiGate-VM64-KV1 |

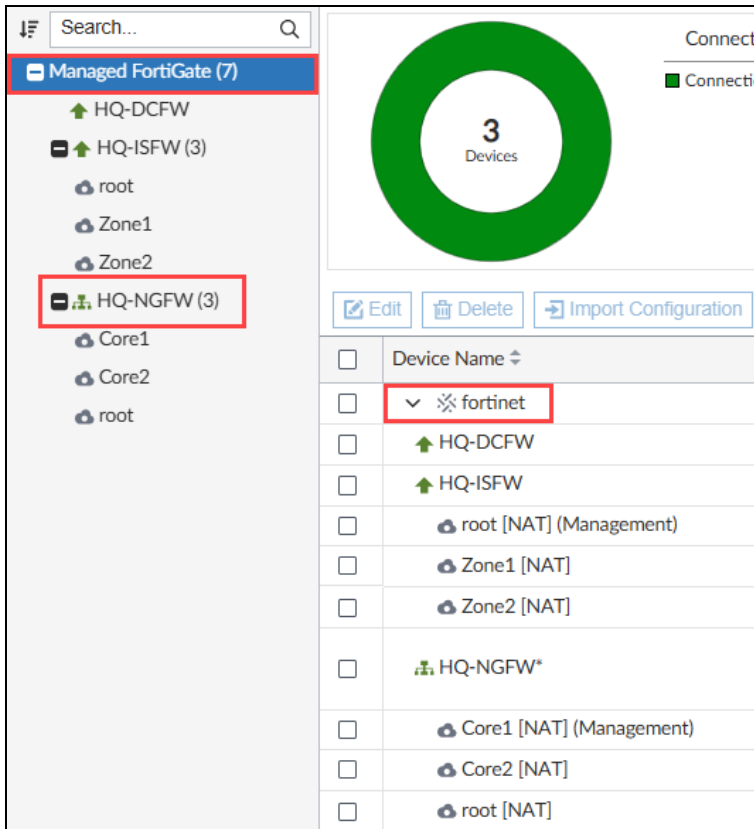
7. Click **OK**.

## Configure an Automation on FortiManager

On FortiManager, you will configure the trigger and action to create an automatic configuration backup stitch.

### To configure the trigger

1. Continuing on the FortiManager GUI, click **Fabric\_ADOM**.
2. Click **Device Manager > Device & Groups > Managed FortiGate**.



You may notice that **fortinet** is listed in the **Device Name** column. This confirms that all the devices nested under **fortinet** belong to the same Security Fabric environment.

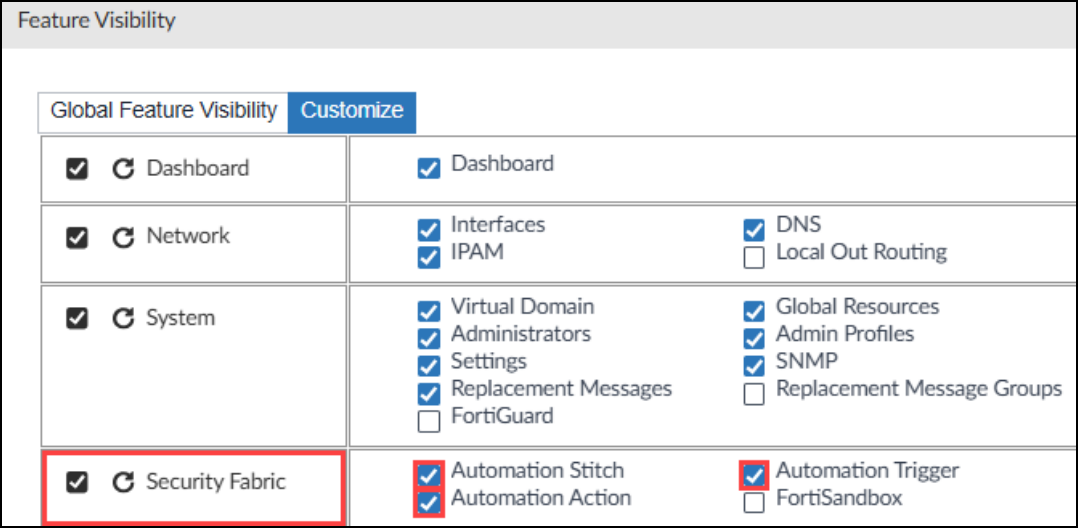
If **fortinet** is not listed, you must retrieve the HQ-NGFW configuration to update FortiManager with the Security Fabric environment.

3. Click **Managed FortiGate > HQ-NGFW > Feature Visibility**.

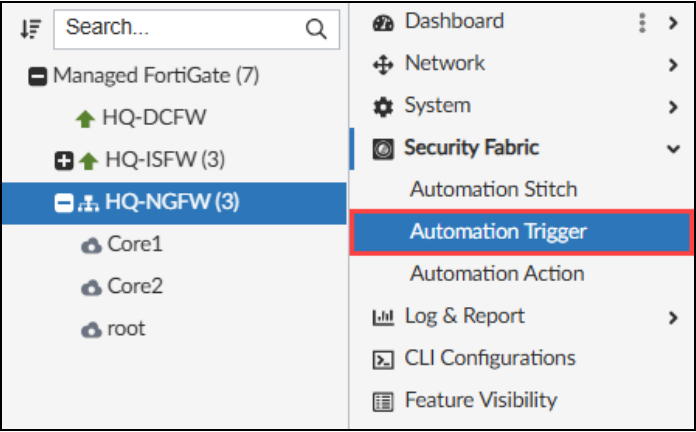


By default, the Security Fabric options are not available. You must enable them in the **Feature Visibility** section.

4. On the **Customize** tab, in the **Security Fabric** section, select the checkboxes for **Automation Stitch**, **Automation Trigger**, and **Automation Action**.



- Click **OK**.
- Click **Managed FortiGate > HQ-NGFW > Security Fabric**, and then select **Automation Trigger**.



- Click **Create New**, and then configure the following settings:

| Field     | Value                                                                                                                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name      | Frequency                                                                                                                                                                                                       |
| Type      | Schedule                                                                                                                                                                                                        |
| Frequency | Hourly                                                                                                                                                                                                          |
| Minute    | Type the minutes of your current time plus 10 minutes.<br><br>For example, if the current time is 12:35 PM, you would add 10 minutes, and then type 45 in the <b>Minute</b> field (12:35 + 10 minutes = 12:45). |

Your configuration should look similar to the following image (with a value in the **Minute** field that is based on your current time):

Create New Automation Trigger

Name

Frequency

Description

Type

Schedule

Schedule

Frequency

Hourly

Minute

45

Advanced Options >



When you set the **Frequency** field to **Hourly**, the stitch is triggered every hour when the minutes match the number that is configured in the **Minute** field. You then have 10 minutes before the next stitch occurrence for the purposes of this lab.

In a production environment, you would configure the frequency according to the frequency of configuration changes.

8. Click **OK**.

To configure the action

- 1. Click **Security Fabric**, and then select **Automation Action**.
- 2. Click **Create New**, and then configure the following settings:

| Field  | Value                |
|--------|----------------------|
| Name   | Automatic_Backup     |
| Type   | System Action        |
| Action | Backup Configuration |

Your configuration should look like the following image:

Create New Automation Action

|             |                      |
|-------------|----------------------|
| Name        | Automatic_Backup     |
| Description |                      |
| Type        | System Action        |
| Action      | Backup Configuration |

3. Click **OK**.

To configure the stitch

- 1. Click **Security Fabric**, and then select **Automation Stitch**.
- 2. Click **Create New**, and then configure the following settings:

| Field         | Value            |
|---------------|------------------|
| Name          | Backup_Stitch    |
| Trigger       | Frequency        |
| Action Object | Automatic_Backup |

Your configuration should look like the following image:

Create New Automation Stitch

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| Name             | Backup_Stitch                                                                    |
| Status           | <input checked="" type="checkbox"/> Enabled <input type="checkbox"/> Disabled    |
| FortiGate        | <div><div>Q</div><div>All FortiGates</div><div>1 entry selected</div></div>      |
| Action Execution | <input checked="" type="checkbox"/> Sequential <input type="checkbox"/> Parallel |
| Description      |                                                                                  |

Stitch

|               |                                                                                                                                           |               |                  |       |   |        |  |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------|-------|---|--------|--|
| Trigger       | Frequency                                                                                                                                 |               |                  |       |   |        |  |
| Action        | <table><tr><td>Action Object</td><td>Automatic_Backup</td></tr><tr><td>Delay</td><td>0</td></tr><tr><td>Action</td><td></td></tr></table> | Action Object | Automatic_Backup | Delay | 0 | Action |  |
| Action Object | Automatic_Backup                                                                                                                          |               |                  |       |   |        |  |
| Delay         | 0                                                                                                                                         |               |                  |       |   |        |  |
| Action        |                                                                                                                                           |               |                  |       |   |        |  |

3. Click **OK**.



## Verify the Stitch on the Security Fabric Devices

You will install the stitch on the root FortiGate and verify the automatic configuration backup.

### To install the stitch

1. In the menu at the top, click **Install Wizard**, and then select **Install Device Settings (only)**.
2. Click **Next**.
3. Confirm that **HQ-NGFW** is selected, and then click **Next**.
4. Click **Install**.
5. Wait until the installation finishes, and then click **Finish**.

### To verify the stitch installation on the root FortiGate

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **Login Read-Write**, and then click **Yes**.
3. Click **Security Fabric > Automation > Stitch**.

Your configuration should look like the following image:

Stitch

Trigger

Action

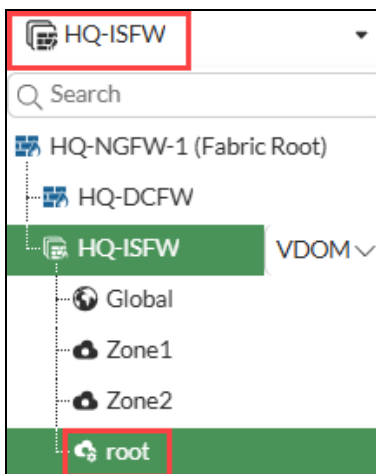
+ Create new

+ Q Search

| <input type="checkbox"/>            | Name                | Status  | Trigger   | Actions          | FortiGate(s)   |
|-------------------------------------|---------------------|---------|-----------|------------------|----------------|
| <input checked="" type="checkbox"/> | Compromised Host 1  |         |           |                  |                |
| <input checked="" type="checkbox"/> | FortiOS Event Log 3 |         |           |                  |                |
| <input checked="" type="checkbox"/> | HA Failover 1       |         |           |                  |                |
| <input checked="" type="checkbox"/> | Incoming Webhook 1  |         |           |                  |                |
| <input checked="" type="checkbox"/> | License Expiry 1    |         |           |                  |                |
| <input checked="" type="checkbox"/> | Reboot 1            |         |           |                  |                |
| <input checked="" type="checkbox"/> | Schedule 1          |         |           |                  |                |
| <input type="checkbox"/>            | Backup_Stitch       | Enabled | Frequency | Automatic Backup | All FortiGates |

### To verify the stitch occurrence

1. Connect to the HQ-ISFW GUI, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
2. Click **Login Read-Write**, and then click **Yes**.
3. In the upper-left corner, click **HQ-ISFW**, and then click **root**.



4. Click **Log & Report > System Events**.
5. Click the **Logs** tab.

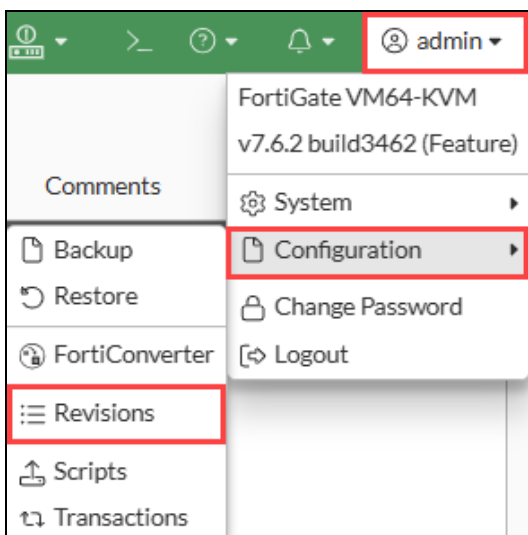
The result should be similar to the following image:

| Summary             |        |      |                                    |                             |
|---------------------|--------|------|------------------------------------|-----------------------------|
| Logs                |        |      |                                    |                             |
| Search              |        |      |                                    |                             |
| Date/Time           | Level  | User | Message                            | Log Description             |
| 2025/03/10 00:45:28 | Notice |      | stitch:Backup_Stitch is triggered. | Automation stitch triggered |



In the **Date/Time** column, you can see that the stitch is triggered a few seconds after the minutes configured in the automation trigger.

6. In the upper-right corner, click **admin > Configuration > Revisions**.



7. In the **7.6.2 build 3462** row, click **+** to expand the list.

| Config ID            | Username     | Date                | Comments                                     |
|----------------------|--------------|---------------------|----------------------------------------------|
| + 7.6.0 build 3401 2 |              |                     |                                              |
| - 7.6.2 build 3462 2 |              |                     |                                              |
| 4                    | daemon_admin | 2025/03/10 00:45:29 | Autod backup config by stitch: Backup_Stitch |



The automatic configuration backup is available as a revision.

The same automatic configuration backups are available on the other downstream FortiGate devices because you configured the stitch to **all FortiGates**.

## Exercise 3: Configuring the Automation With a Script

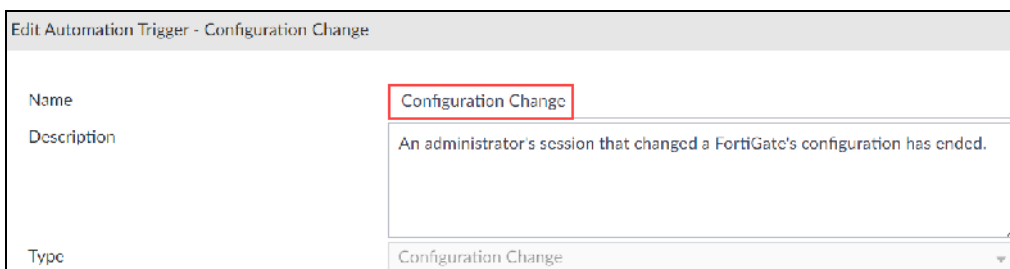
In this exercise, you will configure the automation stitch on FortiManager to take CLI commands from a script and email the results.

### Configure the Automation

On FortiManager, you will verify the preconfigured trigger based on configuration changes, create an action to take CLI commands with a script, create an action to email the script results, and then add them to a stitch.

#### To verify the preconfigured automation trigger on FortiManager

1. Log in to the FortiManager GUI with the following credentials:
    - Username: admin
    - Password: Fortinet1!
  2. Click **Fabric\_ADOM**.
  3. Click **Device Manager > Device & Groups**.
  4. Click **Managed FortiGate > HQ-NGFW**.
  5. Click **Security Fabric**, and then select **Automation Trigger**.
  6. Select **Configuration Change**, and then click **Edit**.
- The configuration should look like the following image:



The screenshot shows the 'Edit Automation Trigger - Configuration Change' window. It has three main fields: 'Name' with the value 'Configuration Change', 'Description' with the text 'An administrator's session that changed a FortiGate's configuration has ended.', and 'Type' with the value 'Configuration Change'.

7. Click **Cancel**.

#### To configure a CLI script automation action on FortiManager

1. Click **Security Fabric**, and then select **Automation Action**.
2. Click **Create New**, and then configure the following settings:

| Field | Value        |
|-------|--------------|
| Name  | Alert_Script |
| Type  | CLI Script   |

| Field                      | Value                                                                                |
|----------------------------|--------------------------------------------------------------------------------------|
| Script                     | config global<br>exec time<br>get system admin list<br>get system performance status |
| Administrator Profile      | super_admin                                                                          |
| Execute on Security Fabric | Enable                                                                               |

Your configuration should look similar to the following image:

Edit Automation Action - Alert\_Script

Name

Alert\_Script

Description

Type

CLI Script

Minimum Interval ⓘ

0

second

CLI Script

Script

config global  
exec time  
get system admin list  
get system performance status

Upload

Administrator Profile ⓘ

super\_admin

Execute on Security Fabric ⓘ

☒

3. Click **OK**.

To configure an email notification automation action on FortiManager

- 1. Click **Security Fabric > Automation Action**.
- 2. Click **Create New**.
- 3. Configure the automation action using the following settings:

| Field | Value                      |
|-------|----------------------------|
| Name  | Configuration_Change_Alert |
| Type  | Email                      |
| From  | fortinet@training.lab      |

| Field                   | Value                      |
|-------------------------|----------------------------|
| Send to FortiCare Email | Disabled                   |
| To                      | admin@training.lab         |
| Subject                 | Configuration Change Alert |
| Body                    | %%results%%                |

Your configuration should look like the following image:

4. Click **OK**.

### To configure the automation stitch on FortiManager

1. Continuing on the FortiManager GUI, click **Security Fabric > Automation Stitch**.
2. Click **Create New**, and then configure the following settings:

| Field         | Value                      |
|---------------|----------------------------|
| Name          | Config_Change_Alert        |
| Trigger       | Configuration Change       |
| Action Object | Alert_Script               |
| Delay         | 10                         |
| Action Object | Configuration_Change_Alert |



You must click **+** in the **Action** column to create a second action object.

Your configuration should look like the following image:

Create New Automation Stitch

Name: Config\_Change\_Alert

Status: ☒ Enabled ☐ Disabled

FortiGate: All FortiGates (1 entry selected)

Action Execution: Sequential Parallel

Description:

Stitch

Trigger: Configuration Change

| Action                     | Action Object | Delay | Action                                                            |
|----------------------------|---------------|-------|-------------------------------------------------------------------|
| Alert_Script               |               | 10    | <input type="button" value="x"/> <input type="button" value="+"/> |
| Configuration_Change_Alert |               | 0     | <input type="button" value="x"/> <input type="button" value="+"/> |

3. Click **OK**.

### To install the automation stitch on the Security Fabric devices

1. In the menu at the top, click **Install Wizard**, and then select **Install Device Settings (only)**.
2. Click **Next**.
3. Confirm that **HQ-NGFW** is selected, and then click **Next**.
4. Click **Install**.
5. Wait until the installation finishes, and then click **Finish**.

## Test the Automation Stitch and View the Email Alert

You will test the automation stitch by forcing a configuration change on HQ-NGFW-1, and then view the email alert.

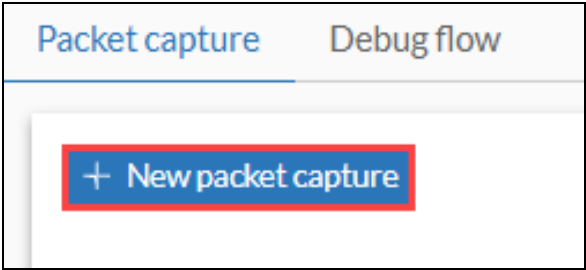


An SMTP mail server is required for email alerts to operate. Because configuring a mail server is out of scope for this lab, one was configured for you. You can view the email service configuration on the HQ-NGFW-1 GUI by clicking **System > Settings**, and then scrolling down to the **Email Service** configuration.

### To force a configuration change and view the email alert

1. On the HQ-PC-1 VM, open a browser.
2. Connect to the HQ-NGFW-1 GUI.
3. Log in with the following credentials:

- Username: admin
  - Password: Fortinet1!
4. Click **Login Read-Write**, and then click **Yes**.
  5. Click **VDOM: Global**, and then click **Core1**.
  6. Click **Network > Diagnostics**.
  7. In the **Packet capture** window, click **New packet capture**.



8. Configure the new packet capture using the following settings:

| Field                    | Value        |
|--------------------------|--------------|
| Interface                | port2        |
| Name                     | Email        |
| Maximum captured packets | 50           |
| Filters                  | Enabled      |
| Host                     | 100.65.0.254 |
| Port                     | 25           |

Your configuration should look like the following image:



Core1: New Capture

NPU hardware acceleration must be disabled on the respective firewall policy to see all packets. To do so, set "auto-asic-offload" to "disable" in the CLI.

Interface: port2

Name: Email

Maximum captured packets: 50

Filters

Filtering syntax: Basic Advanced

Include non-IP packets: ☐

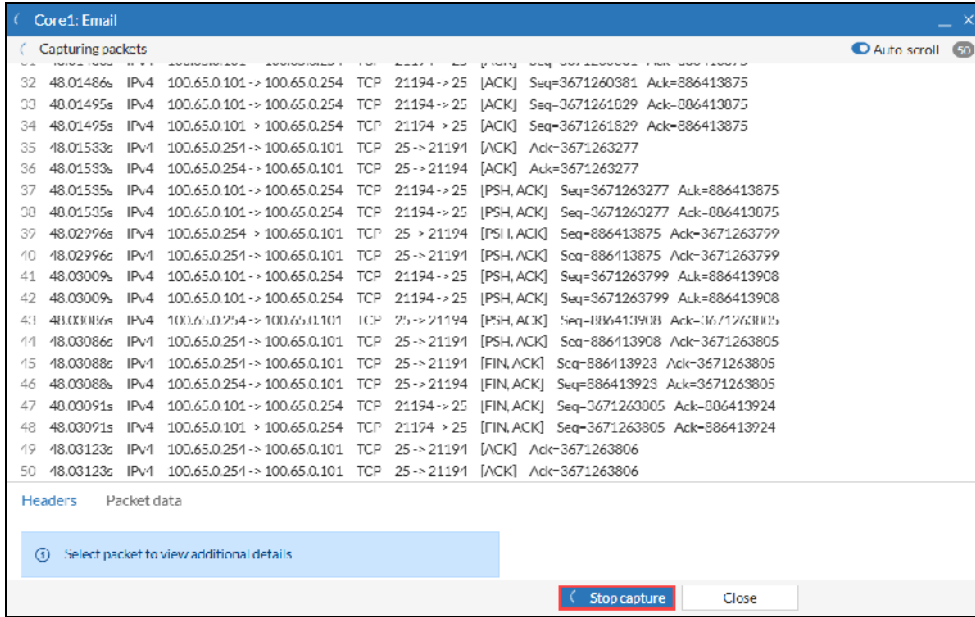
Host: 100.65.0.254

Port: 25

Protocol number:

9. Click **Start capture**.
10. Connect over SSH to HQ-NGFW-1, and then log in with the following credentials:
  - Username: admin
  - Password: Fortinet1!
11. Enter the following commands:

```
config global
  config system interface
    edit port4
      unset allowaccess
    end
  end
end
exit
```
12. On the HQ-PC-1 VM, the output should look like the following image:

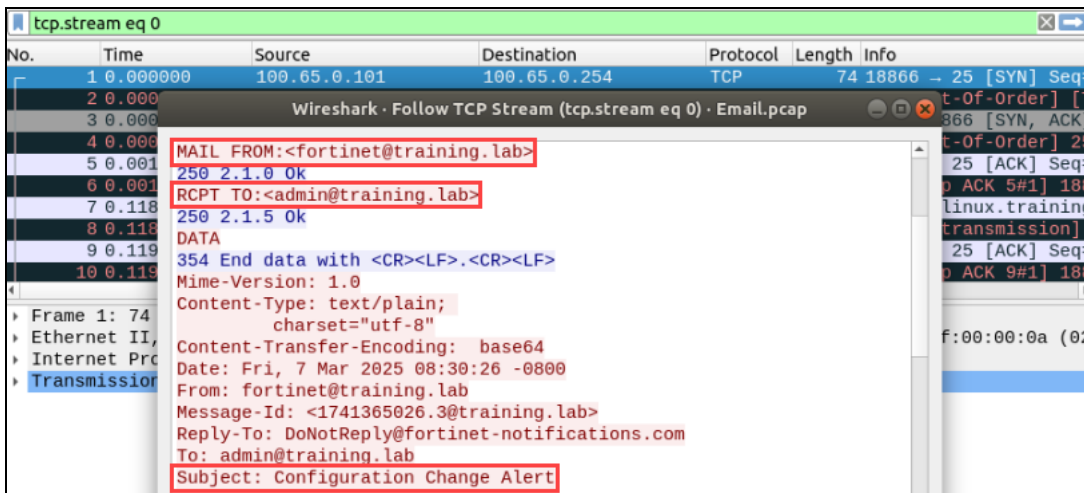


13. Click **Stop capture**, and then click **Save as .pcap**.
14. On the HQ-PC-1 VM, in the **Downloads** folder, open the `Email.pcap` file.



On the HQ-PC-1 VM, Wireshark allows you to open the `Email.pcap` file.

15. Right-click the first SYN packet, and then click **Follow > TCP Stream**.  
The email header should look similar to the following example:



This lab is one example. To get a more comprehensive understanding of the management of all the devices integrated with the Security Fabric environment, you can investigate the automation options further.

## Lab 10: Use Cases

In this lab, you will configure Fortinet devices based on requirements that a customer has provided. The lab is preconfigured with IP addresses.

### Objectives

- Complete all tasks to configure the network based on the customer requirements

### Time to Complete

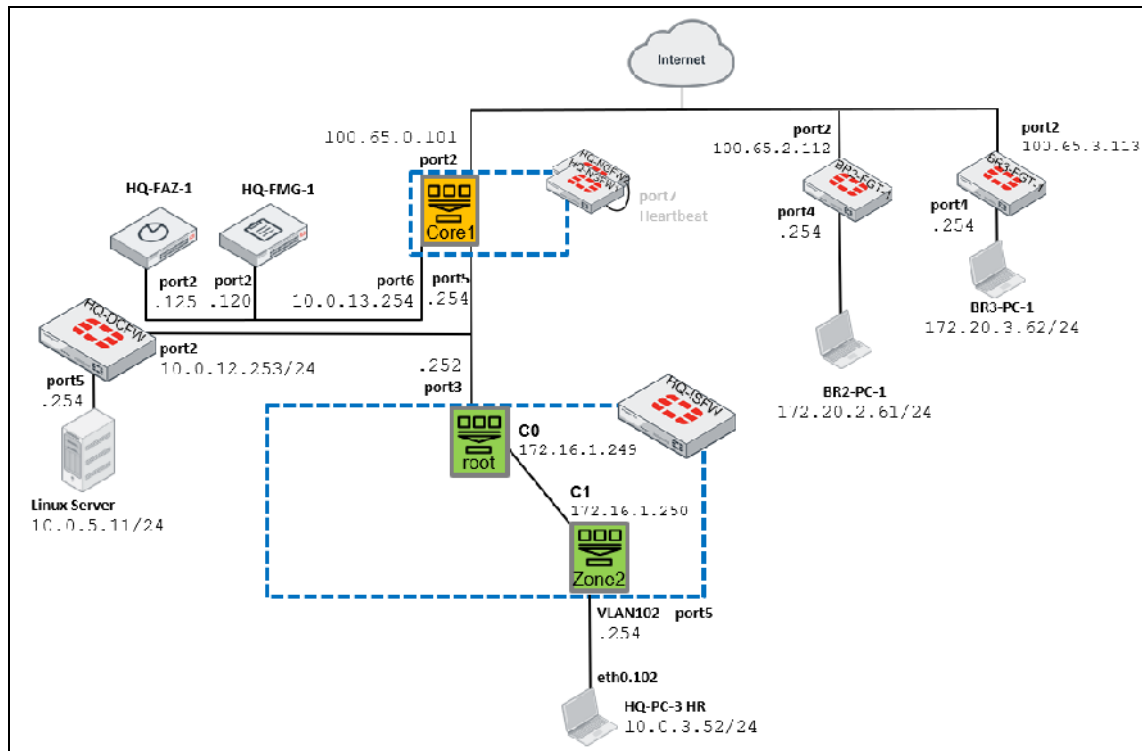
Estimated: 150 minutes

### Which Network Segment Will You Work On?

In the first exercise, you will segment the HR network and connect its OSPF area to the backbone.

In the second exercise, you will implement ADVPN on HQ-NGFW with BR2-FGT-1 and BR3-FGT-1 as spokes.

In the third exercise, you will implement the Fortinet Security Fabric to create an automatic backup.



## Prerequisites

Before you begin this lab, you must restore the initial configuration files on the Fortinet devices. The configuration files are located on the desktop of HQ-PC-1.

### To restore the HQ-DCFW configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the HQ-DCFW GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
4. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
5. Click **Local PC**, and then click **Upload**.
6. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `HQ-DCFW_Use_Case_initial.conf`, and then click **Select**.
7. Click **OK**.
8. Click **OK** to reboot.

### To restore the HQ-NGFW-1 configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the HQ-NGFW-1 GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
4. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
5. Click **Local PC**, and then click **Upload**.
6. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `HQ-NGFW-1_Use_Case_initial.conf`, and then click **Select**.
7. Click **OK**.
8. Click **OK** to reboot.



After the reboot, you can adjust the override setting and the HA priority on HQ-NGFW-1 and HQ-NGFW-2, so HQ-NGFW-1 remains the primary FortiGate.

### To restore the HQ-ISFW configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the HQ-ISFW GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`

4. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
5. Click **Local PC**, and then click **Upload**.
6. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `HQ-ISFW_Use_Case_initial.conf`, and then click **Select**.
7. Click **OK**.
8. Click **OK** to reboot.

#### To restore the BR2-FGT-1 configuration file

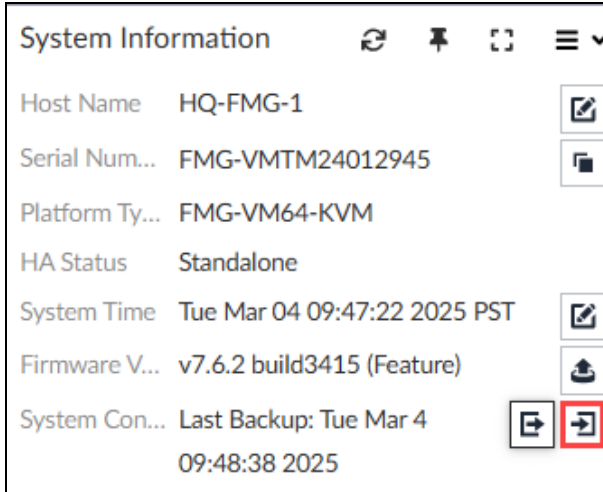
1. On HQ-PC-1, open a browser.
2. Connect to the BR2-FGT-1 GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
4. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
5. Click **Local PC**, and then click **Upload**.
6. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `BR2-FGT-1_Use_Case_initial.conf`, and then click **Select**.
7. Click **OK**.
8. Click **OK** to reboot.

#### To restore the BR3-FGT-1 configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the BR3-FGT-1 GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
4. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.
5. Click **Local PC**, and then click **Upload**.
6. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `BR3-FGT-1_Use_Case_initial.conf`, and then click **Select**.
7. Click **OK**.
8. Click **OK** to reboot.

#### To restore the FortiManager configuration file

1. On HQ-PC-1, open a browser.
2. Connect to the HQ-FMG-1 GUI.
3. Log in with the following credentials:
  - Username: `admin`
  - Password: `Fortinet1!`
4. Click **root**.
5. In the **System Information** widget, in the **System Configuration** field, click the **Restore** icon.



| System Information |                                      |
|--------------------|--------------------------------------|
| Host Name          | HQ-FMG-1                             |
| Serial Num...      | FMG-VMTM24012945                     |
| Platform Ty...     | FMG-VM64-KVM                         |
| HA Status          | Standalone                           |
| System Time        | Tue Mar 04 09:47:22 2025 PST         |
| Firmware V...      | v7.6.2 build3415 (Feature)           |
| System Con...      | Last Backup: Tue Mar 4 09:48:38 2025 |

6. In the **Backup File** field, click **Add Files**.
7. Click **Desktop > Resources > Enterprise-FW > Use\_Case**, select `HQ-FMG-1_Use_Case_initial.dat`, and then click **Select**.
8. In the **Password** field, type `Fortinet1!`.
9. Click **OK**.



After you restore the database on HQ-FMG-1, you must disable offline mode in **System Settings > Advanced > Misc Settings**.

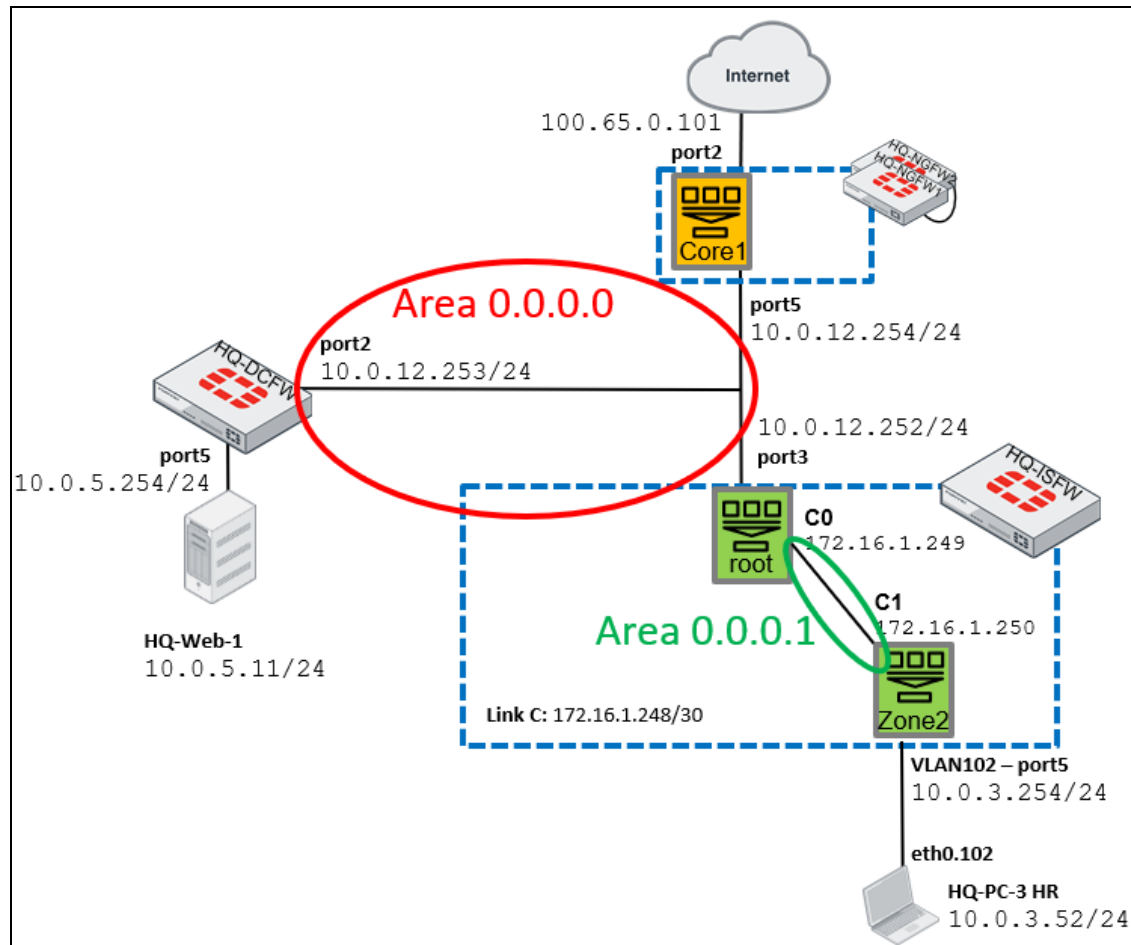
## Exercise 1: Configuring the HR Network

In this exercise, you will configure the enterprise network based on the following basic customer requirements:

- Implement segmentation on the HQ-PC-3 HR network.
- Add dynamic routing.
- Allow the HR segment to access the internet and HQ-Web-1.

### Network Topology

Review the current configuration, as shown in the following image, before you proceed to the next section:



The OSPF area 0.0.0.0 is already configured and includes HQ-DCFW, the Core1 VDOM, and the HQ-ISFW root VDOM.

## Requirements

### To implement segmentation in the HR network

- On HQ-ISFW, create the Zone2 VDOM and VLAN102 interface with the **PING** option.
- Create the inter-VDOM link between the root VDOM and Zone2 VDOM.

### To add dynamic routing

- On HQ-ISFW, configure the 0.0.0.1 OSPF area between the root VDOM and new Zone2 VDOM.
- Allow the OSPF protocol to inject the 10.0.3.0/24 subnet into the backbone area.

### To allow the HR segment to access HQ-Web-1 and the internet

- Allow only ping from the HR segment to HQ-Web-1.
- Allow only ping and web traffic from the HR segment to the internet.



You may use the **HQ-ISFW\_root** and **HQ-ISFW\_Zone2** scripts to configure the corresponding policy packages.

---

## Test the Configuration

Make sure you complete all the configuration steps before you test the configuration.

### To test basic connectivity

You must be able to ping the VLAN102 interface from the HQ-PC-3 HR network.

### To verify dynamic routing

- In the HQ-ISFW root VDOM, you must see the 0.0.0.1 area in the OSPF status with one fully adjacent neighbor.

The output should be similar to the following example:



```
HQ-ISFW (root) # get router info ospf status
Routing Process "ospf 0" with ID 0.0.0.3
Process uptime is 1 hour 12 minutes
Process bound to VRF default
Conforms to RFC2328, and RFC1583Compatibility flag is disabled
Supports only single TOS(TOS0) routes
Supports opaque LSA
Do not support Restarting
This router is an ABR, ABR Type is Standard (RFC2328)
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Refresh timer 10 secs
Number of incoming current DD exchange neighbors 0/5
Number of outgoing current DD exchange neighbors 0/5
Number of external LSA 0. Checksum 0x000000
Number of opaque AS LSA 0. Checksum 0x000000
Number of non-default external LSA 0
External LSA database is unlimited.
Number of LSA originated 7
Number of LSA received 15
Number of areas attached to this router: 2
  Area 0.0.0.0 (BACKBONE)
    Number of interfaces in this area is 1(1)
    Number of fully adjacent neighbors in this area is 2
    Area has no authentication
    SPF algorithm last executed 00:23:32.830 ago
    SPF algorithm executed 3 times
    Number of LSA 6. Checksum 0x01deba
  Area 0.0.0.1 (Stub)
    Number of interfaces in this area is 1(1)
    Number of fully adjacent neighbors in this area is 1
    Number of fully adjacent virtual neighbors through this area is 0
    Area has no authentication
    SPF algorithm last executed 00:22:57.830 ago
    SPF algorithm executed 3 times
    Number of LSA 5. Checksum 0x02fd7
    Shortcutting mode: Disabled, S-bit consensus: ok
```

- On HQ-DCFW, you must have the 10.0.3.0/24 subnet learned with the OSPF protocol in the routing table.

The output should be similar to the following example:

```
HQ-DCFW # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [1/0] via 10.0.12.254, port2, [1/0]
O IA 10.0.3.0/24 [110/102] via 10.0.12.252, port2, 00:33:04, [1/0]
C 10.0.4.0/24 is directly connected, port4
C 10.0.5.0/24 is directly connected, port5
C 10.0.12.0/24 is directly connected, port2
O IA 172.16.1.248/30 [110/101] via 10.0.12.252, port2, 00:33:39, [1/0]
C 192.168.0.0/16 is directly connected, port1
```

### To test the HR segmentation

- You must be able to ping HQ-Web-1 from the HQ-PC-3 HR network.
- You must be able to browse the internet from the HQ-PC-3 HR network.
- You must not be able to ping the HQ-PC-3 HR network from HQ-DCFW.

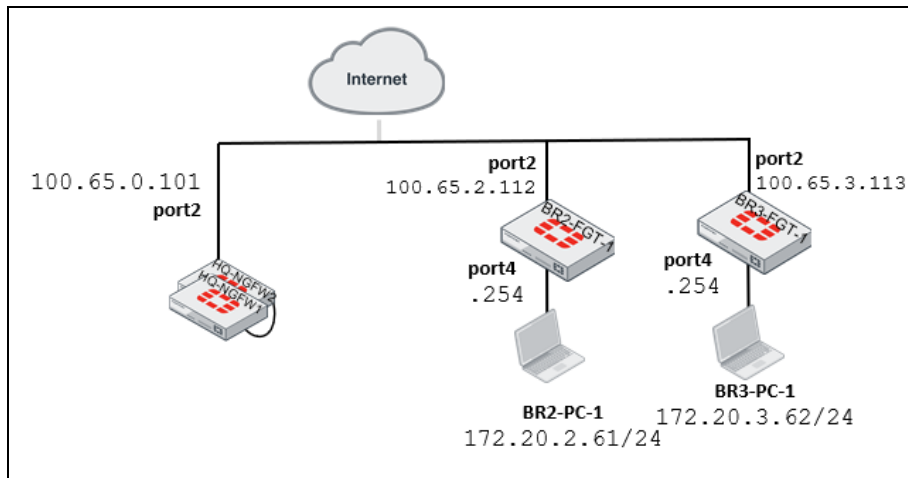
## Exercise 2: Configuring ADVPN

In this exercise, you will configure the enterprise network based on the following basic customer requirements:

- Implement a VPN with HQ-NGFW as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes.
- Implement ADVPN on the hub and spokes.
- Allow traffic between the hub and spokes.
- Implement IBGP on the hub and spokes.

### Network Topology

Review the current configuration, as shown in the following image, before you proceed to the next section:



### Requirements

#### To implement a VPN with HQ-NGFW as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes

- Create a VPN tunnel as a hub on the Core1 VDOM.
- Create the corresponding VPN tunnels as spokes on BR2-FGT-1 and BR3-FGT-1.

#### To implement ADVPN on the hub and spokes

- Configure the tunnel interfaces with the overlay local IP addresses and remote subnet.
- Enable **auto-discovery-sender** on the hub.
- Enable **auto-discovery-receiver** on the spokes.

### To allow traffic on the hub

- Allow traffic from the spokes to the Core1 VDOM.
- Allow traffic from the Core1 VDOM to the spokes.
- Allow traffic between the spokes.

### To allow traffic on the spokes

- Allow traffic from the spokes to the hub.
- Allow traffic from the hub to the spokes.

### To implement IBGP on the hub and spokes

- Use the same BGP autonomous system (AS).
- Configure the hub as a BGP route reflector.
- Configure the hub with neighbor groups and ranges.
- Configure the spokes with their corresponding network prefixes.

## Test the Configuration

Make sure you complete all the configuration steps before you test the configuration.

### To check the status of the VPN tunnels

- Make sure that you bring up the tunnels on the spokes with the following command:  

```
diagnose vpn tunnel up <phase2_tunnel_name>
```
- Connect over SSH to the Core1 VDOM, BR2-FGT-1, and BR3-FGT-1, and get the tunnel list.

The output should look similar to the following example in the hub:

```
HQ-NGFW-1 (Core1) # diagnose vpn tunnel list
list all ipsec tunnel in vd 2
-----
name=VPN1_0 ver=2 serial=5 100.65.0.101:0->100.65.3.113:0 nexthop=100.65.0.254 tun_id=172.16.1.2 tun_id6=
bound_if=4 real_if=4 lgwy=static/1 tun=intf mode=dial_inst/3 encap=none/74408 options[122a8]=npu rgwy-chg

parent=VPN1 index=0
proxyid_num=1 child_num=0 refcnt=5 ilast=27 olast=27 ad=s/1
stat: rxp=19 txp=18 rxb=1243 txb=2772
dpd: mode=on-idle on=1 status=ok idle=60000ms retry=3 count=0 seqno=7
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
proxyid=VPN1 proto=0 sa=1 ref=3 serial=1 ads
src: 0:0.0.0.0-255.255.255.255:0
dst: 0:0.0.0.0-255.255.255.255:0
SA: ref=3 options=20a02 type=00 soft=0 mtu=1438 expire=42446/08 replaywin=2048
seqno=13 esn=0 replaywin_lastseq=00000014 qat=0 rekey=0 hash_search_len=1
life: type=01 bytes=0/0 timeout=43187/43200
dec: spi=a1e5480b esp=aes key=32 5bd0928dfb028355e71929573a566cdaf408f2d5d7c983a2687d0027f42285a5
ah=sha256 key=32 48aa6fab697eaf0f723ce14ee2200db9a2bd1c6d8af4e9483a70da7db0a2ef3c
enc: spi=27691f93 esp=aes key=32 98930232b06051ab82770a2a589e2249439bde21efe159715efe34028bfbf63e
ah=sha256 key=32 766662e6154b3f8cd889a0cef0bf21407468ef61c2daf5c446df7d192d17455b
dec:pkts/bytes=19/1243, enc:pkts/bytes=18/2520
npu_flag=00 npu_rgwy=100.65.3.113 npu_lgwy=100.65.0.101 npu_selid=3 dec_npuid=0 enc_npuid=0
-----
name=VPN1_1 ver=2 serial=4 100.65.0.101:0->100.65.2.112:0 nexthop=100.65.0.254 tun_id=172.16.1.3 tun_id6=
bound_if=4 real_if=4 lgwy=static/1 tun=intf mode=dial_inst/3 encap=none/74408 options[122a8]=npu rgwy-chg

parent=VPN1 index=1
proxyid_num=1 child_num=0 refcnt=5 ilast=41 olast=41 ad=s/1
stat: rxp=19 txp=18 rxb=1243 txb=2772
dpd: mode=on-idle on=1 status=ok idle=60000ms retry=3 count=0 seqno=6
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
proxyid=VPN1 proto=0 sa=1 ref=3 serial=1 ads
src: 0:0.0.0.0-255.255.255.255:0
dst: 0:0.0.0.0-255.255.255.255:0
SA: ref=3 options=20a02 type=00 soft=0 mtu=1438 expire=42436/08 replaywin=2048
seqno=13 esn=0 replaywin_lastseq=00000014 qat=0 rekey=0 hash_search_len=1
life: type=01 bytes=0/0 timeout=43190/43200
dec: spi=a1e5480a esp=aes key=32 b5e0462bbd2a766aa802c5f8dfb30d6c55bfcc9c0e7ff3b8f3b85e4bc96a05f
ah=sha256 key=32 0e17867178b27c94b8ba96139a65cde41e026a37b58cd70c5020669a41b25409
enc: spi=2b097774 esp=aes key=32 d7baac080f775cc4b44f1490852bcfd99db500b1f308a48ecfd86cb1df1fde8
ah=sha256 key=32 f652c53f72cd36ba0ea095804c41ab76f30714c3493da3125ade9ff0bbcc577b
dec:pkts/bytes=19/1243, enc:pkts/bytes=18/2520
npu_flag=00 npu_rgwy=100.65.2.112 npu_lgwy=100.65.0.101 npu_selid=2 dec_npuid=0 enc_npuid=0
```

## To verify the IBGP implementation

Log in to the HQ-NGFW-1 GUI and get the routing table in the Core1 VDOM.

The output should look similar to the following example:

| Network       | Gateway IP | Interfaces | Distance | Type |
|---------------|------------|------------|----------|------|
| 172.20.2.0/24 |            | VPN1       | 200      | BGP  |
| 172.20.3.0/24 |            | VPN1       | 200      | BGP  |

## To validate the ADVPN implementation

- Enter the following commands on BR2-FGT-1:

```
execute ping-options source 172.20.2.254
execute ping 172.20.3.254
```



With the ping option, you generate a ping from the BR2-FGT-1 internal interface to the BR3-FGT-1 internal interface to bring up the on-demand tunnel.

- On BR3-FGT-1, in the routing table, you must see the BR2-FGT-1 internal subnet `172.20.2.0/24` directly connected with the on-demand tunnel.

The output should look similar to the following example:

```
BR3-FGT-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*    0.0.0.0/0 [10/0] via 100.65.3.254, port2, [1/0]
B     10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:03:25, [1/0]
C     100.65.3.0/24 is directly connected, port2
S     172.16.1.0/24 [5/0] via HUB1-VPN1 tunnel 100.65.0.101, [1/0]
S     172.16.1.1/32 [15/0] via HUB1-VPN1 tunnel 100.65.0.101, [1/0]
C     172.16.1.2/32 is directly connected, HUB1-VPN1_0
C     172.16.1.3/32 is directly connected, HUB1-VPN1
                        is directly connected, HUB1-VPN1_0
B     172.20.2.0/24 [200/0] via 172.16.1.2 (recursive is directly connected, HUB1-VPN1_0), 00:02:56, [1/0]
C     172.20.3.0/24 is directly connected, port4
C     192.168.0.0/16 is directly connected, port1
```

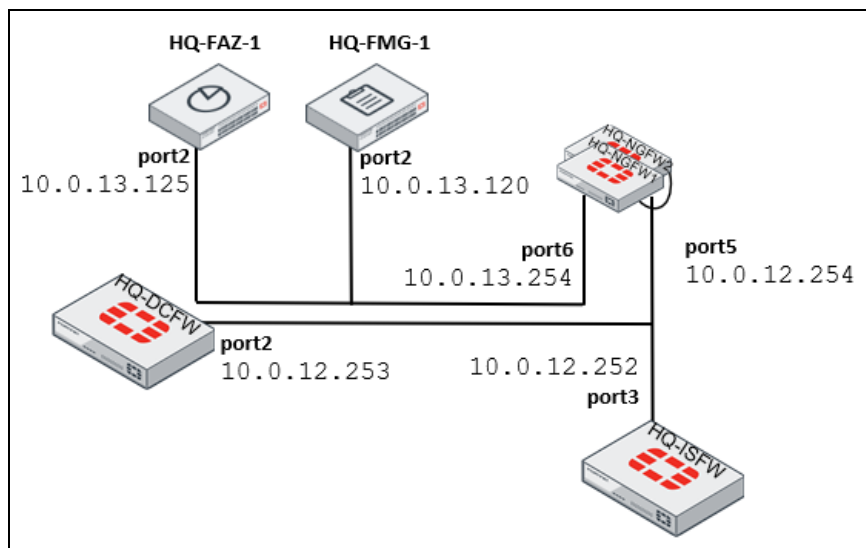
## Exercise 3: Configuring Automatic Backups

In this exercise, you will configure the enterprise network based on the following basic customer requirements:

- Implement the Security Fabric within the LAN FortiGate devices.
- Implement the automatic configuration backup of the devices in the Security Fabric environment.

### Network Topology

Review the current configuration, as shown in the following image, before you proceed to the next section:



### Requirements

#### To configure the Security Fabric

- Configure HQ-NGFW to be the root FortiGate.
- Configure HQ-DCFW and HQ-ISFW to join the Security Fabric environment.



Log in to the FortiManager GUI and ensure that HQ-NGFW, HQ-DCFW, and HQ-ISFW are in the **EFW\_Use\_Case** ADOM.

### To achieve automatic configuration backup

- Select the automation trigger based on configuration changes.
- Configure the automation action configuration backup.
- Create the stitch based on the automation trigger and action that you configured.

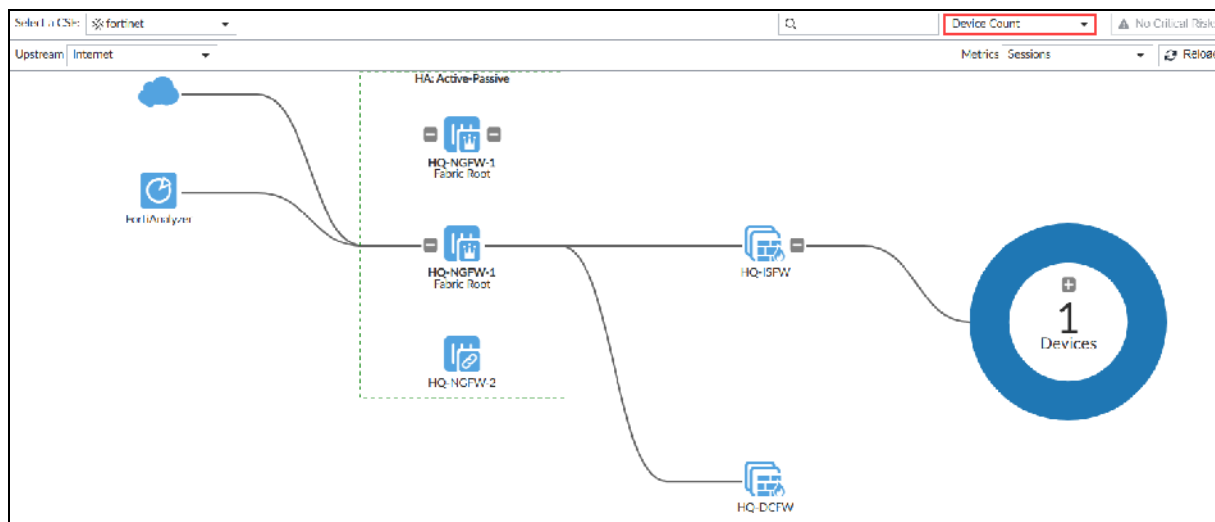
## Test the Configuration

Make sure you complete all the configuration steps before you test the configuration.

### To verify the Security Fabric configuration

In the FortiManager **EFW\_Use\_Case** ADOM, click **Fabric View > Physical Topology**.

The output should be similar to the following image:



### To test the automatic configuration backup

1. Connect over SSH to HQ-DCFW, and then log in with the following credentials:
  - Username: admin
  - Password: password
2. Enter the following commands:

```
config system interface
edit port2
set description modification
next
end
exit
```
3. On HQ-DCFW, in the upper-right corner, click **admin > Configuration > Revisions**.
4. In the **7.6.2 build 3462** row, click **+** to expand the list.

The output should be similar to the following image:

| Config ID          | Username     | Date                | Comments                                     |
|--------------------|--------------|---------------------|----------------------------------------------|
| 7.6.2 build 3462 2 |              |                     |                                              |
| 2                  | daemon_admin | 2025/03/12 00:44:16 | Autod backup config by stitch: Config_Change |

5. On HQ-DCFW, click **Log & Report > System Events > Logs**.
6. Confirm that you see the **Configuration changed** stitch and **Automation stitch triggered** event logs.  
The output should be similar to the following image:

|                     |                    |       |                                                         |                             |
|---------------------|--------------------|-------|---------------------------------------------------------|-----------------------------|
| 2025/03/12 00:44:16 | Notice             |       | stitch:Config_Change Is triggered.                      | Automation stitch triggered |
| 2025/03/12 00:44:15 | Alert Notification | admin | Configuration is changed in the admin session           | Configuration changed       |
| 2025/03/12 00:44:15 | Information        | admin | Administrator admin logged out from console             | Admin logout successful     |
| 2025/03/12 00:44:15 | Information        | admin | Edit system:interface port2                             | Object attribute configured |
| 2025/03/12 00:44:06 | Information        | admin | Administrator admin logged in successfully from console | Admin login successful      |



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